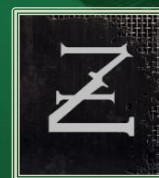
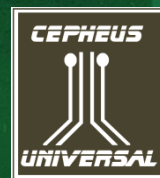


# PLAYER'S BOOK



CEPHEUS UNIVERSAL





**Player's Book** is a supplement for the Cepheus Universal sci-fi roleplaying game

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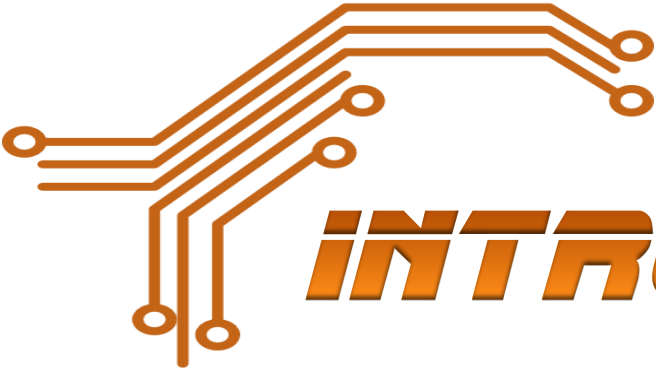


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# INTRODUCTION

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This **Player's Book** is a supplement for **CEPHEUS UNIVERSAL**, which is a science fiction roleplaying game for adventuring in many different kinds of science fiction settings! Everything you need as a player to create your character is contained within this book. It takes all of the key, player-focussed rules from that game and packages them into a single, useable volume. The **Player's Book** also adds NEW features to the character creation process, including a term-by-term random method, familiar to players of most other Cepheus rules-sets; there is a group-creation system, quick-play gear loadouts for each career type, as well as random character name lists. Using the resources within this book, you can create a character ready for play in ten minutes, or go detailed and create a more three-dimensional and believable character in half an hour. Create your character randomly or by design – the choice is yours. Included are **CEPHEUS UNIVERSAL'S** task resolution and combat rules, to aid in player familiarisation.

## WHAT IS ROLEPLAYING?

What is roleplaying? If you already know the answer to this, please skip this section and go straight to the next chapter, entitled Campaign Concepts, where you can see what kinds of game and character possibilities exist. For a new player, this brief discussion will help you to understand the difference between this roleplaying game and a more traditional board game.

Roleplaying is a hobby that has been developing for fifty years, and although initially dominated by the fantasy genre, it now spans almost every genre of adventure fiction that you can imagine, including, of course science fiction. It is probably one of the most wonderful pastimes available, plunging the players into a fictional universe, in which they are able to make decisions, solve crimes, explore new planets, defeat evil monsters, and follow their own audacious plans and enterprises. Each player takes on the role of a character in this setting, much as an actor takes on a role, but there is no acting required. Each player describes the intentions of their character, what they plan to do, and how they plan to do it. A roleplaying game is a game most often played with 2 to 6 players, around a table using notebooks and pencils, as well as dice. A set of rules is also required. The **Player's Book** is just such a set of rules.

## THE PLAYERS

What role do the players have in the game? The players might all be star commandoes on an expedition to kidnap a high-ranking scientist from an outpost on a desert world. The player whose character is skilled in navigation will get the characters across the wilderness safely. One of the commandoes may be a recon expert, and detect concealed bunkers and fortifications near the outpost. The rest of the players will have a spread of skills, so that each has a role to play: the demolitions guy, the heavy weapons guy, the computer expert, and so on. All of their skills will be crucial in penetrating the outpost, defeating the guards and getting the scientist out of there. The players will work together to achieve the game's goal, there is no winner or loser.

This isn't just an exercise in storytelling however, it really is a game, because the outcome is never guaranteed. Firstly, dice are rolled when the outcome of any task is in doubt. Just because a player declares their character is going to knock out the guard with a blow to the head, doesn't mean it will succeed. The player rolls the required dice and considers the skills of their character - do they succeed? The dice will decide! This means plans can fail and the players must sometimes try new strategies or explore other avenues to reach their goal. It is an exciting experience, in which players must use ingenuity, imagination and co-operation to get the job done and achieve the goal.

## THE GAME MASTER

Who sets the goal and creates the situation? Who decides if a task needs a dice roll, or will succeed automatically? That is the Game Master's job. The Game Master is one player who does not have a character, instead they act as the 'play maker' creating goals for the players to achieve as well as a range of obstacles, challenges and upsets. The players get their enjoyment from overcoming the challenges put in front of them, working together and achieving the game's objectives. The Game Master, meanwhile, gets their enjoyment from providing entertaining challenges, creating memorable moments of excitement and tension, and in discovering how the players achieve their characters' goal.

Usually, the players will be expected to have a copy of the rules. If not they should consult with their Game Master to familiarise themselves with the game mechanics, and help in creating their characters. This ensures that everyone at the table has an idea of how to play and is ready to begin. It is similar to the owner of a new board game opening it up, reading the rules, and then explaining those rules briefly to the players so that the game can get under way. The owner probably doesn't explain all of the rules, just the ones to get the players' started, along with the general objective of the board game. None of the other players typically need to read the instructions of the board game; they tend to leave it to the game's owner. In many ways this is similar to the way in which many role-playing games are administered.



## WHAT PLAY IS LIKE

Anyone new to roleplaying may be struggling to understand what this all looks like in play. Picture four players and a Game Master sitting around a table; each player has a sheet of paper with their character's name and skills written on it. The Game Master has the rule book for reference, as well as his or her notes, and these detail the story concept that will form the adventure ... a beginning that leads to a middle ... and various notes about possible outcomes and challenges. The story has no ending, because the players haven't decided as to how it is going to end.

The Game Master begins the game by explaining what the starting situation is, where the players' characters are and what they are doing. The players will explain, one-by-one the kinds of things their characters are doing. As the Game Master adds more detail and introduces encounters and events, the players react appropriately as their characters. The GM has a dust storm written in his notes, and wants to know how the players are going to handle the problem. Dice are rolled, the rules consulted and the characters fail to stay on course, and are suffering damage from the storm. Now the Game Master asks the players what their characters will do now. They discuss their options, and decide their characters will find shelter in some nearby caves. The Game Master has some notes about these caves, and the tunnels that lead away into the ground, as well desert creatures and even a tribe of desert-dwelling aliens living in the area.

Where-ever the players go, the Game Master can drop some of these encounters in, to create a challenge for the characters. Will they head into the tunnels? Will they meet the nomads and barter with them for information? Might they ask to be 'smuggled' into the outpost in disguise? The GM uses his own ingenuity to be flexible, and to respond with appropriate challenges, tasks and mysteries, to whatever the players decide their characters will do. Will they get to the outpost alive? Once there, will they have the strength and resourcefulness to locate and kidnap the scientist? If they fail, then the players have not 'lost', since they have hopefully enjoyed the game's challenges, laughs, tense moments and tragedies. If everyone who is sitting around the table had fun, no matter what the outcome, then everyone around the table 'won'.



# THE SCIENCE FICTION GAME

**CEPHEUS UNIVERSAL** represents the types of settings and activities that take place in the future, rather than a fantasy or historical setting. Cepheus is a family of science fiction roleplaying games that first appeared in 2016 when the talented game designer Jason Kemp published the Cepheus Engine RPG, a 'clone' of the world's most famous science fiction roleplaying game from the 1970s. So popular has been Jason's Cepheus Engine, that other versions have been published since its introduction, including Cepheus Light, Clement Sector, Retro Sci-Fi Rules, Hostile and Cepheus Deluxe. This book is the latest version of Jason's game, and it includes all kinds of rules for many different types of science fiction settings, made famous in novels, TV shows or movies. Science fiction gaming has a couple of major differences from fantasy roleplaying, but these do not make the genre more difficult to play, instead they are simply differences that the Game Master, in particular, should keep in mind.

## TECHNOLOGY

The first is technology - it can get in the way of plot and mystery. Technology can make it harder to isolate a group of characters who are in peril, and isolation can be extremely useful in forcing them to deal with the situation themselves, no matter how dangerous, rather than call for the help of the authorities. Communications in science fiction are typically ubiquitous, as they are today. Automation also has a risk of replacing character action. Robots and sophisticated computers can do many of the critical and sometimes dangerous jobs that are typically carried out by the brave and clever heroes in most science fiction. Commanding a fleet of robots to fix every problem from the comfort of a command chair, does not make for a very interesting game.

In science fiction roleplaying, it is best to see technology as a crutch which humans (and the players' characters) rely on. And then start taking that crutch away. Isolation might be hard to achieve on a planet like Earth, but instead we have the depths of space and remote star systems as our settings, where help can be weeks away (depending on the speed of travel and the existence of interstellar communication). Get into trouble, or meet the wrong people, out here - and you're on your own. And those robots are great when they work, but they can go wrong. What's worse, they themselves can be the problem. Perhaps they are malfunctioning and damaging the ship, or (worse) are under the control of some other entity, and are now attacking the characters. Or the robots work fine, but whatever they do, it's just not quite as good or effective as if a character had carried out the task themselves.

The James Cameron movie *Aliens* (1986) perfectly illustrated this philosophy. The heroes have a heavily armed starship and shuttle, an armed and armoured personnel vehicle, lots of guns, flamethrowers, sentry guns and grenades. They even have an android with them. But they nearly all get wiped out by creatures that are probably as physically dangerous as an adult tiger. Cameron strips their assets away from them. An ambush causes chaos and several soldiers are killed straight away. Then they are cut off from their starship when the shuttle is sabotaged, destroying the APC in the process. Now they have to restrict how much ammo they use up as the aliens use wave tactics to wear the heroes down. A traitor within has them expending even more energy and

ammo. They are trapped and even if they had those over-powered vehicles and ships with them, it wouldn't make a difference. Once the Game Master knows what kind of technology the players' characters will have with them, he can build that into the plot as an asset, have it be part of the problem, or at least, a useless crutch.

## SCIENCE

The second difference is science. This might sound abstract, but science fiction deals, of necessity, with some basic scientific principles. If the characters decide to leave a ship's airlock in their space suits to cross over to a space wreck, there are various scientific principles that come into play that don't factor in any fantasy adventure. The suits have air for six hours, the characters might float away from the ship, the suits might get torn or punctured - which might kill the wearer. Is there any power on the wreck? What about lighting and air?

Do the players and Game Master need to possess a high level of scientific education to get all of the details right in circumstances like this? Certainly not. Very few science fiction roleplaying games (if any) attempt to emulate cold, hard scientific fact. The group is advised to always treat their game as a TV show or a clever and exciting science fiction movie, and NOT a simulation of reality. Use the scientific tropes you are familiar with from those sources and have fun, try to keep it feeling realistic, but don't 'sweat the details'. This is particularly important if one of the players has qualifications in engineering or astrophysics (and the author has played with one of the latter).

The Game Master, for example, might describe a character's spacesuit helmet being smashed, and describe how the head of that character explodes, like it did in the 1981 movie *Outland*. One of the players might take issue with the science behind this effect, but everyone around the table should remember that this is a game, and like a movie, we sometimes throw in some things just for dramatic effect, because they're cool, or because we didn't really know the science behind something. But that's the way it's done and we move on. It's no big deal. It's always best to respond to that player with something like "thanks for letting us know, we'll remember that next time it happens" rather than say "OK, I didn't know that, forget everything that just happened, let's roleplay that bit again". There is an almost unwritten rule in roleplaying that once players or Game Master declare an action out loud, it has happened, and there are rarely any 'do overs'.

Sometimes scientific knowledge might have a crucial bearing on the plot, in which case, a bit of to-and-fro discussion between players or Game Master can be used to make a change or two that keeps the game flowing, but that still accommodates that new piece of knowledge. Players should always keep the tone of these discussions upbeat and friendly, and any discussion should be brief, and not a distraction from the game. Sometimes movies make mistakes, but are extremely cool and entertaining nonetheless! This kind of thing happens very rarely, however, with much of science fiction roleplaying actually taking the form of science fiction *adventure* roleplaying.

## GAME TERMINOLOGY

The **CEPHEUS UNIVERSAL** game uses certain words and abbreviations throughout the rules system. The following serves as a guide:

**2D6:** Two six-sided dice, used to resolve most actions.

**Action:** A character activity.

**Adventure:** A story created by the referee and players, comprised of a series of related scenes or encounters. Also referred to as a 'scenario'.

**Attack roll:** A skill check used to determine whether an attack hits during combat.

**Attack:** Any of numerous actions intended to harm, disable, or neutralize an opponent.

**Bonus:** A positive modifier to a die roll.

**Campaign:** A series of linked adventures or scenarios, often featuring the same set of player characters.

**Character:** A fictional individual in the game. The players control characters, whilst the GM controls non-player characters.

**Characteristic modifier:** These are modifiers to some task rolls that are based on the characteristics of the player character. A high Intelligence characteristic, for example, benefits from a positive characteristic modifier which is applied to an intelligence-based task, as required.

**Characteristic score:** One of the six basic character traits -- Strength (Str), Dexterity (Dex), Endurance (End), Intelligence (Int), Education (Edu) and Social Influence (Soc).

**Credit (Cr):** The primary unit of currency used in the Cepheus Universal rulebook. In your setting, you can easily change the name to some other name, if desired.

**D66:** A special dice roll generated by rolling one die twice, and reading the result as a two-digit number between 11

and 66. Often used when rolling to find an entry on a long list of things.

**Damage:** Harm caused to a character by injury, illness, or some other source. Often noted as a number of six-sided dice.

**Dice modifier (DM):** A modifier applied to a check or roll.

**Die (plural is dice):** A small polyhedron, typically a cube, with each side having a different number on it, ranging from one to the number of sides of the polyhedron, thrown and used in gambling and other games involving chance. This book uses six-sided dice exclusively to create random results during play.

**Difficulty:** A modifier applied to a check that is assigned by the referee, reflecting the relative ease or difficulty of a given action.

**Durability Points:** A measure of how much damage an object or robot can take before being disabled.

**Dying:** Unconscious and near death, a dying character can take no actions.

**Effect:** The difference between a roll result and the target of 8 (i.e. how much higher or lower the result is).

**Encounter:** An unexpected or casual meeting with someone or something. A large part of the Game Master's job is the administration of encounters.

**Exceptional failure:** Any check that fails by 6 or greater (i.e. has an Effect of -6 or worse).

**Exceptional success:** Any check that succeeds by 6 or greater (i.e. has an Effect of +6 or better).

**Extended action:** An action in combat that takes longer than a single combat round to complete.



**Grapple:** An unarmed attack involving a wrestling move.

**Game Master:** The one person who portrays characters not controlled by the other players, arbitrates the rules and makes up the story and setting for the game.

**Homeworld:** The world that serves as a character's place of origin, usually the world on which the character was raised and which had the most impact on their development during their pre-adult life.

**Hull Points:** A measure of how much damage a vehicle, building or spacecraft can take before being disabled.

**Mainworld:** The primary world of a star system; the world represented by the UWP in a list of worlds for a given region of space.

**Melee attack:** A physical attack in close combat – either hand-to-hand, or with a weapon.

**Melee weapon:** A handheld weapon designed for close combat.

**Minor action:** An action so fleeting that it does not require a full combat round to complete, such as checking the time, looking through a window, turning around, falling prone, shouting a warning, etc.

**Modifier:** Any bonus or penalty applied to a die roll.

**Natural:** A natural result on a roll is the actual number appearing on the dice, not the modified result obtained by adding bonuses or subtracting penalties.

**Non-lethal damage:** Damage that can potentially stun or knock out a target, but does no permanent harm.

**Non-player character (NPC):** A character controlled by the GM (as opposed to a character controlled by a player).

**Non-starship:** A spaceship without a Faster-Than-Light drive and thus incapable of interstellar travel on its own.

**Player character (PC):** A character controlled by a player, one of the

protagonists of an adventure or campaign.

**Patron:** A non-player character who gives financial or other support to a person, organization, cause, or activity. GMs often use patrons as a tool to attempt to engage player characters in adventures.

**Penalty:** A negative modifier to a die roll.

**Ranged attack:** Any attack made at a distance, including gunfire and thrown attacks.

**Ranged weapon:** A firearm or thrown weapon designed for attacking at a distance.

**Roll:** A method of deciding the result of a character's action. Rolls are based on a relevant ability, skill, or other trait. To make a roll, roll 2D6 and add any relevant modifiers. If the roll result equals or exceeds a target of 8 or the result of an opponent's roll, it succeeds.

**Round:** A six-second unit of game time used to manage actions, usually in combat.

**Significant action:** An action intended to do something within about 3 or 4 seconds. You can perform a single significant action per round.

**Seriously wounded:** If you have two Characteristics reduced to 0, then your character is seriously wounded suffering bleeding or other trauma.

**Skill:** An ability to perform a set action, such as navigating a starship, operating a rifle, or programming a computer. Skills are attained in levels (Navigation-1, Computer-2, etc.); the higher the level of a skill, the more expertise a character has in that area. Level-3 is considered to be quite an expert. Many different individual skills are available to characters.

**Small Craft:** A vessel under 100 tons, capable of only interplanetary, not interstellar travel.

**Standard Day:** A unit of time that is 24 hours long.

**Standard Year:** A unit of time that is 365 Standard Days in length.

**Starport:** A port where interstellar and interplanetary vessels load or unload, are repaired and refuelled. Sometimes also referred to as a spaceport, landing field or downport.

**Starship:** A spaceship with a Faster-Than-Light drive, capable of interstellar travel on its own.

**Structure Points:** A measure of how much damage a disabled vehicle, building or spacecraft can take before being destroyed.

**Target:** The intended recipient of an attack, action, or effect.

**Tons and Tonnes:** A standard 'ton' is a unit of *volume or displacement*, approximately 500 cubic feet or 14 cubic meters in size (sometimes abbreviated to dton). A tonne is a metric unit of *mass*, equal to 1,000 kilogrammes.

**Trained:** Having knowledge of, and therefore levels in, a skill.

**Unarmed attack:** A melee or hand-to-hand attack made without a weapon.

**Untrained:** Having no ranks in a skill. Some skills cannot be used if untrained. Unskilled skill checks usually suffer a DM-3 penalty.

**UPP:** Universal Personality Profile – a single line alphanumeric entry describing the most important attributes of a character, accompanied by a list of skills they possess.

**UWP:** The Universal World Profile, a single line alphanumeric entry describing the most important attributes of a world.

**Vessel:** General term used for starships, small craft, or vehicles as a general inclusive group. Most commonly, it refers to any vehicle or ship capable of interplanetary or interstellar travel.

**World:** A generic term in the rules for any asteroid, moon or planet represented by a Universal World Profile.



# TECHNOLOGICAL LEVELS

Technology Levels (TL) can be assigned to a vehicle or device, or society, to give you some idea of the society that produced or uses it.

## **TL 0      Primitive**

*Stone Age.*

## **TL 1      Primitive**

*Roughly on a par with Bronze or Iron age technology. Earth 3300 BC-1400 AD*

## **TL 2      Primitive**

*Renaissance technology, scientific method and gunpowder. Earth 1400-1700.*

## **TL 3      Primitive**

*Beginnings of mass production allows for product standardization, urbanisation and steam power. Earth 1700-1860.*

## **TL 4      Industrial**

*Transition to steam-powered industrialisation is complete, bringing electric generators, telephones and other such inventions. Earth 1860-1900.*

## **TL 5      Industrial**

*Widespread electrification, telephones and internal combustion for cars and aeroplanes. Medical advancement. Earth 1900-1939.*

## **TL 6      Industrial**

*Development of nuclear power and more advanced computing, as well as rockets and jet engines. Earth 1940-1969.*

## **TL 7      Pre-Stellar**

*Supersonic passenger planes, hovercraft, telecommunications satellites, reliable access to space, portable computers. Earth 1970-2000.*

## **TL 8      Pre-Stellar**

*Modern Day. The internet, smartphones, and an integrated network society. Possible to reach other worlds in the same star system with rocket technology. Basic robots now available. Earth 2000 onwards.*

### **TL 9 Pre-Stellar**

*Cybernetics in use. Slower-than-light drives mature and a star system can be fully explored and colonised. Fusion power plants are developed. Cyberpunk tropes fit this TL.*

### **TL 10 (A) Early Stellar**

*Development of gravity manipulation, as well as faster-than-light travel, nearby systems are opened up. Use of robots becomes widespread. The first primitive artificial intelligences become possible.*

### **TL 11 (B) Early Stellar**

*Sealed combat armour is in use. Grav vehicles begin to replace other types of transport. Holograms now in regular use.*

### **TL 12 (C) Average Stellar**

*Weather control revolutionizes terraforming and agriculture. Terraforming becomes possible.*

### **TL 13 (D) Average Stellar**

*Powered armour called 'battle dress' appears on the battlefield. Full artificial intelligence is perfected. Cloning and DNA modification is now common.*

### **TL 14 (E) Average Stellar**

*Portable plasma weapons feasible. Gravity attractor beams as well as repulsor shields are now in use.*

### **TL 15 (F) High Stellar**

*Floating cities are now possible, although rare and expensive.*

### **TL 16 (G) Beyond Stellar**

*Disintegrator weapons are in use, as is food and material replication technology. Matter transport becomes possible. Anti-matter powerplants now standard.*

### **TL 17 (H) Beyond Stellar**

*Transhumanist society, minds easily uploaded and downloaded, blurring the definition of android, human and sentience. Ringworlds now possible.*

### **TL 18 (J) Quantum**

*Energy transfer is now perfected. Sentience can travel without physical form, and snap in and out of planes of existence. Indistinguishable from magic.*



# CAMPAIGN CONCEPTS

These science fiction roleplaying rules allow many kinds of futuristic sci-fi adventures, in a range of settings. The players should decide on the type of game or campaign they want to play, with the help of the Game Master. Will they be the crew of a salvage ship, looking to make a big score? Or a squad of star marines who are dropped into one warzone after another? How about playing out the first mission to Mars, to find intelligent (and perhaps hostile) life there? Game or campaign concepts might be on-going, multi-session affairs. If so, the concept must support the episodic TV style of play. Features of episodic play include a broad spread of character skills ready for future potential situations, characters without local ties that are free to move from one adventure location to another, and threats that aren't going to incapacitate or kill too many of the player characters (they will be needed for the next scenario in the campaign!).



Concepts can also revolve around a single event or encounter, with player characters designed specifically for that unique adventure. It may take one, two or more sessions to play through at the gaming table, but once the story has reached its finale, the game comes to an end. This is the movie style of play, and it benefits from not having to include the features of episodic play. Challenges can be more brutal, skill choices can be more focussed on the situation at hand, and the finality of the game's ending can be extremely satisfying, for all concerned.

## CAMPAIGN SUGGESTIONS

There are a number of campaign concepts, activities that players can engage in, that are already associated with the science fiction genre.

### AGENTS

Player characters are secret agents or law enforcement officers, who are dispatched to other planets in order to investigate and then solve problems. Each mission is different from the last, each involves travel to alien worlds and each mission invariably involves danger. They might be interstellar marshals, star cops, agents of a secret intelligence service, or perhaps even the corporate agents of a massive megacorporation.

**Careers** – Most of the characters will be Agents, but a Fixer or a Rogue might also prove useful.

### COLONISTS

The player characters are the key personnel of a small colony. It might be newly established on a wild and alien planet, with the characters facing many challenges ahead in order maintain the colony and stop it from failing. All the while trying to expand its infrastructure and explore the world around them. Alternatively, the colony might be long-established, but an unexpected threat has appeared (civil unrest, alien organism, rival human faction?) and the PCs must attempt to deal with it and prevent the colony from being ripped apart.

**Careers** – Most of the PCs will be Citizens, although one or two might be a Scientist or Technician. It would be helpful if at least one of the player characters possessed the Leader skill, in order to get the colonists to follow a course of action.

### FREE TRADERS

This commercial starship crew have a ship and hire it, and themselves, out to undertake charters, salvage operations and special one-of-a-kind missions. Sometimes it's just hauling cargo because the regular freighter is unserviceable. More often the cargo is hazardous, or has to go somewhere off the beaten track, or it is dangerous or illegal. The crew don't ask questions and their lives (and the missions themselves) are varied, interesting and not without danger.

**Careers** – Merchants, with engineering and repairs handled by a Technician. A Fixer might be perfect for all those trade talks and business deals.

### INFANTRY SQUAD

Marines are shipboard troops that carry out ship-to-ship boardings, customs searches and security details. In times of war they are dropped on to planet to seize installations, carry out commando strikes and rescue hostages. They are closely linked with the interstellar navy, and serve as an elite fast-reaction force, often going in ahead of conventional forces. Interstellar hijackings, terror situations, uprisings, colonial insurgencies and inter-colony strife may all result in Marine intervention. Meanwhile, planets often have their own military forces, with armies that can defend the population. The PCs might be on-planet members of one such army, about to face some great threat to the colonists, such as local alien organisms, or indeed a human foe. Some of these infantry forces might actually be mercenary units, which make great vehicles for roleplaying. The PC mercenary unit can move from planet to planet, following the rise and fall of conflicts and strife, skirting legality and butting heads with the conventional forces they are there to supplement.

**Careers** – An infantry campaign will naturally feature either all-Marines, or if a colonial army or mercenary campaign, will feature all-Mercenaries.

### SURVEY SCOUTS

The player characters are deep space explorers, on their own, and far from home. Are they mapping new star systems? Are they exploring a planet ready for colonization? Have they been ordered to make contact with a new alien race? Some exploration teams are only interested in finding something that will turn a profit, rare minerals, and exotic gems, etc. Explorers must be tough, resourceful and skilled.

**Careers** – Most of the crew will be Explorers, but variety can be added with a Scientist or Technician.

### INTERSTELLAR NAVAL OFFICERS

The player characters are the bridge crew of a powerful naval warship, and their missions will run the gamut, from fleet exercises that go wrong, to search and rescue operations, patrolling the space lanes, interception of enemy or pirate vessels, fleet actions against enemy worlds or installations, and plenty of diplomatic missions. Each character takes up a key role on the bridge, and the space combat rules are written to facilitate this character-led approach. Although one of the group must play the captain, he will only act as a spokesman at the gaming table, the players as a group decide what to do in any particular circumstance. The captain is not really in charge, the players co-operate as they would in any other game. There are rules for capital ships and space combat within this rulebook.

**Careers** – Everyone will be a Spacer, although the engineer might be a Technician (just assign him a Spacer rank). The Game Master might want to let each player have two characters, one serving on the bridge, and a second 'below decks' or 'away team' character that can get into situations that members of the bridge crew should not.

### PIRATES

Who wants to follow interstellar law?! The player characters may want the thrill of surviving on the wrong side of the law, dodging the interstellar navy and avoiding custom patrols, all in order to get rich by plundering cargo ships and the yachts of the elite, as well as carrying out salvage operations, and smash-and-grab raids on remote installations and space stations. The pirate ship must be disguised as something else in order to move freely through space, and land at starports without arousing suspicion. Perhaps it is a converted cargo hauler, mining vessel or scout ship. There must be a cover story, and the pirates must have a cover occupation in order to mask their dark deeds ... and dark hearts.

**Careers** – Most of the pirate crew will be Spacers, Merchants or Rogues.

### REBELS

Some interstellar states are tyrannical and authoritarian. When that occurs there may be freedom fighters, rebels and guerrilla forces that are waging a star spanning insurgency against the forces of the state. These fighters will be a rag-tag bunch, arming themselves however they can, and desperately trying to tell friend from foe as they attempt to recruit allies in their fight against tyranny. The rebellion might be small in scale, limited to one key planet, or it may be star spanning in nature, which means the player characters will most likely have a ship. This ship is their base of operations, their mode of travel, and their chief weapon against the forces of the state. In this case the campaign will closely resemble the Pirates campaign (see above), with the ship adopting a false identity and the PCs adopting a cover occupation.

**Careers** – Rebels may be of any career, people from many walks of life are drawn to resist the forces of tyranny. Realistically, however, given some experience fighting against the interstellar state, most of the PCs will resemble Mercenaries (or Spacers, if they have their own ship). Rogues make great rebels, and it is always useful to have at least one on the team.

### TIME TRAVELLERS

The player characters have been selected for their talents and skills to travel in time periods of the past and future. The missions they carry out will vary depending on the nature of the campaign and the intentions of the builders of the time machine. Time travelling teams must be resourceful, independent, fearless, and possess a wide variety of skills.

**Careers** – Many careers fit the bill for this type of campaign, including Scientist, Technician, and Agent. A soldier or ex-cop, such as a Mercenary, will prove useful for security, and characters good at stealth, breaking-and-entering and social skills, will also come in useful (such as the Agent or Rogue). Finally, if a character from a TL 0-3 society joins the team, that character would use the Primitive career.



### POST-APOCALYPSE SURVIVORS

The setting might be Earth in the near (or even distant) future, where a catastrophe has plunged civilisation into chaos. Cities are destroyed, and society has broken down into small, defended townships or fortress cities. In the barren wild lands are dangers that will only be faced by the bravest, and hardest of travellers: radiation, genetic mutation, malevolent cyborgs, alien invaders, human raiding gangs, zombies ... or whatever else the setting includes as part of the reason behind the destruction of civilisation.

**Careers** – The Scavenger is a prime career for this campaign, but the Technician, Mercenary, Primitive, Citizen and Rogue could all play their part in this type of game.

### SALVAGE TEAM

The campaign is based around a group of player character salvors, men and women who recover lost or derelict spacecraft for a fee. They also strip out components from wrecks that they can sell on the open market. The salvage teams will have a ship, perhaps owned by their employers, or perhaps they are paying it off with their profits. The characters operate in deep space, without back up, and engage in difficult and dangerous work. They are a tough, no nonsense breed of people. And salvors are known to skirt the law, sometimes claiming wrecks or debris they have no right too. Some flirt with piracy. And their skills often come in useful as a rescue crew, so that local authorities sometimes call on them to perform rescues in deep space ... for a fee, of course.

**Careers** – The most likely careers for a salvage crew are Spacers, Technicians and perhaps a Belter. A Fixer will be useful in selling the goods, negotiating contracts and dealing with clients.







## CHARACTER SHEET

| NAME |
|------|
|------|

## Psych Evaluation

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| <b>STR</b>  |
| <b>Mod:</b> |

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| <b>DEX</b> |
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| <b>END</b> |
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| Career |
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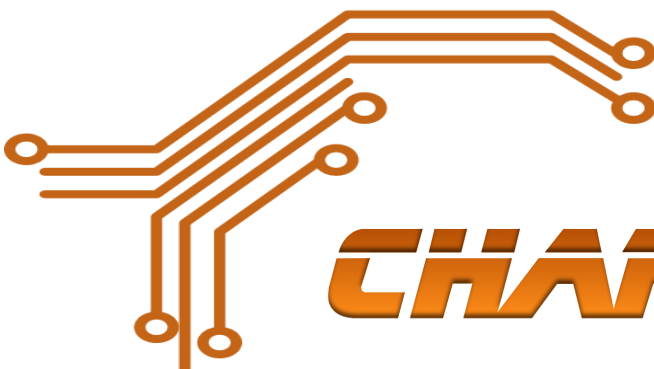
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| Life Events |                                |
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| 1           | Death of a spouse              |
| 2           | Divorce                        |
| 3           | Marriage                       |
| 4           | Death of a close family member |
| 5           | Personal illness               |
| 6           | Family illness                 |
| 7           | Death of a friend              |
| 8           | Change of residence            |
| 9           | Change of job                  |
| 10          | Financial difficulties         |
| 11          | Change of family structure     |
| 12          | Change of school               |
| 13          | Change of religion             |
| 14          | Change of community            |
| 15          | Change of lifestyle            |
| 16          | Change of diet                 |
| 17          | Change of exercise routine     |
| 18          | Change of social life          |
| 19          | Change of hobbies              |
| 20          | Change of interests            |
| 21          | Change of beliefs              |
| 22          | Change of values               |
| 23          | Change of goals                |
| 24          | Change of priorities           |
| 25          | Change of responsibilities     |
| 26          | Change of roles                |
| 27          | Change of relationships        |
| 28          | Change of friendships          |
| 29          | Change of family members       |
| 30          | Change of pets                 |
| 31          | Change of environment          |
| 32          | Change of climate              |
| 33          | Change of season               |
| 34          | Change of time of day          |
| 35          | Change of weather              |
| 36          | Change of location             |
| 37          | Change of direction            |
| 38          | Change of path                 |
| 39          | Change of route                |
| 40          | Change of journey              |
| 41          | Change of destination          |
| 42          | Change of purpose              |
| 43          | Change of meaning              |
| 44          | Change of significance         |
| 45          | Change of importance           |
| 46          | Change of value                |
| 47          | Change of worth                |
| 48          | Change of price                |
| 49          | Change of cost                 |
| 50          | Change of expense              |
| 51          | Change of income               |
| 52          | Change of profit               |
| 53          | Change of loss                 |
| 54          | Change of gain                 |
| 55          | Change of benefit              |
| 56          | Change of advantage            |
| 57          | Change of disadvantage         |
| 58          | Change of opportunity          |
| 59          | Change of risk                 |
| 60          | Change of reward               |
| 61          | Change of punishment           |
| 62          | Change of consequence          |
| 63          | Change of result               |
| 64          | Change of effect               |
| 65          | Change of impact               |
| 66          | Change of influence            |
| 67          | Change of power                |
| 68          | Change of control              |
| 69          | Change of authority            |
| 70          | Change of responsibility       |
| 71          | Change of obligation           |
| 72          | Change of duty                 |
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| 74          | Change of wrong                |
| 75          | Change of justice              |
| 76          | Change of fairness             |
| 77          | Change of equity               |
| 78          | Change of balance              |
| 79          | Change of harmony              |
| 80          | Change of peace                |
| 81          | Change of war                  |
| 82          | Change of conflict             |
| 83          | Change of cooperation          |
| 84          | Change of competition          |
| 85          | Change of collaboration        |
| 86          | Change of partnership          |
| 87          | Change of alliance             |
| 88          | Change of coalition            |
| 89          | Change of union                |
| 90          | Change of separation           |
| 91          | Change of division             |
| 92          | Change of unity                |
| 93          | Change of diversity            |
| 94          | Change of similarity           |
| 95          | Change of difference           |
| 96          | Change of sameness             |
| 97          | Change of uniqueness           |
| 98          | Change of commonality          |
| 99          | Change of individuality        |
| 100         | Change of identity             |

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[illegible]



# CHARACTERS

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## CHARACTERS

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Player characters (PCs) in **CEPHEUS UNIVERSAL** will face dangerous situations with the odds stacked against them, they will need to use their resourcefulness and skills to overcome the challenges that they might face. In this chapter we create our heroes, the player characters, and create first their physical and mental characteristics, and then determine their skills.

Who are the player characters? What do they do? What have they been up to in the past? All of that depends on the type of science fiction setting that you intend to run. The PCs might be the crew of a trading ship, making money where they can, and skirting the edges of legality. They might be mercenaries, moving from warzone to warzone in search of lucrative military contracts. They could be explorers, opening up new star systems for colonisation, or perhaps scientists and adventurers, travelling through time to set things right. Look at the campaign concepts in the previous chapter; what type of campaign will you be playing? Each suggestion provides ideas for character careers that will provide the character with skills appropriate to that role.

Because the scope of adventure and possibility is so wide, these character creation rules are likewise wide in scope and provide the player with a variety of character creation methods. Design your character based on a Career that fits the GM's setting and the players' campaign type. There are three character creation options; it is best if all of the players in the game use the same creation method.

### 1 – DESIGNED CREATION

No dice are involved. The player has a concept that fits in with the campaign suggestion, and then makes choices or picks attributes or skills from lists, to build the character that he or she wants.

### 2 – FULLY RANDOM CREATION

This is the quickest creation method, and relies on random rolls to create a character that could come from virtually any walk of life. An option allows the player to instead 'set the parameters' by starting with a rough concept, and then letting the dice fill in the rest of the details.

### 3 – TERM-BY-TERM CREATION

In this more detailed character creation method, the player uses dice to track the character's career 'term-by-term' (a term being a block of four years). The player has some choices to make, but generally the character slowly emerges through random rolls from his own experiences and challenges. The final character enters play with a prior history and events that have shaped his personality and world-view.



## THE CHARACTER

A completed player character will have the following attributes:

**BASICS** – A Name, Age and Sex.

**CHARACTERISTICS** – Strength, Dexterity, Endurance, Intelligence, Education and Social Influence; all given a value between 2 and 12 (with an average of 7).

**APPEARANCE** – A key facet (or facets) of the character's appearance.

**HEIGHT & WEIGHT** – Determined by the player, or by reference to the character's Strength, Dexterity and Endurance.

**PSYCH EVALUATION** – A personality label.

**CAREER** – A broad description of the character's past occupation or activities.

**TERMS** – How many 4 year 'terms' the character has completed in his career (only relevant when using the Term-By-Term creation method).

**RANK** – The character's seniority within his occupation; it is determined differently based on the creation method used.

**SKILLS** – How competent the character is in specialised areas of expertise (see SKILLS, below).

**RELATIONSHIPS** – Non-Player characters that have some connection with the character's current life, either FRIENDS (NPCs actively assisting the character), ENEMIES (NPCs actively working against the character) or CONTACTS (helpful NPCs).

**CASH** – The amount of money (in credits) the character owns.

### RECORDING CHARACTER INFORMATION

This game utilizes a concise format to record a character's characteristic scores, age, skills, psych evaluation, height, weight cash and personal equipment in a manner that, with a little practice, can be quickly and easily read. The string of six numbers represent the individual's characteristic scores. These scores are, in order: *Strength* (Str), *Dexterity* (Dex), *Endurance* (End), *Intelligence* (Int), *Education* (Edu), *Social Influence* (Soc). Below is an example of a dedicated starship pilot:

|                                                    |        |        |
|----------------------------------------------------|--------|--------|
| <b>Second Officer Lisa Nyanda (Ship Pilot)</b>     | 567984 | Age 34 |
| Vacc-suit-1, Pilot-3, Leader-1, Comms-2            |        |        |
| Cr10,000, Stubborn, Height: 155 cm, Weight: 52 kg. |        |        |

This method does not record all of the information, for a more detailed method or recording, use the **CEPHEUS UNIVERSAL** Character Sheet (page 22).

# CHARACTERISTICS

Characteristics measure a character's most basic abilities: how strong, agile, educated or clever they are. Characters have six abilities: Strength (Str), Dexterity (Dex), Endurance (End), Intelligence (Int), Education (Edu), and Social Influence (Soc). Strength, Dexterity, and Endurance are called physical abilities, whereas Intelligence, Education, and Social Influence are loosely termed mental abilities.

**Strength (Str):** Physical strength, fitness and forcefulness.

**Dexterity (Dex):** Physical co-ordination, agility and reflexes.

**Endurance (End):** Determination and stamina.

**Intelligence (Int):** Awareness and problem solving.

**Education (Edu):** Learning and general knowledge.

**Social Influence (Soc):** A character's ability to influence others.

## CHARACTERISTIC MODIFIERS

Characteristic modifiers are applied to any task roll where a characteristic will play an important part. Climbing an uplink tower, for example, might require a characteristic roll on 2D6 for 6+, with the player adding in his Dexterity modifier. The GM will indicate when a characteristic modifier (and which one) will be required for a roll. Not all task rolls will require a characteristic modifier to be used.

| CHARACTERISTIC MODIFIERS |     |     |     |      |       |     |
|--------------------------|-----|-----|-----|------|-------|-----|
| Characteristic           | 0-2 | 3-5 | 6-8 | 9-11 | 12-14 | 15+ |
| Modifier                 | -2  | -1  | -   | +1   | +2    | +3  |

## HEXADECIMAL NOTATION

These rules use a form of pseudo-hexadecimal notation as a type of shorthand in noting specific values of characteristic scores. The 'pseudo' hexadecimal notation proceeds as normal for values from 0 to 9, but extends beyond 9 with letters replacing two digit numbers A for 10, B for 11, C for 12, etc. For example, a character with Str 7, Dex 7, End 10, Int 6, Edu 12 and Soc 2, would look like this: 77A6C2

**Characteristic Score Limit** – For player characters, a characteristic score may not typically exceed a maximum of 15, nor may a score drop permanently below 1 except under certain circumstances.

**Altering Characteristic Scores** – Over the course of play, your character's characteristic scores may change for the following reasons:

- Physical damage, such as from combat, falling, disease or poison.
- Mental trauma, such as head injuries, panic and drug problems.

## WHAT DO CHARACTERISTICS REPRESENT?

The characteristics range from 2 to 12, and will average 7. The values can tell you a lot about your character – perhaps more than you might think. The trick is not to be too literal but also not to push the facts too often. There is no universal truth in those numbers, they are suggestions and spurs to imagination.

- **Strength** – Strength can represent muscle, size and physical presence. High strength may indicate an intimidating character, without necessarily meaning a huge, stocky physique. Low strength, conversely, may reflect a slight, skinny figure or perhaps one with very little presence – a quiet character.
- **Dexterity** – Dexterity is associated with agility, co-ordination and speed, but also may indicate self-control and willpower. Low Dex may mean lack of self-control, perhaps not just physically, but also mentally.
- **Endurance** – Toughness and stamina. Endurance also indicates a pain threshold. Does this indicate a character with a past filled with hard knocks and hard living? Low End may mean a pampered lifestyle, a low tolerance of stress, pain and discomfort.
- **Intelligence** – The classic measurement of logic and problem solving, but one that probably also defines memory, quick-thinking and adaptability. A high Int might indicate high levels of perception, or someone who is always thinking two steps ahead, planning the future, perhaps expecting the worst. Low Int might well be indicative of someone oblivious to the subtle signs around them, someone who looks at the here-and-now, and does not plan ahead. They may see the world in black and white terms. Low Int doesn't just mean 'stupid'.
- **Education** – Traditionally a measure of book-learning. A high Edu probably indicates a highly educated character that has gone through college and university. But there is more to it than that – Education should include information gathering and general knowledge. Edu should give you an indication of how interesting a PC is to talk to, what kind of a conversationalist he or she is, weaving interesting facts with stories and personal anecdotes. In a way it represents experience that has been absorbed outside of the normal skill system.
- **Social Influence** – How does the character interact with other people? A high Social Influence might mean a character who lights up the room, is friends with everyone, loud, witty and fun. But it just as easily might mean a quiet schemer, capable to manipulating people by telling them what they want to hear, pressing their buttons with lies or suggestions. A low score can mean a quiet wallflower with low confidence, or an arrogant thinker who keeps their thoughts to themselves. The low score might even be that loud, brash person who dominates a room, but whom everyone avoids and ridicules behind their back (effectively they have bags of personality, but it has a negative effect!).

You don't need to carry out a slavish analysis of every characteristic, but pay a bit of attention to any value higher than 8 or lower than 6.

**Example:** Let's roll some dice to create a set of characteristic values as an example. We roll 679995 for our character. Physically she is quite average, although her Endurance of 9 might mean she is determined and doesn't easily give up. Both her Int and Edu are 9, which could indicate a clever girl who has studied hard and gained the equivalent of a degree. This all fits together nicely. Her Soc of 5 is lower than average. Perhaps she has no patience with people who don't share the same work ethic as her: 'time wasters', 'layabouts', 'lazy people', and so comes across as insular, private and 'stuck up'.

**Example:** We roll characteristics for another character: 846385. Those low scores should be interesting! Physically, he is strong and healthy, but his Dex of 4 means he is either clumsy, slow, lazy or undisciplined. I like lazy. Despite his low (3) Intelligence, his Education is good and he left school with a good educational attainment, or ... his education was more unorthodox. Perhaps the low Int reflects very low mental resilience, he *is* lazy, after all, and can't be bothered to go through the steps of planning things out, considering the consequences, etc. He just acts without thinking to get things done quickly. He may have picked up his knowledge working a lot of different jobs, or his family moving house frequently. His Social Influence is below average, again perhaps his impatience comes across as an off-hand, flippant manner.





# SKILLS

A skill is rated from 1 to 6, with a low number representing a basic qualification or equivalent, and a higher number representing much more advanced expertise in that skill. A character with Medical-2 might be a paramedic, while one with Medical-3 might be a doctor or a surgeon. The skill value encompasses not just training, but experience, competence, awareness of experts in the field, common practices, things happening in the industry and so on. Most skills are received as level-1, with additional gains adding +1 to that skill level. Level-0 is a familiarisation course or some basic training, often given to someone outside of the profession just to 'get by'.

## WHAT DO SKILL LEVELS REPRESENT?

Many of the skills and levels can be easily equated to actual licenses, ratings and military proficiencies within the Game Master's setting. Civilian licences usually include grades or certifications of competence, such as Class 1 or Class 2, that could equate to this skill level system. Often, a level 3 skill rating represents an Advanced Licence, so a character with Broker-3 has an *Advanced* commercial Brokerage Licence. In many settings (such as a post-apocalyptic wasteland), most skills will be picked up unofficially; Broker skill, for example, would be a talent picked up over time, rather than an the result of some official training program.

| SKILL LEVELS |                                                                     |
|--------------|---------------------------------------------------------------------|
| Skill Level  | Description                                                         |
| Level-0      | Familiarisation, brief, or partial training.                        |
| Level-1      | Professional rating, enough to be hired for a suitable job.         |
| Level-2      | Experienced, possessed by senior personnel, leaders, etc.           |
| Level-3      | Advanced level, or mastery of a skill; top of the game.             |
| Level-4-6    | Renowned as an expert in the field, instructor, lecturer, innovator |

| LEVEL EXAMPLES (1) |                                     |                                          |
|--------------------|-------------------------------------|------------------------------------------|
| Skill Level        | Carousing                           | Leader                                   |
| Level-0            | Guidance from a friend              | 'A taste of command'                     |
| Level-1            | Confident and outgoing              | Established leader                       |
| Level-2            | Making friends made easy            | Seasoned leader                          |
| Level-3            | You live for nightlife & good times | Leader of complex organisation           |
| Level-4-6          | Party-master and social fixer       | Brilliant, inspirational maverick leader |

| LEVEL EXAMPLES (2) |                            |                                      |
|--------------------|----------------------------|--------------------------------------|
| Skill Level        | Gun Cbt                    | Mechanical                           |
| Level-0            | Weapon handling course     | Introduction to mechanics course     |
| Level-1            | Military basic training    | Professional mechanics qualification |
| Level-2            | Marksmanship training      | Experienced mechanical skill         |
| Level-3            | Rangemaster                | Advanced mechanical knowledge        |
| Level-4-6          | Special weapons instructor | Inventor, custom-builder             |

## SKILL ROLLS

Characters in **Cepheus Universal** games engage in a variety of activities, using their various skills to accomplish the challenges that confront them. Skills and their usage are described below. Skill rolls use the core task resolution system to resolve actions. Whenever your character attempts any action with a chance of failure, roll 2D6, add the appropriate skill level and the relevant difficulty DM. If the result equals or exceeds 8, the action succeeds. If the result is lower than 8, the action fails. The basics of the task resolution system can be found in the Game System chapter.

## TASK FORMAT

Skill rolls are noted as comprising the difficulty level followed by a required skill. For example, repairing damage to a starship in mid-combat might be an Average (0) Engineering task; docking with a tumbling space station will be noted as a Difficult (-2) Pilot task. The GM gets to choose the time increment involved, if it is not already obvious from the context. If the GM decides, a suitable characteristic modifier can be added to the player's roll. That space station docking roll might qualify for a Dex modifier to be included, for example, based on the character's Dexterity characteristic.

## UNTRAINED & ZERO-LEVEL SKILLS

With GM approval, characters can perform some tasks without any training in a skill. Common skills, such as Vacc Suit and Carousing might entail a -3 DM penalty for being unskilled, whilst skills requiring extensive training such as Navigation and Engineering, will entail a penalty of -5 DM.

If a character has zero level in a skill (level-0), then he only has some surface-level familiarity. He does not get any bonus from that skill but at least he avoids the penalty for being untrained. It is common to give all player characters level-0 in Vacc Suit, Ground Vehicle and the combat skills of Gun Combat and Melee Combat, giving them more options when faced with challenging situations.



## SKILL DESCRIPTIONS

---

**Administration:** Admin, paper-work, dealing with rules, regulations and officials, as well as law and legal personnel.

**Agriculture:** Growing and harvesting crops and raising animals, including hydroponics and aquaculture (on an ocean world).

**Aircraft:** Operation and control of atmospheric craft, including transports, helicopters, fast jets and jump jets. Aircraft expertise can also be used as Grav Vehicle-0.

**Archery:** Allows use of bows and crossbows, useful on TL 0-3 worlds.

**Bribery:** Offering bribes to circumvent local law, or to influence someone's decision. The cash bribe must be appropriate to the situation.

**Broker:** Locating suppliers and buyers, and facilitating the purchase and resale of commercial goods, haggling, bartering and so forth.

**Carousing:** Social skills, including picking up gossip or rumours, making friends and reading people's body language.

**Comms:** Operating drones, sensors and radio equipment. Skilled characters can boost an incoming or outgoing signal, create or break a secure channel, detect signals and anomalies, hide or piggyback on another signal, jam local communications, locate and assess potential threats, and analyse complex sensor data.

**Computer:** Operation and programming of computers, including creating or breaking data encryption; mining data effectively; creating or breaking data and network security protocols; overriding a computer protocol, as well as other general programming tasks.

**Demolitions:** Using demolition charges and other explosive devices, including assembling or disarming bombs.

**Electronics:** Operating and repairing complex electronic devices.

**Engineering:** Use and maintenance of powerplants, reactors and starship drives.

**Forgery:** Faking documents, currencies, and identification badges in order to deceive officials, government agents and security forces.

**Grav Vehicle:** Operation and control of TL 10+ anti-grav vehicles of all kinds. Grav expertise can also be used as Aircraft-0.

**Ground Vehicle:** Operation and control of wheeled and tracked vehicles, including ATVs and Armoured Fighting Vehicles.

**Gun Combat:** Using and maintaining small arms including pistols and rifles, SMGs, lasers, machineguns and shotguns.

**Gunnery:** Using starship-mounted weaponry.

**Heavy Weapons:** Using military support weapons including rocket launchers, tactical missile launchers, artillery, grenade launchers, energy cannons and tank guns.

**Investigate:** Scientific analysis and the use of complex and accurate scientific tools and equipment to gather clues at a crime scene or scientific location.

**Leader:** Motivating others in times of crisis or stress, particularly Non-Player Characters who might be reluctant to carry out an action.

**Liaison:** Skill in negotiations, protocol, manners of address and codes of conduct.

**Loader:** Operation and control of all heavy machinery, everything from fork-lift trucks to cranes, augmented power loaders, diggers and all forms of heavy machinery.

**Mechanical:** Operating and repairing mechanical devices, from truck engines to airlock doors, hydraulic lifters to life support machinery.

**Melee Combat:** Fighting hand-to-hand, with fists, feet, wrestling or grappling moves, or any number of melee weapons (such as knives, swords and clubs).

**Medical:** Training and skill in the medical arts and sciences, from diagnosis and triage to surgery and other corrective treatments. This skill represents a character's ability to provide emergency care, short term care, long-term care, and specialized treatment for diseases, poisons and debilitating injuries.

**Mining:** The character has experience and training in prospecting and mining, both on a world surface and in a zero-G environment. It includes the use of mining equipment.

**Navigation:** Plotting courses on planets, and in space and using instruments, maps and beacons to determine exact location.

**Pilot:** Operation and control of small craft, as well as interplanetary, and interstellar spacecraft. Pilot expertise can also be used as Aircraft-0 and Grav Vehicle-0.

**Recon:** Scouting out dangers and spotting threats, ambushes, booby traps, unusual objects or out of place people. Characters are also adept at silent movement and in camouflage techniques.

**Security:** Installing and also bypassing or dismantling security measures, from mechanical locks to swipe-card locks, keypad locks, surveillance cameras and various types of alarms and their triggers.

**Steward:** The care and serving of passengers and other guests, including customer service.

**Streetwise:** Familiarity with underworld society, its rules, personalities, groups and customs.

**Survival:** Staying alive in the wild, including hunting or trapping animals, avoiding exposure, locating sources of food and fresh water, producing fires, finding shelter, avoiding dangerous flora and fauna and dealing with the dangers of hazardous climates (arctic, desert, etc.).

**Tactics:** This skill covers tactical military planning and decision, as well as calling in fire support correctly.

**Vacc Suit:** Use and training in all types of vacuum suits, hostile envirosuits and combat armour, as well as operating in a zero-G environment.

**Vehicle (Cascade Skill):** The various specialties of this skill cover different types of planetary transportation. When this skill is received, the character must immediately select one of the following: Aircraft, Grav Vehicle, Ground Vehicle or Watercraft.

**Watercraft:** Operation and control of ocean or water-going craft including motorboats, hydrofoils, hovercraft, submarines and large commercial ships.

## GROUP CREATION

There are two methods the players can adopt in order to create their characters 'as a group'.

1) Look at the Campaign Concept before character creation, and decide together which player will take what type of character role. This way, the group gets a good spread of careers and skills, useful during play. Then create those characters.

2) The GM and players define the group's concept and how it fits into the setting. Next, they create their characters. Then in a random order, the first player states his name and role in the group; the second player states his name, role and his relationship to the first character. The third player states his name, role and relationship to the first two characters, etc, going all the way around to the first player, who gets to create a relationship with one or more of the other characters. Not all the relationships need to be good ones, but their intention is always to bind the characters together in order to have a stake in one another's past, present and future.



# 1 – DESIGNED CREATION

Start the process of building a character with a basic concept, “a devious scientist with a mysterious past” and then go from there. We start with the basics: name, sex, personality, etc. and then move on to his characteristics and skills.

## 1 – NAME

At the very least, every character needs a name, preferably one that fits the style of your gaming group and the setting. Consider choosing a name based on actual ‘real world’ names. If you are stuck, look up your favourite movie on IMDb and pick a name from the acting or production credits.

## 2 – SEX

In most games, this means choosing either male or female. There’s nothing that says you have to play a character of your own sex.

## 3 – APPEARANCE

Players are free to detail their character’s appearance, such as hair colour, clothing style and so forth. This makes it easier to describe your character, and helps everyone at the table visualize them in-play. Distinctive features often give your character a way of standing out. The Notable Features table indicates some notable feature of a character’s appearance, and can be used as a list to pick from, or can be rolled on randomly. Rolling the entry ‘boots’ for example, indicates the character’s boots are non-standard, non-regulation or flamboyant. Serving military characters will have less leeway in this regards than their civilian counterparts.

| NOTABLE FEATURES |                               |     |                                   |
|------------------|-------------------------------|-----|-----------------------------------|
| 1D6              | TABLE A                       | 1D6 | TABLE B                           |
| 11               | Wild hair                     | 11  | Scar                              |
| 12               | Stiff body posture            | 12  | Taps foot, drums fingers          |
| 13               | Brand logo                    | 13  | Jewellery                         |
| 14               | Boots                         | 14  | Cap or hat                        |
| 15               | Bored expression              | 15  | Sits or leans, rather than stands |
| 16               | Hair, short or tied back      | 16  | Shirt                             |
| 1D6              | TABLE C                       | 1D6 | TABLE D                           |
| 11               | Hair, long                    | 11  | Customised uniform                |
| 12               | Military surplus              | 12  | Fashionable clothing              |
| 13               | Clears throat before speaking | 13  | Accessory Item                    |
| 14               | Shades or Glasses             | 14  | Hands shoved in pockets           |
| 15               | Prominent tattoo              | 15  | Cocky smirk                       |
| 16               | Jacket                        | 16  | Bald                              |

## 4 – PSYCH EVALUATION

The player is free to play the character however they wish. Optionally, choose a general psych evaluation file (a personality trait) from the table below that might fit the character, or roll 2D6 for a random determination. The evaluations do not need to be followed slavishly, they are intended to suggest possible character responses to certain situations. Player freedom is paramount.

| PSYCH EVALUATION FILE |                                                          |
|-----------------------|----------------------------------------------------------|
| 2D6                   | Trait                                                    |
| 2                     | Talks tough but is scared inside.                        |
| 3                     | Joker who does not take life (or death) seriously.       |
| 4                     | Grim and quiet, they let actions speak for them.         |
| 5                     | Stubborn and strong-willed.                              |
| 6                     | Comically pessimistic and a touch paranoid.              |
| 7                     | Rebellious in nature which often gets them into trouble. |
| 8                     | Easy going and always relaxed ... right?                 |
| 9                     | Optimistic and determined.                               |
| 10                    | Selfish, always thinking about reward and compensation.  |
| 11                    | Overconfident, a cover for fear of failure.              |
| 12                    | Cynical, downbeat and despondent.                        |

## 5 – DETERMINE CHARACTERISTICS

### THE CHARACTERISTIC STRING

Select one of the following strings of scores, and assign these numbers in any way you see fit between your six characteristics:

String A: 7, 8, 7, 5, 8, 7  
String B: 10, 7, 4, 6, 6, 9  
String C: 3, 8, 9, 8, 5, 9  
String D: 2, 12, 9, 7, 5, 7

Example: I use String A, and allocate the numbers for my Marine like this: Str 8, Dex 7, End 8, Int 7, Edu 5, Soc 8.

## 6 – EDUCATION

Education represents knowledge and understanding about the world, with high levels standing for broad general knowledge or advanced academic interests such as geology, genetics or material science. As a very general guide, the Education value can be equated to standard educational milestones (see table). Allow Education rolls if a character's career and backstory fit the subject; if it's general knowledge then anyone

| Education | Educational Experience        |
|-----------|-------------------------------|
| 5+        | High School                   |
| 9+        | Bachelors' Degree (BA or BSc) |
| A         | Masters' Degree (MA)          |
| B         | Doctorate (PhD.)              |
| C+        | Doctoral Fellowship           |

can make a roll. If it's quite specific academic information, then a roll should probably only be open to someone whose background can justify it.

**Languages** – A character can speak their own language with fluency 5. For every point of Edu above 5, that character gains one fluency level in another language. Level 1 represents useful phrases and words. Use fluency level as a skill in rolls to communicate something unusual or understand jargon or obscure texts. Not all settings will suit this approach, in which case ignore this rule.

## 7 – HEIGHT & WEIGHT

As with a character's appearance, the player can set their height and weight at any value they deem suitable for the characteristic values. Alternatively, the player may calculate them using the characteristics as a basis. The results are broadly representative and not intended to cover every conceivable body type. Treat Str over 10 as 10.

**HEIGHT:** Multiply either Str or Dex by 3 (player's choice). Add the result to 154cm (male) or 144cm (female).

*Example: Dex 9 Male (9x3) = 27, add 27 to 154cm = 181cm (5'11')*

*Example: Str 7 Female (7x3) = 21, add 21 to 144cm = 165cm (5'5")*

**WEIGHT:** If *all three* characteristics are under 7 then the character is Out-of-Shape; see the relevant section, below. For other characters, add Str to End and multiply by 4. Add the total to 24 kg (if male) or 14 kg (if female). For Out-of-Shape characters, subtract first Str and then Dex from 7. Add them together and multiply the total by 10. Finally add 50 kg (if male) or 40 kg (if female).

*Example: Female (6 + A) = 16x4 = 64 + 14 = 78 kg (12st 4lb)*

*Example: Out-of-Shape Male (7-3, 7-4) = 4 + 3 = 7x10 = 70 + 50 = 120kg (18st 12lb)*

## 8 - AGE

Decide how old your character is, or roll on the table. Most characters will be experienced, anywhere from 25-50; this has no real game effect. As an option, however, young characters might get fewer skill levels.

| 1D6 | Age                          | Optional Skill Level Total |
|-----|------------------------------|----------------------------|
| 1   | Early Twenties               | 4                          |
| 2,3 | Late Twenties-Early Thirties | 5                          |
| 4,5 | Late Thirties-Early Forties  | 6                          |
| 6   | Late Forties-Early Fifties   | 6                          |

9 - CAREER SKILLS

Divide up **SIX** skill levels between the skills offered by your character’s chosen career, occupation or lifestyle. No skill level may start at more than 3. You *must* select the Automatic Skill as one of your skill level choices (think of it as a basic skill received during initial training in the career). Other skills are chosen at your discretion. If Vehicle is a skill within the character’s career, decide which type of vehicle you will choose: Ground Vehicle, Watercraft or Aircraft. When you gain a skill and you do not already have levels in that skill, take it at level 1. If you already have the skill, increase your skill by one level.

The careers listed below are broadly representative of the types of occupations and activities that feature in science fiction movies, TV shows and novels.

- Agent:** A law enforcer, secret agent, spy or police officer.
- Belter:** An asteroid miner, pioneer, prospector or colonial roughneck.
- Citizen:** A colonist or civilian, representing one of many occupations.
- Explorer:** A survey scout, deep space explorer, first-in mission specialist.
- Fixer:** A deal-maker, middle-man, executive, sleazy lawyer, street fixer.
- Marine:** Mobile infantry, space marines, ship’s troops or star commandos.
- Mercenary:** Ground-based infantry, either regular army or mercenaries.
- Merchant:** Interstellar truckers, haulage crews, making money between the stars.
- Rogue:** A criminal, thief, gang-member or saboteur.
- Primitive:** Inhabitant of a low tech, primitive world, barbarian, savage.
- Spacer:** Military crewman or officer, member of the interstellar navy or space force.
- Scavenger:** A post-apocalyptic road warrior, survivor or resistance fighter.
- Scientist:** Anything from a doctor to a geologist, physicist to bioweapon expert.
- Technician:** A gearhead, engineer, hacker, mechanic, cyborg technician, etc.

|                                                                                                                                                                                                                |                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>AGENT</b></p> <p>Automatic Skill</p> <p>Investigate</p> <p>Other Skills</p> <p>Gun Combat</p> <p>Melee Combat</p> <p>Computer</p> <p>Streetwise</p> <p>Medical</p> <p>Security</p> <p>Ground Vehicle</p> | <p><b>BELTER</b></p> <p>Automatic Skill</p> <p>Vacc Suit</p> <p>Other Skills</p> <p>Mining</p> <p>Mechanical</p> <p>Comms</p> <p>Ground Vehicle</p> <p>Loader</p> <p>Navigation</p> <p>Melee Combat</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                           |                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>CITIZEN</b></p> <p>Automatic Skill<br/>Vehicle</p> <p>Other Skills<br/>Mechanical<br/>Agriculture<br/>Carousing<br/>Leader<br/>Medical<br/>Computer<br/>Loader</p>                  | <p><b>EXPLORER</b></p> <p>Automatic Skill<br/>Survival</p> <p>Other Skills<br/>Pilot<br/>Navigation<br/>Vehicle<br/>Comms<br/>Investigate<br/>Leader<br/>Gun Combat</p>                  |
| <p><b>FIXER</b></p> <p>Automatic Skill<br/>Broker</p> <p>Other Skills<br/>Admin<br/>Computer<br/>Carousing<br/>Forgery<br/>Bribery<br/>Leader<br/>Liaison</p>                             | <p><b>MARINE</b></p> <p>Automatic Skill<br/>Gun Combat</p> <p>Other Skills<br/>Melee Combat<br/>Vacc Suit<br/>Ground Vehicle<br/>Recon<br/>Heavy Weapons<br/>Demolitions<br/>Tactics</p> |
| <p><b>MERCENARY</b></p> <p>Automatic Skill<br/>Gun Combat</p> <p>Other Skills<br/>Melee Combat<br/>Bribery<br/>Ground Vehicle<br/>Recon<br/>Heavy Weapons<br/>Demolitions<br/>Tactics</p> | <p><b>MERCHANT</b></p> <p>Automatic Skill<br/>Vacc Suit</p> <p>Other Skills<br/>Pilot<br/>Navigation<br/>Engineering<br/>Computer<br/>Comms<br/>Loader<br/>Broker</p>                    |



|                                                                                                                                                                                       |                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>ROGUE</b></p> <p>Automatic Skill<br/>Streetwise</p> <p>Other Skills<br/>Security<br/>Gun Combat<br/>Melee Combat<br/>Bribery<br/>Forgery<br/>Ground Vehicle<br/>Demolitions</p> | <p><b>SPACER</b></p> <p>Automatic Skill<br/>Vacc Suit</p> <p>Other Skills<br/>Pilot<br/>Gunnery<br/>Navigation<br/>Engineering<br/>Computer<br/>Comms<br/>Medical</p>                  |
| <p><b>PRIMITIVE</b></p> <p>Automatic Skill<br/>Survival</p> <p>Other Skills<br/>Riding<br/>Leader<br/>Archery<br/>Melee Combat<br/>Recon<br/>Tactics<br/>Carousing</p>                | <p><b>SCAVENGER</b></p> <p>Automatic Skill<br/>Ground Vehicle</p> <p>Other Skills<br/>Navigation<br/>Leader<br/>Gun Combat<br/>Melee Combat<br/>Survival<br/>Broker<br/>Mechanical</p> |
| <p><b>TECHNICIAN</b></p> <p>Automatic Skill<br/>Computer</p> <p>Other Skills<br/>Mechanical<br/>Electronics<br/>Engineering<br/>Comms<br/>Vehicle<br/>Vacc Suit<br/>Security</p>      | <p><b>SCIENTIST</b></p> <p>Automatic Skill<br/>Investigate</p> <p>Other Skills<br/>Computer<br/>Vehicle<br/>Admin<br/>Comms<br/>Survival<br/>Navigation<br/>Medical</p>                |

## 10 - BONUS SKILL

Your character gains a bonus level 1 skill. Select a skill from Table A or Table B. If the character already has that skill, increase its skill level by +1 (if it pushes the skill level above 3, then select a different skill).

| 1D6 | TABLE A      |
|-----|--------------|
| 1   | Carousing    |
| 2   | Leader       |
| 3   | Melee Combat |
| 4   | Gun Combat   |
| 5   | Vehicle      |
| 6   | Streetwise   |

| 1D6 | TABLE B     |
|-----|-------------|
| 1   | Medical     |
| 2   | Electronics |
| 3   | Admin       |
| 4   | Vacc Suit   |
| 5   | Steward     |
| 6   | Computer    |

## 11 - RANK

Use your Social Influence score to determine your character's status within his career. Since we assume characters have left their careers ('mustered out') in order to begin the game, these ranks are mostly for flavour. The ranks below are representative. Specific agencies and organisations in your setting may have different rank titles.

| Soc   | Rank | Agent            | Belter         | Civilian        | Explorer             | Fixer            | Marine     |
|-------|------|------------------|----------------|-----------------|----------------------|------------------|------------|
| 2-4   | 1    | Detective        | 'Miner         | 'Worker'        | Scout                | Hustler          | Marine     |
| 5-6   | 2    | Senior Detective | Senior Miner   | Senior 'Worker' | Senior Scout         | Sales Rep        | Sergeant   |
| 7     | 3    | Lieutenant       | Supervisor     | Supervisor      | Lieutenant           | Senior Sales Rep | Lieutenant |
| 8     | 4    | Captain          | Team Leader    | Asst. Chief     | Lieutenant Commander | Manager          | Captain    |
| 9     | 5    | Commander        | Superintendent | Chief 'Worker'  | Commander            | Executive        | Major      |
| 10-12 | 6    | Chief            | Manager        | Manager         | Commodore            | Senior Exec      | Colonel    |

| Soc   | Rank | Mercenary  | Merchant                | Rogue      | Scientist        | Spacer               | Technician        |
|-------|------|------------|-------------------------|------------|------------------|----------------------|-------------------|
| 2-4   | 1    | Private    | Crewman                 | Soldier    | Researcher       | Crewman              | Junior Technician |
| 5-6   | 2    | Sergeant   | Bosun                   | Enforcer   | Team Lead        | Petty Officer        | Senior Technician |
| 7     | 3    | Lieutenant | 3 <sup>rd</sup> Officer | Lieutenant | Scientist        | Lieutenant           | Supervisor        |
| 8     | 4    | Captain    | 2 <sup>nd</sup> Officer | Captain    | Senior Scientist | Lieutenant Commander | Asst. Chief       |
| 9     | 5    | Major      | Chief Officer           | Deputy     | Chief Scientist  | Commander            | Chief             |
| 10-12 | 6    | Colonel    | Captain                 | Boss       | Project Manager  | Captain              | Manager           |

### 12 – SPECIAL POWERS

Some campaign concepts may come with the expectation that characters have either retrogenics, cybernetics or mind powers. Here are a few examples:

**CYBERCOPS** – Cops brought back from the dead and rebuilt with cybernetics to enforce the law (and the secret goals of the cyber-tech corporation). Characters are Agents. All characters begin with 20 points of Cybernetics.

**MUTANT BOUNTY HUNTERS** – Mutant scum that have only one job available to them: interstellar bounty hunter. Characters are a mix of Rogues, Mercenaries and Agents. All characters begin with 20 points of Retrogenics, and a physical deformity or unusual appearance.

**FIRST EARTH BATTALION** – Suited to the *Bad Moon Rising* campaign setting in **Cepheus Universal**, the unit (actually platoon-sized) is composed of US special forces soldiers in a psionic mental powers unit. All are fluent Russian speakers. The team-members act as international superspies and commandos, waging a secret war against the Soviets in an alternate 1970s when a limited war is being fought between the superpowers on the Moon's surface and both the Moon's and the Earth's orbit. Characters are Mercenaries or Agents. All characters begin with 20 points to spend on psionic talents and levels of Psionic Strength.

### 13 – FINAL DETAILS

#### CASH

Every character begins with some starting cash (the generic interstellar currency in these rules is the Credit, with a million credits represent by 'MCr'). Simply multiply Soc by  $1D6 \times 500$ . My scientist, with Soc 6, rolls 4, and receives  $6 \times 2000$  for a total for Cr12,000. If the characters have Mustered Out (officially left their career to work freelance) then they must buy any equipment they need. If the player characters are still in their careers, then they will have access to equipment used in that job, although they will not own it.

#### OWNING A STARSHIP

Some campaign concepts may require the characters to own, or command, a starship. If so, the GM should provide a suitable vessel in order that the player characters can trade, travel or explore. If the GM would rather leave it to Fate, then any Rank 3 Merchant or Explorer (or higher) can roll 2D6. On an unmodified 10+ that character owns outright a 200-ton frontier trader (or equivalent).

#### BUYING EQUIPMENT

Weapons and all kinds of equipment can be bought using the lists provided in the Equipment chapter.

## CARRYING GEAR

Strength determines how much weight that characters can lift and how much they are slowed down. Characters carrying more than their Light Load suffer a penalty to climbing, throwing, jumping and other purely physical activities (including hand-to-hand combat). Most tasks such as shooting, driving, navigating or fixing machinery will not be affected. Use these rules when they matter, not all of the time. Medium and Heavily Loaded characters will also suffer fatigue faster than Lightly Loaded characters – see the Fatigue rules on page 104.

- **Light Load (Up to 2x Strength):** A character can lift and carry up to twice their Strength characteristic score in kilogrammes without any penalties. Speed is unaffected.
- **Medium Load (Up to 4x Strength):** This is four times their Strength characteristic score in kilogrammes. Characters carrying a medium load suffer a DM-1 to all physically based checks. In addition, they move at 75% of their base speed.
- **Heavy Load (Up to 6x Strength):** This is six times their Strength characteristic score in kilogrammes. Characters carrying a heavy load suffer a DM-2 to all physically based checks. In addition, they move at 75% of their base speed.

## 14 – DESIGNED LIFEPAATH

On page 42 is a randomised Lifepath system, in which the player rolls dice to create turning points in their character's life: good events, tragedies, love affairs, friendships and animosities. Although it is best used as a random, dice-based tool, the player wanting to design all aspects of their character can use it to add more depth and backstory to their character. To use the Lifepath for a designed character, follow the steps below:

**1 – Select Homeworld** (page 42)

**2 – Select Parents and Standing.** Use the tables as directed, make your selection.

**3 – Select Siblings.** Use the tables as directed, make your selection.

**4 – Life Events.** There are a variety of random 'Lucky Breaks' and 'Tragedies' in the Lifepath. Here you will choose up to three Good Events from the table below, but each Good Event results in having to choose a Bad Event. You are under no obligation to choose a Good Event. You may select the same type of Good or Bad Event more than once, although you can only choose 'gain wealth' only once, and it should come with some kind of disadvantage or limitation. The Good and Bad Events are described in the Lifepath system in the Lucky Breaks, Tragedies, Tragic Love Affair and Romance sections (page 45).

| LIFEPAATH EVENTS |                    |
|------------------|--------------------|
| GOOD EVENTS      | BAD EVENTS         |
| Friendship       | Enemy              |
| Luck Break       | Tragedy            |
| Romance          | Tragic Love Affair |

## 2 – FULLY RANDOM CREATION

You may well prefer a fully random, or semi-random, method of creating your characters. In this section, we provide an optional randomised system, with two approaches: 1) Roll everything randomly and explain what or who the resultant character is; or 2) Decide on a profession or role, then select the Skill Category tables to roll on.

### DETAILS

Determine Steps 1-4 (pages 32-33), and Step 11 (page 38).

### CHARACTERISTICS

For random creation, roll each characteristic with 2D6 and keep the result. For semi-random, roll randomly, then swap the values of two characteristics, if desired.

### SKILLS

A player will have **six rolls** to divide up between TWO Skill Category tables, as the player sees fit. For random creation, roll to determine those two tables on 1D6 (re-roll if you get the same table twice). When it comes to rolling for the actual skills, the player can roll odd or evens for each skill roll, to determine which table to roll on. For semi-random, decide on your character's career or role, then pick two Skill Categories that best fit that concept. Roll randomly on those tables as desired. If the same skill is rolled again, this indicates an increase in skill level. We limit the player to only two tables to try and prevent every character ending up with 1 level in every skill he possesses.

| SKILL CATEGORIES |              |
|------------------|--------------|
| 1D6              | Table to Use |
| 1                | COMBAT       |
| 2                | PEOPLE       |
| 3                | SPACE OPS    |
| 4                | TECHNICAL    |
| 5                | CIVIL        |
| 6                | WILDERNESS   |

| 1 - COMBAT |               |
|------------|---------------|
| 1D6        | Skill         |
| 1          | Tactics       |
| 2          | Gun Cbt       |
| 3          | Gun Cbt       |
| 4          | Melee Cbt     |
| 5          | Heavy Weapons |
| 6          | Demolitions   |

| 2 - PEOPLE |            |
|------------|------------|
| 1D6        | Skill      |
| 1          | Bribery    |
| 2          | Leader     |
| 3          | Carousing  |
| 4          | Admin      |
| 5          | Streetwise |
| 6          | Forgery    |

| 3 – SPACE OPS |                   |
|---------------|-------------------|
| 1D6           | Skill             |
| 1             | Gunnery or Loader |
| 2             | Pilot             |
| 3             | Navigation        |
| 4             | Vacc Suit         |
| 5             | Computer          |
| 6             | Medical           |

| 4 - TECHNICAL |             |
|---------------|-------------|
| 1D6           | Skill       |
| 1             | Comms       |
| 2             | Mechanical  |
| 3             | Electronics |
| 4             | Computer    |
| 5             | Security    |
| 6             | Engineering |

| 5 - CIVIL |                       |
|-----------|-----------------------|
| 1D6       | Skill                 |
| 1         | Vehicle               |
| 2         | Computer              |
| 3         | Mining or Agriculture |
| 4         | Medical               |
| 5         | Investigate           |
| 6         | Loader                |

| 6 - WILDERNESS |                      |
|----------------|----------------------|
| 1D6            | Skill                |
| 1              | Survival             |
| 2              | Navigation           |
| 3              | Leader               |
| 4              | Recon                |
| 5              | Medical              |
| 6              | Vehicle or Vacc Suit |

**Bonus Skill** – Also, remember to roll for a random Bonus Skill (page 38).



## LIFEPATH

If the player wants to expand their character's back story, then they might want to use the following lifepath system, inspired by a classic cyberpunk roleplaying game. It begins by establishing the type of planet the character originated from, looks at their family background, and then goes on to suggest key events from the character's life so far.

Follow the numbered tables, 1, 2, 3, 4 and 5, and roll on sub-tables as directed.

### 1 - HOMEWORLD

Roll 2D6 to determine the type of world that your character grew up on. This will determine the type of environment and Tech Level (TL) that they will be most familiar with. Roll the Tech Level using a 1D3, 1D6 or 2D6 as directed. For those campaigns set on some future Earth, then instead roll on the Future Earth table to determine the type of environment your character was brought up in.

| HOMEWORLD |                                  |
|-----------|----------------------------------|
| 2D6       | Result                           |
| 2-3       | Garden – Low Tech Level (TL 1D3) |
| 4         | Asteroid Belt (TL 1D6+9)         |
| 5         | Desert (TL 2D6+3)                |
| 6         | Garden – Mid Tech (TL 1D6+3)     |
| 7-8       | Garden – High Tech (TL 1D6+9)    |
| 9         | Waterworld (TL 2D6+3)            |
| 10-12     | Hellhole (TL 1D6+9)              |

| FUTURE EARTH |                        |
|--------------|------------------------|
| 2D6          | Result                 |
| 2-3          | Harsh off-world colony |
| 4            | Orbital habitat        |
| 5-7          | Metropolitan sprawl    |
| 8-9          | Crowded arcology       |
| 10-11        | Rural town             |
| 12           | Seabed community       |

### 2 - PARENTS

Roll 2D6 to determine the status of your parents.

| PARENTS |                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------|
| 2D6     | Result                                                                                               |
| 2-3     | Orphan (go to SURROGATE)                                                                             |
| 4-5     | Abandoned (go to SEPARATED, then SURROGATE)                                                          |
| 6-8     | Alive (go to STANDING)                                                                               |
| 9-12    | One is Absent (roll randomly for which); roll 1D6 to determine reason: 1-3 CASUALTY<br>4-6 SEPARATED |

### STANDING

How has your family coped. What are they like?

| STANDING |                                                     |
|----------|-----------------------------------------------------|
| 2D6      | Result                                              |
| 2        | Parent/parents were difficult and nasty individuals |
| 3        | They are strangers in a strange land                |
| 4        | Constantly on the move to earn a living             |
| 5        | Dirt poor and struggling                            |
| 6        | Incredibly wealthy                                  |
| 7        | Average, they get by just fine                      |
| 8        | Life was really easy, too easy                      |
| 9        | A poor family made good                             |
| 10       | High status family, but lost its wealth             |
| 11-12    | Bound into a life of crime and trouble              |

## CASUALTY

Somebody has died. How?

| CASUALTY |                                    |
|----------|------------------------------------|
| 2D6      | Result                             |
| 2        | Assassinated by the government     |
| 3        | Assassinated by other organisation |
| 4-5      | Killed in an accident              |
| 6        | Innocent casualty of war           |
| 7        | Died of illness                    |
| 8        | Murdered in random act             |
| 9        | Targeted murder                    |
| 10       | Killed in terror attack            |
| 11       | Died of disease/outbreak           |
| 12       | Suicide                            |

## 3 – SIBLINGS

How many brothers & sisters do you have?

| SIBLINGS         |        |
|------------------|--------|
| 2D6              | Result |
| 2-4              | None   |
| 5-6              | 1      |
| 7-8              | 2      |
| 9-10             | 3      |
| 11-12            | 4      |
| 13+              | 5      |
| Homeworld TL 1-3 | +2     |
| Homeworld TL 10+ | - 2    |

## SIBLING RELATIONS

Roll for relations with each sibling.

| SIBLING RELATIONS |                                    |
|-------------------|------------------------------------|
| 2D6               | Result                             |
| 2-6               | Traditional sibling love & rivalry |
| 7                 | Bitter rival                       |
| 8                 | Sibling looks up to you as a hero  |
| 9                 | Sibling holds a grudge             |
| 10                | Sibling is dead (go to CASUALTY)   |
| 11                | Sibling hates you                  |
| 12                | Sibling absent ( go to SEPARATED)  |

## SEPARATED

Somebody is no longer there. Why?

| SEPARATED |                                    |
|-----------|------------------------------------|
| 2D6       | Result                             |
| 2         | Amnesia                            |
| 3-4       | Vanished without a trace           |
| 5         | Kidnapped                          |
| 6         | In hiding – why?                   |
| 7-8       | Separation by work or disagreement |
| 9-10      | Consumed by addition/obsession     |
| 11-12     | Psychological problems             |

## SURROGATE

Who brought you up?

| SURROGATE |                                     |
|-----------|-------------------------------------|
| 2D6       | Result                              |
| 2         | Raised in peaceful religion         |
| 3         | Raised by street gang               |
| 4-5       | Raised by the state                 |
| 6-8       | Raised by relatives (go to PARENTS) |
| 9-10      | Adopted (go to PARENTS)             |
| 11        | Entered a military organisation     |
| 12        | Raised in unusual circumstance      |

## 4 – LIFE EVENTS

Roll for 1D6+2 Life Events.

| LIFE EVENTS |                   |
|-------------|-------------------|
| 2D6         | Result            |
| 2-5         | Go to LUCKY BREAK |
| 6           | Go to ROMANCES    |
| 7           | Go to FRIENDSHIPS |
| 8           | Go to ENEMIES     |
| 9-12        | Go to TRAGEDIES   |

## ENEMIES

Roll for an Enemy.

| ENEMY |                           |
|-------|---------------------------|
| 2D6   | Result                    |
| 2-3   | Old friend                |
| 4     | Criminal                  |
| 5     | Relative                  |
| 6     | Ex-lover                  |
| 7     | Co-worker                 |
| 8     | One of your bosses        |
| 9-10  | Powerful official         |
| 11-12 | A victim of your employer |

## WHO IS UPSET?

Who is angry with the other?

| WHO IS UPSET? |             |
|---------------|-------------|
| 2D6           | Result      |
| 2-5           | You are     |
| 6-8           | They are    |
| 9-12          | It's mutual |

## LUCKY BREAK

Roll for a Lucky Break.

| LUCKY BREAK |                                                                            |
|-------------|----------------------------------------------------------------------------|
| 2D6         | Result                                                                     |
| 2-4         | Gain recognition, fame and a name for yourself.                            |
| 5           | Gain some unusual training, not normally available to you. One level.      |
| 6           | You are owed a great favour by a contact, willingly or reluctantly.        |
| 7           | You gain wealth from an unexpected source (Cr20,000, plus 1D6 x Cr10,000). |
| 8           | Gain an increase in status, rank or position of authority.                 |
| 9           | Gain a useful contact in an organisation that can supply information.      |
| 10-12       | Gain a powerful friend who can trade assistance, for a favour in return.   |

## FRIENDSHIPS

Roll for a Friendship.

| FRIENDSHIPS |                               |
|-------------|-------------------------------|
| 2D6         | Result                        |
| 2-3         | Relative                      |
| 4           | Childhood friend              |
| 5-6         | Share common history/interest |
| 7-8         | Co-worker                     |
| 9           | Past romantic interest        |
| 10          | Mentor, teacher or old boss   |
| 11-12       | Old enemy or rival            |

## CAUSE of ENMITY

Roll for the cause of the Enmity.

| CAUSE of ENMITY |                                     |
|-----------------|-------------------------------------|
| 2D6             | Result                              |
| 2               | Opposed the other's plan            |
| 3               | Caused other a disability or scar   |
| 4               | Caused other to be humiliated       |
| 5               | Caused other to lose relative/lover |
| 6               | Caused other lose status            |
| 7               | Betrayed or deserted the other      |
| 8-9             | Caused other to lose their job      |
| 10              | Spurned offer of job or love        |
| 11-12           | Caused other to be imprisoned       |

## HOW UPSET?

How upset is the character?

| HOW UPSET? |                                    |
|------------|------------------------------------|
| 2D6        | Result                             |
| 2-5        | Avoid and ignore one another       |
| 6-7        | Verbal put downs/arguments         |
| 8-9        | Sabotage their life and livelihood |
| 10-12      | Cause them physical harm           |

## TRAGEDIES

Roll for a personal Tragedy.

| TRAGEDIES |                                    |
|-----------|------------------------------------|
| 2D6       | Result                             |
| 2-3       | Hunted by criminals                |
| 4         | Falsely accused                    |
| 5         | Bout of illness or addiction       |
| 6         | Debt or financial ruin             |
| 7         | Psychological trauma               |
| 8         | Lost a career or status in society |
| 9         | Responsible for a tragedy          |
| 10        | Injured in an accident             |
| 11-12     | Hunted by the authorities          |

## TRAGIC AFFAIR

Why was the affair so tragic?

| TRAGIC AFFAIR |                                              |
|---------------|----------------------------------------------|
| 2D6           | Result                                       |
| 2-5           | They died (go to CASUALTY)                   |
| 6-8           | It was intense - but didn't work out         |
| 9-10          | Vanished (go to SEPARATED)                   |
| 11-12         | A 3 <sup>rd</sup> person broke up the affair |

## HOW DO YOU FEEL?

If you had issues, or the affair was tragic and the other person is still around, how do you feel about one another?

| HOW DO YOU FEEL? |                              |
|------------------|------------------------------|
| 2D6              | Result                       |
| 2-3              | They hate you                |
| 4                | They like you, you hate them |
| 5                | You still love them          |
| 6                | You are distant friends      |
| 7                | No feelings, it's over       |
| 8                | You hate each other          |
| 9                | They still love you          |
| 10               | You like them, they hate you |
| 11-12            | You hate them                |

## ROMANCES

Roll for a romantic relationship.

| ROMANCES |                    |
|----------|--------------------|
| 2D6      | Result             |
| 2-5      | TRAGIC AFFAIR      |
| 4-6      | Happy Affair       |
| 7-8      | Affair with ISSUES |
| 9-12     | Brief Fling        |

## ISSUES

What issues did your 'Affair with Issues' have?

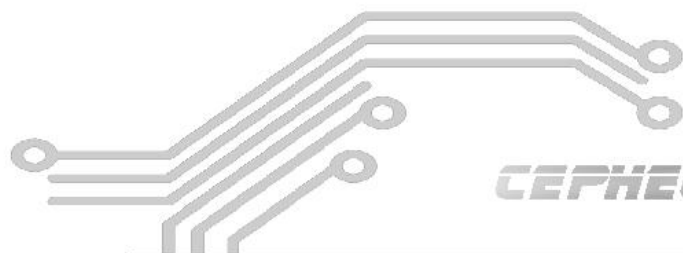
| ISSUES |                                     |
|--------|-------------------------------------|
| 2D6    | Result                              |
| 2-4    | Had conflicting lives/backgrounds   |
| 5      | Your friends/family hate your lover |
| 6      | Lover's friends/family hate you     |
| 7      | You were professional rivals        |
| 8-9    | You fought constantly               |
| 10     | One of you had an affair            |
| 11-12  | There was a lot of jealousy         |

## 5 – REASON FOR LEAVING HOMEWORLD

Why has the character sought a life amongst the stars?

| REASON FOR LEAVING HOMEWORLD |                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------|
| D66                          | The Reason                                                                                                       |
| 11                           | Attracted by tales of adventure and daring-do amongst the stars.                                                 |
| 12-13                        | Learnt something you shouldn't know – you fled for your own safety.                                              |
| 14                           | You are following your father's or mother's footsteps.                                                           |
| 15-16                        | Earn money for your impoverished family.                                                                         |
| 21                           | Escape a crushing family situation.                                                                              |
| 22                           | Your partner moved off-world years ago, and you are desperate to find them.                                      |
| 23                           | You are punishing yourself with hardship, privation and danger. Why?                                             |
| 24                           | Your opinions are not popular in your home nation and so you fled to the colonies where you can speak your mind. |
| 25                           | Escape a tragic love affair – or a vindictive stalker!                                                           |
| 26                           | You want to make your parents proud of you after several screw-ups.                                              |
| 31-33                        | The job pays good, steady money, plus there's fewer people and a lot of wilderness.                              |
| 34-35                        | Attempting to discover the fate of a missing relative or lover on the Frontier.                                  |
| 36-41                        | Trying to find an off-worlder who committed a crime against your family.                                         |
| 42-43                        | Home city or community devastated by pandemic, eco-crisis or social unrest.                                      |
| 44                           | Trying to pay off bad debts of Cr50,000                                                                          |
| 45-51                        | Caught up in bad political turmoil back on a Core world.                                                         |
| 52                           | Spent time in prison – now making a new start where freedom is guaranteed. Or is it?                             |
| 53-54                        | Hiding in the colonies from the authorities under a false ID.                                                    |
| 55-56                        | Pursued by agents of a crime gang who seek revenge. For what?                                                    |
| 61                           | Tricked into working for a crime gang – fleeing to escape their hold over you.                                   |
| 62-63                        | Committed serious crime, so you now operate amongst the colonies.                                                |
| 64-66                        | Attracted by the rumours of huge fortunes to be made by opportunistic risk-takers.                               |





# CEPHEUS UNIVERSAL

## TERM WORKSHEET 1/2

NAME

|        |      |      |      |      |      |      |
|--------|------|------|------|------|------|------|
| Career | STR  | DEX  | END  | INT  | EDU  | SOC  |
|        | Mod: | Mod: | Mod: | Mod: | Mod: | Mod: |

|         |     |
|---------|-----|
| Species | Sex |
|         |     |

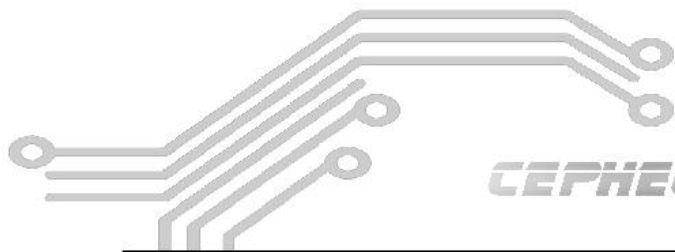
|              |  |
|--------------|--|
| Skill Tables |  |
|              |  |

|        |        |
|--------|--------|
| Height | Weight |
| cm     | kg     |

|            |
|------------|
| Origins    |
| Homeworld: |
| Family:    |
| Siblings:  |

|                 |                       |
|-----------------|-----------------------|
| TERM 1          |                       |
| Career          | Commission (y/n)      |
| Risk (y/n)      |                       |
| Promotion (y/n) |                       |
| Skill Gains     |                       |
| Notes           |                       |
| Remain? (y/n)   | Rank (Number/Title) / |

|                 |                       |
|-----------------|-----------------------|
| TERM 2          |                       |
| Career          | Commission (y/n)      |
| Risk (y/n)      |                       |
| Promotion (y/n) |                       |
| Skill Gains     |                       |
| Notes           |                       |
| Remain? (y/n)   | Rank (Number/Title) / |



**CEPHEUS UNIVERSAL**

**TERM WORKSHEET 2/2**

**NAME**

|                                  |                       |
|----------------------------------|-----------------------|
| <b>TERM</b> <input type="text"/> |                       |
| Career                           | Commission (y/n)      |
| Risk (y/n)                       |                       |
| Promotion (y/n)                  |                       |
| Skill Gains                      |                       |
| Notes                            |                       |
| Remain? (y/n)                    | Rank (Number/Title) / |

|                                  |                       |
|----------------------------------|-----------------------|
| <b>TERM</b> <input type="text"/> |                       |
| Career                           | Commission (y/n)      |
| Risk (y/n)                       |                       |
| Promotion (y/n)                  |                       |
| Skill Gains                      |                       |
| Notes                            |                       |
| Remain? (y/n)                    | Rank (Number/Title) / |

|                                  |                       |
|----------------------------------|-----------------------|
| <b>TERM</b> <input type="text"/> |                       |
| Career                           | Commission (y/n)      |
| Risk (y/n)                       |                       |
| Promotion (y/n)                  |                       |
| Skill Gains                      |                       |
| Notes                            |                       |
| Remain? (y/n)                    | Rank (Number/Title) / |

**BENEFITS**

**Tally:**

## 3 – TERM BY TERM CREATION

The term-by-term creation method in this section is essentially a character creation mini game. Although you as the player make some basic decisions for your character, the dice and the character's Fate will ultimately create the character's past. Every four year period (a 'term'), dice are rolled to overcome Risk to the character, to seek Promotion and to Remain in the career that has been chosen. Each completed term will provide one or more skill levels that are pertinent to that career, but continued service has its risks. Failing a Risk roll might force the character to leave the career under a cloud, or risk injury or some other negative consequence. Remaining in the career beyond the third term also means checking for aging effects on the character. At some point the player decides to pull the character out of the creation process and begin play, or the character is ejected from the career due to failing to Remain.

This method of character does not suit everyone. You need to begin the process with the open-minded attitude of "I wonder what my character will be like at the end of all this". It is almost a little solo game, an experience to enjoy, but it takes more time than methods 1 and 2, and requires both patience and imagination, as well as a willingness to join the dots, take the clues provided and build a narrative that explains the character's past and how that might manifest in the present-day character at the game table.

### 1 – DETERMINE CHARACTERISTICS

Roll 2D6 six times, once each for Strength, Dexterity, Endurance, Intelligence, Education and Social Influence. You may swap around two values with one another, if desired, or leave the six values as they stand. Look at these and understand what kind of character you have. Use the 'What Do Characteristics Represent' guidance on page 26 to help you. Example: *My characteristics are 664C55. I decide to swap the 12 in Int for the 4 in End, and so now his characteristics are: 66C455.*

### 2 – NAME & SEX

Determine these two fundamental points of identity for your character at this stage.

### 3 – ORIGINS

Randomly determine the character's homeworld, family situation and siblings (Steps 1-3 of the Lifepath featured on page 42). We will determine the character's Psychic Evaluation later, once we understand the kind of life he or she has led.

### 4 – CAREER CHOICE

Based on the setting and also the campaign concept that this character is being created for, select one of the careers from those listed below. The careers are broadly representative of the types of occupations and activities that feature in science fiction movies, TV shows and novels. Note that some ranks have been expanded or renamed in this procedure to better represent a progression.

CAREER FORMAT

Each career listing follows the same format. Following its title and description, there are six listings.

**Preferred Characteristic:** Having this at 10+ provides a +1 to all Risk rolls.  
**Auto Skill:** Gain an automatic level-1 in this skill upon joining the career.  
**Risk:** Either Safe (5+), Standard (6+) or Hazardous (7+); this roll is made once per term.  
**Promotion:** Either Standard (6+), Competitive (8+) or Cut-throat (9+); this roll is made once per term.  
**Cash:** A base amount, used when calculating cash benefits at the end of the career.  
**Skill Tables:** Two skill tables are listed here. When a skill is acquired, roll on one of these tables, as desired.  
**Ranks:** A table of six ranks is included. Military careers will have two rank tables, one for Enlisted personnel, another for Commissioned.

AGENT

A law enforcer, secret agent, spy or police officer.

**Preferred Characteristic:** Soc

**Auto Skill:** Investigate

**Risk:** Hazardous 7+

**Promotion:** Competitive 8+

**Cash:** Cr10,000

**Skill Tables:** Military, Crime

| Ranks |                        |
|-------|------------------------|
| 1     | Detective/Agent        |
| 2     | Senior Detective/Agent |
| 3     | Lieutenant             |
| 4     | Captain                |
| 5     | Commander              |
| 6     | Chief                  |

BELTER

An asteroid miner, pioneer, prospector or colonial roughneck.

**Preferred Characteristic:** Str

**Auto Skill:** Mining

**Risk:** Hazardous 7+

**Promotion:** Standard 6+

**Cash:** Cr20,000

**Skill Tables:** Space Ops, Civil

| Ranks |                 |
|-------|-----------------|
| 1     | Miner           |
| 2     | Senior Miner    |
| 3     | Supervisor      |
| 4     | Team Leader     |
| 5     | Superintendent  |
| 6     | General Manager |

CIVILIAN

A colonist or civilian, representing one of many occupations.

**Preferred Characteristic:** Edu

**Auto Skill:** Vehicle

**Risk:** Safe 5+

**Promotion:** Standard 6+

**Cash:** Cr6,000

**Skill Tables:** People, Civil

| Ranks |                 |
|-------|-----------------|
| 1     | 'Worker'        |
| 2     | Senior 'Worker' |
| 3     | Supervisor      |
| 4     | Assistant Chief |
| 5     | Chief 'Worker'  |
| 6     | Manager         |

EXPLORER

A survey scout, deep space explorer, first-in mission specialist.

**Preferred Characteristic:** End

**Auto Skill:** Survival

**Risk:** Hazardous 7+

**Promotion:** Competitive 8+

**Cash:** Cr12,000

**Skill Tables:** Wilderness, Space Ops

| Ranks |                      |
|-------|----------------------|
| 1     | Scout                |
| 2     | Senior Scout         |
| 3     | Lieutenant           |
| 4     | Lieutenant Commander |
| 5     | Commander            |
| 6     | Commodore            |

FIXER

A deal-maker, middle-man, executive, sleazy lawyer, street fixer.

**Preferred Characteristic:** Soc

**Auto Skill:** Broker

**Risk:** Safe 5+

**Promotion:** Cut-Throat 9+

**Cash:** Cr24,000

**Skill Tables:** Business, People

| Ranks |                  |
|-------|------------------|
| 1     | Hustler          |
| 2     | Sales Rep        |
| 3     | Senior Sales Rep |
| 4     | Manager          |
| 5     | Executive        |
| 6     | Senior Executive |

MARINE

Mobile infantry, space marines, ship's troops or star commandos.

**Preferred Characteristic:** Str

**Auto Skill:** Gun Combat

**Risk:** Hazardous 7+

**Promotion:** Competitive 8+

**Cash:** Cr10,000 (officer Cr20,000)

**Skill Tables:** Combat, Wilderness

| Enlisted Ranks     |                    |
|--------------------|--------------------|
| 1                  | Marine             |
| 2                  | Corporal           |
| 3                  | Sergeant           |
| 4                  | Staff Sergeant     |
| 5                  | Gunnery Sergeant   |
| 6                  | Sergeant Major     |
| Commissioned Ranks |                    |
| 1                  | Lieutenant         |
| 2                  | Captain            |
| 3                  | Major              |
| 4                  | Lieutenant Colonel |
| 5                  | Colonel            |
| 6                  | Brigadier          |



MERCHANT

Interstellar truckers, haulage crews, making money between the stars.  
**Preferred Characteristic:** Edu  
**Auto Skill:** Vacc Suit  
**Risk:** Standard 6+  
**Promotion:** Cut-Throat 9+  
**Cash:** Cr10,000  
**Skill Tables:** Space Ops, Business

| Ranks |                                |
|-------|--------------------------------|
| 1     | Crewman                        |
| 2     | Fourth Officer                 |
| 3     | Third Officer                  |
| 4     | Second Officer                 |
| 5     | First Officer or Chief Officer |
| 6     | Captain                        |

PRIMITIVE

Inhabitant of a low tech, primitive world, barbarian, savage.  
**Preferred Characteristic:** End  
**Auto Skill:** Survival  
**Risk:** Hazardous 7+  
**Promotion:** Competitive 8+  
**Cash:** Cr2,000  
**Skill Tables:** Low Tech, Wilderness OR People (select at start of career)

| Ranks |           |
|-------|-----------|
| 1     | Tribesman |
| 2     | Warrior   |
| 3     | Veteran   |
| 4     | Hero      |
| 5     | Chief     |
| 6     | War Chief |

ROGUE

A criminal, thief, gang-member or saboteur.  
**Preferred Characteristic:** Dex  
**Auto Skill:** Streetwise  
**Risk:** Hazardous 7+  
**Promotion:** Competitive 8+  
**Cash:** Cr7,000  
**Skill Tables:** Crime, Business

| Ranks |            |
|-------|------------|
| 1     | Soldier    |
| 2     | Enforcer   |
| 3     | Lieutenant |
| 4     | Captain    |
| 5     | Deputy     |
| 6     | Boss       |

MERCENARY

Ground-based infantry, either regular army or mercenaries.  
**Preferred Characteristic:** Str  
**Auto Skill:** Gun Combat  
**Risk:** Hazardous 7+  
**Promotion:** Standard 6+  
**Cash:** Cr6,000 (officer Cr12,000)  
**Skill Tables:** Combat, Wilderness

| Enlisted Ranks     |                    |
|--------------------|--------------------|
| 1                  | Private            |
| 2                  | Corporal           |
| 3                  | Sergeant           |
| 4                  | Staff Sergeant     |
| 5                  | Master Sergeant    |
| 6                  | Sergeant Major     |
| Commissioned Ranks |                    |
| 1                  | Lieutenant         |
| 2                  | Captain            |
| 3                  | Major              |
| 4                  | Lieutenant Colonel |
| 5                  | Colonel            |
| 6                  | General            |

SCAVENGER

A post-apocalyptic road warrior, survivor or resistance fighter.

**Preferred Characteristic:** Str

**Auto Skill:** Ground Vehicle

**Risk:** Hazardous 7+

**Promotion:** Competitive 8+

**Cash:** Cr2,000

**Skill Tables:** Combat, Wilderness

| Ranks |           |
|-------|-----------|
| 1     | Scavenger |
| 2     | Warrior   |
| 3     | Veteran   |
| 4     | Hero      |
| 5     | Legend    |
| 6     | -         |

SCIENTIST

Anything from a doctor to a geologist, physicist to bioweapon expert.

**Preferred Characteristic:** Edu

**Auto Skill:** Investigate

**Risk:** Safe 5+

**Promotion:** Standard 6+

**Cash:** Cr10,000

**Skill Tables:** Science, Technical OR People (select at start of career)

| Ranks |                             |
|-------|-----------------------------|
| 1     | Researcher                  |
| 2     | Team Leader                 |
| 3     | Scientist                   |
| 4     | Senior Scientist            |
| 5     | Chief Scientist             |
| 6     | Project Manager/ Dept. Head |

TECHNICIAN

A gearhead, engineer, hacker, mechanic, cyborg technician, etc.

**Preferred Characteristic:** Int

**Auto Skill:** Computer

**Risk:** Safe 5+

**Promotion:** Standard 6+

**Cash:** Cr8,000

**Skill Tables:** Technical, Civil

| Ranks |                         |
|-------|-------------------------|
| 1     | Technician Third Class  |
| 2     | Technician Second Class |
| 3     | Technician First Class  |
| 4     | Assistant Chief         |
| 5     | Chief                   |
| 6     | Technical Manager       |

SPACER

Military crewman or officer, member of the interstellar navy or space force.

**Preferred Characteristic:** Int

**Auto Skill:** Vacc Suit

**Risk:** Hazardous 7+

**Promotion:** Competitive 8+

**Cash:** Cr10,000 (officer Cr20,000)

**Skill Tables:** Space Ops, Technical

| Enlisted Ranks     |                            |
|--------------------|----------------------------|
| 1                  | Crewman                    |
| 2                  | Senior Crewman             |
| 3                  | Petty Officer              |
| 4                  | Senior Petty Officer       |
| 5                  | Chief Petty Officer        |
| 6                  | Master Chief Petty Officer |
| Commissioned Ranks |                            |
| 1                  | Ensign                     |
| 2                  | Lieutenant                 |
| 3                  | Lieutenant Commander       |
| 4                  | Commander                  |
| 5                  | Captain                    |
| 6                  | Admiral                    |

5 - GAIN AUTO SKILL

Once the character enters a career, he or she gains the auto skill that is listed, treat this as a basic training qualification. Gain one level of the skill.

6 – ROLL RISK

Each four year term, Risk is rolled on 2D6. This is a measure of both physical and professional challenge, remember that if the character has the career’s Preferred Characteristic at 10+, then they get a DM +1 on the Risk roll every term. Risk has three difficulty levels that vary between careers.

|           |    |
|-----------|----|
| Safe      | 5+ |
| Standard  | 6+ |
| Hazardous | 7+ |

Whether you made the roll or failed it, pay careful attention to the amount you succeeded or failed (we call this the ‘Effect’). Did you fail by -3 points? Did you succeed by +1 point? Now go and roll 2D6 on the Risk table and add the Effect (whether positive or negative). If you failed by 3 points, roll 2D6+3 on the Risk *Fail* table; if you succeeded by 4 points, roll 2D6+4 on the Risk *Success* table. Generally, failure means something bad happened and success means something good happened. You may gain a friend or enemy, and you will need to use your imagination a little to apply the result to your specific career. Be flexible, particularly if you roll the same result again on a subsequent term. Example: *For a Marine, "Fast dates and partying. Gain Carousing-1", might be interpreted as a memorable shore leave on some pleasure world with his comrades.*

7 – ROLL COMMISSION

Most careers can skip this phase, since it only applies to the military careers of Mercenary, Marine and Spacer. Only those careers have TWO rank tracks (enlisted and commissioned). The default rank track to use is that of the enlisted ranks. Every term, including the first, however, a character with Education 8+ can choose to make a Commission roll of 9+ in an attempt to be selected for Officer Candidate School, and then switch immediately to the commissioned rank track. If successful, gain an extra skill roll in the term you were commissioned, and become a Rank 1 officer. Your Cash Bonus value is doubled. Once commissioned you no longer roll for commission, but *do* continue to make Promotion rolls as normal (including in the term you were commissioned).

8 – ROLL PROMOTION

All careers have the opportunity for advancement. This promotion provides status, but also a free skill gain that term, and it also contributes to the character’s Benefits total once the character ends their career. Each four year term, Promotion is rolled on 2D6. Promotion has three difficulty levels that vary between careers. Just as with the Risk roll, pay careful attention to the amount you succeeded or failed (the ‘Effect’). Next, roll 2D6 on the Risk table, and add the Effect (whether positive or negative). If you failed by 3 points, roll 2D6+3 on the Promotion *Fail* table; if you succeeded by 4 points, roll 2D6+4 on the Promotion *Success* table. Generally, failure means no promotion, with reason or consequence, and success means you were

|             |    |
|-------------|----|
| Standard    | 6+ |
| Competitive | 8+ |
| Cut-throat  | 9+ |

promoted, often with an added bonus. You may even gain a friend or enemy. As with Risk, you may have to use your imagination to apply the result to your specific career.

9 – GAIN SKILLS

The character gains one skill at the end of each term, except the first where TWO skills are rolled for (in addition to the Auto Skill). A successful promotion and a successful commission also provide one additional skill each. Roll for these skill gains at the end of the term in which they were received; divide the rolls as you desire between the two skill tables listed for your career. The first time a skill is received it is taken at level-1. Subsequent gains of that same skill add one to the skill level. Rolling the Pilot skill three times over several terms,for example, results in the character having Pilot-3.

| SKILL ELIGIBILITY       |   |
|-------------------------|---|
| Initial term of service | 2 |
| Each additional term    | 1 |
| Receiving commission    | 1 |
| Receiving promotion     | 1 |

10 – REMAIN

At the end of each term, the player rolls to see if the character continues in the career or leaves it, to begin play at the roleplaying table. Roll UNDER 12 on 2D6, but add the number of terms already served (including the current one!). A character ending his fourth term, for example, would roll 2D6+4 and hope for a result under 12, we roll 8 for a total of 12, which means that character leaves the career ready for play. If the roll is failed, go to 11-LEAVING THE CAREER. Otherwise begin a new term and return to 6-ROLL RISK. If you failed your roll, check the Reasons for Leaving table to discover why you no longer work for your employer, follow that career path, or have moved on to greener pastures. Or, you may already have some idea based on the events and mishaps that occurred during the character’s career.

You might spend a few moments before moving on to reflect on the term just completed, what happened? Look at the results of the Risk and Promotion rolls. What skills were acquired, and why? Were there any catastrophes? Any great achievements? What activities might the character have been pursuing in those four years, and what led to those new friends or new enemies? Part of the solo game aspect is adding imaginative ‘meat’ to the bones that the dice results provide. Have fun!

11 – AGING

The downside to surviving term after term, is the aging process. This is intended to give players who want very old, highly skilled characters, pause for thought. Aging begins at the end of the fourth term, and is checked for at the end of every subsequent term. If term 4 to 6, roll 2D6 and lose 2 characteristic points on 4 or less, or 1 characteristic point on 8 or less. If term 7 or more, roll 2D6 and lose 2 characteristic points on 6 or less, or 1 characteristic point on 10 or less. The specific physical characteristic that is lowered by 1 or 2 points is chosen by the player. For example, a sixth term Belter rolls 5 on 2D6, indicating the loss of 2 characteristic points. The player could lower Strength by 2, or perhaps Strength and Dexterity by one each, as he prefers.

| AGING ROLLS (2D6) |                              |
|-------------------|------------------------------|
| Term 4-6          | 4- Lose 2 pts; 8- Lose 1 pt  |
| Term 7+           | 6- Lose 2 pts; 10- Lose 1 pt |

Should a characteristic be lowered to 0, the character suffers an aging crisis and is very ill. Roll 8+ and apply the character's End modifier to avoid death. Success means that the characteristic reduced to zero is set at 1 and the character must leave his career at the end of that term.

## 12 – LEAVING THE CAREER

The character is ready to play. Whether or not the character has actually left their career is based on the Campaign Concept. If the campaign is focused on a mercenary squad in action, and everyone is a mercenary, then they are still working as mercenaries. 'Leaving the Career' essentially means 'leaving the character creation process' to begin play.

The character begins play with Cr1000. The character is also eligible for end-of-career bonuses. Roll on the Bonus table ONCE per term served. Characters who achieved Rank 1-2 gain +1 roll; characters who achieved Rank 3-4 gain +2 rolls; and characters who achieved Rank 5-6 gain +3 rolls.

### BENEFIT ELIGIBILITY

|                     |   |
|---------------------|---|
| Per term of service | 1 |
| If rank 1 or 2      | 1 |
| If rank 3 or 4      | 2 |
| If rank 5 or 6      | 3 |

| BENEFITS |                                                         |
|----------|---------------------------------------------------------|
| 2D6      | Benefit Gained                                          |
| 2        | +1 Dex                                                  |
| 3        | +1 Str                                                  |
| 4        | +1 Int                                                  |
| 5        | +1 Soc                                                  |
| 6        | CASH                                                    |
| 7        | CASH                                                    |
| 8        | +1 Edu                                                  |
| 9        | Weapon, or rare or unique or custom, piece of equipment |
| 10       | CASH                                                    |
| 11       | Augment 5 points (or CASH)                              |
| 12       | Augment 5 points (or +1 End)                            |

**Characteristic points:** These are improvements made during the course of the career.

**Cash:** Receive a cash amount equal to the career's cash value.

**Weapon/Equipment:** This item should be in-line with the career, and nothing worth more than Cr5,000. The item may be sentimental, or customized with a cool feature by the character.

**Augment:** If these are allowed in the setting, each Augment results in a gain of 5 points of retrogenics or cybernetics (check with the GM to discover which is more appropriate).

| REASON FOR LEAVING (1D6) |                                                |               |                                        |
|--------------------------|------------------------------------------------|---------------|----------------------------------------|
| TABLE A (1-3)            |                                                | TABLE B (4-6) |                                        |
| 1D6                      | Reason                                         | 1D6           | Reason                                 |
| 1                        | They don't need you anymore.                   | 1             | You are bored.                         |
| 2                        | You seek adventure!                            | 2             | You seek fortune!                      |
| 3                        | You know a dark secret – and fled.             | 3             | The job isn't what it was.             |
| 4                        | The career was crushing you.                   | 4             | Competition getting harder every year. |
| 5                        | You seek a missing friend or relative.         | 5             | You screwed up and bailed out.         |
| 6                        | New identity to hide from agents or criminals. | 6             | At war with your colleagues too much.  |

## 13 – PSYCH EVALUATION

At this point the player may have a good idea of what the character's personality is like (their Psych Evaluation). If not, roll on the table provided on page 33.

## SKILL TABLES

| COMBAT |               |
|--------|---------------|
| 1D6    | Skill         |
| 1      | Tactics       |
| 2      | Gun Cbt       |
| 3      | Gun Cbt       |
| 4      | Melee Cbt     |
| 5      | Heavy Weapons |
| 6      | Demolitions   |

| PEOPLE |            |
|--------|------------|
| 1D6    | Skill      |
| 1      | Bribery    |
| 2      | Leader     |
| 3      | Carousing  |
| 4      | Admin      |
| 5      | Streetwise |
| 6      | Forgery    |

| SPACE OPS |                   |
|-----------|-------------------|
| 1D6       | Skill             |
| 1         | Gunnery or Loader |
| 2         | Pilot             |
| 3         | Navigation        |
| 4         | Vacc Suit         |
| 5         | Computer          |
| 6         | Medical           |

| TECHNICAL |             |
|-----------|-------------|
| 1D6       | Skill       |
| 1         | Comms       |
| 2         | Mechanical  |
| 3         | Electronics |
| 4         | Computer    |
| 5         | Security    |
| 6         | Engineering |

| CIVIL |                       |
|-------|-----------------------|
| 1D6   | Skill                 |
| 1     | Vehicle               |
| 2     | Computer              |
| 3     | Mining or Agriculture |
| 4     | Medical               |
| 5     | Investigate           |
| 6     | Loader                |

| WILDERNESS |                      |
|------------|----------------------|
| 1D6        | Skill                |
| 1          | Survival             |
| 2          | Navigation           |
| 3          | Leader               |
| 4          | Recon                |
| 5          | Medical              |
| 6          | Vehicle or Vacc Suit |

| CRIME |              |
|-------|--------------|
| 1D6   | Skill        |
| 1     | Streetwise   |
| 2     | Forgery      |
| 3     | Melee Combat |
| 4     | Gun Combat   |
| 5     | Bribery      |
| 6     | Security     |

| BUSINESS |           |
|----------|-----------|
| 1D6      | Skill     |
| 1        | Admin     |
| 2        | Broker    |
| 3        | Carousing |
| 4        | Forgery   |
| 5        | Broker    |
| 6        | Bribery   |

| SCIENCE |             |
|---------|-------------|
| 1D6     | Skill       |
| 1       | Navigation  |
| 2       | Vehicle     |
| 3       | Vacc Suit   |
| 4       | Computer    |
| 5       | Investigate |
| 6       | Computer    |

| LOW TECH |              |
|----------|--------------|
| 1D6      | Skill        |
| 1        | Riding       |
| 2        | Archery      |
| 3        | Melee Combat |
| 4        | Recon        |
| 5        | Tactics      |
| 6        | Artisan      |

## A SECOND CAREER?

As an option, the GM might allow a character to start another career after leaving his or her first career. To enter a second career, make a Remain roll, with +1 if the PC has the Preferred Characteristic at 10+, and +1 if the PC has any levels in the Automatic skill. If using Load Outs, take the load out for the second career.

PCs using Designed Creation might also be allowed to add in a second career. Simply divide the 6 skill level between the two careers, with the caveat that one of the skills from the 'second' career must be taken at level-0.



## RISK TABLE

Roll 2D6 and add the Effect (whether positive or negative).

| FAIL  |                                                                                       |
|-------|---------------------------------------------------------------------------------------|
| 2D6   | Result                                                                                |
| 3     | You broke the rules to help out an innocent person. Gain a Contact. What did you do?  |
| 4     | You saved lives, but it had to be covered up. You cannot talk about it. Ever.         |
| 5     | Revolution, war or disaster turns your world and relationships upside down.           |
| 6     | People in power have been meddling with you, your team and your organisation.         |
| 7     | Your achievements were sabotaged by a colleague. Gain an Enemy                        |
| 8     | Get into debt of 4x Career Cash, by turning to criminal world. Why did this happen?   |
| 9     | Stressful situations cause a mental collapse. Roll 2D6, on 7 or less, lower Int by 1. |
| 10    | Great achievements stolen by a rival group or organisation, taking all the credit.    |
| 11    | You get the blame for a disaster that killed or injured several people.               |
| 12-13 | Badly injured. Lower Str, Dex or End by 1. How did it happen? Leave the Career.       |
| 14+   | Died in service.                                                                      |

| SUCCEED |                                                                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2D6     | Result                                                                                                                                                                                                                                     |
| 2       | Betrayed by a close friend, now an Enemy.                                                                                                                                                                                                  |
| 3       | Gain a useful 'friend-of-a-friend' Contact.                                                                                                                                                                                                |
| 4       | Romance that continues today, or ended on a good note. Gain a Friend.                                                                                                                                                                      |
| 5       | Uncover a dangerous secret about a person or organisation. Gain Bribery-1.                                                                                                                                                                 |
| 6       | Gain fame and recognition in your career for deserved actions. Gain +1 Soc.                                                                                                                                                                |
| 7       | An intense situation was survived with a colleague. Make a Friend.                                                                                                                                                                         |
| 8       | Fast dates and partying. Gain Carousing-1.                                                                                                                                                                                                 |
| 9       | Conflict became a part of your life. Gain +1 Str. What was going on?                                                                                                                                                                       |
| 10      | Make a good Friend in your own career or organization.                                                                                                                                                                                     |
| 11      | Tragic love affair. Roll 1D6: 1-2 They died; 3-4 They Vanished; 5-6 They are still around.                                                                                                                                                 |
| 12      | After an incident you are pushed into a backroom job. Gain Computer-1.                                                                                                                                                                     |
| 13      | Gain a powerful Contact in the organisation. What did you do to gain this contact?                                                                                                                                                         |
| 14      | A romance expanded your horizons, +1 Edu. Are you still in love, or is it over?                                                                                                                                                            |
| 15      | You scored big this term. Gain Cr20,000. How did you get this money?                                                                                                                                                                       |
| 16      | You are thrown into a crisis and take charge. Gain Leader-1.                                                                                                                                                                               |
| 17      | Caused someone to be humiliated or demoted. Gain an Enemy.                                                                                                                                                                                 |
| 18      | Chance to make it big! Roll 2D6, on 4+ gain automatic promotion this year and one extra skill roll. On 2-3, you spend this term in prison. Lower Soc by 1 and gain a criminal Contact. You do not have to make this roll if you so choose. |

# PROMOTION TABLE

Roll 2D6 and add the Effect (whether positive or negative).

| FAIL |                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------|
| 2D6  | Result                                                                                                                               |
| 3    | You were second choice, they like you! Gain a Contact and +2 on next term's attempt.                                                 |
| 4    | Something about your personality means you got overlooked. What is it?                                                               |
| 5    | Your superior sabotaged your chances, he needs you where you are.                                                                    |
| 6    | A rival got the position you were after, now making your life hell. Gain an Enemy.                                                   |
| 7    | You tried for promotion but failed. C'est la vie.                                                                                    |
| 8    | Too many small screw ups this term. What were they?                                                                                  |
| 9    | You're too involved with romance. Gain a romantic Friend.                                                                            |
| 10   | People around you messed up and made you look bad. What did they do?                                                                 |
| 11   | You broke a lot of rules to get the job done. Too many rules. What was the job?                                                      |
| 12+  | You presided over a disaster and the stigma will stay with you for years. What disaster? Gain a -1 on next term's promotion attempt. |

| SUCCEED |                                                                                       |
|---------|---------------------------------------------------------------------------------------|
| 2D6     | Result                                                                                |
| 2       | A colleague colluded to make you look good. Now this Enemy has a hold over you.       |
| 3       | Your promotion was built on lies and false achievements. Only you know this secret.   |
| 4       | Promotion means a transfer. Lose one Friend or Contacts.                              |
| 5       | You beat a friend to the promotion, they are bitter. Make an Enemy.                   |
| 6       | You now owe a favour to a senior officer. Make a note of this.                        |
| 7       | You scraped past, it doesn't look good. You spend this term proving yourself.         |
| 8       | You did great work, and you've been recognized with promotion! Gain +1 Edu.           |
| 9       | You won promotion and an award, accolade or fame. What great thing did you do?        |
| 10      | A senior officer acts as a patron, ally and Contact.                                  |
| 11      | You broke the rules to win big. They had to promote you when they wanted to fire you. |
| 12+     | Gain an extra Benefit roll when you leave the Career.                                 |

## PLAYING ALIEN CHARACTERS

Humans are assumed to be the standard species in these rules. However, this section offers several examples of humanoid alien archetypes that frequently occur in science fiction. Some settings are populated only by humans, whilst others might have dozens or even hundreds of alien species for the characters to encounter. The Game Master should decide which (if any) of the alien races in his setting he allows to be used as player characters. Not all alien species are suitable for the types of situations (and the types of locations) that player characters frequently find themselves in.

### AMPHIBIAN

These humanoid amphibians are descended from aquatic hunters and are around 1.5m in height and weight 60 kg. Their eyes are large and round, with enlarged lenses, adapted for focusing on light that is refracted upon entering water. The lenses are not as well-adapted for sight in air. They prefer to breathe Thin air but can also breathe Standard air without difficulty. Naturally they are fantastic swimmers and can hold their breath for up to 15 minutes. Their thick skin gives them a natural Armour Value of 1.

Str -1, Dex -1, End +2

### MYSTIC

The mystic species is a humanoid alien race that has an innate telepathic ability. Members of the species are known for their total self-control, lack of emotional outbursts, and wise and thoughtful philosophy. Some (or all) of the species has mastered telepathy. Depending on the setting, the talent may have to be brought out by long periods of training, or it might be a power that all possess from birth. Regardless, the player character has the talent of telepathy, and a Psionic Strength (Psi) rating of 1D6+6. See page 83. for more details.

Soc +1, End -1

### SAURIAN

Bearing a strong lizard-like appearance, the saurians descend from homeothermic carnivorous chaser stock. Their scales tend to be patterned, particularly among the males of the species. Standing approximately 1.9m tall, with a mass of 85 kg, the saurians still bear many of the features of their ancestors. Saurians are driven by a strong sense of territoriality and a primal need to hunt, making the bounty hunter career a popular one for them in interstellar society. After discovering FTL travel, the saurians found themselves with entire new worlds to claim. They can be untrustworthy. They are able to tolerate Hot conditions without suffering damage. However, in Cold conditions saurians are sluggish, reduce Dex by half (round down).

Str +1, Dex +1, Edu-1

## HIVE

Descended from carnivore chaser stock, the hive is an insect-like race that is hard for most other races to understand. Averaging 1.8m in height, and massing around 90 kg, the hive are very community-conscious, putting the needs of their colony before the needs of the individual. They have a very limited concept of individual identity, and more often consider themselves as an extension of the queen. With improved nutrition impacting intelligence, as well as exposure to other cultures, a growing minority of hive have begun to establish a stronger sense of a personal self, but remain heavily influenced by their communal instincts. Hive often serve the greater whole of interstellar society in service-oriented roles, depending on their particular gender, those being 'worker' (sterile female), 'soldier' (sterile male), 'drone' (fertile male), or 'queen' (fertile female). While there is little in the way of gross anatomical differences between the genders, carapace coloration and bearing help non-hive distinguish one from another. Their carapace gives them a natural Armour Value of 2.

Dex +2

## WARRIOR

This warrior species is tall (2m) and powerfully built. The males may dominate the species, with their territorial nature and warlike culture. Heroism, bravery, truth and sacrifice are often highly prized. The alien race may follow strict ritual behavior to prevent dangerous aggression between individual warriors. Both male and female have a high pain threshold, with a +2 to ignore debilitating pain from torture or injury.

*Males: Str+3, Soc-2; Females: Str+2*

## ANDROID

Some settings may have cyborgs which can pass as human, and the Game Master may allow a player to take one as a character, along with a suitable back story. In settings where androids are used as slave labour, or are treated as machines, the PC may find significant problems operating freely in human society. Other settings may allow AI androids full or partial citizenship. And in some settings there may be an entire world, or an entire society of androids, perhaps governing themselves, or governed by a centralised artificial intelligence. In many settings, androids are incapable of causing harm to humans, or allowing them to be harmed; consider this the downside to the upside of being stronger and more capable than humans. The player should not put an android through the Life Path system. The android described below is built at Tech Level 14.

Str 10, Dex 7, Durability Points 36, Int 8, Edu 5, Soc 2, Vacc Suit-0, Vehicle-0.

Runs for six months on internal power. Select 3 skill levels from the following: Flight, Medical, Science, Technical, Translator (Technical is Electronics, Mechanical *and* Engineering; Flight is Pilot *and* Navigation).

## EXAMPLE CHARACTERS

### COMMANDER KALLEN ALTURN (RANDOM)

|                                                              |        |        |
|--------------------------------------------------------------|--------|--------|
| <b>Commander Kallen Alturn</b>                               | 679746 | Age 52 |
| Leader-3, Bribery-1, Recon-1, Admin-1, Medical-1             |        |        |
| Cr3000, Easy Going & Relaxed, Height: 175 cm, Weight: 74 kg. |        |        |

We will follow the creation of a character carried out completely randomly, serving as an example for players and Game Master.

#### 1 – CHARACTERISTICS

We roll the numbers 6, 7, 9, 7, 4, 6 and allocate them to the six characteristics in that order.

#### 2 – SKILLS

For Skill Categories, we roll People and then Wilderness. And since we have 6 skill rolls, we roll a D6 to see how we should divide up those skill rolls. We roll 3 ... that's 3 rolls on the first category, leaving the rest (another 3) of the rolls for Wilderness. Our skill rolls give us: Leader-3, Bribery-1, Recon-1 and Medical-1. All characters get a free roll on the Bonus Skill tables, we roll on Table B and gain Admin-1.

#### 3 – SKILL INTERPRETATION

So, what does this character do? He is a natural-born leader, but in what? He has no combat skills, so perhaps he leads exploration or survey missions. That would explain his medical and recon training. He is easy-going, so maybe he has a lot of experience and is very capable, his administration skills suggest that he is a good organiser too, so he doesn't just order people about, he looks like a very good team leader. Let's say he is a Commander in an Exploration or Scout Service, this doesn't line up if we use his Social characteristic to determine his Rank, but that's just optional, anyway.

#### 4 – FINAL DETAILS

Before we explore Kallen's Life Path, we finish up his Final Details. He is bald, suggesting an older man, he measures 175 cm in height, and weighs 74 kg. He only has Cr3000 to his name.

#### 5 – HOMEWORLD & FAMILY

We move on to his Life Path information, and find out he grew up on a TL 15 Garden world, but was orphaned as a child and was brought up within a military academy. This might explain his excellent leadership qualities! He has no brothers or sisters.

#### 6 – LIFE EVENTS

We rolled for five Life Events: two Romances, a Friendship, a Lucky Break and a Tragedy:

**Lucky Break:** Gained fame and recognition.

**First and Second Romances:** Both were brief flings without consequences.

**Tragedy:** He was, or is, hunted by the authorities.

**Friendship:** A co-worker, probably another senior exploration officer.

7 – REASON FOR LEAVING HOMEWORLD

We roll to find out why Kallen left his homeworld and discover that he left it in order to make his parents proud of him. Because he was brought up in a military academy, we instead change that to 'make his teachers (or perhaps a single instructor or mentor) proud', instead.

SUMMARY

So, what happened? With no long term romances, we can say Kallen is a good and charismatic leader, but otherwise withdrawn and quite private (that also fits his Social Influence score of 6). He has a buddy in the service with whom he has shared some challenging missions, and one of those missions propelled him to fame, perhaps gaining him an award or medal for bravery. He might have led a mission to rescue the passengers of a civilian starliner that had crashed on a hostile planet, bringing them all out alive. Only after the dust had settled and Kallen had received his recognition, did Kallen piece together the facts that pointed to the crippling of the liner not by pirates, but a Star Navy patrol ship on a training mission. The Navy has denied this, and aggressively attempted to destroy Kallen's reputation and livelihood. Maybe that's why he has left (or been kicked out of) the service, and why he only has Cr3000 to his name!

LT. CDR VANCE CODINGER (DESIGNED)

|                                                                  |        |        |
|------------------------------------------------------------------|--------|--------|
| <b>Chief Technician Vance Coder</b>                              | A74669 | Age 37 |
| Mechanical-2, Computer-1, Engineering-1, Grav Vehicle-2, Comms-1 |        |        |
| Cr9000, Joker, Height: 184 cm, Weight: 80 kg.                    |        |        |

We create Kallen's explorer buddy using the design method, although the Life Path will still be rolled randomly.

1 – SKILLS

We select the Technician career, and divide up our six skill levels this way: Computer-1, Mechanical-2, Engineering-1, Grav Vehicle-1 and Comms-1. All characters get a free roll or choice on the Bonus Skill tables, we select Vehicle/Grav Vehicle from Table B, to push Vance's Grav Vehicle skill up to Grav Vehicle-2.

2 – CHARACTERISTICS

We select the characteristic string B (10, 7, 4, 6, 6, 9), and decide to leave them in that order.

3 – SKILL INTERPRETATION

So, what does this character do? He is a starship engineer on a survey ship, with some skills (Grav Vehicle and Comms) for use on planetary surveys.

4 – FINAL DETAILS

Before we explore Vance's Life Path, we finish up his Final Details. He has short, spiky hair, measures 184 cm in height, and weighs 80 kg. He has Cr9000.



### 5 – HOMEWORLD & FAMILY

We move on to his Life Path information, and find out he grew up on a TL 7 Desert world. His father was absent, in hiding. The family had been wealthy, but had lost its wealth. Perhaps this explains why his father was in hiding. Was he massively in debt? He has a single sister, and has always had a rivalry with her.

### 6 – LIFE EVENTS

We rolled for six Life Events: two Lucky Breaks, a Friendship and two Tragedies:

***First Lucky Break:*** Gain a powerful friend.

***Second Lucky Break:*** Gain another powerful friend.

***Friendship:*** A co-worker, in this case Commander Kallen.

***First Tragedy:*** Responsible for a tragedy.

***Second Tragedy:*** Debt or ruin.

### 7 – REASON FOR LEAVING HOMEWORLD

We roll to find out why Vance left his homeworld and discover that he left because he learnt something he shouldn't know, and fled for his own safety.

### SUMMARY

What is Vance's story? It might start with his father in hiding; perhaps he owed millions of credits to a crime syndicate, perhaps he was their accountant and got caught embezzling the funds. That story might account for Vance, as his son, suddenly leaving the planet 'for his own safety'. One of the Life Path tragedies was 'Debt' so we can deduce that the syndicate want the money from Vance if they can't find his father. Let's say Vance Codinger is an assumed name, to hide his family connections. The other tragedy suggested that, whatever it was, Vance was responsible for it. Maybe he let people die ... for his own benefit. I am thinking that a law enforcement agency forced him to cut the power to a colony, in order to starve terrorists of oxygen. It turned out there were hostages who also died. The agency paid him off with a false identity in order to hide him from the crime gangs who were hunting him, if he kept quiet. His contact at the law enforcement agency remains a powerful ally that he can occasionally call on. Finally, he has one more powerful ally. I will pick something at random: a starport official at the subsector's capital world.

## GAINING EXPERIENCE

Many roleplaying games feature an Experience Point (XP) system, allowing characters to slowly advance in abilities and competency over the course of several game sessions. **Cepheus Universal** does not lend itself too easily to this style of play, primarily because a skill of 1 is itself influential on gameplay, whilst a skill level of 3 is very powerful. A skill of 1 is essentially a qualification – to fly an aircraft, to work as a mechanic and so on. It is a professional level of skill. Level 3 in a skill represents a highly educated, or highly experienced, professional (a doctor or lawyer, a firearms instructor or chief engineer). These skill levels take years of training and on-the-job experience to gain, and so increasing these levels every couple of game sessions would transform the player characters into advanced experts in all of their skills within the year.

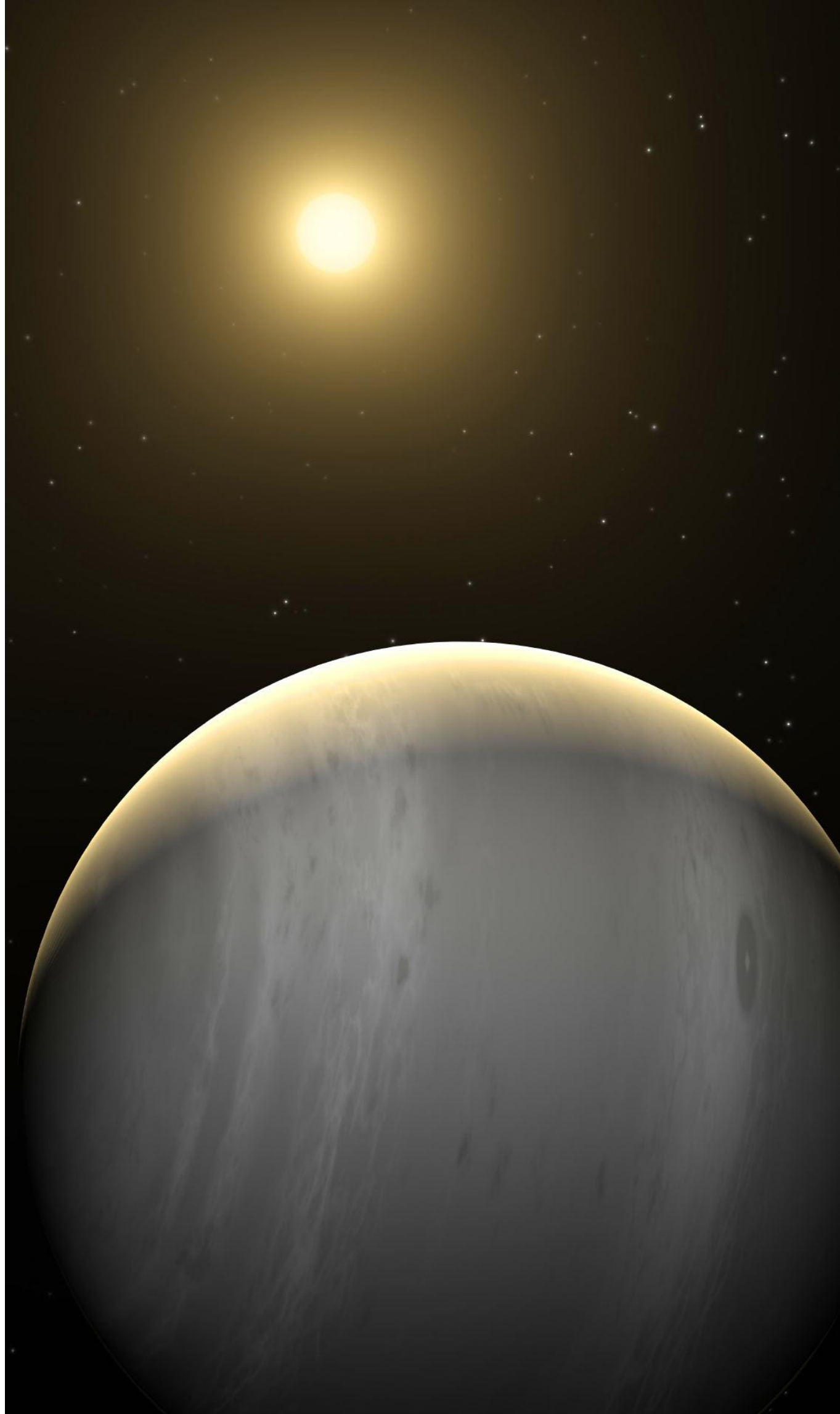
## TANGIBLE EXPERIENCE

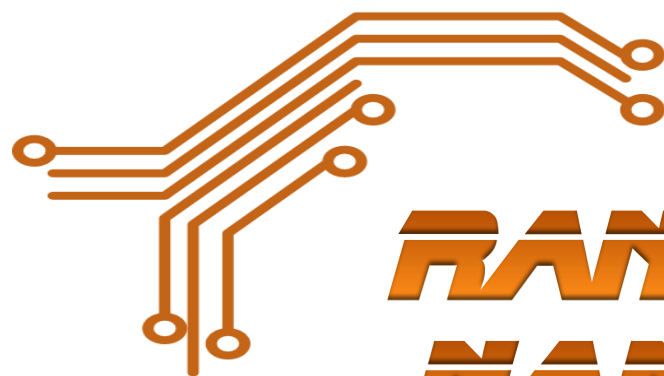
The way that Cepheus Engine, and other 2D6 sci-fi games, represent experience is through tangible rewards. As they play in session after session, the characters will make valuable contacts, they will, amass wealth, perhaps gain political power, spend their money on customising their starship, add more starships to their inventory, set up a business, or a mercenary unit, or a criminal organisation ... in short, they live their lives and advance within the game setting through their tangible actions.

## ABSTRACT EXPERIENCE

However, some players crave that feeling of advancement, particularly if they are used to such systems in the other games that they play. This is understandable. An abstract system of Experience Points (XP) is presented here, and this works best over a limited duration campaign, or for groups of players who meet up once every three or four weeks for a game session, not every week. In the end the Game Master will have to decide if this system suits his game and his players. It is GM controlled, though, so he does have the ability to throttle the amount of Experience Points that are earned.

- **Gaining Experience Points** – Each character gains 1 XP per adventure, or 2 XP if that character, or the group as a whole, carried out some spectacular or dramatic act. Experience Points can be spent either after, or before, a game session to increase skills and/or characteristics.
- **Using XP To Gain a Level-0 Skill** – Cost 1 XP. The maximum number of level-0 skills is equal to ½ Int, round up.
- **Using XP To Increase Skills** – Cost 5 XP multiplied by the desired skill level. The character must already possess the skill level below it. To gain Medical-4 costs 20 XP, but only after Medical-1, Medical-2 and Medical-3 has already been gained, either during character generation, or through the expenditure of XP. Possession of level-0 in a skill is not required.
- **Using XP To Increase a Characteristic** – Cost 3 XP multiplied by the desired characteristic level. The character must already possess the characteristic level below it. To gain Strength 10 costs 30 XP, for example, but to do so, the character must already have a Str of 9. Only Str, Dex, End and Edu can be increased in this way.
- **Using XP To Succeed In-Game** – With the Game Master's approval, 1 XP can be spent to automatically succeed in a non-combat task. The task must be one that is realistically achievable by one of the characters – opening a locked airlock, for example, not walking through a depressurized cargo bay without suffering any injury. Spending 1 XP is not invoking magic, it is that whispered prayer, or crossing of fingers, it is Fate smiling kindly on the player character. If the player character can verbally encourage another character or NPC, then that player character can 'gift' 1 XP to that other character in need, who can use it to automatically succeed.





# RANDOM NAMES

Select a character name from one of the following six tables. Alternatively, roll 1D6 to determine the table and D66 to randomly determine the name. Of course you can roll twice and mix and match first names and surnames, as you prefer.



| RANDOM NAME (1) |                   |     |                  |
|-----------------|-------------------|-----|------------------|
| D66             | Name              | D66 | Name             |
| 11              | Martina Larsen    | 41  | Scarlett Foster  |
| 12              | Chloe Brooks      | 42  | Isaac Bennett    |
| 13              | Sebastian Powell  | 43  | Mason Wright     |
| 14              | Sofia Andersen    | 44  | Ava Simmons      |
| 15              | Gabriel Hughes    | 45  | Harper King      |
| 16              | Isabelle Lopez    | 46  | Daniel Solberg   |
| 21              | Corbin Lyons      | 51  | Valeria Keller   |
| 22              | Laurits Alscher   | 52  | Lucia Garcia     |
| 23              | Zane Michaels     | 53  | Barbel Ortiz     |
| 24              | Diego Rodriguez   | 54  | Mia González     |
| 25              | Newton Iñíguez    | 55  | Isabella Hansen  |
| 26              | Emma Müller       | 56  | Diego Rodriguez  |
| 31              | Buddy Dawson      | 61  | Zoe Anderson     |
| 32              | Sophia Turner     | 62  | Noah Parker      |
| 33              | Trevelyan Lorenzo | 63  | Caleb Phillips   |
| 34              | Valentina Haugen  | 64  | Irmtraud Harland |
| 35              | Oliver Schmidt    | 65  | Amelia Braun     |
| 36              | Liam Wagner       | 66  | Krystine Welch   |

| RANDOM NAME (2) |                      |     |                          |
|-----------------|----------------------|-----|--------------------------|
| D66             | Name                 | D66 | Name                     |
| 11              | Rahul Sharma         | 41  | Avni Sharma              |
| 12              | Diya Joshi           | 42  | Anyu Dashdorj            |
| 13              | Zhanbolat Langeurver | 43  | Harsh Desai              |
| 14              | Sanya Neuer          | 44  | Zarina Riya              |
| 15              | Zhanna Abylay        | 45  | Ashwin Singh             |
| 16              | Arnav Mehta          | 46  | Varun Thapa              |
| 21              | Talgat Karimov       | 51  | Dias Gulov               |
| 22              | Narantungalag Dorj   | 52  | Kavita Thapa             |
| 23              | Nikhil Shah          | 53  | Alisher Jumabek          |
| 24              | Batbold Jargal       | 54  | Aruzhan Doszhan          |
| 25              | Delgerma Boldbatar   | 55  | Boldsaikhan Bayarsaikhan |
| 26              | Yuvraj Khatri        | 56  | Gankhuyag Batmunkh       |
| 31              | Naranbaatar Tuvshin  | 61  | Arava Thapa              |
| 32              | Daniyar Khamzina     | 62  | Marat Zhandosov          |
| 33              | Saran Bataar         | 63  | Tuvshinbayar Ochir       |
| 34              | Oyundari Gan-Ulzii   | 64  | Rohan Sharma             |
| 35              | Trisha Mehta         | 65  | Batzorig Bayasgalan      |
| 36              | Oyuntuya Damdinsuren | 66  | Amarbold Davaa           |

| RANDOM NAME (3) |                     |     |                   |
|-----------------|---------------------|-----|-------------------|
| D66             | Name                | D66 | Name              |
| 11              | Van Petrov          | 41  | Amina Hassan      |
| 12              | Irina Kirov         | 42  | Noura Khalidi     |
| 13              | Ibrahim Al-Amin     | 43  | Sanaa Al-Khatib   |
| 14              | Ahmed Al-Hadi       | 44  | Dmitry Kuznetsov  |
| 15              | Samira Al-Farsi     | 45  | Alexei Popov      |
| 16              | Viktor Smirnov      | 46  | Omar Al-Kazemi    |
| 21              | Valeria Smirnova    | 51  | Sergei Nikitin    |
| 22              | Rami Saidov         | 52  | Farid Abbasov     |
| 23              | Dmitriev Fedorov    | 53  | Karim Al-Farid    |
| 24              | Yelena Romnova      | 54  | Tatyana Baranova  |
| 25              | Pavel Ivanovich     | 55  | Amir Akhmatov     |
| 26              | Aisha Al-Sabah      | 56  | Anatoly Kuznetsov |
| 31              | Fatima Al-Mansoori  | 61  | Yasmin Rostov     |
| 32              | Max Al-Mazroui      | 62  | Zainab Aziz       |
| 33              | Kirill Ivanov       | 63  | Elena Gorbazal    |
| 34              | Antonova Suleimanov | 64  | Leila Kareem      |
| 35              | Nikolai Volkov      | 65  | Natalia Fedorova  |
| 36              | Layla Fahd          | 66  | Mikhail Sokolov   |

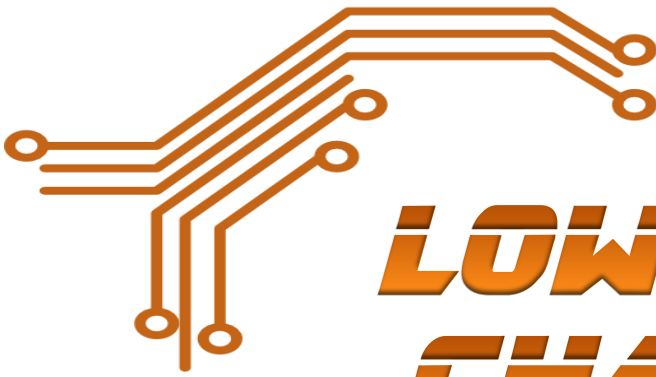
| RANDOM NAME (4) |                      |     |                        |
|-----------------|----------------------|-----|------------------------|
| D66             | Name                 | D66 | Name                   |
| 11              | Castor Severian      | 41  | Andromeda Appius       |
| 12              | Clio Lepidian        | 42  | Diogenes Maximinus     |
| 13              | Thalassa Proculus    | 43  | Hyperion Tullian       |
| 14              | Philoctetes Cassian  | 44  | Orestes Sertorian      |
| 15              | Leander Varian       | 45  | Ariadne Senecio        |
| 16              | Theron Serapion      | 46  | Pan Lentulia           |
| 21              | Calliope Gallius     | 51  | Electra Horatian       |
| 22              | Thais Corvinian      | 52  | Philostratus Verus     |
| 23              | Aeneas Priscian      | 53  | Persephone Quintilian  |
| 24              | Arion Celerian       | 54  | Melaina Augustinus     |
| 25              | Penthesilea Nerva    | 55  | Bellerophon Aurelianus |
| 26              | Orion Drusian        | 56  | Iphigenia Apollonius   |
| 31              | Calchas Tiberius     | 61  | Zenobia Lucilian       |
| 32              | Odysseus Geminus     | 62  | Hippolyta Valerian     |
| 33              | Atalanta Octavianus  | 63  | Antigonos Tacitus      |
| 34              | Xanthe Aurelian      | 64  | Lycomedes Cassian      |
| 35              | Theocritus Cassander | 65  | Heliodorus Vespasianus |
| 36              | Xanthe Livianus      | 66  | Cassandra Quinctius    |

| RANDOM NAME (5) |                   |     |                     |
|-----------------|-------------------|-----|---------------------|
| D66             | Name              | D66 | Name                |
| 11              | Amina Okonkwo     | 41  | Nomthandazo Khumalo |
| 12              | Zanele Vilakazi   | 42  | Sipho Dimmini       |
| 13              | Aisha Nkosi       | 43  | Rukayya Antigonus   |
| 14              | Kabelo Khoza      | 44  | Kabelo Khoza        |
| 15              | Ayanda Bester     | 45  | Kabiru Nkosi        |
| 16              | Onke Jantjies     | 46  | Tumelo Steyn        |
| 21              | Mariam Nkemakolam | 51  | Umaru Chitalu       |
| 22              | Zintle Jacobs     | 52  | Ibrahim Mwangi      |
| 23              | Jabulani Adams    | 53  | Siyanda Mabuza      |
| 24              | Kagiso Du Preez   | 54  | Themba Joubert      |
| 25              | Halima Ngozi      | 55  | Zanele Vilakazi     |
| 26              | Mosa Moloi        | 56  | Abubakar Nkosi      |
| 31              | Tarima Periander  | 61  | Cleopatra Nyandoro  |
| 32              | Hassan Chanda     | 62  | Farida Nkandu       |
| 33              | Usman Chukwu      | 63  | Nana Arnavor        |
| 34              | Zainabu Odhiambo  | 64  | Zainabu Moyo        |
| 35              | Shehu Mulembo     | 65  | Cassander Maiyanga  |
| 36              | Tijani Chisenga   | 66  | Aisha Chilufya      |



| RANDOM NAME (6) |            |     |           |
|-----------------|------------|-----|-----------|
| D66             | Name       | D66 | Name      |
| 11              | Sunarr     | 41  | Adhikari  |
| 12              | Sirubari   | 42  | Nagarkot  |
| 13              | Marpha     | 43  | Lumbini   |
| 14              | Kagbeni    | 44  | Tansen    |
| 15              | Panauti    | 45  | Bandipur  |
| 16              | Khumjung   | 46  | Balthali  |
| 21              | Dhampus    | 51  | Korala    |
| 22              | Gosaikunda | 52  | Ghandruk  |
| 23              | Chitlang   | 53  | Jomsom    |
| 24              | Daman      | 54  | Gandak    |
| 25              | Syangja    | 55  | Dhankuta  |
| 26              | Gokyo      | 56  | Beni      |
| 31              | Nuwakot    | 61  | Tansen    |
| 32              | Myagdi     | 62  | Dhulikhel |
| 33              | Badimalika | 63  | Dharan    |
| 34              | Karnali    | 64  | Narayani  |
| 35              | Lamjung    | 65  | Trishuli  |
| 36              | Khokana    | 66  | Karnali   |





# LOW TECH CHARACTERS

The Character chapter is focussed on technically advanced civilisations of Tech Level 5 or greater. Characters from TL 4 societies should use the Low Tech Skills and Bonus Skills, while those from TL 1-3 worlds should use all the rules in this chapter. TL 4 societies will use the standard method for determining Education.

## LOW TECH SKILLS

**Administration:** Dealing with rules, regulations, law and officialdom.

**Agriculture:** Growing and harvesting crops and raising animals.

**Archery:** Allows use of bows and crossbows.

**Artillery:** Using torsion catapults or gunpowder cannon, according to Tech Level.

**Artisan:** Select from one of Baker, Leatherworker, Bronzesmith, Potter, Cobbler, Mosaic-maker, Plasterer, Tilemaker, Glassblower, Fuller, Tailor, Cutler or other similar trade. Note that Stonemason, Carpenter and Blacksmith are important trades with their own skills.

**Blacksmith:** Fabricating tools, weapons and other items out of iron.

**Bribery:** Offering bribes to circumvent local law, or to influence someone's decision.

**Broker:** Skill in buying, selling, haggling and locating suppliers and buyers.

**Carousing:** Picking up gossip or rumour, making friends and reading body language.

**Carpenter:** Creating useful objects in wood including furniture, wagons and houses.

**Civil Engineering:** Designing and organising the construction of buildings, etc.

**Deception:** Lying, acting and misdirecting.

**Driving:** Controlling beasts of burden to pull wagons, chariots, carts or ploughs.

**Evaluate:** Assessing the relative value of goods, treasures, heirlooms and jewels.

**Forgery:** Faking documents and currencies.

**Gunpowder:** Knowledge and use of primitive gunpowder firearms.

**Hunting:** Tracking in the wilderness, hunting beasts, preparing hides and meat. This was previously part of the Survival skill.

**Investigate:** Conducting an investigation, gathering evidence and clues.

**Leader:** Motivating others in times of crisis or stress.

**Liaison:** Skill in negotiations, protocol, manners of address and codes of conduct.

**Magic:** Knowledge of arcane lore, traditions and practices.

**Melee Combat:** Fighting hand-to-hand, with fists or hand weapons.

**Medical:** Training and skill in the medical arts and sciences.

**Mining:** The character has experience and training in prospecting and mining.

**Navigation:** Plotting courses on planets, and using instruments and maps.

## LOW TECH CHARACTERS

**Recon:** Scouting out dangers threats, ambushes and booby traps. NOT including stealth.

**Riding:** Training in the riding of animal mounts, in their behaviour and care.

**Siege Combat:** Skill in siege warfare, siege mines, towers, rams and siege defences.

**Stealth:** Characters are adept at silent movement and in camouflage techniques.

**Steward:** The care and serving of wealthy customers or guests.

**Stonemason:** Skill in carving stone to make houses, buildings, bridges and statues.

**Streetwise:** Familiarity with underworld society, its rules, groups and customs.

**Survival:** Staying alive in the wild, NOT including hunting or trapping animals.

**Tactics:** This skill covers tactical military planning and decision.

**Watercraft:** The use and maintenance of boats, sailing ships, and galleys.

### Education in Low Tech Societies

Low Tech societies often have very low levels of literacy, and literacy is required in order gain high scores in the Education characteristic. Consequently, when creating a character randomly from a Low Tech world, roll Education on 1D6+1. Otherwise, allocate the lowest value on a Characteristic String to Education when designing a character.

## LOW TECH CAREERS

The following list of careers is representative of those that might be available on a Low Tech world (TL 1-3). The skill in brackets is the career's automatic skill, at least one skill level *must* be allocated to it.

### Aristocrat

[Admin] Liaison, Bribery, Leader, Deceive, Carousing, Riding

### Hunter

[Hunting] Survival, Archery, Artisan, Navigation, Melee Combat, Stealth, Recon

### Mercenary

[Melee Cbt] Siege Cbt, Artillery, Tactics, Leader, Archery, Riding, Gunpowder

### Merchant

[Broker] Admin, Carousing, Evaluate, Bribery, Leader, Driving, Deception

### Nomad

[Riding] Melee Combat, Archery, Driving, Artisan, Navigation, Recon

### Sailor

[Watercraft] Artisan, Carpenter, Carousing, Melee Cbt, Artillery, Navigation, Recon

### Scribe

[Admin] Steward, Magic, Investigate, Navigation, Forgery, Evaluate, Deception

### Servant

[Steward] Deception, Bribery, Driving, Broker, Artisan, Medical, Carousing

### Sorcerer or Mystic

[Magic] Admin, Deception, Navigation, Artisan, Medical, Investigate

### Thief

[Deception] Melee Cbt, Streetwise, Driving, Evaluate, Stealth, Recon, Bribery

### Warrior Noble

[Melee Cbt] Riding, Leader, Navigation, Recon, Tactics,

## BONUS SKILL

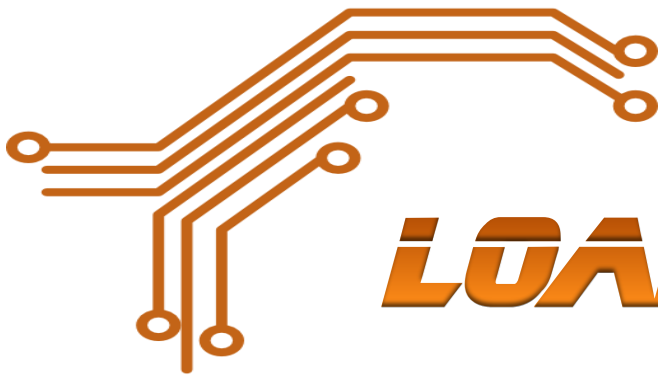
The Low Tech character also gains a bonus level 1 skill. The Game Master can allow the player to select from any skill on the skill list, or roll on the Social Standing table (below) where a selection of skill choices are provided.

| SOCIAL STANDING |              |                                                                                                                                                                      |
|-----------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2D6             | Social Class | Available Skills                                                                                                                                                     |
| 2-7             | Peasant      | Survival, Agriculture, Driving, Navigation, Melee Combat, Archery, Bribery, Carousing, Deception, Mining, Recon, Stealth, Watercraft.                                |
| 8-10            | Townsman     | Artisan, Bribery, Admin, Stonemason, Carpenter, Blacksmith, Broker, Steward, Melee Combat, Watercraft, Carousing, Deception, Gunpowder, Medical, Stealth, Streetwise |
| 11-12           | Noble        | Riding, Leader, Liaison, Hunting, Bribery, Carousing, Deception, Investigate, Admin, Medical, Navigation, Tactics, Melee Combat, Watercraft                          |

## MONEY & GOODS

The credit is worth much less on a low tech world. Every character begins with some starting cash: peasants 1D6 x Cr10, townsmen 1D6 x Cr50, nobles 1D6 x Cr2000.

| SAMPLE LOW TECH PRICES |           |                        |           |
|------------------------|-----------|------------------------|-----------|
| Item                   | Cost (Cr) | Item                   | Cost (Cr) |
| Beast of Burden        | 800       | Rations for 1 week     | 5         |
| Slave, Common          | 600       | Blanket                | 10        |
| Slave, Skilled         | 1200      | Lantern/Tool/Rope(30m) | 40        |
| Sailing Ship           | 10,000    | Mess Pan               | 30        |
| Rowboat                | 200       | Set of Clothes         | 80        |
| War Galley             | 20,000    | Wooden Bowl/Plate/Cup  | 5         |
| Wagon                  | 500       | Torch                  | 1         |



# LOAD OUTS

This chapter provides a series of pre-set equipment lists for various careers. They aren't balanced in any way, a mercenary will be far more heavily equipped than a scientist, for example. Use them for NPCs, for players unsure of where to begin with equipping their characters, or as basic guidelines for the inexperienced roleplayer. Most of the equipment mentioned comes from the Equipment chapter, *other items* should be self-explanatory.

**AGENT**

Pistol (plus one reload), Concealed Vest (AV 6), Hand Radio.

**BELTER**

Vacc Suit, Multi-Tool, Flashlight, Hand Radio.

**CITIZEN**

Hand Computer, Multi-Tool, *one job related item*.

**EXPLORER**

Vacc Suit, Multi-Sensor, Camera, Survival Mask, Combat or 'Survival' Knife.

**FIXER**

Portable Computer, *Briefcase, Sensitive Papers*, Bug Unit.

**MARINE**

Rifle (type based on setting and TL, plus ten reloads), Combat Armour, Combat Knife, 2 Grenades, EITHER Motion Tracker, Grenade Launcher (with 12 grenades) or Disposable Missile.

**MERCENARY**

Rifle (type based on setting and TL, plus ten reloads), Rigid Battle Armour, Combat Knife, 2 Grenades, EITHER Pistol (plus one reload), Night Scope or Grenade Launcher (with 12 grenades).

**MERCHANT**

Hand Computer, EITHER Combat Knife, Pistol (plus one reload) or Tech Scanner.

**ROGUE**

Combat Knife, EITHER Security Kit, Disguise Kit, or Pistol (plus one reload).

**PRIMITIVE**

Leather Jack Armour (AV3), Melee Weapon (worth Cr150 or less), Tarpaulin, Frame Pack, Flint Sparker, *Amulet*.

**SPACER**

Hand Computer, Multi-Tool, EITHER Pistol (plus one reload) or Tech Scanner.

**SCAVENGER**

Leather Jack Armour (AV3), Melee Weapon (worth Cr150 or less), Flashlight, Slugthrower Firearm (worth Cr500 or less, with 4 reloads).

**SCIENTIST**

Portable Computer, Scientific Toolkit.

**DOCTOR/SCIENTIST**

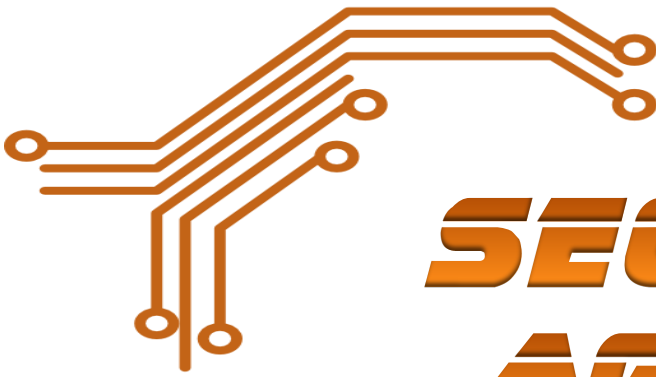
Portable Computer, Doctor's Kit.

**TECHNICIAN**

Multi-Tool, Tech Scanner, Flashlight.







# SECRET AGENDAS

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Some settings, and some campaign concepts, might benefit from the use of 'secret agendas'. These might be personal issues that the character is pursuing that do not align with, or even go against, the mission goals. They might also be hidden goals that a character is trying to pursue for a secret patron (a political leader, a corporation, agency or other faction). The use of secret agendas is entirely optional. If used, each player selects a faction with which to ally his character at the start of the game, from a list provided by the Game Master.

## HOW THEY WORK

As an example, a newly created character might select the GBX corporation as his ally. It is assumed that characters are in touch with their faction, or usually act in the ally's best interests. On *some* of the missions the character undertakes, GBX will give him some extra task, often minor, to accomplish. It may be nothing more than 'report to us - tell us what the mission achieved', it may be more, such as 'pass us blueprints of the cargo', right up to the ultimate 'your navigator works for a rival - discredit her'. Tasks and objectives such as murder and assassination should not be very common, and might not be allowed. Although that kind of extreme secret agenda might work once or twice within a cyberpunk or espionage game, it will destroy any team cohesion, since the supposed secret and covert actions of a character have been flaunted in the open, for all to see. Secret agendas work best when they are so subtle that none of the players can really detect that they are being carried out. The aim of their inclusion isn't to trick, embarrass, frustrate or confuse the other players, it is simply a neat little roleplaying 'side-quest' to make the game more interesting for all of the players sat around the table. Factions, secret agendas and Status (see overlead) are not recorded on the character sheet, to prevent other characters seeing them.

## AGENDA DOs & DON'Ts

- Each character should have a secret agenda (decided at the start of the campaign).
- A secret mission should never aim to kill or harm, another player character.
- Only one or two characters per adventure are given a secret mission.
- Characters successfully completing a secret mission gain 1 XP.
- A character exposed and his faction identified, loses that faction's support.

## **STATUS WITHIN A FACTION**

Each time a secret agenda is completed, the character gains 1 XP. He also gains 1 point of Status. Status is held in the eyes of the faction; the character is now 'on their radar' and may be given other secret missions. Status will increase slowly as more missions are completed. A failed secret mission will garner no status increase, and a botched secret agenda mission will probably result in a status being lowered by 1 point.

### **PERKS**

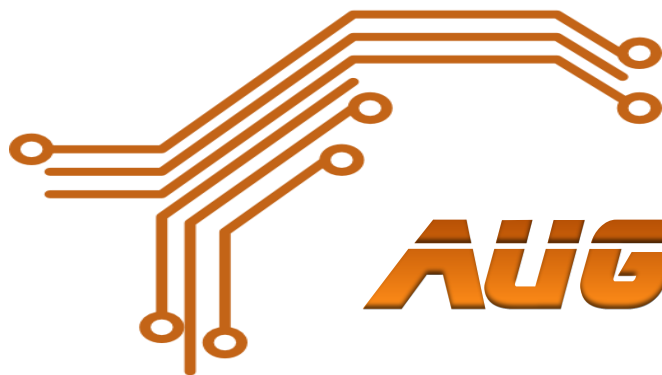
Status forms a whole new fund of perks and resources that a player can tap. These Perks will be very tightly constricted to that organisation's aims and activities. Having Status gets the character access to:

- Useful Contacts
- Equipment
- Information
- Resources
- Funds (up to Cr1,000 per point of Status)
- Influence

All these things make life easier and often turn a difficult secret (or even an official) mission into a much less difficult one! In a scenario where the player's character has been given a secret agenda mission to carry out, that player can request one of the secret Perks from his faction. There must be a realistic way of making contact with the faction in the context of the adventure. A character who is part of a team of exploration scouts on an alien world, cannot call for funds or equipment. However, once back at base or a space station, there might realistically be an agent of the character's secret faction there, that can provide assistance or some other benefit to the character. Be warned that the acquisition of Status can foster a very competitive style of play.

## **JOINING FORCES**

By now it should be pretty obvious where this kind of secret agenda game is heading. The situation puts all of the PCs together and forces them to co-operate to succeed in their mission as well as give them competing secret agendas to pursue for their ally faction. If two player characters find they share the same ally faction or secret agenda, then the GM has lost a point of friction (and therefore interest). Or has he? It might not be the end of the world if both players find out that their characters are both allied with the same military intelligence service. Yes, they may begin to work together, but this trust can readily be exploited by a diligent GM. Chances are they know different military contacts, and will be issued with slightly different orders. In fact, the chances are that each contact will represent a separate (and perhaps rival) faction *within* the military secret service. Imagine the player's surprise when his character is ordered to secretly erase all of his secret service comrade's files. Conflict always equates to story.



# AUGMENTS

Some settings may feature human augmentations that generally fall into two broad categories: cybernetics and retrogenics. The Game Master's setting might not include either, include one or the other, or both. Each augmentation is given a points rating, and the GM can use these points as a currency to allow PCs to gain them during character creation. If the campaign concept involves the PCs working as cyber cops in a near future megacity, for example, the GM might allow each starting player to choose 20 points of cybernetics (or whatever value he deems appropriate). Alternatively, the GM might allow these augmentations to be purchased freely, and assign a monetary price to each (such as Cr1000, Cr5000, or perhaps Cr10,000, per point).

## RETROGENICS

These are genetic modifications carried out on a patient from TL 13 onwards. Similar to a standard virus, the DNA Symbiot as it is called, is purpose-built to enter the DNA of human cells and make subtle changes to the coding. When this is achieved the DNAs die off. From there the human cells carry-on replicating, but according to the new DNA gene pattern (whatever the DNAs was 'programmed' to do). All DNAs also instruct the human cells to reproduce quickly until the adaptation is complete, much like a controlled cancer. It takes one week for a symbiot to do its job; meanwhile the patient is kept sedated.

## RETROGENIC ADAPTATIONS

**Ambidextrous (10):** Self-explanatory, a character will not get the -3 penalty for using the off-hand.

**Amphibious (5):** Includes gill-lung adaptations as well as fatty-layer under the skin for depth to 2 atmospheres, and an underwater membrane over the eyes. Watch out for 'the bends!'

**Animal Empathy (5):** The PC can charm creatures (such as guard-dogs and ravenous mutated test creatures) automatically.

**Cat's-Eyes (10):** Night vision provided some light is available, even a tiny LED.

**Eidetic Memory (5):** Photographic memory. A visual memory only.

**Enhanced Respiration System (10):** The user has a greatly improved respiratory system that can filter out the effects of harmful toxins and gases. In addition it can retain oxygen for up to 20 minutes for an inactive character, or 5 minutes for an active one. Popular with firefighters, divers and assault teams.

**Flexibility (5):** Double-jointed, gives +3 to climb rolls and the ability to climb into very, very tiny spaces.

**High Pain Threshold (10):** The PC is virtually invulnerable to pain. Ignore penalties from wounds. However, extreme actions while suffering from an unnoticed wound will cause a further 1D6 damage.

**Inner-Ear (5):** Superb balance and co-ordination. No rolls are required for walking along narrow ropes, ledges, etc. Running does, however. Reduce distance from a fall by 4m.

**Insect Eye (10):** Independently focused eyes, able to look in two different directions! A PC can fire at two targets simultaneously (best used with Ambidextrous).

**Macro-vision (5):** Telescopic sight just by concentrating. The character can spot a human-sized figure up to 5km away!

**Metabolic Accelerator (15):** Genetic-level metabolic accelerator. A radical rewire of the central nervous system, jacking up the user's physical reflexes to an almost inhuman level. Popular with hitmen who need that split second advantage. For 12 combat rounds, allows the user to act **twice** in every round. Note that machinery (such as cars) will not act at double speed. Cannot be reused within one hour. If injured (reduced to 0 Endurance) the acceleration will automatically shut off.

**Metabolic Decelerator (5):** Can reduce a character's metabolism to the minimum required for life. The character appears dead, and he will have a very distorted vision of what is going on, all in high speed. Will take four times as long to die from wounds, will breathe a quarter as much air, etc. The character must make an Endurance roll every minute if he wants to come out of this. Minimum deceleration time 1 minute.

**Micro-vision (5):** Microscopic vision, for checking fibers, marks or clues.

**Mimicry (10):** If the PC spends a minute listening he can mimic a voice, but only repeat what he heard. If he spends ten minutes listening to the voice and makes an Intelligence roll he can imitate the voice in general conversation.

**Muscle Matrix (15):** Toughened tissue fibers around the body, strengthened bones and joints. Raises the Endurance characteristic by +2.

**Parabolic Hearing (5):** Can zoom in on sounds within 10m if the character concentrates, and hear things just out of the range of normal hearing, +2 on hearing rolls (usually Intelligence rolls).

**Pheromone Sense (5):** Acts as a person detector out to 5m. Can also be used to identify people within 5m if the character concentrates.

**Poison Fangs (10):** Retractable snake-like fangs used to bite (inflicting 2 points of damage – a feat only possible in combat if the victim is first grappled). Also can be filled with poison! A typical poison does 2D6 damage if an Average Endurance roll is failed, and 1-3 points if it is made.

**Poison Immunity (5):** Immunity from most toxins entering the bloodstream; but not acids!

**Regeneration (15):** An enhanced cell-regeneration system that allows a Seriously Wounded character to stave off deterioration rolls for two hours, not one. Characters gain an extra 2 points per day during Natural Recovery.

**Retractable Claws (10):** These claws are in the human fist and can be available for use in an instant. They act as a melee weapon (2D6 damage) and require Melee Combat skill for effective use.

**Second Skin (10):** The PC is immune to fire and napalm attacks by virtue of a second isothermic outer skin. Within minutes this skin sheds, leaving the PC stunned for ten minutes. A new second skin will regrow after a night's sleep.

**Subdermal Body Armor (10):** Toughened skin layering and subcutaneous tissue enhancements turn skin into 4-point ballistic armor!

**Subdermal Pouch (5):** A secret sub-skin carry pouch for small items.

**Superarms (10):** Very strong, tough arms with strengthened muscle and bone. Allows fantastic feats of arm strength, bending and breaking, gripping and crushing. A punch does 1D6+2 points of damage. But useless for lifting etc. without Muscle Matrix to back it up.

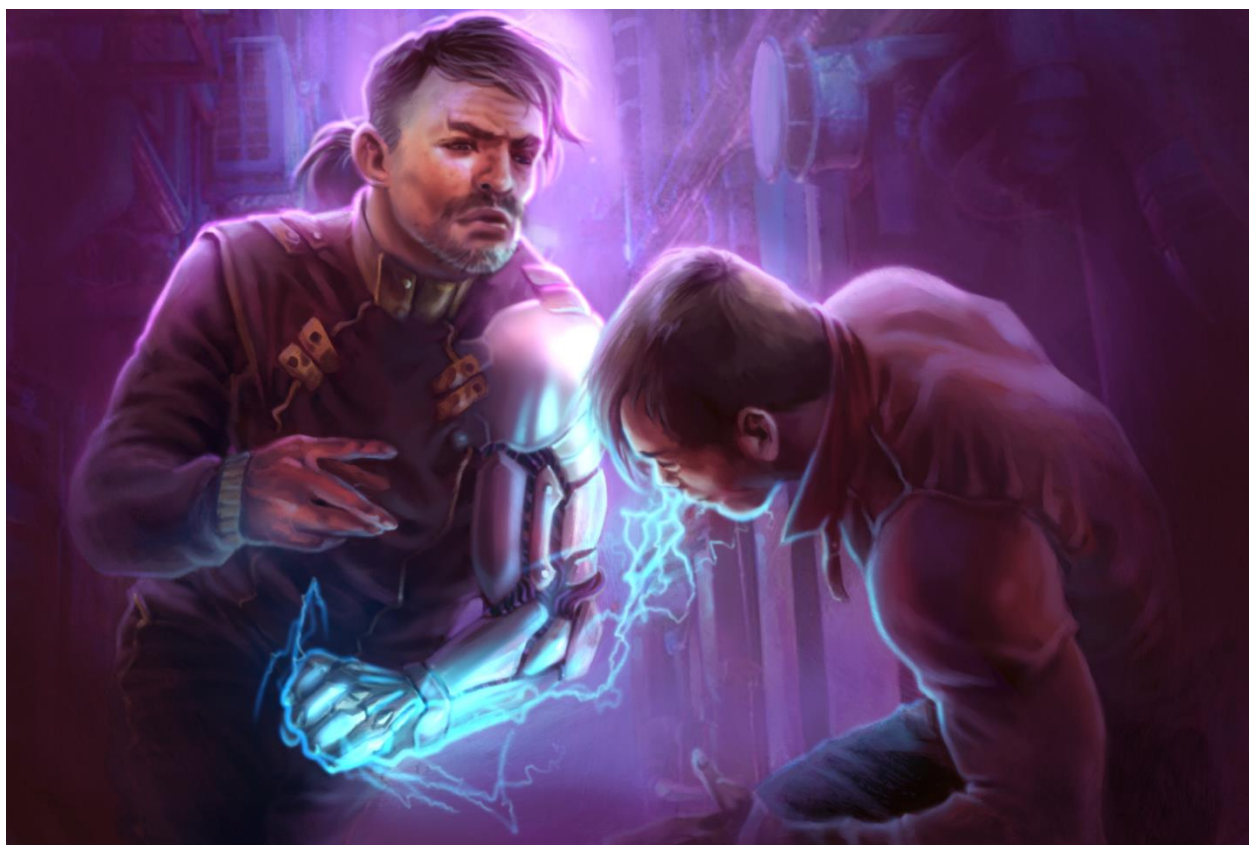
**Superlegs (10):** Allows running at double speed for 5 minutes, and jumps of great distance (6m up, 8m across with a running jump). Any kick inflicted with Superlegs does 1D6+2 points of damage.

**Thermograph (5):** Heat vision, allowing the discernment of hot spots and varying heat signatures. It does obscure fine perception and long range vision.

**Transmorph (20):** Major adaptation involving facial and body muscles, air sacs and follicle implants. The PC can alter his body (requires 1 minute and a mirror) to change his appearance in order to carry out a disguise to perfection. A minute is needed to copy someone, an hour or more to fool their acquaintances. Height, build, facial features, hair color and length are all variable! Can only be kept up for 1D6+1 hours.

**Vat Grown Eyes (5):** Vat grown eyes are perfectly cloned designs with vivid bright colors, with the tiny logo of the lens design visible in the iris as a badge of quality. Users of vat-grown eyes gain a +1 bonus on all sight rolls (an Intelligence roll or Recon roll).

**Waking Sleep (5):** The PC never has to sleep, but instead meditates for 10 minutes, nullifying a - 2 penalty for staying awake longer than 16 hours.





## CYBERNETICS

Synthetic adaptations and implants are a signal trope of the cyberpunk genre, but work for other science fiction settings as well. Advances in computing, material science and power systems allows these implants and enhancements to become available from TL 9 onwards. In many settings, cybernetics are made obsolete by retrogenics at TL 13. As a general rule, a character may only have a number of cybernetics implanted up to his Endurance characteristic. A range of cybernetics are listed here, together with their TL of introduction and a cost, in points. Game Masters may want to pick and choose which cybernetics they allow in their setting.

**Artificial Gill (5):** see Amphibious retrogenic. TL 9.

**Body Plating (20):** This is a jointed exo-covering, that toughens and protects the user, giving him an Armour Value of +8. Often used in conjunction with Power Frame and a set of cyberarms and legs. TL 10.

**Cortex Bomb (5):** Two pinhead explosives that are implanted in the neck, and that detonate when their coating dissolves after several hours (set by the implanter). At TL 12, the bombs can be detonated remotely from a distance of 5 km, or from an unlimited range if connected to a global data net. Death is instant. TL 10.

**Cyberaudio (5):** see Parabolic Hearing retrogenic. TL 9.

**Cybernetic Arms (10):** see Super Arms retrogenic. Two arms count as two implants. TL 9.

**Cybernetic Legs (10):** see Super Legs retrogenic. Two legs count as two implants. TL 9.

**Cyberoptic (5):** A pair of cyberoptic eyes provide a +1 bonus on all sight rolls (an Intelligence roll or Recon roll). In addition, for double the cost, the eyes may have an additional feature, choose from the following types – Cat's Eyes, Thermograph, Insect Eyes, Micro-vision or Macro-vision. See retrogenic descriptions. TL 9.

**Cyberweapon (10):** A retractable melee weapon, such as a long blade or spike (2D6+1 damage). A retractable firearm may be fitted, either pistol (10), SMG or shotgun (15). These firearms will be capable of only 4 shots (or bursts of fire). TL 9.

**Cyberhood (10):** A complete cranial fitting, protecting the head and covering the face. It comes with cyberoptics, as well as two of the cyberoptic options. TL 10.

**Datalink (5):** A neural connection allowing the user to jack directly into a computer system or piece of electronic hardware to interface with, and control, it using only thought. TL 9.

**Power Frame (20):** A full body exoskeleton, with some internal organs replaced with artificial organs. Raises the Endurance characteristic by +2, provides Armour Value 2, and when combined with cyberlegs and cyberarms, allows for superhuman feats of lifting and power. TL 11.

**Puppet Processor (10):** A neural implant that allows an outside source to take control of the character's actions, including movement, speech, eye movement and everything else except thought. No-one has a puppet processor implanted voluntarily! TL 12.

**Retractable Claws (10):** see the retrogenic description (TL 9).

**Skin Weave (20):** see Subdermal Body Armour retrogenic. TL 14.

**Subdermal Fashion (5):** A radical under-skin effect, from pulsing lights to moving tattoos to colour change, can be fitted. Create your effect when buying this augment. TL 9.



**Tracker (2):** A basic radio transponder giving out the location of the user to a distance of 5 km, or unlimited if connected to a global data net.

**Wetware Socket (5, 10 or 15):** A wetware socket is implanted into the cranium behind an ear and allows the user's brain to interface with sophisticated wetware program chips that can be slotted into the socket. One, two or three wetware sockets can be fitted (costing 5, 10 or 15 points respectively). They allow the user of a program chip to gain that skill temporarily while the chip is slotted into the wetware socket. Wetware skill programs are limited to level-0 at TL 12, level-1 at TL 14 and level-2 at TL 16. If the user rolls a double '1' result when attempting a wetware task, he suffers a seizure and will collapse (or be paralysed) for 1D x 1D minutes.

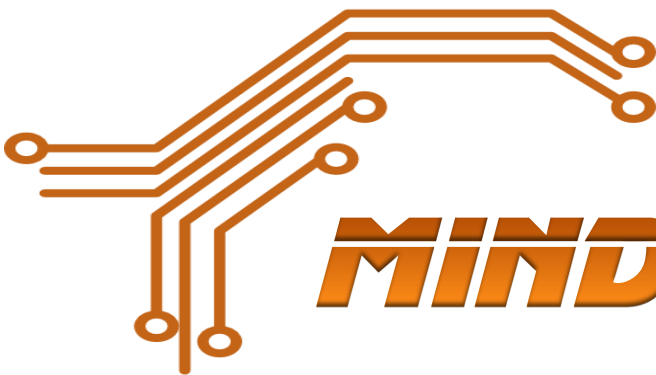
**Wetware Software Chip (no point cost):** A variety of wetware skill chips are available on technical matters, and cost around Cr5000 (level-0), Cr10,000 (level-1) and Cr20,000 (level 2). TL 12. Those skills involving a social component, or that are used in highly fluid stressful environments (such as Gun Combat) are not available as wetware. Common wetware includes: *Agriculture, Aircraft, Comms, Computer, Demolitions, Electronics, Mechanical, Engineering, Ground Vehicle, Gunnery, Navigation, Pilot, Security, Watercraft.*

## **HAVING CYBERWEAR FITTED**

Implantation requires a consultation followed by surgery and then recovery, at least one surgeon with Dex 8+ and Medical-3 capability, and a fully equipped clinic of the correct Tech Level. The time period from consultation to walking out of the clinic is a number of days x double the points cost. A datalink, for example, can be fully fitted in 10 days.

Repair takes a number of hours equal to the points cost, as does removal of the cyberwear. Note that some cyberwear cannot be removed and the body restored intact (such as the power frame or body plating. The costs of these procedures follows the same formula as the cost of fitting (Cr1000, Cr5000 or Cr10,000 or whatever the GM set the cost at) per point. A repair is 5% of that cost, a removal is 25% of that cost.

For implantation or removal, the patient must make an Average Endurance roll, or have the period extended by 50%. If the clinic's Tech Level, or the surgeon's skill, is below that required, then the patient will also suffer 3D6 points of damage if the Endurance roll is failed.



# MIND POWERS

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Psionics are latent mental powers that defy the laws of physics. The GM's setting might include them in one form or another. Psionics are divided into six different categories, called talents. These talents are: Awareness, Clairvoyance, Telekinesis, Telepathy, Teleportation and Special. The setting might have a humanoid alien race with Telepathy, or there might be a government intelligence agency that trains psions as secret agents with the Telepathy and Clairvoyance talents. One famous science-fiction setting has an elite order of warrior monks who are trained in the Telekinesis and Telepathy talents. Use these rules as required, they are wholly optional.

## PSIONIC STRENGTH

Psionics are powered by the Psionic Strength characteristic (Psi), typically rated 2-12. However, if the character has been trained to use psionic talents either by his alien upbringing, or is a gifted individual trained by some organization, then his Psi should be rolled on 1D6+6. To activate a talent, the psion must spend the listed number of Psi points, which are taken temporarily from the Psi characteristic. If this cost brings them below zero Psionic Strength, then any excess points are applied to their Endurance score as damage. A character with no Psionic Strength points remaining cannot activate a talent. Each talent notes a Psi level required for its activation, which is also the cost in Psi points to activate. It also notes the talent's range. Characters regain their full Psi points after an 8-hour rest.

## GAINING TALENTS

As already discussed, gaining a talent might depend on the details of the psionics that appear within the setting. Only one or two might be available, and there is probably no choice. Or in some settings, *all* of the possible talents are available to be learned. In that case, allow a psionic 'student' to select ONE talent. Ask your Game Master. He may also attempt to gain further talents. For each additional talent, make a Psi characteristic roll to receive that talent, DM-1 per talent the character already has. His training may cost as much as Cr100,000 and take 4 months, there may be other difficulties and challenges in finding a teacher. Talents listed include Psi cost to use and maximum range. The talents are:

**Awareness** – Powers that allow the psion's mind to control his own body.

**Clairvoyance** – These powers permit perception at a distance.

**Telekinesis** – Mind over matter.

**Telepathy** – Involving the reading of minds and mental communication.

**Teleportation** – Moving from one point to another instantly.

**Special** – Some other power, unique to the GM's setting.

## POINT CREATION

If multiple talents exist in the setting, the Game Master can use an optional point system as a currency to allow PCs to gain mind powers during character creation (just as with Augmentations). Begin with 20 points and a Psionic Strength of 6. Use points to buy talents (points listed with each talent), as well as additional levels of Psi. Each extra level of Psi costs 3 points.

## AWARENESS (3)

Awareness is the psionic talent which allows greater control over one's own body.

**Enhanced Characteristic (Psi 1+, Personal)** – The character may add psionic points to their Str, End, or Dex as desired. This costs 1 Psi point per characteristic point added and lasts for 10 minutes.

**Suspended Animation (Psi 3, Personal)** – Enter a meditative state for 7 continuous days without requiring food or water or a breathable atmosphere. The psion may terminate this talent at will during this period.

**Regeneration (Psi 9 but see below for cost; Personal)** – The psion can immediately (within 1 combat round) heal their own Str, Dex, or End points lost due to combat damage. This requires 1 Psi point per characteristic point healed but requires a minimum of Psi 9 to use.

## CLAIRVOYANCE (5)

Clairvoyance allows the character to sense events at a location away from the viewer. It usually cannot be blocked.

**Sense (Psi 1, Medium)** – The character learns the rudimentary characteristics of a nearby location when applying this ability.

**Clairvoyance/Clairaudience (Psi 5, Very Distant)** – This ability allows the character to view (clairvoyance) or to hear (clairaudience) at a specific displaced point up to 25 km range.

**Clairsentience (Psi 7, Very Distant)** – The psion can both hear and see from any point within 25 km range.

## TELEKINESIS (6)

Telekinesis allows the psion to manipulate objects without physically touching them.

**Lift 1 gram (Psi 1, Medium)** – Lift and manoeuvre up to a 1 g object for 1 minute (10 combat rounds)

**Lift 1 kg (Psi 2, Medium)** – Lift and manoeuvre up to a 1 kg object for 1D6 combat rounds. Can cause 1D6 damage if hurled at a target within Short range.

**Lift 100 kg (Psi 5, Medium)** – Lift and manoeuvre up to a 100 kg object for 1D6 combat rounds. This will cause 4D6 damage if hurled at a target within Short range.

**Choke (Psi 8, Medium)** – Grab a single humanoid by its throat, or equivalent, and suffocate them for 1D6 damage per round, for a maximum number of rounds equal to the character's full Psi score.

**Microkinesis (Psi 10 but see below for cost, Close)** – Manipulate microscopic objects. Microkinesis may be used for subtle, but highly effective attacks on internal organs, causing 1D6 damage per 2 Psi points invested. Alternatively, this power may aid microsurgery by DM+1 per 3 Psi points invested. Finally, the psion may assist Engineering and other repair skill throws by DM+1 per 3 Psi points invested in microkinesis.

## TELEPATHY (5)

Telepathy is the talent of establishing mind-to-mind contact. It is usually subtle but can also be used to bluntly crush the wills of those who oppose the telepath. Note that thoughts are universal, and thus psionic communications ignores language barriers. However, psionic scan or communication with a target significantly different from the user (such as an insectoid alien scanned by a human) incurs double Psi cost.

**Life Detection (Psi 1, Short)** – Detect the presence of other minds, their general type (human, alien, animal, etc.), and location.

**Telempathy (Psi 2, Short)** – Read and communicate basic emotions; may influence the behavior of others, both sentient beings and aliens, subject to GM discretion.

**Read Surface Thoughts (Psi 3, Short)** – Read active, current thoughts. A non-Psionic target cannot notice this talent being used on them, but a Telepathic Shield (see below) blocks the attempt.

**Communicate (Psi 5, Distant)** – Communicate telepathically with one individual. The target does not have to be a telepath to communicate with the psychic.

**Emotion (Psi 6, Medium)** – Causes one target to feel a very strong emotion (such as fear) for 1D6 combat rounds.

**Suggestion (Psi 7, Distant)** – Plants a simple hypnotic suggestion in the target. Note that targets will not obey suggestions which may physically harm themselves; if the suggestion is that the character will harm a friend, they may make an Int roll to resist the effect.

**Probe (Psi 9, Close)** – Reads deep thoughts and long-term memories.

**Assault (Psi 10, Long)** – A brute-force telepathic assault, causing 4D6 damage to the target. Apply damage first to the target's Psionic Strength, then Int and finally End. Characters who are reduced to 0 in all these characteristics die from brain hemorrhage. Lost Int regenerates at a rate of one point per day.

**Domination (Psi 12, Distant)** – Mind-control one target for 1D6 minutes. The control is complete and may include harming the target's friends. Only when the dominated target is forced to harm itself, it may make an Int roll to shake off the mind control.

**Shields and Telepathic Combat** – All telepaths can shield themselves, and other individuals in a 3m radius from telepathic intrusion. Attempting to use a telepathic talent against a shielded target requires an Opposed Psi roll. The defender wins on a tie. If the intruder wins, they may use their talent as desired at the usual cost, and the defender loses one point of Psionic Strength. If the intruder loses, they fail to penetrate the shield and expend one point of Psionic Strength.

## TELEPORTATION (7)

Teleportation allows instant travel between locations. The psion may only teleport their body, clothes, and personal equipment. Characters cannot teleport over a height differential of over a kilometre, as the difference in potential energy heats or cools the user, and beyond one kilometre, the user's body temperature will change to a lethal degree.

**Blink (1 Psi, Close)** – Teleport 3m in any direction while carrying up to a light load.

**Teleport Self (Psi 7, Continental)** – Teleport up to 5000 km with up to a double load.

**Teleport Extra (Psi 11, Continental)** – As teleport self, but allows the teleporter to teleport while carrying over a double load (including carrying another living being).

## PSIONIC EQUIPMENT

The following psionic equipment is available in settings using psionics.

| Item                   | TL | Cost   |
|------------------------|----|--------|
| Psionic Booster        | 9  | Cr500  |
| Psionic Inhibitor      | 9  | Cr4000 |
| Psionic Shield Helmet  | 8  | Cr8000 |
| Psionic Shield Implant | 13 | MCr1   |



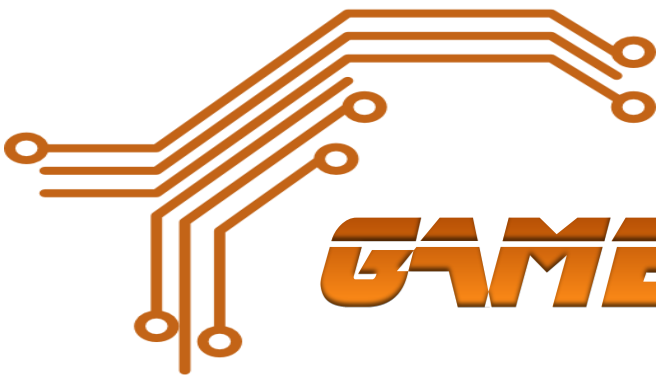
**Psionic Booster** – This drug stores up to 6 points of Psi. If taken at full Psi, it temporarily increases Psi by 4. Whenever taking more than one dose per week, make an End roll or suffer 3D6 damage and permanently lose one point of Psi.

**Psionic Inhibitor** – This drug forces the subject to make a Psi roll to use any psionic talent and prevents Psi regeneration. This drug lasts for 4D6 hours, but the character can make an End roll each hour to shrug off its influence.

**Psionic Shield Helmet** – This fully protects a non-psion wearer from telepathic intrusion. However, a character wearing this helmet cannot themselves use psionics. This helmet does not protect its wearer from the effects of other psionic talents, such as telekinetic attacks.

**Psionic Shield Implant** – Like the psionic shield helmet (above) but implanted and invisible to the casual observer.





# GAME SYSTEM

The characters will face dangerous situations with the odds stacked against them and no help expected from the authorities. Decisions made and actions taken might well be prove fatal. We don't let the players themselves decide if their characters succeed or fail, neither do we give this power to the GM. We often let the dice decide – it's the only way to be sure of some impartiality. Right? For tasks that could be considered easy or routine or without stress (cycling an airlock, for example), then GM may just declare the attempt to be a success. Where tasks are carried out under stress, with haste, under trying conditions or opposed by some other force, then we turn to the dice.

## TASK RESOLUTION

When you have to resolve a task using dice, it will typically involve a 2D6 roll plus modifiers, against a target of 8+ (read that as eight or higher). You always want to roll high on a task roll. It is the GM who will generally ask a player to 'roll the dice' to resolve a situation. We use 2D6 to make task resolution rolls, there are four kinds:

- **Skill Rolls** (patch a leaking spacesuit using a skill bonus)
- **Characteristic Rolls** (climb that cliff using a characteristic bonus)
- **Attack Rolls** (attack with a weapon using a weapon skill bonus)
- **Perception Rolls** (spot something unusual or hidden using Recon skill and Int bonus)

## SKILL ROLL

A skill roll determines what your character can accomplish with a particular skill. It is a roll of 2D6, with your skill level (if possessed) added on, increasing your chance of success. Finally, a Die Modifier reflecting the difficulty of the task will be added or subtracted. The aim is to gain a roll of 8 or more. That is a success. Skill rolls sometimes have gradations of success and failure based on the Effect of your check result (how much higher or lower your total result was when compared to the target of 8). See the section titled Difficulty, below. If a character does not have any levels in the required skill, then he or she suffers a -3 penalty for being unskilled. If the GM decides that a specific characteristic might have an influence on the skill roll, then add the appropriate characteristic score modifier. Characteristic modifiers are explained in the Character chapter.

8+



CHARACTERISTIC ROLL

A characteristic roll is very like a skill roll but instead it measures raw ability, like strength, endurance, or intelligence. These rolls are used when the task is one not covered by an obvious skill, or where the character’s innate abilities are the most important influence on the result. To make a Characteristic check, you must roll 2D6 and add the appropriate characteristic score modifier. Characteristic modifiers are explained in the Character chapter.

6+

| CHARACTERISTIC MODIFIERS |     |     |     |      |       |     |
|--------------------------|-----|-----|-----|------|-------|-----|
| Characteristic           | 0-2 | 3-5 | 6-8 | 9-11 | 12-14 | 15+ |
| Modifier                 | -2  | -1  | -   | +1   | +2    | +3  |

ATTACK ROLLS

An attack roll determines whether or not you hit an opponent in combat. An attack roll is essentially a skill roll, using your skill levels in a relevant combat skill as a modifier. There are some additional rules involved in combat, and these are found further on in the Combat chapter.

8+

PERCEPTION ROLL

The GM may offer a character the chance to notice something useful or potentially dangerous in or out of combat. This calls for an Average (0) Recon roll, adding the Int characteristic modifier, and will vary in difficulty based on lighting, distance to the thing being spotted and so on. Characters sprinting will not be able to make perception checks. In combat, the initial combat encounter will be determined by a group perception roll, see Initiative in the Combat chapter.

8+

DIFFICULTY

A task’s Difficulty is a number set by the GM that modifies your task result. When no Difficulty is given, the assumed Difficulty is +0. So, for a task with a Difficulty of -2, you must subtract -2 from the roll. You succeed on a total of 8 or better. The list of Difficulty ratings can be found below:

| Difficulty     | DM | Example: Driving a Car                   |
|----------------|----|------------------------------------------|
| Easy           | +4 | Starting it while under stress           |
| Routine        | +2 | Reversing at speed                       |
| Average        | +0 | Swerving to avoid a crash                |
| Difficult      | -2 | Handbrake turn to throw off pursuer      |
| Very Difficult | -4 | Putting the car on two wheels            |
| Formidable     | -6 | Jumping the car from one roof to another |

## EFFECT

In some cases, the consequences of a roll vary based on how much the result is above or below the target of 8. The difference between the result and the target of 8 is called the Effect. If the Effect is 6 or higher, the check is considered an Exceptional Success. When the Effect is -6 or lower, the check is considered an Exceptional Failure.

| Effect              | Degree of Success   |
|---------------------|---------------------|
| <b>-6 or lower</b>  | Exceptional Failure |
| <b>-1 to -5</b>     | Failure             |
| <b>0 to +5</b>      | Success             |
| <b>+6 or higher</b> | Exceptional Success |

In combat, damage inflicted is increased by the Effect of the attack. The surgery rules, for example, use the Effect to determine how much healing takes place (see the Combat chapter). The GM, however, may come across other situations in which the Effect can play a crucial part. Perhaps a Recon roll gains several points of Effect, providing the player character with extra pieces of information – likewise with the Investigation skill, or any technical skill being used to diagnose a problem. In any Opposed roll (see below), the character with the highest Effect wins that contest, whether it be arm wrestling, fast drawing, vehicle racing or whatever. In a similar way to Aiding Another (see below), there are some circumstances where one character's task result may influence the task of another character (such as a character with Streetwise trying to set up a deal for another character with Broker skill). The Effect rolled can, in some non-combat circumstances, be added to the second character's roll as a DM (whether positive or negative!).

## OTHER SITUATIONS

### AIDING ANOTHER

Sometimes characters work together and help each other out, carrying a stretcher or battering down a door. The GM imposes a minimum number of people required for the task. The task might be impossible if that number isn't met or the GM may impose a negative DM.

A standard one-man task might benefit from assistance from another character if the GM allows. In this case, one character (usually the one with the highest skill) is considered the leader of the effort and makes the check normally, while each helper makes the same check. The Effect of a helper's check result can provide either a bonus (DM +1 if Success, DM +2 with an Exceptional Success), no bonus (failed roll), or even a penalty (DM -2 with an Exceptional Failure) to the leader's check result. In many cases, outside help isn't beneficial, or only a limited number of helpers can aid someone at once. The GM limits aid as appropriate for the task and conditions.

## CIRCUMSTANCE MODIFIERS

Some circumstances make a check easier or harder, resulting in a bonus or penalty that is added to the check result. A character might benefit from useful gear or tools (+1 or +2), or suffer a penalty based on lack of tools (-1 or -2). There may be other hampering factors such as a sandstorm, darkness, fog, heavy rain, etc. which will impose a negative DM up to -5.

## MULTIPLE ACTIONS

A character can try to do two or more things at once, like firing a spacecraft's weapons while also flying, or disarming a bomb while hiding from guards. For every extra thing that the character is doing, he suffers a -2 DM to all skill checks.

## OPPOSED ROLLS

If two characters are opposing each other directly in a task, then the character who obtains the highest Effect wins. For ties on opposed rolls, the character with the highest relevant characteristic score wins. If the characters tie on characteristic scores, they reroll.

## TIME TAKEN

Outside of stressful situations such as combat, performing a particular task often takes a set period of time, depending on circumstances. The GM should let the player know how long it will take to carry out the task. At other times the GM can let the dice decide, using a D6 to represent 1-6 seconds, minutes or hours...

You can choose, before you roll, to move up or down one or two rows on the Time Frame table. Moving up (reducing the time increment) gives you a -1 DM for every row you move; moving down and increasing the time taken gives you a +1 DM for every row you move. Your GM will help adjudicate any issues that might arise from a change in the time frame.

| Time Frame             | Base Increment                                        |
|------------------------|-------------------------------------------------------|
| <b>1D6 seconds</b>     | One second                                            |
| <b>1D6 rounds</b>      | One personal combat round (6 seconds)                 |
| <b>1D6 minutes</b>     | One minute (60 seconds, or 10 personal combat rounds) |
| <b>1D6 kiloseconds</b> | One kilosecond (~16.67 minutes)                       |
| <b>1D6 hours</b>       | One hour (60 minutes)                                 |
| <b>1D6 days</b>        | One day (24 hours)                                    |
| <b>1D6 weeks</b>       | One week (7 days)                                     |
| <b>1D6 months</b>      | One common month (30-31 days)                         |
| <b>1D6 quarter</b>     | One quarter (3 common months)                         |

## **TRYING AGAIN**

In general, you can try a task again if you fail and keep trying indefinitely. Some tasks, however, have consequences for failure. For example, failing an Engineering roll while stopping a fusion reactor going critical probably means it cannot be stopped and you need to get to minimum safe distance. Such tasks just can't be attempted again once a roll has failed. For most tasks, once you've succeeded, additional successes are meaningless. (Once you've fixed a broken radio using the Comms skill, for instance, there's no further benefit from additional Comms checks.)

## **ADDING DRAMA!**

There are ways to inject added drama into a situation through the use of certain types of dice roll.

### **ESCALATING PERIL**

The outcome of a task has some great importance attached to it. And the player fails the roll. The Game Master might allow a second roll, but only with the understanding that a failed roll this time, will have far more dire consequences. Then, let the player decide whether or not to attempt that second roll. This only works with those tasks that can ordinarily only be attempted once.

### **COUNTDOWN TO DISASTER**

Sometimes the player characters might be attempting a task over time, with a deadline. Perhaps an escape pod is hanging off a cliff, and the PCs need to get out of it with as much survival equipment as they can salvage before it slips off the edge. The GM sets a figure, such as 14, and reduces that number by 1D6 for each attempt made (or each round that passes, depending on the situation). Of course the target number can be changed to 10, for a shorter period. When it reaches 0, the expected catastrophe occurs. In this case the life pod finally slips off the edge of the cliff, taking with it anyone who was still inside. Another example might be driving a vehicle across a frozen river, the ice gives way when 0 is reached. This type of thing is a standard movie trope, from *Battle of the Five Armies*, to *Mission: Impossible – Dead Reckoning* to *The Italian Job*.

### **RANDOM DISASTER**

The single D6 is versatile and extremely useful. At a moment of tension or crisis the Game Master can ask himself: 'what is the worst thing that could happen right now?' Then, roll a D6. On a '1' that terrible calamity occurs. This certainly creates tension and ramps up the stakes if a '1' comes up.

Example: *The characters are wading across a rapidly-flowing river, near some semi-submerged rocks. They carry their kit above their heads. The characters make a quick skill roll and succeed - but I wonder, what is the worst thing that could happen right now? Well ... wouldn't it be awful if the local ambush predators (six legged critters that are a cross between crocodiles and spider crabs) suddenly attacked the characters? Let's roll a D6...*

# STRESS

People can only handle so much stress before falling apart. A rapid number of terrors and psychological stresses will soon drive even the strongest to the brink of sanity. Players are always in control of their characters, but there are times – typically when faced with overwhelming and indescribable horror – that the character may freeze or even be temporarily removed from play due to extreme psychological stress. These should be rare and the character is always returned back to the player’s control.

## STRESS EVENTS

Characters suffer stress in a number of ways. These are called Stress Events. These include:

- In a breached vacc suit, or a vehicle, or ship that takes damage for the first time.
- Each time a characteristic reaches zero (such as from wounds or hunger).
- Seeing anyone in your party injured, poisoned, or fail a Panic roll.
- Catching fire
- Encounters with terrifying creatures or situations.
- Losing control in zero-G during a space-walk.
- Firearm malfunction in combat
- Facing violence or witnessing a violent act

## THE STRESS ROLL

Each time a character experiences a Stress Event, he makes a Stress roll, which is an Average (0) Intelligence roll. Succeeding a Stress roll indicates the character is able to deal with the stress and can continue to function normally. Failing a stress check always results in the temporary loss of 1 or more points of Int ('Overstress') and the character being stunned for one full combat round per Int point lost. Typically 1 point will be lost, but truly awful horrors may dial up that Int loss to 2, 3 or even 4. Lowering Intelligence represents a character becoming less rational, less focussed and more troubled. Points lost to Overstress are restored to its initial score at the end of the adventure. A lowered Intelligence characteristic may have repercussions:

8+

***First point of Int lost during a game*** – Stunned for one round.

***Int reduced to Half (round up)*** – Make an Average (0) Int Panic check; failure indicates that the player must roll on the Panic Effects table.

***Int reduced to 0*** – The player makes a Difficult (-2) Int Panic check. If successful, Int is returned to 1. If failed, roll on the Panic Effects table with a +2 DM.

## PANIC ROLL

When a character’s stress overwhelms them, they must struggle to retain control of their sanity. The Panic check is an Int roll, and if failed, requires the player to roll on the Panic Effects table. Success indicates that the character fought back the terror with grim determination.

| PANIC EFFECTS |                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2D6           | Description                                                                                                                                                                                                                             |
| 2-4           | <b>Stunned</b> – 1D6+2 rounds                                                                                                                                                                                                           |
| 5-6           | <b>Flee</b> – Overwhelming drive to escape the situation for 1D6+2 rounds                                                                                                                                                               |
| 7-8           | <b>Stunned</b> – 1D6+2 minutes                                                                                                                                                                                                          |
| 9             | <b>Phobia</b> – Stunned for 1D6+2 rounds, and also develop a phobia. Pick something appropriate; each time you are exposed to it take a -2 to all Stress checks.                                                                        |
| 10            | <b>Flee</b> – Overwhelming drive to escape the situation for 2D6 minutes                                                                                                                                                                |
| 11            | <b>Stupefied</b> – Blank, uncommunicative, unresponsive and oblivious for 1D3 days                                                                                                                                                      |
| 12            | <b>Psychotic Break</b> – Reality has no meaning, the character turns illogically on his comrades by sabotaging equipment and hampering any chance of escape or survival, even if it means the character's own death. Lasts until cured. |

Stunned characters may babble incoherently, suffer total paralysis, stand there screaming, curl up into a foetal position, throw up, cower in abject fear ... and so on, depending on the situation. Characters stunned for days are stupefied, staring blindly into space, unable to communicate and capable of being tamely led around. Phobias suitable for those who receive them include fear of blood, fear of the dead, fear of being left alone and fear of enclosed spaces (including autodocs, vacc suits and cryoberths). Characters who flee will do so blindly and heedless of any danger they may be putting both themselves, and their comrades, into. Psychotic characters require 1D3 weeks of counselling to recover, with a dose of polyphenol-z administered each week.

## RECOVERING FROM STRESS

Overstress can be reduced under the right circumstances. The character requires safety and security, plus ways to cope such as little comforts or frivolous escapes. These are called Recovery Events and they can reduce Overstress:

- Defeating or escaping from an immediate physical threat.
- Successfully performing a heroic feat, or seeing one occur.
- Motivated by a comrade (an Average Leader check).
- Being treated for shock by a medic (an Average Medical check).
- Rest for 8 hours (minus the PC's End modifier) in a safe, dry, warm place.
- A comfort: hot meal, a warm bed, a mug of hot tea or coffee, a shot of whiskey, a heart-to-heart with a buddy, etc.
- An escape: reading a book, a favourite song or movie, time to lift weights or run, sex, recreational drugs, etc.
- Receiving 1D3 days of counselling from a trained medic (Medical-1+), or a mild dose of truth drug administered with a Routine (+2) Medical check.

Each time a character experiences a Recovery Event he makes a Routine (+2) Int check and if successful regains a point of Int, in effect becoming a little calmer and more capable, returning to normal bit-by-bit. Failure has no effect. Psychotic characters require 1D3 weeks of counselling from a trained medic (Medical-1+) to recover, with a mild dose of truth drug administered each week and a Routine (+2) Medical check.



## THE HORROR!

Note that characters may encounter situations, or entities such as sanity-smashing alien beings, that can rip right through this process. This may include forcing an immediate Stress Roll at a penalty, or forcing a player character to make Panic roll regardless of their Stress. The GM is cautioned to use such 'cosmic horror' encounters sparingly!

## ACID

Corrosive acids deal 1D6 damage per round of exposure, except in the case of total immersion (such as into an industrial vat of acid), which deals 10D6 damage per round. An attack with acid, such as from a hurled vial or an animal's acidic spittle, counts as a round of exposure.

The fumes from most acids are poisonous. Those who come close enough to a large body of acid to dunk a creature in it must make an Average (+0) Endurance check or take 1D3 point of damage. All such characters must make a second Average (+0) Endurance check one minute later or take another 1D3 point of damage. Creatures that are immune to acid's caustic properties might still drown in it if they are totally immersed and need to breathe. For more details, see Suffocation.

## FALLING

A character who falls on a 1-G (one gravity, or Earth-like) world suffers 1D6 damage per two metres fallen. Optionally, high or low gravity worlds will increase or decrease the damage per die. Check the size code for the world and apply the damage DM.

-2 if Size 3-  
-1 if Size 5-  
+1 if Size 10+  
+2 if Size 12+

## FIRE

Characters exposed to a fire source might find their clothes, hair, or equipment on fire. Those at risk of catching fire are allowed a Dexterity check to avoid this fate. If a character's clothes or hair catch fire, he takes 2D6 damage immediately. In each subsequent round, the burning character must make another Dexterity check. Failure means he takes 2D6 damage that round. Success means the fire has gone out. Being on fire forces an instant Panic Roll. A character on fire may automatically extinguish the flames by jumping into enough water to douse himself, spraying himself down with a fire extinguisher or otherwise smothering the flames. If the character has no such means, rolling on the ground or smothering the fire with cloaks or the like permits the character a DM+2 on his next Dexterity check. Putting out a fire with a fire extinguisher or extinguishing a fire on another person requires a successful Easy (+4) Dexterity roll.

## DISEASES

Diseases reduce a character's characteristics, usually Endurance. The character must make an Average (0) Endurance roll with the listed DM to resist the effects of the disease. If the character fails the Endurance roll then he takes the listed damage after the rolled interval and must make repeated Endurance rolls at the same interval. Once an Endurance roll has been passed the character has fought off the disease.

| SAMPLE DISEASES |    |        |           |                                       |
|-----------------|----|--------|-----------|---------------------------------------|
| Disease         | DM | Damage | Interval  | Symptoms                              |
| Wound Infection | +4 | 1D3    | 1 day     | Swelling, pain, fever                 |
| Pneumonia       | +2 | 1D6+4  | 1D6 weeks | Cough, fever, chest pain              |
| Anthrax         | -2 | 1D6+2  | 1D6 days  | Cough, fever, tiredness, chest pain   |
| Cholera         | +1 | 1D6-2  | 2D6 hours | Diarrhoea, vomiting, thirst           |
| Ebola           | -4 | 1D6+4  | 2D6 days  | Fever, muscle pain, internal bleeding |

## POISONS

Poisons reduce a character's characteristics, usually Endurance. The character must make an Average (0) Endurance roll with the listed DM to resist the effects of the poison. If the character fails the Endurance roll then he takes the listed damage within 1-6 combat rounds.

| SAMPLE POISONS |    |                                      |
|----------------|----|--------------------------------------|
| Poison         | DM | Damage                               |
| Nerve Agent    | 0  | 6D6 (or 2D6 damage if End roll made) |
| Toxin          | -2 | 4D6 (or 1D6 if End roll made)        |
| Sedative       | -4 | Unconsciousness for 1D6 x 10 mins    |
| Paralytic      | -2 | Paralysis for 2D6 mins               |
| Cyanobacteria  | 0  | Mild insanity for 1D6 hours          |

## SUFFOCATION

In an area where sufficient oxygen is not long available, such as on board a starship without life support, a character begins to suffocate, suffering 1D6 damage each minute. A character who is utterly without air (such as one who is being smothered or strangled) suffers 1D6 damage each round, instead.

## HOSTILE ATMOSPHERES

Many atmospheres are tainted (Type 4, 7, 9) or unbreathable (Type 2, 3 or A) ; but some are vacuum (Type 0 and 1), whilst others are extremely toxic to humans (Type B and C). Taints may be severe, doing damage every minute, or more long term, doing damage every hour.

| DAMAGE FROM ATMOSPHERE                   |                         |
|------------------------------------------|-------------------------|
| Atmosphere Type                          | Damage                  |
| <b>BREATHABLE</b> (Type 5, 6, 8)         | Fine to breathe         |
| <b>TAINTED</b> (Type 4, 7, 9)            | 1D6 each minute or hour |
| <b>UNBREATHABLE</b> (Type 2,3, A)        | 1D6 each minute         |
| <b>VACUUM</b> (Type 0, 1)                | 3D6 each round          |
| <b>CORROSIVE/INSIDIOUS</b> (Type B or C) | 1D6 each round          |

## TEMPERATURE

The following labels cover a temperature band that can be equated with an Earth-like climate or the climatic conditions on some moon or planet within the solar system. As a planetary label, it is simply a generalization

### **Frozen [-100°C or lower]**

Too cold for human habitation, protective gear required: Moon in shade, Pluto.

### **Cold [-15°C or lower]**

Lower limits of human habitability: Yellowknife in winter, Mars.

### **Cool [-10°C to -1°C]**

Habitable zone, but cooler on average than Earth: Toronto in winter.

### **Temperate [0°C to 24°C]**

Habitable zone, Earth average climate, similar to Boston or Paris.

### **Warm [25°C to 34°C]**

Habitable zone, but warmer on average than Earth: Los Angeles in summer.

### **Hot [35 °C or higher]**

Upper limits of human habitability: Mojave Desert in summer.

### **Inferno: [100°C or higher]**

Too hot for humans - protective gear required: Moon in sunlight, Mercury in daylight.



Each damage value below is paired with a time increment – how often that damage is incurred. This time increment will decide the scenario’s time scale; the characters might be moving week by week across an ice field on a Frozen world, in an ATV, but if a character one day smashes his suit helmet whilst digging the ATV out of a ditch, then the scale of time moves instantly from weekly to minutes (the character is inside a sealed suit, but since it is no longer sealed we can treat it as ‘Suitable Clothing’ and begin inflicting 4 damage points every minute). The GM must decide on the temperature of the terrain being crossed. The dune seas of Mars’ Herschel Impact Basin are Cold. Those of Saudi Arabia’s Empty Quarter are Hot. Polar regions are either Cold, or on icy planets and moons like Ganymede, can be Frozen. Cold and Hot environments are dangerous and can kill if not adequately equipped.

| DAMAGE FROM EXTREME TEMPERATURE |                     |                   |                    |            |
|---------------------------------|---------------------|-------------------|--------------------|------------|
| Surface Temperature             | Unsuitable Clothing | Suitable Clothing | Sealed Suits       |            |
| <b>FROZEN</b>                   | 8 pts /min          | 4 pts /min        | Survival Suit      | 1 pt/hour  |
|                                 |                     |                   | Vacc Suit          | 2 pts/day  |
|                                 |                     |                   | Hostile Envirosuit | 0 pts/day  |
| <b>COLD</b>                     | 12 pts/day          | 6 pts/day         | -                  |            |
| <b>HOT</b>                      | 8 pts/day           | 6 pts/day         | Sand Suit          | 1 pt/day   |
| <b>INFERNO</b>                  | 6 pts /min          | 6 pts /min        | Survival Suit      | 6 pts/hour |
|                                 |                     |                   | Vacc Suit          | 2 pts/hour |
|                                 |                     |                   | Hostile Envirosuit | 0 pts/hour |

**HOT:** Assumes zero water to drink; -6 from damage if minimal water available; -8 if plentiful water available.

**COLD:** Assumes unheated tent/shelter; -4 from damage if heated shelter; -2 if eating hot daily meal.

## RECOVERING TEMPERATURE DAMAGE

Characteristic points lost due to extremes of high or low temperature can only begin to be restored if the character can get into heated (or cooled) shelter. If the character is not Seriously Wounded, then natural recovery can begin.

| Resources                       | Daily Recovery of Characteristic Points |
|---------------------------------|-----------------------------------------|
| Shelter                         | 1 + Endurance DM                        |
| + Hot Food/Drinking Water       | 2 + Endurance DM                        |
| + Medical care/autodoc/hospital | 3 + Endurance DM                        |

A Seriously Wounded character who has found shelter is still deteriorating. Surgery is not appropriate. Instead, they can be treated by a medic once per hour, who must make an Average (0) Medical roll. This restores 3 characteristic points, and allows the character to begin recovery, as above.

After 2 days of 50°C desert temperatures, an explorer (777675) dressed for warm weather but without any water, is reduced to 270675. Two more days of this will kill him. He finds a cool deep cave with a pool of water at the end. This is shelter, with drinking water, and so he gains 2 points each day. Now his problem is food ...

## RADIATION

Radiation can be real killer in space. Solar storms bathe interplanetary space with energized protons that can reach deadly levels very quickly and some stars and gas giants have particularly deadly radiation belts which should making prolonged travel nearby equally unhealthy. Radiation in **Cepheus Universal** is measured in Rads.

### RADIATION EXPOSURE

| Radiation Source                                         | Exposure Level      |
|----------------------------------------------------------|---------------------|
| Hot Zone (close to white dwarf or radioactive gas giant) | High Level per hour |
| Danger Zone (close to gas giant, star)                   | Low Level per hour  |
| Minor Solar Flare (duration 1-6 hours)                   | Low Level per hour  |
| Major Solar Flare (duration 1-6 hours)                   | High Level per hour |
| Nuclear Blast                                            | Extreme Level       |
| Crew Radiation Hit (Minor)                               | Low Level           |
| Crew Radiation Hit (Critical)                            | High Level          |
| Irradiated Area (per hour)                               | Low/High/Extreme    |
| Inside Chamber of Leaking Reactor                        | Extreme             |
| Exposure Level                                           | Rad Exposure        |
| Low Level                                                | 2D6 x 5             |
| High Level                                               | 3D6 x 20            |
| Extreme Level                                            | 1D6 x 1000          |

## RADIATION PROTECTION

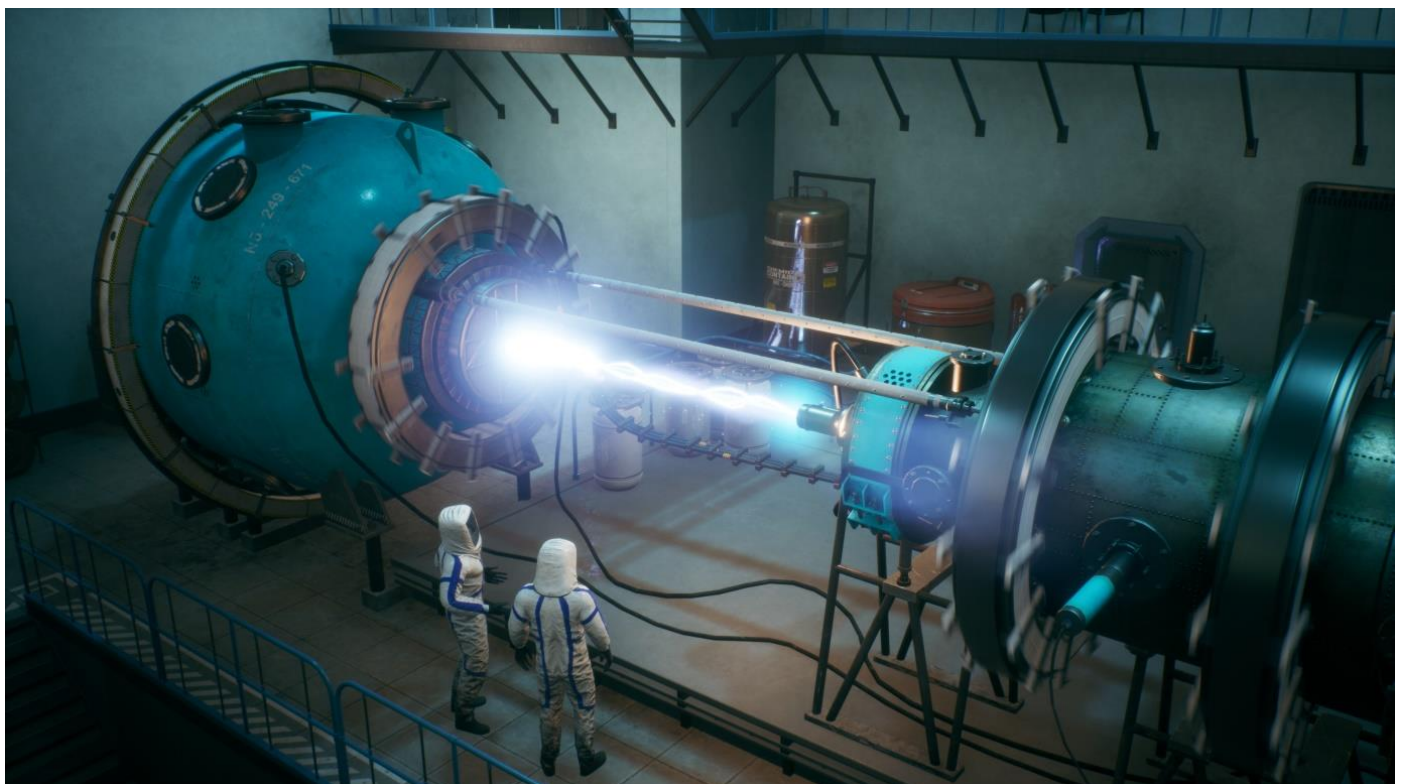
Subtract the relevant value from the Rad Exposure before totalling up the character's Rad dosage. Roll each hour if the exposure is extended.

| Protection                         | Reduction in Rads |
|------------------------------------|-------------------|
| Hazmat Suit                        | 25                |
| Soft Suit                          | 20                |
| Vacc Suit                          | 40                |
| Hostile Envirosuit                 | 90                |
| Hardened Spacecraft Bridge         | 280               |
| Spacecraft Hull                    | 180               |
| Radiation Shielded Spacecraft Hull | 400               |

## DAMAGE FROM RADIATION

Inflict the rolled damage on the character (starting with Endurance). The rads accumulated are then discarded.

| Total Rad Exposure | Damage |
|--------------------|--------|
| 0-24               | None   |
| 25-99              | 1D6    |
| 100-1000           | 2D6    |
| 1000               | 3D6    |
| 2000               | 6D6    |
| 5000+              | 9D6    |





## EFFECTS OF RADIATION DAMAGE

Damage from radiation is not quite like that inflicted by a knife or bullet, its effects manifest over time. The severity of the sickness is indicated by the loss of characteristic points. While the rules here are quite clinical and clean, radiation sickness is not although such unpleasantness may be glossed over as the GM sees fit.

**Endurance Remains Above 0:** No appreciable affects, the character naturally recovers 1 characteristic point per day.

**Endurance 0 (Very Ill):** After a number of hours equal to Endurance the character suffers nausea and sickness. Make an Average (0) End check to recover, or suffer another 1D6 damage and be very ill for 1-3 days. After this time has passed the character naturally recovers 1 characteristic point per day. A medic can assist with a lysophosphatidic acid (LPA) treatment making a Routine (+2) Medical check to assist and provide a +2 DM to the patient's roll.

**Two Characteristics at Zero (Severely Ill):** After a number of hours equal to Endurance the character suffers nausea and sickness, internal bleeding and some hair loss. Make a Difficult (-2) End check or suffer an additional 1D6 damage. Failure means the character is feverish and nauseous, and along with diarrhoea and vomiting, they cannot face the thought of food. Roll each week until the check is successful, at which point recovery begins and the character recovers 1 characteristic point per week. A medic can assist each week with a lysophosphatidic acid (LPA) treatment, making a Routine (+2) Medical roll to assist and provide a +2 DM to the patient's roll. Severely ill characters lose 1 point of Endurance permanently.

## PRESSURE LOSS

On many small moons, on asteroids and in deep space, humans are battling against lack of pressure, in most cases a vacuum or a near vacuum. This is always the greatest danger to any space traveller.

## PUNCTURING THE HULL

An attack from outside the hull may puncture it if the roll to hit is successful. Inside the hull, stray bullets or other missiles may strike furniture or equipment instead of the hull. If a gun is discharged and does not hit a human target, roll its damage dice to determine its trajectory. On 2-9 the bullet lodges in something else, on 10+ the bullet strikes the hull, or just ploughs through the piece of furniture to hit the hull anyway. Lasers and plasma guns will destroy whatever they hit, so the chance of a hull strike for a laser is a basic 50% (roll 2-6 on a 2d6 for a hull strike).

| Hull of a ...                             | Armour Value |
|-------------------------------------------|--------------|
| Vehicle hull                              | 4            |
| Spacecraft or Space Station (TL 6-9) hull | 10*          |
| Spacecraft or Space Station (TL 10+) hull | 20*          |
| Space Habitat hull                        | 30           |
| Colony Base hull                          | 20           |
| *Each point of Spacecraft Armour adds +2  |              |

If a weapon hits an outer hull, roll for weapon damage as normal and subtract from this the armour rating of the target. If penetration occurs, a 1 cm diameter hole is punched into the hull and decompression begins.

Explosives of any kind automatically strike the hull if detonated internally. The fact that the atmospheric pressure of the cabin is exerting great force on the skin of the habitat or vehicle adds a considerable punch to any internal explosion, increasing the chance of a puncture. Roll explosive damage and subtract the relevant armour modifier. Every point of explosive damage beyond that needed to puncture the hull widens the hole by 20 cm. A frag grenade tossed into a TL 8 orbital lab goes off and does 24 points of damage, lowered to 14 after the lab's hull is considered: that constitutes a breach. The 14 points cause explosive decompression with a catastrophic hull rupture 2.8 m in diameter... This combat rule can serve as a guideline for other projectiles, from micrometeoroids striking the hull, to vehicles crashing into a surface habitat or fuel pumps exploding. See the full rules for Defeating Barriers on page 142.

## **THE SPEED OF DECOMPRESSION**

Designers do not build huge open-spaced colony buildings or spacecraft, they are divided up into smaller compartments separated by pressure bulkheads. Decompression of one section will not affect the rest of the vehicle or habitat. Puncturing the skin of a spacecraft or habitat is easy and great efforts must be made to avoid situations where such a catastrophe might occur. But what happens when such a situation arises?

For a 1 ton cabin, a 1 cm diameter hole will cause pressure to drop by 50% in 5 minutes and to vacuum in 10 minutes. The half-way point marks the level at which asphyxiation begins (see below) and the character suffers 1D6 damage per minute. This basic value can be extrapolated to determine pressure loss for other sized-compartments. Should a bullet punch a hole through the skin of a 20 ton cargo hold, for example, pressure loss will take 20x longer (1 hr 40 mins to reach half an atmosphere, 3 hrs 20 minutes to be reduced to vacuum). The real danger is in hatches popping open and in ruptures caused by explosions. Using the previous example of a 10 ton lab module damaged by a grenade, the breach created was 2.8 m in diameter. Increasing the affected volume increases the time to asphyxia to 50 minutes, but the breach was also 280x larger:

*Time to Asphyxia:*  $(5 \text{ mins} \times 10) / 280 = 0.17 \text{ mins}$  (10 seconds, or around one round)

If a 1 ton airlock blew its hatch and exposed unprotected individuals to vacuum, the effects of asphyxia would occur much faster. Assume the hatchway is a 1.2m breach:

*Time to Asphyxia:*  $(5 \text{ mins} \times 1) / 120 = 0.042 \text{ mins}$  (3 seconds or half a combat round)

Note that since full vacuum is reached after *twice* this duration, the people in the airlock will suffer the perils of vacuum at the end of the six-second combat round. What exactly does this do to the characters?

## **EFFECTS OF VACUUM ON THE HUMAN BODY**

Exposure to vacuum results in 3D6 damage per combat round. To save someone suffering catastrophic haemorrhage and embolism requires that pressure be restored to at least 50% or more of atmospheric norm. This will prevent further damage. Recovery requires a trauma kit with a defibrillator to restart the heart, an oxygen bottle and specific drugs (namely pentoxifylline, prostaglandins and calcium channel blockers), all of which are found within spacecraft and habitat trauma kits.

*To stabilize a patient exposed to vacuum:* Difficult (-2) Medical, 1D6 mins. Failure indicates the patient continues to deteriorate at 1D6 pts per hour. The medic can try again each hour. Success indicates the patient regains 3 characteristic points and can begin recovery as normal.

## **OPERATING VACC SUITS**

Sealed spacesuits, such as vacc suits or hostile envirosuits, may be crucial in staying alive in environments where there is no air or that is poisonous to humans. Their drawback is endurance – most have a limited time on surface of around 6 hours. Once a suit's Armour Value is breached, the wearer will take damage: in the first round the wearer suffers no damage, in the second round he suffers 1 pt of damage and in the third and subsequent rounds he suffers 2 pts. Patching a leak is an Average Vacc Suit roll, taking two rounds. At TL 10 suits self-seal in two rounds. For major breaches, the damage is difficult to repair. Being inside a breached suit is a Stress Event.

*To repair a major breach:* Difficult (-2) Vacc Suit, 1D3+2 rounds; or Very Difficult (-4) if repairing one's own suit. All suits include a suit repair kit.



## ZERO GRAVITY

At TL 10 and up, spacecraft and large space stations use grav plates to create artificial gravity, allowing crew to walk around as if they were on an airliner. At TL 7-9, these habitats must be spun to create the impression of gravity. Outside of those comforting environments, and away from the surface of a moon or planet, zero-G is king. Simply moving around carefully requires no roll but carrying out some tricky manoeuvre or task certainly will. If using a firearm in zero-G, resolve the attack roll first, then whether the shot hit or not, attempt to retain control. Losing control in zero-G during a space-walk is a Stress Event.

*Retain Control in Zero-Gravity:* Average (0) Vacc Suit

Apply the following DMs: Using a tool to repair/construct -2; striking with tool or melee weapon; or pushing/pulling -4; using a handhold or brace +2

*Retain Control Whilst Firing a Gun With Recoil:* the player rolls the Recoil number or higher using 2D6.

Apply the following DMs: Dex Modifier; Vacc Suit skill; using a handhold or brace +2; firing a 4-rd burst -4; firing a 10-rd burst -6.

Losing control means the character is tumbling and if a task was being attempted, it is abandoned. Roll again to regain control, but this time there are no DM's, either positive or negative, except for those derived from Vacc Suit skill and the Dexterity characteristic.

## HUNGER & FATIGUE

### HUNGER

Characters might find themselves without food or water and without any means to obtain them. In a survival situation in a temperate climate, a character needs at least 1 litre of water and about 0.5 kg of high protein food (meat, etc.) per day to avoid starvation. In very hot climates, characters need up to five times as much water to avoid dehydration.

**Food Units:** To make rationing food easier, we refer to food 'units', with 1 kg of meat equating to 10 units and 1 kg of plant food equating to 5 units. In a wilderness scenario, we can assume that a character requires **5 units of food** each day to survive (while a contented colonist enjoys 10 food units per day). A 2 kg ration pack provides 5 food units, for three days (15 units in total).

**Starving:** Without any food at all, a character deducts 1 point from Strength, Dexterity or Endurance each *day*. With rationed food (1-3 food units per day) a character deducts 1 point from Strength, Dexterity or Endurance every *week*. This starvation damage cannot be restored by normal means: 2 characteristic points are recovered each day that a full 5 food units are consumed.

**Thirst:** For each day water is rationed, reduce all three characteristics by 1 point if only 0.5 litre is drunk; by 2 points if 0.25 litre is drunk; and by 3 points if no water is drunk.. Similar to hunger, damage sustained from thirst is recovered at the rate of 2 points for *every* characteristic each day they drink 1 litre of water.

## FATIGUE

Player characters require a Sleep Period in every 24, equal to 8 hours, although characters can reduce this by their End modifier for up to a week, followed by at least a week of regular sleep. Sleep must be had as a full Sleep Period, or as two 4 hour Sleep Periods. Fatigue always begins 16 hours after a Sleep Period has ended (14 hours if the character has been hiking with a Medium Load and 12 hours if with a Heavy Load). When this fatigue begins, the character must make an Average (0) End check to avoid the need for sleep. Failure results in a -2 DM to all skill and characteristic checks until he rests. The fatigue roll is made again two hours later, but the DM increases now to -3. After that the character starts making Average (0) End checks every 30 minutes and if he fails, will fall asleep for an hour (minimum), totally overcome by exhaustion. Fatigue may play a part in a wilderness adventure, particularly if the scenario involves a dramatic chase across country, where stopping to rest may prove fatal.

## CLIMBING

Climbing anything more difficult than a ladder might involve a Routine (+2) test of Strength at roughly 3m per round. Failure indicates lack of progress, the climber can try again next combat round. Three consecutive failures indicates you are stuck. To continue make a Str roll at the next difficulty level. If this is failed the climber falls. Sheer surfaces, such as the vertical sides of buildings, dams, bridges, and rock cliffs follow the same rules, but require an Average (0) Strength roll. Long climbs, progressing at 1m per round, will require multiple rolls. Obviously, climbing ropes and climbing gear will help immeasurably in such a climb.

## RAPPELLING & FAST ROPING

Rappelling is a controlled descent off a vertical drop, such as a rock face or building, using a rope, climbing harness and a belaying device, like a descender. Rappelling can be used to descend from the top of a building to reach a target window. A technique called 'fast roping' is similar, but the character is not attached to the rope and carries no climbing gear, he is able to slide down a thickly woven rope, controlling his descent using gloved hands and the pressure of his legs and boots. Rappelling or fast roping requires Routine (+2) Dexterity. Failure to rappel means the character has gotten stuck, and must spend one full combat round untangling the rope from the descender. Failure to fast rope results in the character landing heavily and suffering 1D6+1 damage.

# COMMUNICATIONS

Typical short range communications are through well-known hand signals and verbal communications. Radio communications are affected by the terrain and atmospheric conditions. Stellar flares stop radio communications. To make contact with another radio set within range, make a Comms roll. The GM creates a difficulty by rolling 2D6. Apply the Radio Contact DMs. Failure indicates that conditions are not conducive. Try again somewhere else, or in 15 minutes. Radio comms are often frustratingly hit and miss – just when you need them most ...

| RADIO CONTACT DMs         |         |
|---------------------------|---------|
| Comms skill               | + level |
| Within half radio's range | +2      |
| Inside building           | -3      |
| Built-up area             | -2      |
| Mountains/ thick jungle   | -2      |
| Dry, desert terrain       | -1      |
| Radio TL 8-10             | +1      |
| Radio TL 11-13            | +2      |
| Radio TL 14+              | +3      |

## UNDERWATER

Radios have a restricted range of 20-30m underwater, so instead hardwired cables are used to connect seabed habitats with one another or the surface. Maser communicators are sometimes used to communicate with work groups, subs and ROVs within 300m.

## IN ORBIT

Some radios are able to contact a ship in orbit, but once it sinks below the horizon, the ship is out of contact to all but TL 16 communicators (which can transmit through solid objects). How long is the ship above the horizon? This doesn't depend on the speed of the ship, but its orbital altitude. Use these approximate values below. Note that a ship in geostationary orbit always remains above the same location on the surface.

| World Size | Low Orbit  | High Orbit      | Geostationary Orbit |
|------------|------------|-----------------|---------------------|
| 0          | 1 minute   | 24 hours        | 24 hours            |
| 1,2        | 2 minutes  | 30 mins         | 24 hours            |
| 3,4        | 3 minutes  | 1 hour          | 24 hours            |
| 5,6        | 5 minutes  | 1 hour 30 mins  | 24 hours            |
| 7,8        | 7 minutes  | 2 hours         | 24 hours            |
| 9,10       | 9 minutes  | 2 hours 30 mins | 24 hours            |
| 11+        | 15 minutes | 3 hours         | 24 hours            |

**Note:** On Earth, a low orbit might be 400 km, a high orbit might be 20,000 km and geostationary orbit is 35,786 km. On larger worlds the same orbits will be further out than that.

## INTER-SYSTEM COMMUNICATION

Some settings may have faster-than-light communications, enabling star systems to talk to one another. The easier this becomes, however, the less reason there is to travel between the stars. Being able to call for help when your starship is stranded takes some of the adventuring potential and isolation away from the player characters. Having no FTL comms is fine, but if they *are* included, make them expensive to use and requiring a time delay. This might be 1 hour per parsec, 1 day per parsec, or 1 week per parsec. In a setting that Goes Where No Man has Gone Before, the speed of communication is roughly 1 parsec per hour.



# MOVEMENT

## RUNNING

Running requires the character to devote a Significant Action to movement. Running can be either a sprint or a jog. ***Sprinting*** allows up to **18m** movement in a combat round, but sprinters cannot make perception rolls, dodge or dive for cover. ***Jogging*** allows up to **9m** movement, it allows perception rolls, the chance to dodge or dive for cover and involves staying low to try and avoid being hit by gunfire. Characters can alternatively crawl 3m per round, helping them to try and remain undetected by an enemy force. Difficult terrain, such as rubble, mud or thick under-brush can halve a character's movement. Anyone can move **2 or 3m** as a Minor Action.

## JUMPING

Characters can jump a distance of 1m easily, or jump over low obstacles with no roll needed. To jump across 1.5 to 2.5m, though, requires a Routine (+2) Dexterity skill roll. This roll could also be used to attempt some other unusual jump manoeuvre.

## SWIMMING

These rules assume that all player characters can swim in calm water, unencumbered by clothing or equipment at 3m per round. Add both the Str *and* Dex characteristic modifiers to this figure. Use of swim fins increase speed by 2m per round. Bursts of fast swimming speed at 6m per round can be attained for a number of minutes equal to Endurance. A Routine (+2) Str roll is required each 30 minutes of swimming, or if the swimmer is in unusual circumstances, such being lightly loaded, is injured, jumps into water from a height, is in turbulent water or is being shot at. Failure indicates either lack of progress or disorientation. Make a second roll, if this is failed, the character suffers 1D6 drowning damage each round, and must attempt a fresh Str roll every round (with a cumulative DM -1) until he or she either falls unconscious and dies, or regains control.

***Cold Water:*** Falling or jumping into cold water (0-10°C) without a survival suit can cause shock in the human body. This is a Stress Event. Make a Routine (+2) End roll or suffer drowning damage, as above. The character then suffers 1D6 points of cold damage every 10 minutes if without a survival suit, or 1D6 points every hour if wearing a survival suit.

***Lifesaving:*** An Average (0) Str roll is required to swim to the aid of a victim and prevent them from drowning. Failure means the victim almost drowns you in their desperation (suffer 1D3 damage). A second failed roll indicates that the lifesaver themselves is now being pulled under and is drowning. Roll Routine (+2) Strength to fight them off and swim away. If a lifesaving roll is successful, the character may be able to tow the victim at half speed to safety if applicable to the situation.

UNDERWATER

Diving is possible using a self-contained underwater breathing apparatus (SCUBA) such as a rebreather or scuba tanks and equipment. Both allow a diver to swim without the need for decompression down to a depth of 20m (on Earth). Characters descend and ascend at their normal swim speed; a Str 10 character with diving fins, for example, descends at 3m +1 + 2 = 6m per round (36m per minute).

PRESSURE AT DEPTH

Atmospheric pressure on the Earth’s surface is 1 ‘atmosphere’ (atm), this pressure on a diver increases the deeper he goes, reducing his air supply and increasing his chance of suffering nitrogen narcosis. See the Pressure & Depth table. The depth a diver can reach on an alien world depends on the planet’s gravity and its atmospheric density. For a Thin atmosphere +5m to the depth; for a Dense atmosphere -10m for the depth. The GM will need to decide the air density himself in atmospheres A, B or C. The final depth can be multiplied by the world’s surface gravity value if additional realism is required. Since the air breathed by the diver will be compressed with depth, more air must be breathed the deeper he goes. Time allowed below the surface is determined by the air tank’s supply divided by the atmospheric pressure at the depth of operations. A diver with six hours of oxygen, diving to 3 atmospheres (20m) will only have 2 hours of air, for example.

| PRESSURE & DEPTH |                    |
|------------------|--------------------|
| Pressure (atm)   | Depth on Earth (m) |
| 1                | 0                  |
| 2                | 10                 |
| 3                | 20                 |
| 4                | 30                 |
| 5                | 40                 |
| 6                | 50                 |

MAXIMUM DIVE TIMES

At depths of 3 atm or more, the water pressure causes a diver's body tissue to absorb more nitrogen gas than normal. This reduces his time at depth and if he does not ascend slowly at the end of his dive, his body cannot eliminate all of the expanding nitrogen that forms bubbles in his tissues and blood causing decompression sickness ('the bends'). This blocks blood flow to various parts of the body, causing strokes, painful paralysis and other life-threatening problems. The table, below, provides an approximate guide to the time spent in decompression stops whilst ascending to the surface (for simplicities sake, either 3 or 5 hours or none needed at all). Often air tanks for the purpose of decompression stops are fixed at set points on a dive line.

| MAXIMUM DIVE TIMES ALLOWED |                             |             |              |
|----------------------------|-----------------------------|-------------|--------------|
| Pressure (atm)             | Decompression Time Required |             |              |
|                            | None                        | Three hours | Five hours   |
| 1                          | <i>surface</i>              |             |              |
| 2                          | no time limit               | -           | -            |
| 3                          | 2 hrs                       | 4 hrs       | 6 hrs        |
| 4                          | 1 hr                        | 2 hrs       | 3 hrs        |
| 5                          | 30 mins                     | 1 hr        | 2 hrs        |
| 6                          | not allowed                 | 40 mins     | 1 hr 20 mins |

**Example:** Wanting to access a disabled ROV at 4 atm depth, our diver looks along the table – he can work on repairing the drone for 1 hr, 2 hrs or 3 hrs, but a 2 hr schedule will mean his ascent must take 3 hrs; a 3 hr schedule means his ascent must take 5 hrs.

## **DECOMPRESSION SICKNESS**

A character who exceeds his allotted time at depth by 25% or more, or who does not spend the required time rising to the surface, will show signs of the bends within 3D6 minutes of surfacing. The character makes a Difficult (-2) End roll and if successful will suffer 1D6 points of damage. If the roll is failed then 1D6 damage is suffered *every hour* until treatment is received. Oxygen can be administered with a Routine (+2) Medic roll, which will reduce damage to 1 pt per hour instead. Treatment and recovery requires 1D3+1 hours within a hyperbaric recompression chamber (common on large diving ships), or an autodoc. A starship airlock could fulfil the same purpose.

## **NITROGEN NARCOSIS**

Nitrogen narcosis or ('rapture of the deep') is a condition that occurs when the diver breathes compressed air. When divers go to 3 atm or greater, the partial pressure of nitrogen produces an altered mental state similar to alcohol intoxication. This can be very dangerous, since the diver himself may not realise he is being affected. At 3 atm depth, a diver should always make an Easy (+4) roll to avoid narcosis, apply both the character's Int and End modifier. At each atmosphere pressure below 3, the roll is repeated with an additional -1 DM. If failed, the GM should impose a – 1D6 penalty on any action taken. Even to carry out his own decisions (such as ascend to a safer depth), the character must make a Routine (+2) Int roll! The effects of narcosis disappear once the diver has ascended above 3 atmospheres.

## **DIVING IN A HOSTILE ENVIROSUIT**

Most diving below 2 atmospheres of pressure is carried out in hostile envirosuits. These are high-tech versions of the 'atmospheric diving suits' used today by salvors, rig divers and explorers, such as the Newtsuit, ADS2000 and the WASP suit. These 40 kg suits can maintain air supply at surface pressure, requiring no need for decompression stops, no risk of narcosis and with the only limit on how long a diver can remain at depth, being the suit's integral air supply. Wearers can be lowered to the sea floor and walk about, but many operators fit a thruster unit (Cr1000, 4 kg) which provides a swimming speed of 5m per round (30m per minute) for a total time of 60 mins.

## BUILDING INTRUSION

At some point, there will be player characters who need to gain access to a secret laboratory, or to an off-limits military complex at the far end of the starport. And getting in to those places, may not be easy...

### BUILDING SECURITY LEVELS

The GM should rate the security measures of a building:

- 0 - **NO SECURITY** - No doormen, no cameras, no checks. Freedom and anonymity. Typically: a street market or park.
- 1 - **LOW SECURITY** - A camera takes pictures at frequent intervals (every 10 seconds). No doorman, no checks. Typically: a cafe, bar or typical restaurant, or shop.
- 2 - **HIGH SECURITY** - Cameras are installed at the entrance and other sensitive areas, and doormen check visitors for guns and knives using detectors. There are alarmed windows, lifts often have cameras. Typical: A laboratory, upmarket restaurant, expensive hotel, corporate office block.
- 3 - **MAXIMUM SECURITY** - Full video surveillance, doormen use a walk-through X-Ray machine to check for concealed weapons. All visitors must check in and out (usually wearing a broadcasting ID badge) or if a club or bar, for example, must gain membership with a credit check. Shatterproof windows, lifts have cameras and some floors may require ID checks. Typical: airport terminal, any major corporate HQ or sensitive military complex.

### GETTING ON TO THE SITE

Buildings often have several layers of physical defense. The external perimeter is usually guarded by a fence or other barrier that must be climbed or cut through.

**Low Security** - 2m high, requires a Dexterity roll to climb over.

**High Security** - 6m high, requires two Dexterity rolls to climb over.

**Maximum Security** - 6m high, requires two Dexterity rolls to climb over, and is electrified and or alarmed – this must be disabled first.

A failure to climb a barrier means the character is stuck halfway-up, roll again. Two failures in a row means a fall. It is usually best to cut through fences so as to not be seen struggling over them, this is an Average (0) Mechanical roll, taking 1D6 mins (if using bolt-cutters, disintegrator or cutting torch). Concrete or steel barriers can only be breached using explosives or disintegrators, although neither are discrete.

**Avoiding Foot/Robot Patrols** – Make a Security roll (add Int modifier) to move around the inside of a compound or to climb a fence unseen. Difficulty is listed on the table below, with the first column on the table being used if the character is attempting a random penetration, the second column is if he is using a timepiece or rota to time the foot or robot patrols. The time-factor is how long-apart the patrols are. Crossing floodlit areas at night means that the character will probably be visible to cameras or patrols. If a character with Security skill leads the penetration of the compound, he makes the rolls (assuming his comrades do as they are told).

| Level of Security | Frequency | Task Difficulty |              |
|-------------------|-----------|-----------------|--------------|
|                   |           | - Random -      | - Timed -    |
| Low Security      | 30 mins   | Routine (+2)    | Easy (+4)    |
| High Security     | 10 mins   | Average (0)     | Routine (+2) |
| Maximum Security  | 2 mins    | Difficult (-2)  | Average (0)  |

## CAMERAS

Cameras cannot be watched 24 hours a day, and are therefore hooked up to a video recorder. A character may be able to get away with being visible to a guard scanning several monitors. For a *permanent system malfunction* (such as disabling the camera) or blatant video evidence (such as a dead guard lying in a corridor) roll 5+ on 2D6 every 5 minutes for it to be detected. For *fleeting system malfunction* or video evidence (such as shooting someone in front of a camera, but pulling the body out of the way) roll 9+ once only, to spot. Players can roll a Routine (+2) Security roll to spot a particular camera before their agents are seen by it. At TL 12, a camera is monitored by computer which watches for unusual activity, or faces that it does not have in its database. It requires a Difficult (-2) Security roll to fool such a surveillance system.

## BYPASSING LOCKS

Lock types vary by Tech Level, and these types provide a DM to defeat them if the character possesses the right equipment. Mechanical locks can be cracked with improvised tools at -2. Others *always* require dedicated security kits.

| LOCK TYPES            |                                             |    |
|-----------------------|---------------------------------------------|----|
| Tech Level Introduced | Type                                        | DM |
| 4                     | Mechanical                                  | +1 |
| 6                     | Passcode/Card                               | +1 |
| 8                     | Biometric (Thumbprint/Retina/DNA/Facial ID) | 0  |

Locks of the same type can be made more difficult to bypass depending on the security classification of the area they protect. The mechanical lock of a bank's vault will be tougher than the front door lock, that in turn will be tougher than a desk drawer lock. The difficulty of the bypass attempt will be based on the area it protects, modified by the Tech Level of the lock. Give a building a single general security rating then decide what locks defend the important locations within.

|                               |                                                                             |
|-------------------------------|-----------------------------------------------------------------------------|
| <b>No Security Building</b>   | Exterior: Routine (+2)<br>Interior: Easy (+4)<br>Secure: Average (0)        |
| <b>Low Security Building</b>  | Exterior: Average (0)<br>Interior: Routine (+2)<br>Secure: Average (-2)     |
| <b>High Security Building</b> | Exterior: Average (0)<br>Interior: Average (0)<br>Secure: Difficult (-2)    |
| <b>Max. Security Building</b> | Exterior: Difficult (-2)<br>Interior: Average (0)<br>Secure: Difficult (-2) |





**Roll To Bypass The Lock:** The character makes a Security roll at the difficulty stated in the table above, modified with a DM based on the lock's type. Each attempt requires 1 minute, and gets more and more difficult (-1 each time). Three failures in a row either activates an alarm or jams the door lock mechanism. Mechanical skill can be substituted for Security when dealing with a mechanical lock, and Electronics can be substituted for Security when dealing with a card/voice/retina lock. There will be a -2 penalty (just like improvising) and each attempt takes 2 minutes, not 1 minute.

**Example:** Marco (Security 1 skill, with a security kit), wants to break into a starport warehouse to look at a bonded cargo shipment. This is a High Security Building at TL 10 (biometric locks). To gain access through the exterior door is Average, with a DM 0. He rolls 6+1 for 7, but he needs an 8 to succeed. He tries again, now with a -1. He rolls 10+1 and -1 for 9, and succeeds on his second attempt.



# WEATHER

## TRAVEL

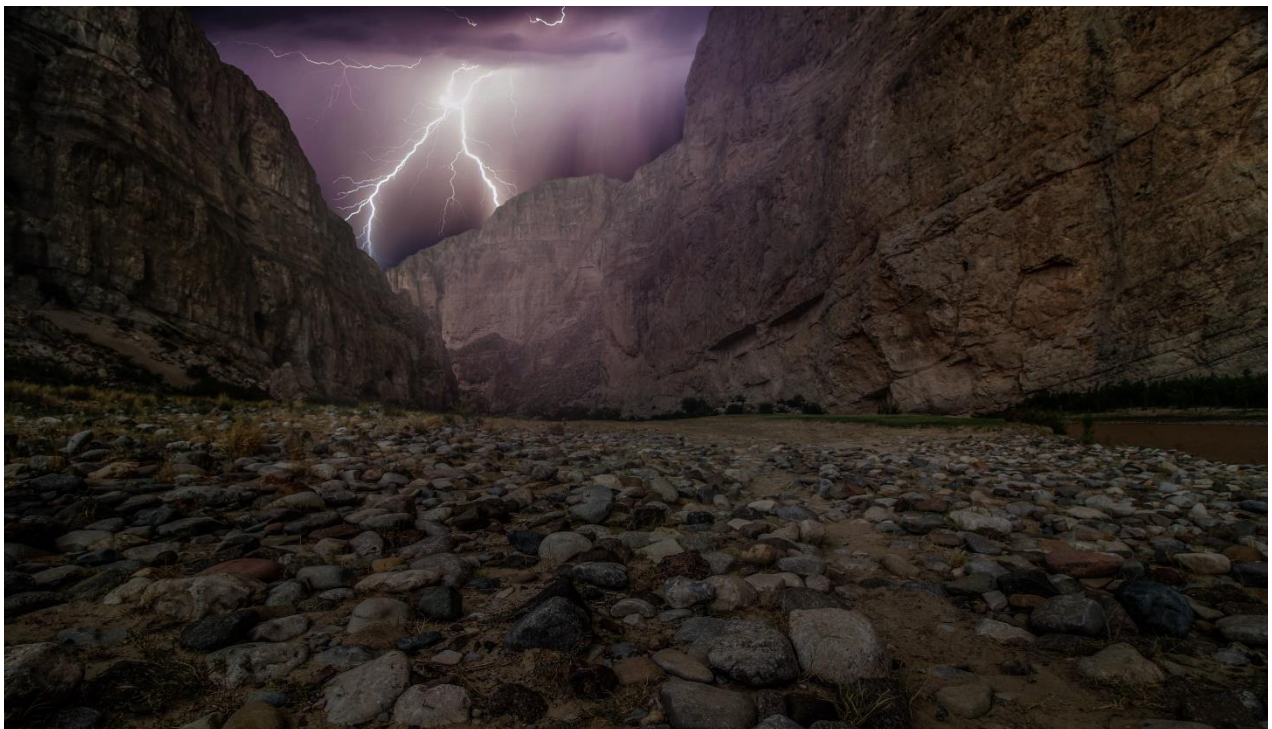
- Deep snow stops wheeled vehicles and halves the speed of tracked vehicles.
- Heavy rain will lower ground vehicle speed by 10%
- Wind, snow or heavy rain will lower hiking speed by 1 kph.
- Hurricane winds fill the air with wind-blown debris, aircraft are stranded and ground vehicles are in danger of being tipped over. Hikers must find shelter immediately or be subjected to 1D3 points of damage (make a Routine Strength roll to avoid). Further checks might be demanded by the GM should characters insist on moving about unprotected.
- Thick mud reduces travel to slow going, a vehicle driver must make an Average (0) drive roll, or get the vehicle mired in the mud. It will take 2D6 hours and a Difficult drive roll to unfree the vehicle, +1 if 3 or more people get out and help, +2 if 6 or more get out and help.

## INITIATIVE

Snowfall counts as visibility restricted (-2 DM) whilst spotting the enemy or making an initiative roll.

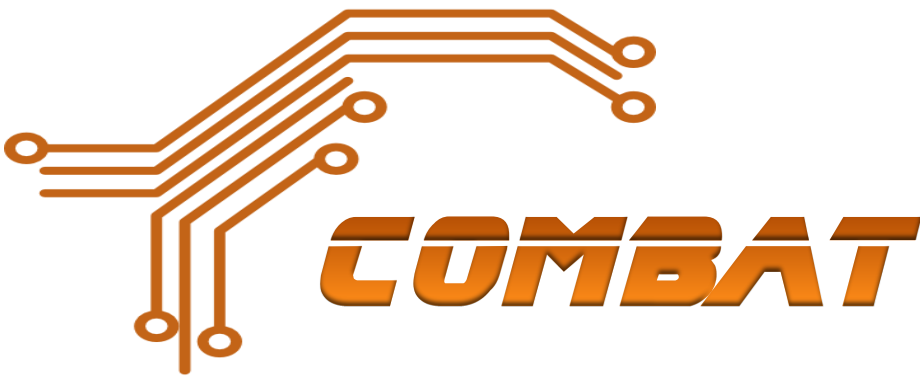
## WEATHER & COMBAT

Driving wind, rain, snowstorms and so forth gives a -1 DM to ranged attacks from poor visibility. Some sensors, such as motion trackers or infra-red scopes, can be used to avoid this visibility penalty. Extremely high winds, torrential rain and floodwaters can inflict a negative Dice Modifier of -1 to -4 to all skill checks, and may even impose Strength rolls to remain upright.









Tense situations, such as combat, are divided up into combat rounds. These are essentially one 'turn' for every participating player and non-player character. It is an arbitrary 'chunk' of time equal to about 6 seconds, roughly equivalent to the time it takes for someone to fire a shot from behind cover then duck back again, or for a character to sprint across a street. In a combat round each player character gets one significant action and one minor action.

**Significant actions** include shooting at the enemy, throwing a grenade, starting a truck, walking 6m, running up to 18 metres, calling for help on the radio, and so on. Some tasks (giving medical aid, jump starting a vehicle) will require more than one significant action (and combat round) to complete.

**Minor actions** can usually be accomplished with little effort: looking around, calling over to someone, dodging into cover, parrying a melee attack, checking the time, moving up to 3m. etc. A character can also change to any one of the three stances – prone, crouched or standing – as a minor action.

## RANGE

Personal combat is divided into a series of range bands. These are identified in the table below:

| COMBAT RANGE BANDS |                                |
|--------------------|--------------------------------|
| Range              | Distance to Target             |
| Personal           | Touching, less than 1.5 metres |
| Close              | 1.5 to 3 metres                |
| Short              | 3 to 12 metres                 |
| Medium             | 12 to 50 metres                |
| Long               | 51 metres to 250 metres        |
| Very Long          | 251 metres to 500 metres       |
| Distant            | 501 metres to 5 km             |
| Very Distant       | 5 km to 25 km                  |
| Extreme            | 25 km to 50 km                 |
| Regional           | 50 km to 500 km                |
| Continental        | 500 km to 5000 km              |

## ENCOUNTER RANGE

When a combat begins, the GM must decide the starting range that separates the combatants. For combat in tight quarters, such as during an ambush, inside a building or within an underground tunnel system, the starting range is usually Short. The range for jungle, woodland or urban encounters is usually Medium and more open areas such as farmland, prairies, hills, mountains, deserts, and open roads make it easier to see at a distance, so Long or even Very Long range would be appropriate. If the GM would rather let the encounter range be randomised, refer to the table below:

### DARKNESS

Total darkness reduces starting range to Short or less. Partial darkness restricts starting range to Medium or less.

| RANDOM ENCOUNTER RANGE |                         |
|------------------------|-------------------------|
| 2D6                    | Range of the Enemy Unit |
| 1                      | Close (3m or less)      |
| 2                      | Close (3m or less)      |
| 3                      | Short (3-12m)           |
| 4                      | Medium (13-50m)         |
| 5                      | Short (3-12m)           |
| 6                      | Medium (13-50m)         |
| 7                      | Long (51-250m)          |
| 8                      | Long (51-250m)          |
| 9                      | Long (51-250m)          |
| 10                     | Very Long (251-500m)    |
| 11                     | Long (51-250m)          |
| 12                     | Very Long (251-500m)    |
| 13+                    | Very Long (251-500m)    |

| Terrain Type            | Modifier |
|-------------------------|----------|
| Interior of a Building  | -5       |
| City, Urban             | -4       |
| Open Colony Settlement  | -2       |
| Thick Jungle            | -1       |
| Swamp, Marsh            | -1       |
| Jungle, Forest          | -        |
| Grassland, Savannah     | +1       |
| Sea Surface             | +4       |
| Hills, Rough Land       | +2       |
| Arctic                  | +2       |
| Mountains, Plains, Open | +3       |
| Desert                  | +4       |

## INITIATIVE

In a combat situation, these rules use a group perception mechanic, where one player rolls for the PCs and the GM rolls for the NPC antagonists.

### SPOTTING THE ENEMY

In any potential conflict situation, it is important to determine if the player characters see their opponent first or vice versa; this is done with a group perception roll. Of course, it is possible that both sides may become aware of one another simultaneously. The side that spots the other without being detected has the option of attempting to attack with the advantage of surprise, or may decide to withdraw to safety. One player makes an Average (0) Int group perception roll for the PCs whilst the GM makes a group perception roll for the antagonists. Animals use their Instinct modifier, rather than Intelligence. In addition, apply the following DMs for each roll:

| INITIATIVE 8+                       |          |
|-------------------------------------|----------|
| Distance                            | Modifier |
| Short                               | +3       |
| Medium                              | 0        |
| Long                                | -3       |
| Very Long                           | -6       |
| Other Factors                       | Modifier |
| Group has 8 or more members         | -1       |
| Resting/distracted/busy/unprepared  | +/-1     |
| One character has Tactics-1 or more | +1       |
| One character has Recon-1 or more   | +1       |
| Riding in a Vehicle                 | +/-3     |
| Ambush                              | +/-3     |
| Visibility restricted               | -2       |

**Example:** The players are moving through a ruined colony, between buildings, and they will meet a gang of six colony scavengers, looting a building, the encounter is at Short range. We roll for both sides, first the players. The roll is 8+ with +3 for Short range, +1 because the scavengers are busy, +1 because one of the PCs has Recon-1. We roll  $9+5 = 14$ . The PCs spot the scavengers. The GM rolls for the scavengers, with +3 for range, -1 for being distracted, and rolls  $9+2 = 11$ . The scavengers simultaneously spot the player characters!

In some cases the decision is made by the GM on the logic of the situation. If an ambush party of player mercenaries has been sat at the road junction all day to catch an unwary patrol in their sights, they have the initiative; but many times the side with initiative should be decided randomly using a group perception roll.



## AMBUSHES

The PCs or their antagonists may be able to set up an ambush in concealment, this ambush must be established in a concealed location likely to be visited by the enemy. When the victims of the ambush arrive within Medium range of the location, a -3 DM is applied to their group perception roll. Should this roll fail, the ambushers gain a full combat round of surprise, catching the victims exposed and out on the open. Alternatively, the ambushers can wait for the victims to enter Short range where a second group perception roll will be required. Conversely, the GM may assume that the ambushers automatically spot the victims walking into their trap, or he may ask for a group perception roll with a +3 DM.



## SPOTTING TRAPS & MINES

In a military scenario, the player characters may approach the location of a concealed booby trap or anti-personnel mine. The lead character should make a Routine (+2) Recon roll. If the roll is successful, then the trap or mine is detected and can be marked and then avoided. If not, ask each player in turn to roll 1D6 for their character, with the first '1' result rolled, triggering the mine or trap. The victim (and only the victim) takes the assigned damage. If no PC activates the mine, then one of the NPCs in the group (if present) will certainly do so. A spike trap inflicts 1D6+1 damage, whilst a typical antipersonnel mine or IED inflicts 4D6 damage.

If at any time a PC declares he is looking at a particular location (a doorway, a walkway, a bunker entrance, a hole in a compound wall, etc.) for a mine or trap, then he will automatically succeed. Mines, traps and improvised explosives are nasty, indiscriminate weapons that are not roleplaying friendly. You cannot outfight a mine or use clever tactics against it. And so the GM is advised to use these weapons very sparingly. If they occur in every scenario, all the time, the game will slow to a snail's pace as the characters constantly 'check for traps'. Their impact in a game is always heightened when they are used sparingly as a shock element in the scenario.



## ORDER OF COMBAT

The order of combat is pretty straight forward: the player characters get to act, then any NPC squad-members on the side of the PCs, and finally the enemy. When it's the turn of the player characters to attack in combat, they have the following options:

**Melee Combat** – A hand-to-hand attack within 3m or less, the opponent can parry and counter-attack

**Direct Fire** – Shooting at opponents that are visible.

**Area Fire** – Shooting at an area that is concealing opponents.

**Non-Combat Action** – Run, reload, radio in, start CPR, unjam a gun, etc.

Melee attacks ALWAYS go before Direct Fire to give them an edge. At close range, the melee fighter strikes before the gunman - does the gunman defend himself with a bayonet, or use his rifle as a club, or risk waiting ... and then attempting Direct Fire?

## MELEE COMBAT

Any character within 3m of an opponent can make a hand-to-hand attack with fists, knife, rifle butt, entrenching tool, bayonet, whatever. The attacker declares his target. The attacker then makes a skill check, and if a successful roll of 8+ is made, deals damage to his target. The defender always gets a free **chance to parry** an attack (if they are holding a weapon or other item with which they might block). The target then gets to make an **immediate counter-attack**, although that becomes his action for the round.

# 8+

- + *Melee Combat skill*
- + *Strength or Dexterity DM (player's choice)*
- + *Dexterity only, if an animal or monster*
- 1 if not fighting at weapon's range

An unarmed human can inflict 1D6 damage when they hit in combat, +1 if Str 9 or more, -1 if Str 5 or less.

## RANGE OF MELEE

There are two ranges at which melee combat can take place, **Close** (1.5-3 metres) and **Personal** (in touching distance, less than 1.5 metres). Melee weapons are listed either as Personal range (such as a dagger) or Close range (such as a fire axe). Fighting at the wrong range incurs a -1 penalty. Changing ranges is a minor action; moving away from Personal to Close can be done freely, but moving inwards from Close to Personal may be contested by your opponent. Both roll their Melee skill, with the winner deciding the outcome of that action.

## PARRYING

Characters attacked in Melee Combat can always attempt to parry if they have a hand weapon or other item with which to block the attack (laptop, spear, sword, rifle, chair, whatever). To parry a blow, simply make an Average (8+) Melee Combat roll, add in the Dex modifier. If failed, the attacker still inflicts damage on the defender. If successful, the Armour Value of the object will block the attack if the incoming damage is equal to or less than the object's AV. If it *exceeds* the AV of the object, then whatever damage exceeds the AV is still inflicted on the defender.

### Is the parrying object damaged?

Virtually all items have an Armour Value that reflects their ruggedness. If used to parry, everyday items will be damaged. The exception to this is melee weapons, which are designed to parry other melee weapons.

- *If the incoming damage exceeded the AV of the parrying weapon:* lower its AV by 1 point permanently, and carry on with the combat. When it reaches 0 AV in combat, the weapon breaks.
- *If an object was used to parry and it is important to the story:* whether or not it blocked the incoming damage or not, subtract the object's AV from the damage inflicted, and whatever is left is used to lower the object's Durability Points (roughly equal to its mass in kilogrammes). When reduced to zero, the object is destroyed. See Damaging Objects on page 139. Note that a rifle used to parry will be considered an object, and may be damaged or destroyed if used in this way.

## BACKSTAB

A successful surprise attack with a blade will inflict 3D6 damage (plus Str modifier AND Melee Combat skill), whether knife, sword, axe or whatever.

## GRAPPLING

A character can attempt to wrestle or grab another person instead of hitting him. The attacker must make a Melee Combat 8+ roll. The opponent will probably try to parry (see above) and can do so, even if unarmed. If the attack succeeds, he has 7 options:

- Continue the grapple with no other effects.
- Disarm his opponent. If he succeeds by 6+ he can take the weapon away; otherwise it just ends up on the floor.
- Drag his opponent up to three meters.
- Escape the grapple and move away (a normal movement action).
- Inflict damage equal to 2 + the Effect.
- Knock his opponent prone.
- Throw his opponent up to three metres for 1D6+2 damage.

Throwing an opponent ends the grapple. With any other option the winner can choose to end or continue the grapple (with a +2) as he sees fit. A character in a grapple cannot move nor do anything other than make opposed hand-to-hand combat skill rolls. He may struggle to break the grapple for a number of rounds equal to his End. Each time an opposed roll is made the winner can choose an option from the above list.

## ANIMALS IN COMBAT

Animals on alien worlds can make hand-to-hand attacks just like humans, although using whatever natural weapons (horns, claws, ram, kick) at their disposal. They fight with skill-0 and apply their Dex modifier to the attack roll. Unless otherwise noted, animals operate just like characters in combat. The range category of an animal's weapons can be found in the Animal Weapon Ranges table.

| ANIMAL WEAPON RANGES |                  |
|----------------------|------------------|
| Weapon               | Range            |
| Claws                | melee (Personal) |
| Hooves               | melee (Close)    |
| Horns                | melee (Close)    |
| Projectile           | ranged (Short)   |
| Stinger              | melee (Close)    |
| Teeth                | melee (Personal) |
| Thrasher             | melee (Close)    |

**Taking Damage** – When an animal takes damage, it is applied to Endurance. If a target is reduced to Endurance 0, then further damage is subtracted from the target's Strength or Dexterity (randomly rolled). If either Strength or Dexterity is reduced to 0, the animal is seriously wounded and may flee or be knocked unconscious. Any further damage is subtracted from the remaining physical characteristic. If all three physical characteristics are reduced to 0, the animal is killed.

## TWO-HANDED FIGHTING

If a character wants to fight with two one-handed weapons simultaneously, then he or she will suffer -2 DM to both attack rolls. Two-handed fighting allows a character to attack two opponents simultaneously, or one opponent twice. It is difficult to attempt, more cinematic than realistic – but certainly attested in history with the popularity of the main gauche, or parrying dagger.

**Example:** Let's consider Chris Ross, a miner sent out to check the perimeter at night before the shutters are closed up. He has a flashlight and a metal bar used to lock the shutters around the mine entrance.

**Chris Ross** 888555 Age 26  
Vacc Suit-1, Mining-1, Melee Cbt -1

| Weapon         | TL | Cost | Dmg   | Wgt | Range    | AV |
|----------------|----|------|-------|-----|----------|----|
| Club, Metallic | 1  | Cr10 | 1D6+2 | 1   | Personal | 9  |

Unknown to him, a nasty scavenger is prowling around the compound, a lion-sized beast with a coat of coarse striped, bristles.

| No. | Size  | Subtype  | Move | UPP                       | Weapons     | AV |
|-----|-------|----------|------|---------------------------|-------------|----|
| 2D6 | 100kg | Hijacker | W x2 | Str9, Dex17, End9, Inst10 | Teeth (1D6) | 2  |

Let's check out the encounter from the beginning. We roll for the range on 2D6, applying a DM -2 (open colony settlement). It's dark, but there are occasional lights on the sides of buildings:  $7 - 2 = 5$ . That is short range, or within 12 metres, 'pistol' range (let's say '8' metres). Who spots who? We roll an Average (0) Int roll for Chris, who has a DM -1 for his Int of 5. We apply the following modifiers: Short Range +3. He makes it and sees the creature nosing behind a dumpster against the colony wall. Does it see *him*? Same roll. We use Instinct, not Intelligence, and the creature has a DM +1 for its Instinct, for a DM total of +4. A resounding success, the creature hears Chris ... (cont'd)

What next? We always ask the PC first. He can't attack (he's not within 3m) but he will look for a doorway into the colony, and turn and run back through the compound for 18m. The creature can't carry out a melee attack either, but it will chase him. Its twice as fast, so moves 36 metres, covering the 8m range plus another 28 metres. At the start of the next combat round, it has easily caught Chris.

In round two, Chris decides to fight it off with the metal bar. This is 8+, DM +1 Melee Combat skill, but no bonus for his Strength or Dexterity. The steel bar is a Personal Range weapon (within 1.5m), so Chris fights at that range. With a 5, he fails to hurt it. The creature can't parry, it needs a weapon or object to do that, and of course it doesn't need to and now gets to fight back. Animals have skill-0 and add their Dex modifier (in this case +3). It's using teeth, which, like the steel bar are best used at Personal range. A total of 13 on the 2D6 roll, that's a success with an Effect of 5. Chris gets a chance to parry with his weapon... but with a total of 4, he fails.

Its teeth do 1D6 damage, with an added Effect of +5, we roll 5+5 for a result of 10. Poor Chris, despite trying to beat the thing off with the metal bar, is knocked to the ground, the huge creature pinning him down and savaging him with a series of bites.

Chris' physical Characteristics are 888, and his Endurance is immediately lowered to 0. But there are two more points of damage, and the player decides to take them from Dex, lowering that to 6. His Characteristics are now 860. According to the Injury and Recovery rules, Chris is stunned for one round (he can defend, but not attack). We also roll 1D6 for an injury: Leg Disabled (move 6m per round maximum).

In the next round, Chris can only defend by parrying, which he desperately prepares to do. It is the creature's turn to act; it attacks again, with a roll of 5+3 = 8. It will hit, Chris successfully makes a parry roll. The creature's damage must overcome the steel bar's AV of 9. Since it only does 1D6 damage with no Effect in this round, that can't happen. The bite was parried.

In the next round, Dave Santos, another miner steps around the corner with a rifle. He sees his friend in trouble and intends to fire at the creature at Short Range. However, melee combat goes first. Chris isn't stunned anymore and tries to hurt the creature with the metal bar. It's still 8+, +1 for his skill, and we roll 5+1 = 6, he can't hurt it. The creature goes next and rolls 4+3 = 7 (Chris must be doing enough with the iron bar that the creature doesn't bite him again).

Now we move to ranged combat. Dave has a rifle, but only Gun Cbt-0, so he needs 8+ to hit. The rifle has a range of Long, so Short range is no problem, and incurs no penalties.

We could invoke the 'Firing Into Combat' rule. But instead I think we'll just roll the 2D6 and see if Dave hits the creature: we roll 8, and hit the scavenger with no additional Effect. It does 3D6 points of damage, we roll 15, subtracting the creature's AV of 2, leaving 13 points of damage. This reduces the creature's End to 0, with 4 points left over, which we roll randomly and determine that Dex is reduced by those 4 extra points to 13. The Game Master takes pity on Chris, and has the alien creature flee into the night, stung by the pain of the bullet wound. Dave runs across to drag his bleeding friend inside the mining complex.

## DIRECT FIRE

Direct Fire is that standard type of roleplaying gunfire used by PCs - the target can be seen, the target can be shot at. An enemy soldier moving around, standing up, stood out of cover or away from concealment can be targeted by Direct Fire. This is the easiest and most direct way to engage any enemy force, as such, Direct Fire will generally be carried out when enemy troops are seen moving, or if caught out in the open, either in a chance encounter (see Ambush, Surprise and Movement) or in a planned ambush. Once enemies drop to the ground and stop moving or get under cover, they can no longer be targeted by Direct Fire and players must switch to Area Fire to continue the firefight. Note that NPC antagonists have their own combat rules (see later). Combat at Short or Close Range is *always* Direct Fire.

**8+**

| EXAMPLE FIREARM DATA |    |      |       |     |        |     |     |           |        |
|----------------------|----|------|-------|-----|--------|-----|-----|-----------|--------|
| Weapon               | TL | Cost | Range | Dmg | Recoil | Wgt | Mag | Ammo Cost | Notes  |
| Assault Rifle        | 6  | 500  | Long  | 3D6 | 7+     | 4   | 30  | 20        | Auto 4 |

**Dmg:** Damage, **TL:** Tech Level, **Wgt:** Weight, **Mag:** Shots available

To attack a visible opponent at range, the attacker declares his target and then makes a skill roll. The difficulty of the shot is based on the weapon's effective range since all ranged weapons (like firearms) have a Combat Range Band. If a target is within that effective range band the player requires an 8+ to hit and, if successful, he deals damage to that target. Each successive range band increases the difficulty.

| MODIFIERS TO GUNFIRE                                        |                           | Modifier to Hit |
|-------------------------------------------------------------|---------------------------|-----------------|
| Advantage                                                   |                           |                 |
| Shooter's Dexterity DM                                      |                           | +DM             |
| Target in Weapon's Range Band                               |                           | 0               |
| Target 1 band beyond Range Band                             |                           | -2              |
| Target 2 bands beyond Range Band                            |                           | -4              |
| Point Blank (3m or less)                                    |                           | +2              |
| Aiming for one round                                        |                           | +1              |
| Aiming for one round with Scope, Tough Scope or Night Scope |                           | +2              |
| Propelled Grenade with burst effect                         |                           | +1              |
| Shotgun                                                     | Short Range (4D6 damage)  | +1              |
|                                                             | Medium Range (2D6 damage) | 0               |
|                                                             | Long Range (1D6 damage)   | 0               |
| Target Moving, Dodging, Sprinting                           |                           | -1              |
| Target in Cover, Partial                                    |                           | -3              |
| Firing into Hand-to-hand Combat                             |                           | -2              |
| Autofire 4 rd burst                                         |                           | x 2             |
| Autofire 10 rd burst                                        |                           | x 3             |

## AIMING

A character that spends a one full round aiming at an enemy target gets a +1 DM to his next attack on the target, as long as the character does nothing except aim until he makes his attack or unless his aim is spoiled. If using a Scope, Night Scope or Tough Scope, this bonus increases to +2.

## AUTOMATIC FIRE

Weapons given a Rate of Fire (ROF) value of 4 or 10 are capable of automatic fire, giving them multiple attacks against a single target. The attacks are made at the

| ROF | Rounds Expended | Attacks     |
|-----|-----------------|-------------|
| -   | 1 shot          | x 1 attack  |
| 4   | 4-round burst   | x 2 attacks |
| 10  | 10-round burst  | x 3 attacks |

same time and expend the amount of ammunition shown. The attacks can be directed all at one target or, if there are adjacent targets, can be divided up between them. The three attacks from a 10-rd burst, for example, directed at a group of three guards, could all be rolled against one of the guards, could be split up with one attack per guard, or any other combination.

## BLIND FIRING

Firing in pitch darkness or while temporarily blinded, works exactly like any other form of firing (including automatic), but it always treats the firer as having Level 0 in that weapon's relative skill. Additionally an extra die is thrown when firing, but before any calculations are made the highest die (or one of the highest, in case of a tie) is removed completely.

## COVER

Should a character take cover behind a wall or other impenetrable object, he cannot be hit by enemy fire. If he wishes to use the cover to fire from, shooting from behind a low wall, or from behind an airlock door, for example, the opponent's chance to hit him is lowered by -3. For outdoor cover, such as general undergrowth, rocks, farmland, crags, buildings or anywhere else the enemy can drop down and stop moving, they become effectively invisible, and must instead be targeted with the Area Fire rules that follow.

## DODGING

A character who is not surprised, or not sprinting, may try to dodge into cover as a minor action, if there is any to be had within 6m. This gives his attacker a -1 DM. Characters can try to manoeuvre closer to their opponents or escape by exposing themselves to the shots of their enemy very briefly in this way.

## EXPLOSIVE BURST RADIUS

Grenades, rockets and other explosives have an area of effect; if aware of the attack make a Difficult (-2) Dexterity roll to avoid damage when caught within this radius. DMs include: going prone (+1), ducking behind hard cover if available (+2) or occupying a trench or bunker (+4). A character that dives for cover ends up prone and loses his next significant action.



## DISINTEGRATORS

Disintegrators have a variable energy output, and a user can set the output to anything from 1D6 up to 10D6. This means it can deal up to 10D6 in one attack. Simply subtract the number of dice done in an attack from the Magazine Score. If used at the level of 6D6 or more, it can even damage vehicles but unlike slugthrower firearms, disintegrators at 6D6 or more, *can* reduce a vehicle's Hull and Structure points.

**Stun Attack:** A disintegrator's 1D6 attack can be used to Stun an opponent *rather than* inflict damage.

**Wide Attack:** A disintegrator's beam width can be adjusted so that it can be used to affect up multiple adjacent targets (people 'standing in a group' for example). Simply split the damage dice up between any adjacent targets. Only one attack roll is necessary. Faced with three opponents, a character could employ a medium-level 6D6 attack, with 2D6 on each target. Or he could go to 9D6 and inflict 3D6 on each target.

## FIRING INTO COMBAT

If a character is firing a weapon at a target that is within arm's reach of another combatant, then the attack suffers a -2 DM as the character tries not to hit the other person. If the attack fails, roll 1D6. On a 4+, the adjacent person is accidentally hit.

## OFF-HAND

If forced to use their off-hand to fire a pistol (or throw or use a melee weapon) the character suffers a DM -2.

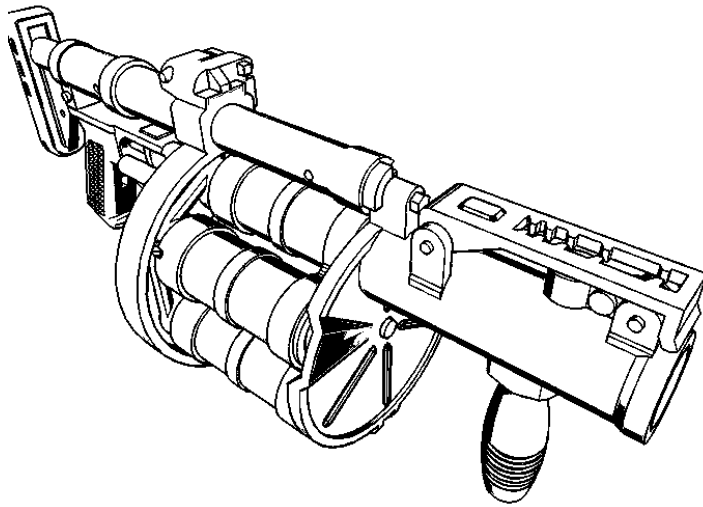
## FRENZY FIRE

Frenzy Fire is what happens when a character 'blasts away' to make sure someone is definitely going down, and it uses all remaining rounds in the weapon. It can be used by anyone with a single-shot weapon, such as a semi-automatic rifle, pistol, shotgun or revolver, who is within Short range of the target. If a hit is scored, additional damage is done. Use the most appropriate row on the table below. Frenzy Fire does not affect vehicles.

| FRENZY FIRE MODIFIERS      |                     |
|----------------------------|---------------------|
| Remaining Ammunition Fired | Attack Modifier     |
| 6 or less                  | +4 points of damage |
| 10 or less                 | +2D6 damage         |
| 20 or less                 | +3D6 damage         |
| 100 or less                | not possible        |

## GRENADE LAUNCHERS

**Direct Fire:** Grenade launchers can direct fire at a target, like a rifle. If so, they receive a DM +1 due to the burst effect giving more chance of the target being caught in the burst radius. All burst effect propelled grenades receive +1 DM to attack a target within range.



**Indirect Fire:** Grenade launchers can also be used for indirect fire, to hit targets in the distance where individual soldiers are not visible. Use Area Fire for this, if employing explosive grenades. Or use this alternate method: the weapon is not aimed at the target, but is fired at a high trajectory to strike the target from above. It has a default DM of -4. If the strike of rounds can be seen, the modifier can be reduced by one each subsequent combat round up to a maximum of the user's Heavy Weapons skill. A character with Heavy Weapons-2, could reduce the penalty to -3 in the second round, and -2 in the third round. All subsequent rounds in which the grenade is fired will have a DM -2. The user himself might not see the fall of shot, but a forward observer who can, could radio the information to the user, allowing him, then, to adjust his fire appropriately.

## RECOIL

Firearms include a Recoil number which indicates the minimum strength required to handle and fire the gun without problems. If the user does not have that Str or greater (even due to wounding), then apply a -1 DM to all shots. Firing a 4-rd burst increases Recoil by +1, while firing a 10-rd burst increases Recoil by +2. Recoil will have a dramatic effect on firearms attacks in zero gravity (page 103).

## SHOTGUN SPREAD

Shotguns are so-called because they fire 'shot', multiple projectiles that spread out as they leave the barrel, increasing the chance to hit. As the shot disperses, the potential damage of that attack reduces. A shotgun gets a +1 at Short range and does 4D6 damage. At Medium range it does 2D6 damage and at Long range 1D6. The damage for shotguns as well as all of the factored range DMs are included in a shotgun section on the table of to-hit Modifiers to Gunfire (page 122).

## STUN

Stun weapons (stun baton, agonizer, etc), are weapons designed to deal non-lethal damage and will incapacitate a living target instead of killing it. On a successful hit, the victim makes a Difficult Endurance roll. If this is failed, the target is incapacitated and unable to perform any actions for 1D6+2 minutes. If the End roll succeeds, the victim is still stunned for one round. Disintegrator hand weapons can also inflict Stun damage.

## THROWN ATTACKS

**Throwing Weapons** - Throwing distance for a primitive ranged weapon is 10m + Strength – weight (kg). The character makes a Difficult (-2) Dex roll; if the target is *under half* the throw distance, apply a +2 DM. Thrown weapons can add the Effect of the attack roll to their rolled damage. Thrown weapons cannot be aimed. Note that grenades and hand-held stones can be thrown much further (see below).

**Throwing Grenades** – Throwing distance for a stone or grenade is 10m + double Strength. To throw a grenade the character makes an Average (0) Dex roll; if the target is *under half* the throw distance, apply a +1 DM. Grenades cannot be aimed.

## SPECIAL DAMAGE

Some weapon damage values, mainly used in vehicle combat, come with a code (HE, AP, AB) which indicate some special damage effect, as explained in the **Cepheus Universal** book. For reasons of space, they are not included here.

## AREA FIRE

Infantry-style firefights in wilderness areas are different to normal RPG gun-play. Just as in most action movies, the more usual RPG combat rules have a hero spot a bad guy, and then roll dice to hit him with gunfire. If the bad guy survives, he may fire back at the hero. This is all well and good and probably simulates many close range urban combats quite well, but in the rock-strewn wilderness of some alien moon, the farmland of a settled colony, or the steaming jungles of some tidally locked eyeball planet, you may not be able to see your enemy at all. Many wartime firefights in military history consist of groups of soldiers firing at a location from which that fire is coming from – **Area Fire**. It may be a rarity to see one of the enemy, and if that is the case, that's probably because they are moving. A method of simulating this 'fog of war' is presented here.

# D6

## THE BEATEN ZONE

Area Fire is directed at a patch of vegetation, a tree-line, rocky outcrop, at a building window or part of a street-front, all roughly 10m across. In military parlance, this area is known as the 'beaten zone'. The enemy is in cover, firing at the players' unit without showing themselves. Any character laying down in vegetation or other cover, whether friendly or hostile, is for combat purposes, invisible and only vulnerable to this type of Area Fire. A character only exposes himself to the risk of observation by moving. Use common sense. If characters hit the deck in the middle of a starport access road then they are still exposed to Direct Fire. At a minimum, there needs to be some cover, such as rocks, vegetation or rubble around. Only when a character moves will he become visible to the enemy and thus susceptible to much more accurate Direct Fire. If the GM is in doubt whether an exchange should be resolved as Direct Fire or Area Fire, use the Area Fire rules. Combat at Short or Close Range is *always* Direct Fire, the combatants are so close that they are visible when they lean out to fire.

## AREA FIRE POOL

Area Fire uses a 'hidden' task resolution system, participants know when their fire is hitting the 'beaten zone' but not if they hit a human target. When player characters use Area Fire, they must nominate a 10m wide patch of vegetation, street front or landscape within range of their weapons. They cannot see individual opponents, perhaps only identifying the beaten zone by the muzzle flashes or the direction of incoming bullets.

Every participating *player character* makes his normal to-hit roll with 2D6, adding a bonus for weapon skill, Dex DM and any range DM. Scopes are not suitable for use in area fire. Auto fire should not provide additional chances to hit – instead add additional dice to the Area Fire Pool as indicated in the Area Fire DMs table (below).

## RESOLVING THE POOL

For every success rolled, the GM takes a D6 and hides it behind his screen to create an Area Fire Pool. If the enemy are heavily entrenched, or dug-into bunkers, the Area Pool is halved. After everyone has made their to-hit rolls, and the infantrymen are still guessing about the effect their volleys had, the GM rolls all the D6 of his Area Fire Pool in secret. His magic number is always a **6**, and each 6 indicates that a member of the enemy unit has been hit and knocked out of the fight, either wounded or killed; multiple 6's mean that more than one of the enemy was hit. Automatic fire, grenades, and rockets can contribute additional D6 to the Area Fire Pool:

| AREA FIRE DMs                          |                                                 |
|----------------------------------------|-------------------------------------------------|
| Weaponry                               | Addition to Area Fire Pool                      |
| Single shot                            | -                                               |
| 4 round burst                          | +1D6                                            |
| 10 round burst                         | +2D6                                            |
| Disintegrator wide-attack (6D6 attack) | +1D6                                            |
| Grenade                                | +2D6                                            |
| Vehicle Support Weapon (see page 181)  | +3D6                                            |
| Vehicle Main Weapon (see page 181)     | +4D6                                            |
| Airstrike/Artillery                    | +8D6<br>+1D6/level of Tactics, of fire director |

Still, the players don't know the effect they've had, they don't see their victims get shot and most likely don't hear any screams. The enemy themselves probably continue firing next turn, or they might pull back, or simply pause to remove the dead or reload. The player characters might want to continue to use Area Fire, or venture out of cover to investigate... exposing themselves to Direct Fire from their waiting enemy... Typically a cautious group will spend several combat rounds pounding a stretch of vegetation or building frontage with Area Fire before moving forward, dodging from cover-to-cover to assault the firing position. Use Area Fire in this way for ***player characters only***. If using the rules for NPC squad members, refer to the section later entitled Rolling For The Enemy Unit.

**Example:** Detective Sergeant Jerry Wagner is a cop on a mission of vengeance; he's tracked down a killer to his lair, an abandoned factory. Wagner carries a police issue TL 14 plasma pistol and wears an AV 6 concealed vest.

|                                                                          |        |        |
|--------------------------------------------------------------------------|--------|--------|
| <b>Jerry Wagner</b>                                                      | 989977 | Age 38 |
| Investigate-3, Security-2, Melee Cbt-2, Gun Cbt-1, Computer-1, Medical-1 |        |        |

| Weapon        | TL | Cost | Range | Dmg | Recoil | Wgt | Mag | Notes |
|---------------|----|------|-------|-----|--------|-----|-----|-------|
| Plasma Pistol | 14 | 1400 | Short | 3D6 | 5+     | 1   | 20  | -     |

His quarry is an unstable killer carrying a small, machine pistol, he knows Wagner is coming, and in the main production hall, is waiting in ambush.

|                    |       |
|--------------------|-------|
| <b>Killer</b>      | 88855 |
| Melee-1, Gun Cbt-0 |       |

| Weapon         | TL | Cost | Range  | Dmg   | Recoil | Wgt | Mag | Notes  |
|----------------|----|------|--------|-------|--------|-----|-----|--------|
| Machine Pistol | 6  | 400  | Medium | 1D6+3 | 7+     | 1   | 15  | Auto 4 |

Wagner moves in stealthily. We roll for first encounter range (Short) then to see if Jerry spots the killer in waiting. He needs 8+, with DM +3 for Short range and DM -3 for an ambush. He rolls a 4 +3 -3, and fails. Now the killer gets the rest of the round to act. He opens fire with a 4-round burst on Wagner, from cover behind machinery. He needs 8+, he has DM 0 skill and two chances to hit due to his 4-round burst. We roll 7 and 6, both miss.

A new round starts with Jerry going first (because he is a PC), he ducks into nearby cover as a Minor Action and returns fire. He needs an 8+, with DM +1 for Gun Cbt-1 and a DM -3 due to the target using cover. We roll a 9 -3 +1 for a 7. He fails to hit with his plasma blaster. Next it's the killer's turn. This time he receives a -3 because Jerry is using cover. We roll a 5 and an 11 for his two chances to hit, and the 11 is high enough ( $11-3 = 8$ ) to hit. Damage is rolled immediately, 8 points, reduced to 2 points by the concealed vest's AV 6. Jerry's End is reduced to 2, he is bruised and winded, but not badly injured.

Round three. To break the stalemate, he risks changing position to get around the side of the machinery to get a shot onto the killer without any cover being in the way. This is a sprint (a major action) to new cover. He cannot fire while sprinting. Now the killer has a chance to shoot Jerry while he is out of cover. The killer gets a DM -1 for Jerry's movement, but of course gets two attacks for his 4 round burst (that will be 12 bullets fired, almost out of ammo). With a 7 and 6, the killer's spray of bullets does not hit Jerry.

It is round four, and from his new position Jerry gets a clear shot at the killer, with a DM +1 for his skill (and no minuses). He rolls 8+1 for a 9, that's a hit with an Effect of 1. His plasma pistol inflicts 14 +1 points of damage, reducing the killer's End to 0, knocking him down and stunning him for 2 full rounds, with the excess 7 points reducing his Dex to 1. Jerry saunters over, holsters his pistol and kicks the machine pistol away.

## MISCELLANEOUS

A character might want to attempt a non-combat action: reload, sprint to a new position, radio for help, hack the airlock door, etc. All of these constitute a significant action. There are a number of special considerations:

### DRAWING & RELOADING

The time taken to draw a weapon depends on its size and ease of use. Most weapons take up just a minor action to unholster, or to ready (if slung on a shoulder strap, for example). A rifle carried ready in two hands can be used immediately. Reloading will typically take one round as the character ducks behind cover, removes the empty magazine, grabs a spare from a pouch and then loads and charges it.

### EXTENDED ACTIONS

Some actions will take longer than a single combat round to complete. The GM will have to decide how long the task will take and then work out how many 6 second combat rounds it will take to complete. A character engaging in an extended action cannot do anything else but can abandon their action at any time and return to the combat at hand. A character who is hit by an attack while undertaking an extended action must make an 8+ roll using the skill in question, with a negative DM equal to the amount of damage the attack caused (after armour). Failure indicates that this combat round's work does not count towards the completion of the task. Failure by six or more (an Exceptional Failure) ruins the task and the character must start again.

### FIRE SUPPORT

In military games, the characters might be able to call on fire support from a grav ship or aircraft tasked to support them, or an artillery fire base in range. However, the troops may have to wait for fire support and might not get any fire support at all. There are times when the unit is out of range of an artillery firebase, or when dropships or jump jet gunships are just not available. If the PCs need some support, the squad leader, or anyone with Tactics-0 or higher, can act as the 'fire director' and call it in on the radio. You can never count on support; the GM must either determine availability, or roll 1D6. On a result of 5-6 fire support or air support is available. To avoid overuse, the GM should allow only **one call** for support per mission in a military scenario. In a multi-day mission the GM may allow more than one call for fire support. He should tell the player characters how many calls they are allowed during the mission briefing. Do not waste them!

Once called, the shells or bombs fall 1D6+3 rounds later. It is an Average (0) Tactics roll, taking 1D6 rounds. If successful, the 10m beaten zone is hit and 8D6 is added to the Area Fire Pool (+1D6 for every level of Tactics the fire director possesses). Remember that every '6' result is a secret enemy casualty. If the enemy is heavily entrenched, or dug-into bunkers, the Area Pool is halved. If unsuccessful, the fire support landed in the wrong area (roll 1D6 on the Fire Mission Failure table). It can be redirected with a new series of Tactics rolls made at +1.



| FIRE MISSION FAILURE |                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------|
| 1D6                  | Failed Fire Support Request                                                                            |
| 1                    | Directly on-top of the player-characters' squad (1-3 random NPC or player characters take 4D6 damage). |
| 2                    | Between the PCs and the enemy (1D6 squad-members take 1D6 damage)                                      |
| 3-6                  | A nearby building, or anywhere else that will cause mayhem.                                            |

## STANCE

A character can be standing, crouched or prone.

- **Standing:** A standing character uses the normal rules.
- **Crouching:** A crouching character moves at a jog (9m per round) but can make better use of cover, and has a better chance of avoiding enemy gunfire whilst moving (enemy gets a DM -1).
- **Prone:** A prone character cannot easily make melee attacks or dodge. He may make full use of any cover. The enemy will find it difficult to hit a prone character.

## UNDERWATER COMBAT

The density of water affects both the use of gunfire and melee attacks underwater. Gunfire suffers a -2 DM, whilst melee attacks suffer -1 DM to hit and damage. Lasers are ineffective beyond close range and also suffer a -1 due to beam scattering. Explosive concussion is doubled in water, but fragmentation is slowed significantly – consequently, explosives should be treated as normal. If a character shoots from the surface at a submerged target, obscuration and refraction result in a -3 DM (and vice versa).

## UNRELIABILITY

This is an optional rule. A character's firearm natural attack roll that equals double-one (a '2' before any modifiers or bonuses), may indicate a sudden malfunction, misfire or reliability issue. The player must immediately roll another 2D6 and roll 8+. If this succeeds, then the result of the '2' was not a malfunction, but was simply an automatic miss. If the roll fails, however, then the weapon has malfunctioned for some reason. This is a Stress Event if it occurs in combat. This does not affect energy weapons.

- If the weapon is a revolver, double-barrelled shotgun or a bolt-action rifle, then the problem is a dud round. It can be cleared easily (if a rifle) or ignored (if a revolver or double-barrelled shotgun) and an attack made as normal in the next combat round.
- If the weapon is pump-action, lever-action, a machinegun, SMG, semi-automatic rifle or assault rifle, then the malfunction is a jam. Dealing effectively with a jammed weapon requires the whole of the next combat round with a successful Average (0) Gun Combat roll. Failure means the user will have to make another attempt to repair the jammed weapon.

## ZERO GRAVITY COMBAT

When fighting in zero gravity refer to the rules for Operating in Zero Gravity on pg. 103.

## NPCS IN COMBAT

### ROLLING FOR FRIENDLY NPCS

During squad-based firefights, the Game Master should roll secretly for the NPC squad, just as he does for the enemy. Once the players have made *their* individual attack rolls, the GM can roll 2D6 secretly for the NPC team shooting at the same target or beaten zone. He is looking at **12+** to tell him if the NPC team or squad members, collectively, scored a hit. Keep this roll secret from the players if this is Area Fire, obviously, unless its standard fire and everyone can see the results.

**12+**

The team or squad kills or disables one enemy soldier on 12+ if conducting Area Fire or Direct Fire, but just as the enemy has more chance of killing enemy soldiers that they can see, so to do friendly NPCs. Not only do they get a +2 DM for Direct Fire, but they may hit additional enemy soldiers out in the open. With Direct Fire, every number over 12 indicates an extra enemy hit, the GM can adjudicate this as a serious wound or a kill etc, just as he would following an Area Dice Pool. Imagine a platoon of guerrillas storming the wire of a starport. With a machinegun and a few assault rifles, the NPC team should rack up several kills in short order.

| NPC TEAM DMs                    |             |
|---------------------------------|-------------|
| Situation                       | NPC Team DM |
| Enemy in the open (Direct Fire) | +2          |
| Enemy prone or in cover         | -           |
| Team has machineguns            | +1          |
| Team has grenade launchers      | +1          |

You will notice that the NPC Team's 12+ roll to hit isn't very effective, that's because the advantage is the player characters and their additional rolls, the ability to split the squad into separate teams sometimes able to attack from different directions, and the fact that this is blind firing – area fire. The idea is for area combat to be slow and mostly ineffectual, with the drive to outflank or assault the enemy in order to get into visual range, using far more effective Direct Fire or even melee combat.

Please note that the use of Area Fire and the GM rolling for NPCs is intended to a) keep players afraid, worried and in the dark, and b) to minimize dice rolling for the GM. On this latter point, the NPC system is a fudge, intended to give an approximation of reality. The rules described here are vague and generalized, but they have to be. As GM, interpret them to suit the situation, but keep dice rolling to a bare minimum. Let players track wounds, positions and ammo of their team members. And remember to try and minimize casualties during Area Fire. We have imposed a limit of one player character casualty per Enemy Rating roll.

## ROLLING FOR THE ENEMY UNIT

If the combat is being conducted with Area Fire and the PCs cannot see the enemy for much of the time, it seems silly to treat the enemy combatants as player characters with individual kit lists and skills. Instead, the GM can decide how many enemy are actually in the unit (1 to 10) and make a single roll for the enemy unit.

### ROLL 2D6 TO HIT

The enemy unit (whether one sniper or ten infantrymen) requires a roll of 12+ to hit one of the players' squad. This casualty should usually be rolled randomly between the NPC squad-members and the player characters. (anyone out in the open or running around should probably be targeted first). There can only ever be a maximum of one squad-member casualty per round using this system.

12+

### ENEMY FIRE DMs

Use the following modifiers based on the movement and position of the player characters. The GM's attack roll is modified by the least concealed type of movement the PCs group is currently engaged in. If combatants are moving in different ways (one team prone, the other charging), then use the DM for the team that is easiest to hit (in this case, those charging).

| ENEMY FIRE DMs                               |          |
|----------------------------------------------|----------|
| Situation                                    | Enemy DM |
| Range is UNDER their weapon range band       | +1       |
| PCs/NPCs in the open, stood or sprinting     | +2       |
| PCs/NPCs in the open, jogging or dodging <6m | +1       |
| PCs/NPCs in the open, prone or crawling      | +1       |
| Enemy have machineguns                       | +1       |
| Enemy have grenade launchers                 | +1       |

### ENEMY MORALE

The enemy is left nebulous and generally unquantified, a force of nature that the GM can play either with cunning and malice, or as limited, ineffectual and cowardly. It is left to the GM, but you might want to assign a Morale skill value to the unit so that its behavior is, ultimately, down to Fate. Either select a troop type or roll the Morale skill randomly on 1D6-1:

| ENEMY TROOP MORALE |                 |                                               |
|--------------------|-----------------|-----------------------------------------------|
| Skill              | Troop Type      | Example                                       |
| 0                  | Non-combatants  | <i>Armed civilians; old men and boys</i>      |
| 1                  | Green           | <i>Fresh recruits or an untested militia</i>  |
| 2                  | Well Trained    | <i>Trained soldiers</i>                       |
| 3                  | Experienced     | <i>Soldiers with combat experience</i>        |
| 4                  | Combat Hardened | <i>Soldiers who are combat veterans</i>       |
| 5                  | Elite           | <i>Experienced and highly motivated units</i> |

When the GM wants to use Morale as a test of courage or skill, he assigns a difficulty to the task, then rolls 2D6, adding the Morale skill.

## Examples of Morale Skill Rolls:

- 8+ Avoid a minefield, conceal positions well, move quietly
- 10+ Morale check at 25% and 50% casualties; flee or surrender.
- 12+ Launch assault over open ground to attack the PC's position.

## NPC CASUALTIES

Enemy casualties are out of the fight, either incapacitated or dead as the GM decides. When an NPC soldier is hit in combat, rather than roll any weapon damage, roll 1D6 (ignore armour): On 1-3 he is injured and stunned for a round; on 4-5 he is seriously wounded; on 6 he is dead. These wound descriptions are explained on page 136.

**Injured:** A PC performing an Average (0) Medical roll will get an injured NPC back in the fight, but he is still classed as wounded. Should he be injured again then he is now seriously wounded.

**Seriously Wounded:** A PC performing a Difficult (-2) Medical roll with a trauma kit can stop a seriously wounded character dying within the hour, but the NPC is incapacitated and must be evacuated to a medbay, autodoc or hospital as soon as possible.

**Example:** The PCs are mercenaries armed with disintegrator carbines and wearing combat armour. They are patrolling a mountain pass, and will come into contact with a squad of 6 rebels, all armed with gauss rifles. One of the rebels also has a grenade launcher. These rebels will roll a basic 12+ as a squad when attacking the PCs.

**PC Mercenaries** 888777  
Gun Cbt-1, Melee Cbt-1, Tactics-1

**Rebels** 777777  
Gun Cbt-0, Melee Cbt-0

| Weapon                | TL | Cost | Range | Dmg  | Recoil | Wgt | Mag  | Notes           |
|-----------------------|----|------|-------|------|--------|-----|------|-----------------|
| Disintegrator Carbine | 16 | 7000 | Long  | 10D6 | -      | 1   | 200R | Variable Energy |

| Weapon  | TL | Cost | Range | Dmg | Recoil | Wgt | Mag | Notes   |
|---------|----|------|-------|-----|--------|-----|-----|---------|
| Railgun | 12 | 1500 | Long  | 4D6 | 6+     | 4   | 60  | Auto 10 |

We roll Medium for the encounter range. Next we roll for initiative, and the PCs roll 9 with the rebels rolling 5, so the player character mercenaries spot the rebels around 30-40 metres away beyond some boulders. The PCs drop into cover, and each pick a target and fire. Since the enemy squad is stood up out in the open, this is Direct Fire. Two of the PCs make their rolls and drop two of the rebels. The surviving four rebels drop into rock-strewn cover, making both opposing sides virtually invisible to one another (*cont'd*).

In the next round the PCs open fire, each making a Gun Cbt roll for 8+ (no penalty for range). They are conserving power so do not use the 'wide attack' option on their disintegrators, opting for a 3D6-damage attack. Two of the four succeed, and so 2D6 is added to the Area Fire Pool. The GM rolls this in secret, the two dice show up as 6 and 5. That's one six, meaning one enemy casualty. The rebels are down to 3 men, but the PCs don't know that, they just see the cloud of energy, dust and rock fragments as they blast the 10 metre area where the rebels took cover.

The rebel squad makes a morale roll to continue fighting. They are experienced (Morale 3), and the difficulty for NPCs to keep fighting after being reduced to 50% casualties is 10+ (add 3 for Morale). We roll  $8 + 3 = 11$ . Success. The enemy squad now need 12+, and gain +1 for being under their weapon's optimal range of Long. We roll a 3. They inflict no PC casualties as they pepper the players' position with fire.

The PCs initiate another Area Fire volley. Two make successful rolls, but the GM rolls only a 3 and 4. No more rebel casualties. It's now the rebels turn, and their roll is 9 ... not enough to inflict casualties on the player characters.

This to and fro combat can continue for a long time, which reflects the reality of this type of combat. Can the stalemate be broken?

The rebels might send out one of their number to crawl out of cover to grab that grenade launcher and give them a bit of an edge (another +1 on that 12+ attack roll). To do that he is moving and therefore becomes visible. Any of the PC mercenaries could opt out of that round's Area Fire and instead conduct Direct Fire on that brave rebel. If the PCs were to try a similar stunt, then the enemy squad would get a +1 on its 12+ roll to hit that brave (or foolhardy) player character!

What could the players do to break the stalemate? If they could spare the power, they could conduct wide-attacks with their disintegrators, committing energy attack of 6D6 to gain a +1D6 to the Area Fire Pool. Or, they might have access to fire support (perhaps an airstrike, or nearby friendly artillery). Since the commander who would call it in has Tactics-1, then if it is succeeded it would add 9D6 to the fire pool. It would be a one off event that would not be able to be used for the rest of the game, so players call in fire support only when there is no alternative. Let's say the roll succeeded; the GM rolls those 9D6 and gets 6,2,4,6,1,2,6,4,3. That's three enemy casualties, wiping out the rebel squad. Of course, when the dust clears, neither the players nor their characters will know that. The rebels may just be stunned or injured. The PCs will either get up out of cover to continue their patrol, or go and investigate. That might be nerve-wracking, being in the open, not knowing the rebels are all dead, or waiting till the PCs are exposed and easy targets for their railguns.







## INJURY AND RECOVERY

Each weapon lists the damage it inflicts as a number of D6. Add the Effect of the attack roll to this damage.

### EFFECTS OF ARMOUR

Armour reduces the total weapon damage by the value of the armour being worn at the time. It will reduce the incoming damage, but often some damage is still suffered through blunt force trauma. If the damage that penetrates is still significant, this indicates the bullet struck some unarmoured part of the soldier's body. It is a generalized defence against a generalized attack. A hit with Effect 6+ always inflicts at least one point of damage, regardless of the target's armor.

### DAMAGE INFLICTED

The first time a character takes damage, it is applied to the target's Endurance.

- If a target is reduced to Endurance 0, he is Injured, further damage is subtracted from the target's Strength or Dexterity (the character's choice). He is stunned for one round, able only to dodge into cover or parry. This is a Stress Event.
- If either Strength or Dexterity is reduced to 0, the character is Unconscious and any further damage is subtracted from the remaining physical characteristic.
- If all three physical characteristics are reduced to 0, the character is Dead.

Once a character has taken this initial damage, then damage from subsequent attacks is allocated to any of the three physical characteristics, as the character so chooses.

**Example:** A rifle bullet hits a character with a roll of 10 when 8+ was required. That is an Effect of 2. This is added to the rolled rifle damage (3D6) for a total of 14 points of damage. Those 14 points cannot be split up when they are being applied to Endurance, Strength or Dexterity. Soon after, the character is hit by a club for 4 points of damage, the player can subtract those 4 points from either Str or Dex his End has been reduced to 0 by the gunshot).

### DAMAGE RESULTS

| DAMAGE RESULTS          |                                                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Characteristic Damage   | Result                                                                                                                           |
| Endurance is damaged    | Wounded; bleeding and bruised                                                                                                    |
| Endurance is 0          | Injured. Stunned one round. 1D6:<br>(1-3) Knocked Down for 1-3 rounds<br>(4) Arm Disabled<br>(5) Leg Disabled<br>(6) Head Struck |
| 2 characteristics are 0 | Unconscious & Seriously Wounded                                                                                                  |
| 3 characteristics are 0 | Dead                                                                                                                             |

**End is Lowered - Wounded:** A wounded character has suffered some minor cuts and bruising, the damage is not enough to lower a characteristic to 0.

**End is 0 - Injured:** The character suffers a minor debilitating injury and is stunned for one round, able only to drop into cover or block or parry in melee combat. Roll 1D6 to determine its details:

- **Arm Disabled** - One arm is hurt and cannot easily be used to carry items or operate a weapon or piece of equipment. Make an Average Str roll. If failed, the arm is broken.
- **Knocked Down** - Knocked to the ground, dazed and stunned for an additional 1-3 combat rounds, after which he may get up and function as before.
- **Leg Disabled** - One leg is hurt. He can no longer sprint and cannot swim, walking speed is at 3m per round (not 6m) unless aided by another. Make an Average Str roll. If failed, the leg is broken.
- **Head Struck** - Dizzy with blurred vision. He suffers a DM -2 on any vision related tasks (including firing a weapon), and -1 on mental tasks. If a helmet is being worn, ignore this injury effect.

**Two characteristics at 0 - Seriously Wounded:** A character is considered seriously wounded if he has two characteristics at zero. He is immediately knocked unconscious and in shock. The character makes an Average (0) End roll every 10 minutes, and if they fail, will lose 1 characteristic point. This deterioration stops when surgery is carried out. Seriously wounded characters who have recovered consciousness can only crawl along at 3m per combat round. If helped by another person holding them up, the two can walk 6m per round.

## UNCONCIOUSNESS

An unconscious character may make an Average (0) Endurance check after every minute of unconsciousness - if successful, he regains consciousness. If he fails he must wait another minute and can then try again with a +1 DM on the check for every check previously failed. Most unconscious characters will revive within 5 minutes or so. When revived, the character is still a little groggy and suffers a -1 to all actions, this penalty remains with him for 24 hours.

## MEDICAL TREATMENT

**Wounded:** For characters who are not injured, anyone with Medical-0 can apply first aid, taking 5 minutes, and restoring 3 points of Endurance.

**Injured:** For characters who are injured, a Routine Medical roll is required (or Average if no medical kit available), taking 5 minutes, and restoring 3 points of Endurance. A broken arm or leg can be put in a splint, but the limb is still debilitating. A head injury can be treated to remove the skill penalties. Failure means that no points are restored, and those head injury penalties are not removed. Further attempts can be made every hour until a success is achieved. A broken limb will start healing when treated in a medical facility by a Medic-1 or greater. It takes 60 days to fully heal a broken limb (TL 6: -20, TL 10: -40, TL 14: -50, TL 16: only requires 6 hours) during which time it will be held immobile in a cast.



**Seriously Wounded:** Requires surgery before any recovery can begin. Surgery requires a 10 minute procedure using a doctor's kit (Difficult Medical roll) or a half-hour procedure in a medical facility or autodoc (Formidable if TL 1-3, Difficult if TL 4-5, Average if TL 6-9, Routine if TL 10+). Assume a Medical skill of 3. The difficulty increases two levels if the surgeon is using improvised tools, or the patient is performing surgery on themselves. After the surgery, a patient will have a number of characteristic points restored equal to the Effect of the surgery roll, and the patient will no longer have to roll for deterioration. Points restored are divided as desired among all damaged physical characteristics. If the roll (and the surgery) is failed the patient loses characteristic points (i.e. suffers more damage) equal to the Effect of the failure. Further half hour-long attempts can be made to extend the surgery, each one incurring an additional -1 to the task roll. Surgery does not benefit characters that are not seriously wounded.

## RECOVERING FROM SURGERY

Once a character's serious wound is stabilized after surgery he can begin to benefit from medical care. Decent medical care restores characteristic points per week equal to 1 for every full 3 points of Tech Level (+ the character's Endurance DM). Divide the points as evenly as possible among all damaged characteristics. Medical care requires a hospital or medbay, and for the patient to undergo full bed rest.

## NATURAL RECOVERY

Any character who has successfully received first aid and has not been seriously wounded, can begin healing naturally. However, characters cannot always afford to stay in one location and wait for an injured comrade to return to full health. Travelling is not conducive to a rapid recovery.

| Travel State  | Daily Recovery of Characteristic Points |
|---------------|-----------------------------------------|
| Hiking        | 0 + Endurance DM                        |
| Being Carried | 1 + Endurance DM                        |
| Resting       | 3 + Endurance DM                        |

## OBJECTS IN COMBAT

Every roleplaying scene is littered with inanimate objects which might need to be damaged or destroyed; a locked door or a radio transmitter, for example. Objects might even be used as improvised weapons, perhaps a chair to break through toughened Plexiglas or a fire extinguisher thrown at an oncoming predator.

### DAMAGING OBJECTS

Up to a limit of 10 kg, an object has a number of Durability Points (DP) equal to its weight in kilogrammes. Over 10 kg, an object has instead 10 DP, plus 1 DP for every 10 kg the object weighs in total. A tall refrigerator weighing 100 kg, will have 10 DP, plus 10 DP for a total of 20 DP. A ceramic toilet weighing 50 kg, will have 10 DP, plus 5 DP for a total of 15 DP. Above 100 kg, use the rules on in **Cepheus Universal** for damaging vehicles, instead. To give your very large object the hull and structure points of a vehicle, assume a car has 3/3, a bus has 12/12, a small commercial fishing boat has 20/20, and a submarine or airliner has 75/75. Buildings have very similar rules to objects.

If the object sustains damage equal to half of more of these points (round up), it is damaged and may not work correctly or reliably. If those Durability Points are equaled or exceeded, then it is effectively rendered useless, disabled or destroyed. Firearms will typically only inflict half their normal damage on inanimate objects (with exceptions for fragile objects). Almost all objects have some inherent Armour Value, decided upon by the GM.

|                                     |                                                     |
|-------------------------------------|-----------------------------------------------------|
| <i>Field radio, 3 kg, AV6</i>       | <i>Damaged after 2 pts, Destroyed after 3 pts.</i>  |
| <i>Autodoc, 10 kg, AV8</i>          | <i>Damaged after 5 pts, Destroyed after 10 pts.</i> |
| <i>Toughened Plexiglas, AV7</i>     | <i>Damaged after 2 pts, Destroyed after 4 pts.</i>  |
| <i>Vacc Suit, 24 kg, AV4</i>        | <i>Damaged after 6 pts, Destroyed after 12 pts.</i> |
| <i>Generator, 70 kg, AV2</i>        | <i>Damaged after 9 pts, Destroyed after 17 pts.</i> |
| <i>Floodlight, 8 kg, AV1</i>        | <i>Damaged after 4 pts, Destroyed after 8 pts.</i>  |
| <i>Portable Computer, 3 kg, AV2</i> | <i>Damaged after 2 pts, Destroyed after 3 pts.</i>  |
| <i>Door Lock, 2 kg, AV6</i>         | <i>Damaged after 1 pts, Destroyed after 2 pts.</i>  |
| <i>Empty Rifle 4 kg, AV10</i>       | <i>Damaged after 2 pts, Destroyed after 4 pts.</i>  |

**Example:** A colonist attempts to break a Plexiglas window with a conference chair in order to escape from a locked room. The chair inflicts 1D6+2 damage, and after three attempts she has not overcome its AR of 7. The chair bounces off the Plexiglas. Her comrade with a Stern 2000 shotgun (4D6 halved) inflicts 5 pts after subtracting the AR of 7, on his second shot completely shattering the window.

## BREAKING THROUGH DOORS

Doors follow these same rules but characters (and creatures) can use their bodily strength to try and force them open. Forcing a door is an Average (0) Str roll. Success indicates that the character inflicts damage against the door equal to his Strength characteristic. If the Effect is -4 or lower, he takes 1 pt of damage and does not damage the door. Firearms generally cannot breach doors, although they can be used to disable some types of door lock (AV 8, 2 kg). Only objects with a cutting blade stand a chance of damaging a door.

**Example:** A starship crewman (Str 8) tries to kick down the sliding door (15 kg) to the computer room, where the crazed captain is trying to shut the mainframe down. His first roll is 3 which is Effect -5, inflicting 1 pt on the crewman. His fourth attempt succeeds and he reduces the 15 points by 8 = 7. His eleventh attempt succeeds, and he reduces the 7 points to 0. He is in – but is he in time? Eleven rounds is just over one minute. If he'd had a fire axe it might have only taken two or three rounds to break through.

## OBJECTS AS WEAPONS

Many objects can be used as improvised weapons, providing a few extra points of damage to be inflicted. The down-side is that these ad hoc weapons are not suited to melee combat and larger items will incur a penalty to hit.

**Penalty:** If the character's Str equals or exceeds the objects weight (in kg) there is no penalty. If the Str is less than the object's weight, then apply a -1 penalty for every 2 points (or part of) that Strength is exceeded.

**Damage Inflicted:** An improvised weapon inflicts 1D6 + Str modifier, +1 for every 5 kg the object weighs.

**Example:** A Str 6 colonist attempts to hit a guard with a 8 kg conference chair in order to escape from a locked room. The chair inflicts 1D6+2 damage and since the colonist's Str is less than the weight of the chair, she will incur a -1 penalty to hit (Str is exceeded by 2 pts).

## THROWING OBJECTS

Of course these improvised weapons can also be thrown if the character is strong enough. Throw distance is 10m + Str – weight (kg). The character makes an Average (0) Dex roll and if the target is over half the throw distance, apply a -2 DM. Note that grenades and hand-held stones can be thrown much further (10m + double Str).

## FALLING OBJECTS

A falling object will suffer 1D6 damage for every 2m it falls, just like humans. If an object is dropped on to a person or creature from a height, it will probably inflict damage. For every full 5 kg that an object weighs, it inflicts 1D6 points of damage per 2m fallen. If the object weighs 1-4 kg, it inflicts only 1 point per 2m fallen. Helmets provide good protection from falling objects equal to the type of armoured suit being worn (concealed and ballistic vests are not provided with helmets).



## DAMAGING BUILDINGS

Buildings have Durability Points (DP) much like other inanimate objects in these rules. If the building sustains damage equal to half of more of its DP points, it is badly damaged, un-liveable or inoperative – windows are smashed, the roof may have collapsed and utilities are disabled. If its total Durability Points are equalled or exceeded, then the building is effectively rubble. Only explosives (bombs, missiles, etc.) can damage vehicles. Like other objects, buildings have an inherent Armour Value that is first subtracted from any incoming damage.

| BUILDING DAMAGE                              |              |                |
|----------------------------------------------|--------------|----------------|
| Target Type                                  | Armour Value | Durability Pts |
| Townhouse                                    | 18           | 20             |
| Villa                                        | 18           | 40             |
| City Building                                | 18           | 80             |
| Factory                                      | 18           | 100            |
| Villa                                        | 18           | 40             |
| Large Compound                               | 18           | 40             |
| SAM/Radar Site                               | 18           | 50             |
| Fuel Storage Tank                            | 18           | 10             |
| One Span of a Bridge                         | 18           | 60             |
| Hanger                                       | 18           | 40             |
| Power Station                                | 18           | 100            |
| Airport/Space Terminal or High-Rise Building | 18           | 150            |
| Squad/Machinegun Bunker                      | 24           | 10             |
| Small Artillery Bunker                       | 28           | 20             |
| Large Artillery Bunker                       | 40           | 40             |
| Communications /Satellite Uplink Tower       | 18           | 20             |
| Small Enclosed Colony Complex                | 18           | 100            |

**Example:** Two heavy bombs (14D6) are successfully dropped on a power station (AV 18, DP 110), in an attempt to cripple it. The first does 51 points of damage (subtract 18 = 33), the second does 46 points of damage (subtract 18 = 28). This is a total damage of 62 points, more than half of the power station's 110 Durability Points, damaging the plant and disabling its power production capabilities. If those same bombs were dropped on a heavily reinforced large artillery bunker, only 17 points would get through, not enough to stop the big gun inside it from firing.

## BUILDING SCREENS

In settings where defensive screens exist, some important buildings may be fitted with them (such as palaces or embassies). These large screens cost anything from Cr100,000 to MCr1, based on the size of the building, and the screen rating of 2D6 only needs to be rolled when hit by a vehicle weapon, heavy weapon, or spacecraft weapon. Such screens have a diameter of 100m per Cr100,000. In some settings these ground-based screens prevent grav vehicle and flying craft entering the field. See Cepheus Universal for more information on how screens work.



**Crashing into Buildings** – A vehicle might be driven at a building or barrier. The GM should assess the size of the object and then its speed. Multiply these numbers together to determine the number of dice damage inflicted. This is a very rough approximation.

| Size                             | Multiplier | Speed                 | Multiplier |
|----------------------------------|------------|-----------------------|------------|
| Tiny (drone)                     | 1          | Slow (< 40 kph)       | 1          |
| Small (car)                      | 4          | Medium (100 kph)      | 3          |
| Medium (bus, truck, fighter jet) | 6          | Fast (300 kph)        | 6          |
| Large (jet bomber)               | 16         | V/Fast (800 kph)      | 10         |
| V/Large (starship, ocean ship)   | 18         | Supersonic (2000 kph) | 18         |
| Massive (capital ship)           | 24         | Hypersonic (8000 kph) | 24         |

**Example:** A truck is used by the PCs to smash through a concrete wall (AV 15) around a villa complex. The truck will inflict 9D6 damage (6 + 3). We roll 32 points of damage, subtract the AV of 15 to leave 17. That's a hole in the wall 1.7m wide!

**Example:** A cargo shuttle (Large) lifting off from the rooftop of an embassy is shot at and disabled, causing it to crash back onto the embassy building (AV 18, Durability 65). It will inflict 19D6 damage (16 + 3). We roll 67 points of damage, subtract the AV of 18 to leave 49. That is more than half (32) of the building's Durability, rendering it damaged and partially collapsed.

## REPAIRING BUILDINGS

Buildings can be hastily repaired to get them operating once again. A work crew equal to the building's total Durability Points can return the building to exactly half DP in 1D3 weeks if an 8+ is made on 2D6. Harsh punishments or large bribes can provide a DM +2. Workers with the skills, tools and materials must be available in the area, however. If the work crew is half the required size, the task is Difficult (-2). Re-roll a failed attempt every 1D3 weeks. An Exceptional Failure indicates the building cannot be repaired and must be demolished or rebuilt taking DP x 4 days. At a more relaxed rate, a quarter-sized work crew can easily repair 1 DP per week.

## DEFEATING BARRIERS

Any door, wall, fence or fortification can be defeated by explosive shells, grenades or demolitions. A hole can be blown through the wall, with increasing damage creating a wider hole. The Armour Value of the barrier is first subtracted from the explosive damage roll. If any damage exceeds the AV, it is multiplied by 10 cm to determine the width of the hole. If the damage equaled the Armour Value exactly, then there is no breach, but the rating is reduced by 1-3 points.

**Armour Piercing** – A weapon doing AP (Armour Piercing) damage only has a multiplier of 1 cm, and its damage goes through that hole to affect anything or anyone directly on the other side of the barrier.

**Pressurised Cabin** – An explosive breach on the outer hull of an aircraft or grav vehicle at high altitude, a spacecraft or submarine, will double the size of the hole.

| BARRIER ARMOUR VALUES   |    |
|-------------------------|----|
| Barrier Type            | AV |
| Wooden                  | 4  |
| Mudbrick, Wooden Gate   | 10 |
| Wooden Palisade         | 12 |
| Brick                   | 13 |
| Concrete                | 15 |
| Stone Wall              | 20 |
| Fortified Stone Wall    | 32 |
| Rock Wall (3m)          | 40 |
| Military Gabion Blocks  | 26 |
| Metal Reinforced        | 20 |
| Earth Bank (2m wide)    | 30 |
| Vehicle Hull            | 4  |
| Spacecraft Hull (TL6-9) | 10 |
| Spacecraft Hull (TL10+) | 20 |
| Space Station Hull      | 20 |
| Space Habitat Hull      | 30 |

**Example:** An insurgent throws a grenade (5D6) at the steel storm wall (AV20) around a colony habitat. The grenade does 19 points of damage, making a hell of a noise, but doing no damage. Later he returns with a 500g (8D6) block of C4 and plants that on the storm wall. When detonated it does 28 points of damage, exceeding the AV by 8 (x 10 cm) creating a hole in the steel wall 80 cm wide.



## LOW TECH FIREARMS

At Tech Levels 2 and 3, gunpowder firearms are available, but are sufficiently more complex and time consuming to use than weapons at higher Tech Levels. In that light, rules are provided here as a guideline for their use. At TL 2, matchlock guns, which are fired by the use of a slow-burning match are dominant, with flintlocks (where a spark ignites the gunpowder) being used at TL 3. Modern metal cartridges are introduced at TL 4.

### RELOADING

Reloading a TL 2 or 3 firearm is a complex and time-consuming business in comparison with the modern ability to simply pull the trigger a second time. With these archaic firearms, the gunpowder has to be seated on the pan, and the rest emptied down the barrel, the musket ball is 'patched' (wadded, to create a tight grip inside the barrel) and then rammed into the barrel with a ramrod. Only then can the gun be raised for firing. Later, TL 3 percussion rifles do not require gunpowder to be sprinkled on the pan, instead they require a brass-cased percussion cap to be fitted to the firing 'nipple'. All this takes time – and with training and experience that time decreases. Check the character's skill level on the following table:

| RELOADING LOW TECH FIREARMS                                                                                                                              |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Gun Combat Skill                                                                                                                                         | Rounds Required |
| 0                                                                                                                                                        | 4 full rounds   |
| 1-2                                                                                                                                                      | 3 full rounds   |
| 3+                                                                                                                                                       | 2 full rounds   |
| <ul style="list-style-type: none"> <li>• Use of paper cartridge assumed</li> <li>• For loose powder, poured from a flask, add +1 combat round</li> </ul> |                 |

### MISFIRES

Gunpowder firearms are notoriously unreliable. A GM wanting to add extra realism (and complexity) should ask a player to roll 1D6 at the moment of shooting. On a '1' the gun misfires and must be reloaded before it can be fired again. The unfortunate player does not have to roll again for a misfire during that combat.

## DUELLING

A duel is a tense hand-to-hand stand-off, with fighters unwilling to charge in, headlong, and more willing to jockey for position, and seek an opening. To simulate this, an alternate melee system is offered here. Game Masters might even want to use this alternate system instead of the standard **Cepheus Universal** melee combat rules.

- 1) Both players roll 2D6 + Melee Combat skill. If the result is greater than 8, then note down the Effect. Repeat this each combat round. Initiative is not used in a duel. Range is always 'Close'.
- 2) If any roll is failed (the result is less than 8) any saved Effect points are lost and the total drops back to 0.
- 3) If the total Effect of any one duellist reaches 6 or more, then he or she makes a successful attack and damage is inflicted. If both duellists reach 6 or more in the same round then the fighter who rolled the highest Effect inflicts damage, preventing the other from scoring a hit. If both rolled the same Effect, then let them both draw blood!

**Example:** A swordmaster defends his honour against an rival officer.

- Swordmaster AA787A Sword-3 +1 Dex DM
- Rival Officer 8699A9 Shortsword-2 0 Dex DM

**Round 1:** Swordmaster rolls 4+3 ... fail. Rival rolls 5+2 (fail).

**Round 2:** Swordmaster rolls 6+3 ...success 1 Effect point. Rival rolls 9+2 ... success 3 Effect points.

**Round 3:** Swordmaster rolls 7+3 .. 3 total Effect points. Rival rolls 7+2 ... 5 total Effect points. He has the Swordmaster at a disadvantage, perhaps forcing the more skilled duellist to commit to a clever feint.

**Round 4:** Swordmaster rolls 7+3 ... 5 total Effect points. Rival rolls 6+2 ... no extra Effect points! Both are on 5 points, both committed, blade to blade with just one slip settling the fight!

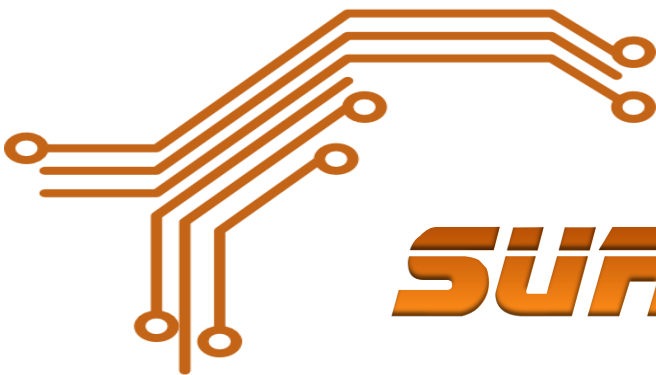
**Round 5:** Swordmaster rolls 9+3 ... 9 total Effect points. Rival rolls 4+2 ... no extra Effect points! The Swordmaster wins, and slices the rival's neck with a nasty cut (3 points) that bleeds profusely. He drops to the ground, beaten.

## USING THIS VARIANT IN MELEE COMBAT

Game Masters might want to use the more 'cinematic' duelling method for resolving all melee combat. Two duelling rounds should take place in a *single* **Cepheus Universal** combat round, this is done to speed up combat and prevent melee fighters from bogging down the entire group's activities.







# SURVIVAL

In the wilderness of an alien planet, flung out from some starship burning up in a rapidly decaying orbit, the characters may have little with them, and desperately need the essentials of survival: tools, clothes, shelter and weaponry with which to catch or kill food. The option exists here, to improvise tools, clothes and equipment, to hunt and gather wild food.

## IMPROVISED EQUIPMENT

It is a fact that the most important piece of survival equipment is a good knife. This useful tool can be used to fabricate an endless number of other items. But if there are sharp stones, such as flint or obsidian (or even scavenged glass, which can be knapped just like flint), then these sharp, hard stones can be carefully shaped to form blades, axes, spear-points, arrowheads and more.

On Earth, wood has been exploited to create all manner of useful objects, from tent poles to fish traps, sleds to arrow shafts. There may or not be a wood-analogue on the wilderness planet - let's hope there is. Likewise, tribes on Earth have used the skins of animals for clothing, shoes, quivers, boat hulls and tent coverings – does such an analogue exist on the character's wilderness planet? In most science fiction settings, wood analogues and leather analogues are fairly well established.

We will assume that anyone can create a useful item, but that Survival skill makes this process much faster – the character with that skill having had plenty of previous experience.

## RAW MATERIALS

To create anything you firstly need the raw materials. Most items will either require leather hide, wood, stone/flint, or bone/horn/ivory. Crash victims will be able to scavenge all sorts of useful raw materials from an escape pod, shuttle or aircraft. Curtains, seat covers, glass ... the possibilities are too numerous to mention. For those without man-made wreckage to exploit, animal hide can be used for an innumerable range of items, from boots to leggings, bags to cordage, bowstrings to tents. A large animal is killed, bled and skinned (then gutted and cooked) and the skin is stretched on a wood frame, the fat and flesh cleaned with a scraper and then cured or dried (either in the sun or over a low fire). If the fur is to be left on, the job is done. If leather is required the cured hide is soaked in water for a couple of days, the fur can then be



pulled out and the hide scraped once again. To tan the hide and make leather, a mixture of the animal's brains and body fat is worked steadily into the skin. Finally the hide is dried over a fire. Total preparation time: 4 days.

Hamstrings and other leg sinews can be dried and used as bindings and threads with which hides can be stitched together or as short rope or bowstrings. Sinews are sticky when wet and dry hard - they make excellent bindings for spear heads. Total preparation time: 3 hours. Bones, antlers and tusks can be turned into sharp spear points, tool handles, toggles, belt buckles and anything else the characters can think of. Terrestrial animal horn can even be heated, flattened and shaped into cups, combs or even lamp-shades! Ice Age tribes used the shoulder blades of cattle as shovels and deer antlers as picks for mining minerals. Be creative!

## MANUFACTURING AN ITEM

Should a player character wish to create something from the list overleaf, make a successful Average (0) Survival skill roll and add the character's Int modifier. The GM *must* apply the -3 penalty for being unskilled, if the task is attempted without possession of Survival skill. Each item has a Time Required to build (in hours).

*To fabricate an improvised item:* Average (0) Survival roll (add Int modifier).

| SPEED OF MANUFACTURE |                 |
|----------------------|-----------------|
| Effect               | Time Reduced By |
| 0 - 2                | -               |
| 3 - 5                | 25%             |
| 6+                   | 50%             |

If the roll is failed, then there was a problem and the task roll must be repeated, following the listed manufacture time. *Three failed rolls* destroy the main components and fresh raw materials must be acquired before the task can be attempted again.

There may be other items that the characters can think of that they might want to fabricate if they have access to the correct raw materials. The Game Master should create a Time Required using those in the table overleaf as a guideline. Err on the side of longer rather than shorter!

**Example:** Mikels has no Survival skill, but is stranded in scrubland, on a dry, desert planet. He fails his first two attempts to make a bow (costing him 30 hours), but on his third try he makes a successful roll (result '9'). He has a bow and it took him 45 hours (roughly 5½ days, or 5½ x 8 hour shifts) to make.

## TEACHING

If a character with Survival skill instructs another person in the techniques required, and models the construction, then that character can attempt to fabricate the item with only a -1 penalty. This resembles a 'workshop' scenario with the expert leading one or more students. If the student successfully constructs the item, he can go on to make more on his own, but without penalty.

| IMPROVISED EQUIPMENT LIST        |               |                               |
|----------------------------------|---------------|-------------------------------|
| Item                             | Time Required | Materials Needed              |
| Animal Trap, Spring-Loaded       | 8             | Wood and 3m thread            |
| Arrow                            | 4             | Wood, fletching, head         |
| Axe, Flint Head - Wooden Haft    | 12            | Sharp Stone, Wood             |
| Bag, Leather                     | 10            | Leather Hide                  |
| Belt, Leather                    | 2             | Leather Hide                  |
| Horn Comb                        | 30            | Horn                          |
| Boots (Foot Wrappings)           | 8             | Leather Hide/Sleeping Fur     |
| Bow                              | 15            | Wood, 3m sinew                |
| Cap/Bonnet, Fur                  | 4             | Sleeping Fur                  |
| Cloak, Sheepskin/Fur             | 4             | Sleeping Fur                  |
| Coracle, Wood & Hide             | 60            | Leather Hides, Wood           |
| Club or Staff, Wood              | 1             | Wood                          |
| Dugout Canoe (3-man)             | 300           | Tree Trunk (large)            |
| Fire Bow, Wooden                 | 6             | Wood                          |
| Fishhook, Bone                   | 3             | Bone                          |
| Fish-trap                        | 16            | Wood                          |
| Flint Sparker                    | 3             | Iron stone, Flint/Obsidian    |
| Flint Tool (Borer, Scraper etc.) | 1             | Flint/Obsidian                |
| Flint Knife - Wooden Handle      | 6             | Flint Tool, Wood              |
| Frame Pack, Wooden               | 12            | Wood, 3m thread               |
| Hammer Stone                     | 1             | Stones                        |
| Healing Herb Remedy              | 4             | Herbs                         |
| Hide of Animal Leather           | 30            | Gutted Animal Carcass         |
| Mittens, Fur                     | 6             | Sleeping Fur                  |
| Needle, Bone                     | 6             | Bone                          |
| Net for Hunting or Fishing       | 6             | 30m of thread                 |
| Pick, Deer Antler                | 1             | Antler                        |
| Pouch, Leather                   | 4             | Leather Hide                  |
| Quiver of Leather                | 8             | Leather Hide                  |
| Raft                             | 8             | Wood, 9m thread               |
| Rations, Dried Meat, per-day     | 4             | Animal carcass                |
| Rope of Plant Fibre (10m)        | 60            | 90m of thread                 |
| Shelter, Lean-To                 | 6             | Wood, Leaves                  |
| Shovel, Shoulder-blade           | 1             | Animal carcass                |
| Sledge, Wooden                   | 10            | Wood, thread                  |
| Sleeping Furs                    | 14            | Animal Carcass                |
| Sling, Leather                   | 4             | Leather Hide                  |
| Spear, Fire-hardened             | 3             | Wood                          |
| Spear, Bone-tipped               | 6             | Wood, Bone                    |
| Spear, Flint-tipped              | 8             | Wood, Flint/Obsidian Tool     |
| Travois                          | 4             | Wood, Thread                  |
| Tent, Leather, per-man           | 22            | 3 Leather Hides, Wood, Thread |
| Thread or Sinew (3m)             | 3             | Suitable woody plant stems    |
| Tree Trunk (large)               | 2             | Axe                           |
| Tree Trunk (small)               | ½             | Axe                           |
| Trousers of Hide                 | 12            | Leather Hide                  |
| Tunic/Jacket of Hide             | 14            | Leather Hide                  |

## MOVING EQUIPMENT

### BEASTS OF BURDEN

It may be that, depending on the nature of the planet, the society living on it (if any) and the particulars of the scenario, that the player characters can use trained herbivores to serve as beasts of burden. Such creatures must be trained by locals. A typical herbivore suited to this task will walk at the human trekking pace, but will be able to carry 200 kg of food, supplies or equipment, or a single wounded character. Some types of creatures (such as horses) will be suitable for riding, but the rider will occupy 100 kg of that creature's load capacity. Beasts can take food and water from the environment unless they are travelling through desert, dune sea, polar, tundra or volcanic terrain types. Beasts of burden may be used as a food source *in extremis*.



### SLED & TRAVOIS

Player characters may be able to construct a wooden sled or travois for dragging large amounts of equipment, these are often better suited to off-road travel than wheeled wagons or carts. The travois is made up of two long poles converging together at the apex, and connected together further down their length by lashing on cross-pieces. The frame is dragged with the narrow end forward and is loaded either by piling goods atop the bare frame and tying them in place, or by first stretching cloth or leather over the frame to hold the load to be dragged. Although a small travois can be dragged via an attached shoulder harness, historically they have also been pulled by dogs, horses or mules. Across packed snow or ice, a traditional toboggan-style sled of bent poles can be constructed that serves the same purpose. A typical hand-pulled sled or travois can be packed with up to 100 kg of equipment, scavenged fuel or prepared kill. It is possible to build a small travois or sled out of scavenged modern or hi-tech materials. The sled is more efficient because of the low friction of ice.

**Sled:** Any weight on the sled is effectively reduced by 70% when determining carrying capacity; 100 kg of supplies feels like pulling 30 kg.

**Travois:** Any weight on the travois is effectively reduced by 50% when determining carrying capacity; 100 kg of supplies feels like pulling 50 kg.

# HUNTING & GATHERING

## HUNTING ANIMALS

Characters may resort to hunting local wildlife either as a source of food, or raw materials for the fabrication of tools or clothing. Each day the group remains in one area, making no forward progress, they may attempt a Difficult (-2) Survival roll. Success will be rewarded with an animal encounter that can either be rolled as normal, or designed by the GM.

- **Is It Edible?** Roll 5+ on 2D6, for the majority of a planet's wildlife to be generally fit for human consumption. Apply a DM of -3 if the world possesses a tainted atmosphere, and DM +2 if it has a human Population of 6 or more. The characters will probably not know the wildlife is poisonous. A successful Difficult Education roll might provide scientific insight and give the characters a warning that there may be dangers based on observable scientific clues.
- **How Much Meat?** Assume an animal killed and butchered in the wild will yield 20% of its mass in kilogrammes; +5% for every level of Survival the butchering character possesses (to a maximum of 40%). So, a 100 kg herbivore, shot and killed by a character with Survival 2 will yield 30% of its mass, equating to 30 kg of meat (or 300 food units, see the Starvation rules, on page 103). Butchery will take a couple of hours.
- **Preserving Meat:** Carrying all this meat might be a problem, however. The raw meat will only remain edible for one week after which time it will be 'off' and make characters sick. The characters may attempt to preserve meat by smoking or drying it, but this is a time consuming process; requiring one full day to preserve up to 20 kg of meat. Characters with Survival skill can add +10 kg to that total for each level of the skill that they possess.

## GATHERING

The gathering of edible plant materials can be done while hunting and over the same time period. Typically 1D6 kg of material can be gathered in one day, except in Steppe, Polar, Mountain, Volcanic, Tundra and Desert terrain types. Steppe and Mountain terrains will only yield 1D-3 kg of plant material. Plant material lasts longer than meat, it will 'go off' after 4 weeks. Winter seasons, if they exist, may not yield *any* edible plant materials.





Player characters can be extremely ingenious and inventive. In a similar vein to the rules for the improvisation of primitive equipment, in the previous chapter, here we look at high tech gadgets and inventions that the technically-minded character might attempt to create. This doesn't represent 'blue sky' thinking, where an inventor creates an entirely new form of power or transport or computing, pushing the local tech into the next level. Instead it represents talented engineers and technicians create devices that utilise the current technology of that world to overcome some challenge. Considering the modern age of TL 7, a character couldn't create an anti-grav drive with these rules, but could build a high-speed, turbojet-powered hydrofoil.

## REQUIREMENTS

Not everyone with a technical qualification has the ingenuity and vision to create new devices and gadgets 'outside-of-the-box'. The minimum required is an Electronics or Mechanical skill at level 3 and an Intelligence of 9 or more. Other skills will come in useful depending on what the device or gadget is going to be. The creator of the hydrofoil mentioned above, for example, will benefit from possession of Watercraft skill. These **supporting skills** may provide additional technical assistance:

- Aircraft
- Computer
- Comms
- Engineering
- Grav Vehicle
- Ground Vehicle
- Gun Combat
- Heavy Weapons
- Security
- Watercraft

## CREATING THE DEVICE

The player comes up with the concept and explains it to the Game Master, who will rate its complexity based on how much work is involved and whether or not parts are available, if not then they must be fabricated, adding complexity, time and difficulty to the build. If unsure, judge the complexity by how long you think it might take to build at the very soonest (failed attempts increase the build time). The GM must also create a cost estimate for the initial build time, perhaps using entries in the Equipment chapter as a guide. If the roll is failed, and a second or third roll must be made, with additional build times, the cost of each attempt is one-half of the original cost.

| Complexity   | Difficulty | Build Time |
|--------------|------------|------------|
| Basic        | 0          | 1D3 days   |
| Average      | -2         | 1D3 weeks  |
| Complex      | -4         | 2D6 weeks  |
| Very Complex | -6         | 2D6 months |
| Amazing      | -8         | 4D6 months |



**TOOLS & PARTS** – The attempt presumes all of the required tools, parts and construction components are available, as well as a suitable shop in which to work.

**SUPPORTING SKILL** – If a supporting skill might prove relevant (such as Computer or Aircraft), then roll 8+ and add its skill level. If the roll is failed, then the main Mechanical or Electronics roll will suffer a DM -2.

**TEAMS OF CLOSE ASSISTANTS** – Skilled (level 2+) assistants may provide a bonus. A gadgeteer can have ONE close assistant, plus one more for each level of Leader he possesses. The GM should make a Difficult (-2) Mechanical or Electronics (as required) roll for each assistant. Should a roll succeed, a DM +1 can be added. If the Effect of a failed skill roll is -6 then that assistant messes up and provides a DM-3!

## THE ROLL

Halfway through the rolled Build Time, the gadgeteer makes an Electronics or Mechanical roll (whichever is most relevant), at the listed difficulty. Add the bonuses listed below. The Int bonus is not added, since having an Int bonus is a prerequisite of being able to utilise the gadgeteering procedure.

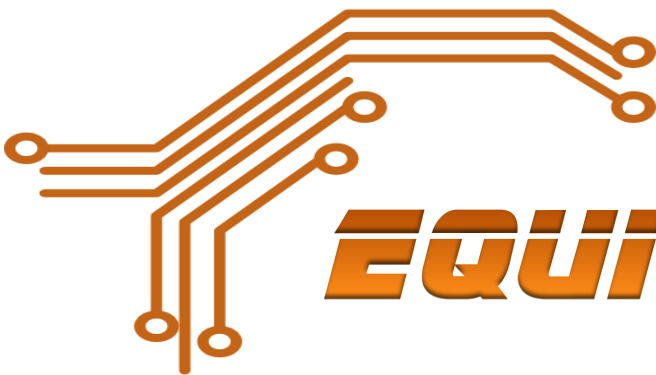
- + Mechanical (or Electronics) skill level
- +1 if an assistant made their Mech or Elec 10+ roll
- Difficulty modifier
- 2 if the supporting skill roll of 8+ was failed
- 3 if an assistant failed their Mech or Elec roll by 6.

**Did it Succeed?** If the roll was successful, then the gadget was finished at the end of the Build Time. The Effect may reduce that Build Time.

| SPEED OF MANUFACTURE |                 |
|----------------------|-----------------|
| Effect               | Time Reduced By |
| 0 - 2                | -               |
| 3 - 5                | 25%             |
| 6+                   | 50%             |

**Did it Fail?** If the roll was failed, then there was a problem and the task roll must be repeated, following the listed Build Time, and adding to the cost by half of the original estimate. *Three failed rolls* destroy the main components and new components must be acquired before the task can be attempted again.

**Example:** A Comms-1, Electronics-3 genius with Int 11, wants to build a TL 14 optic link to his cyborged dog, so he can view what the dog views. This is Average (-2) and costs Cr2000. He fails his Comms roll, then rolls 6 for his Electronics roll (adding +3 for skill, -2 for difficulty and -2 for failed Comms roll). He fails and wastes one week. He tries again, this time making his Comms roll. His main roll is 6 (adding +3 for skill and -2 for difficulty) which is another fail, and another week of work. Again, he makes another Comms roll, but this time his Electronics roll is 9 (adding +3 for skill and -2 for difficulty), which is a total of 10, success. We add another week for 3 weeks Build Time, costing Cr4000 in total.



# EQUIPMENT

A high-tech universe is filled with high-tech equipment, devices and vehicles. A sample of useful equipment is listed here with the Tech Level of its introduction, its weight (in kilogrammes) and the price (in credits). An item can be purchased at a higher TL than its introduction, and may be of better quality and more reliable. Later Tech Level improvements are noted where applicable; these improvements cost the standard price when purchased on a world of that TL, unless stated otherwise.



**Durability of an Object** – As described on page 139, most objects have a number of Durability Points equal to their weight in kilogrammes (minimum 1), as well as an Armour Value. The Armour Value is subjective, but as a guide, use the following parameters:

*The object survives when...*

- |                                                       |        |
|-------------------------------------------------------|--------|
| • Dropped on the floor                                | AV 3+  |
| • Thrown down from a first floor window onto concrete | AV 6+  |
| • Hit with a baseball bat                             | AV 7+  |
| • Driven over with a car wheel                        | AV 10+ |
| • Inside a collapsed building                         | AV 12+ |

**Example:** A TL 13 bioscanner is 1.5 kg in weight, giving it 2 Durability Points. I don't think it would break if I dropped it, but if I threw it down a flight of stairs, that would... so I give it an AV of 4. I hit it with a baseball bat (1D6+2) doing 2 points, so do not damage the scanner. My second whack does 5 points, lowering its Durability Points to 1. It is damaged and may not work properly. It takes two more hits with the bat to destroy the bioscanner. That bout of frustration just cost me Cr900.

## ARMOUR & PROTECTION

Armour protection has an Armour Value that is subtracted from incoming damage. Some of these armour types, such as survival or vacc suits, have other uses, too. Armour can protect from ballistic, energy or melee attacks; some of those; or all of them. See the table below for details.

| ARMOUR              |    |           |              |        |                           |
|---------------------|----|-----------|--------------|--------|---------------------------|
| Armour              | TL | Cost      | Armour Value | Weight | Type of Protection        |
| Leather Jack        | 0  | Cr250     | 3            | 3      | Melee                     |
| Lamellar            | 0  | Cr1000    | 4            | 10     | Melee                     |
| Mail Armour         | 1  | Cr1000    | 5            | 12     | Melee                     |
| Plate Armour        | 2  | Cr5600    | 8            | 25     | Melee                     |
| Ballistic Jacket    | 6  | Cr400     | 6            | 5      | Ballistic, Melee          |
| Concealed Vest      | 7  | Cr1000    | 6            | 3      | Ballistic, Melee          |
| Rigid Battle Armour | 8  | Cr5000    | 8            | 6      | Energy, Ballistic, Melee  |
| Ablative Armour     | 9  | Cr500     | 6            | 3      | Energy, Melee             |
| Combat Armour       | 11 | Cr200,000 | 12           | 10     | Energy, Ballistic, Melee  |
| Power Armour        | 13 | MCr2      | special      | 6      | Energy, Ballistic, Melee  |
| Survival Suit       | 6  | Cr700     | 2            | 5      | Ballistic, Melee          |
| Hazmat Suit         | 6  | Cr1000    | 8            | 5      | Fire                      |
| Soft Suit           | 6  | Cr7000    | 4            | 24     | Ballistic, Melee          |
| Vacc Suit           | 9  | Cr9,000   | 5            | 18     | Ballistic, Melee          |
| Vacc Suit           | 12 | Cr10,000  | 5            | 9      | Ballistic, Melee          |
| Hostile Envirosuit  | 9  | Cr20,000  | 18           | 40     | Energy, Ballistic, Melee  |
| Repulsor Belt       | 16 | Cr1200    | 3            | 0.5    | Ballistic, Lasers, Plasma |

**Leather Jack (TL 0):** A tunic of thick leather armour. It may also include leather greaves and forearm protection. Weight 3 kg. Cr250

**Lamellar (TL 0):** This armour consists of thin metal strips or bands that are held on to a leather backing with rivets. The arrangement of strips can vary tremendously from culture to culture. Metal poor societies can manufacture strips of bone or wood. Also known as splint, banded or studded armour. Japanese samurai armour prior to the 16th century would be classed as lamellar. Damage reduction 4 points from melee damage Weight 10 kg. Cr1000.

**Mail Armour (TL 1):** A shirt of metal rings, or scales backed by a padded lining to protect against hand-held weapons. Includes a metal helmet for the head. Damage reduction 6 points from melee damage Weight 12 kg. Cr1000.

**Plate Armour (TL 2):** Fully-articulated plate armour, providing total coverage for 8 point melee protection. Includes a metal helmet. Weight 25 kg. Cr5600.

**Ballistic Jacket (TL 6):** A bullet-resistant tactical jacket worn by security forces and soldiers. Damage reduction is 6 points from ballistic and melee damage. Weight 5 kg. At TL 9 it weighs 3 kg. Cr400.

**Concealed Vest (TL 7):** A tailored ballistic vest that can be worn under clothing. Damage reduction 6 points from ballistic and melee damage. Weight 3 kg. At TL 10 its weight is negligible. Cr1000.

**Rigid Battle Armour (TL 8):** Military grade ballistic rigid armour, in the form of a helmet; a full-torso, clamshell-style, rigid body shell; and a set of lower leg guards. The helmet features microphone, camera and short-range radio. Weight 6 kg. Cost Cr5000.

**Ablative Armour (TL 9):** Armour that vapourises when hit by a laser or plasma weapon. Damage reduction is 6 points from energy and melee weapons. Weight 3 kg. Cr500.

**Combat Armour (TL 11):** A sealed military combat suit that acts as a vacc suit providing life support for 6 hours. Damage reduction is 12 points from ballistic, energy and melee damage. Weight 10 kg. Cr200,000.

**Power Armour (TL 13):** A 2.3m-tall powered armour suit, also known as 'battle dress'. In combat, treat Power Armour as a robot, with AV and Durability Points. Its internal motors provide great strength, it also acts as a vacc suit providing life support in vacuum and even in hostile environments, for 6 hours. It provides the wearer with Strength 18 as well as night-vision and infra-red sensors, long-range radio and inertial locator. Weight (for the wearer) 6 kg; actual weight 100 kg.

| Armour | Durability | Speed | Endurance | Mass (t) | Strength | Base Cost |
|--------|------------|-------|-----------|----------|----------|-----------|
| 12     | 36         | 6     | 2 days    | 0.1      | 18       | MCr2      |

**Survival Suit (TL 6):** A flexible suit designed to protect the wearer from extreme cold or heat (-50°C to +50°C) and even water immersion. It is not pressurized (although the suit has a hood, gloves and boots). It can be made proof against toxic gas if a survival mask is also worn. Cost Cr700. Weight 5 kg.

**Hazmat Suit (TL 6):** A hazardous materials suit is resistant to toxic spills and airborne viruses, and provides some protection against heat, fire and radiation. It includes boots, hood, oxygen mask and an integrated 3 hour air supply. It has an AV 8 against fire, and reduces rad exposure by 20 per hour. Cost Cr1000. Weight 5 kg.

**Soft Suit (TL 6):** A flexible spacesuit used for protection in toxic environments and as emergency ship suits. Also used at TL 6-8 for work in vacuum environments. Provides 6 hours of air. Weight 24 kg. Cr7,000.

**Vacc Suit (TL 9):** A spacesuit, providing 6 hours of air in unbreathable atmospheres or very hot or cold environments. An insidious atmosphere will breach the seal of a vacc suit in 2D6 hours. Damage reduction is 5 points from ballistic and melee damage. Weight 18 kg, Cr9,000. At TL 12, weight is 9 kg, Cr10,000.

**Hostile Environsuit (TL 8):** A rugged spacesuit with an armoured covering to protect against incredible temperatures, flames, deep-sea pressures, radiation and corrosive or insidious atmospheres. It provides air for 6 hours. Damage reduction is 18 points from ballistic, energy and melee damage. Weight 40kg. At TL 10 it weighs 20 kg. At TL 13 it weighs 12 kg. Cr20,000.

**Repulsor Belt (TL 16):** Some settings may feature the repulsor belt, which, when activated creates a repulsor field around the wearer (out to about 40cm). This stops all ballistic or missile weapons, bends light enough to reduce the accuracy of lasers, and makes it more difficult for melee attacks to reach the user. A society using these will focus almost exclusively on melee combat and dart-firing thud guns. The belt has power for 6 hours of continuous use and must then be recharged, which requires 30 minutes. The repulsor field uses the body's own electrical field to centre itself, touching another person or a large object does not mean the field will grow to envelop that other person or object, it maintains its volume and centres on the wearer's belt unit. The repulsor belt provides an Armour Rating of 3 against incoming melee attacks, and forces an attacker using a laser weapon to take a -3 penalty. Weight 0.5 kg, Cost Cr1200.

## MELEE WEAPONS

Hand-held weapons are still a common feature of combat in most science fiction settings. The following weapons all use the Melee Combat skill.

| MELEE WEAPONS       |    |        |           |      |          |    |
|---------------------|----|--------|-----------|------|----------|----|
| Weapon              | TL | Cost   | Dmg       | Wgt  | Range    | AV |
| Unarmed             | -  | -      | 1D6       | -    | Personal | -  |
| Studded Gauntlet    | 1  | Cr25   | 1D6+1     | -    | Personal | -  |
| Hatchet             | 0  | Cr25   | 2D6       | 1    | Personal | 8  |
| Battle-Axe (1H)     | 1  | Cr300  | 3D6       | 2.5  | Close    | 12 |
| Axe-Mace (2H)       | 1  | Cr80   | 3D6       | 3    | Close    | 10 |
| Great Axe (2H)      | 1  | Cr150  | 3D6+2     | 3.5  | Close    | 10 |
| Club, Wooden        | 0  | Cr5    | 1D6+2     | 0.5  | Personal | 6  |
| Club, Spiked        | 0  | Cr20   | 2D6       | 1    | Personal | 6  |
| Club, Metallic      | 1  | Cr10   | 1D6+2     | 1    | Personal | 9  |
| Mace                | 0  | Cr30   | 2D6+1     | 2    | Personal | 10 |
| War Flail           | 1  | Cr100  | 3D6       | 2    | Close    | 12 |
| Staff (2H)          | 0  | Cr10   | 1D6+2     | 1.5  | Close    | 12 |
| Warhammer (1H)      | 1  | Cr80   | 2D6+2     | 2.0  | Close    | 12 |
| Bayonet, Fixed (2H) | 6  | Cr30   | 3D6       | 0.25 | Close    | 8  |
| Toothed Spear       | 0  | Cr10   | 2D6       | 1.2  | Close    | 11 |
| Spear (1H)          | 1  | Cr15   | 2D6       | 1.5  | Close    | 14 |
| Spear (1H/2H)       | 1  | Cr35   | 2D6+2/+5  | 2.5  | Close    | 14 |
| Broad Spear (2H)    | 1  | Cr60   | 3D6+2     | 2.5  | Close    | 14 |
| Halberd (2H)        | 2  | Cr100  | 4D6       | 3.5  | Close    | 16 |
| War Scythe (2H)     | 1  | Cr100  | 3D6+2     | 2.5  | Close    | 15 |
| Punching Dagger     | 1  | Cr25   | 2D6+1     | 0.5  | Personal | 10 |
| Combat Knife        | 1  | Cr10   | 2D6       | 0.25 | Personal | 8  |
| Long Sword          | 1  | Cr150  | 2D6+2     | 1.2  | Close    | 14 |
| Shortsword          | 1  | Cr90   | 2D6+1     | 1.2  | Personal | 14 |
| Cutlass             | 3  | Cr100  | 2D6+2     | 1.5  | Close    | 15 |
| Toothed Sword       | 0  | Cr60   | 2D6+1     | 1.2  | Close    | 10 |
| Rapier              | 3  | Cr100  | 2D6       | 0.5  | Close    | 10 |
| Telescoping Sword   | 9  | Cr1200 | 2D6+1     | 1    | Close    | 12 |
| Broadsword (1H/2H)  | 2  | Cr180  | 3D6/3D6+1 | 2.5  | Close    | 16 |
| Katana (2H)         | 1  | Cr200  | 3D6       | 1.5  | Close    | 15 |
| Machete             | 1  | Cr75   | 2D6+1     | 1.5  | Personal | 13 |
| Stun Baton          | 9  | Cr300  | Stun      | 1    | Personal | 9  |
| Force Sword         | 14 | Cr5500 | 6D6       | 0.5  | Close    | -  |
| Shield, Small       | 0  | Cr50   | 1D6+1     | 1.5  | Close    | 12 |
| Shield, Medium      | 1  | Cr80   | 1D6+1     | 3    | Close    | 18 |
| Shield, Large       | 1  | Cr150  | 1D6+1     | 6    | Close    | 24 |

**Studded Gauntlets (TL 1):** Thick fighting gauntlets with metal spikes attached.

**Hatchet (TL 1):** A handaxe with a stout, reinforced handle and a metal axe-head suited for close-combat – or as a survival tool. Also known as: *tomahawk*, *boarding axe*, *sagaris*. A stone axehead does 1D6+3 damage.

**Battleaxe (TL 1):** A heavy, one-handed fighting axe with a metal blade. There are many variations on the battle-axe. Also known as: *masakari*.



**Axe-Mace (TL 1):** A long-hafted, double-ended axe weapon. At one end is mounted a flanged axe or half-moon axe blade, at the other a heavy weight, used for punching and bludgeoning.

**Great Axe (TL 1):** A large heavy, metal axe-blade is mounted on a long, 1.4m haft. Some great axes are "double-bitted", meaning double-bladed. Also known as: *Danish axe, broadaxe, sparr axe*.

**Club, Wooden (TL 0):** A handy wooden or polymer club, around 80cm long. Also known as: *nightstick, baton, kosh, tonfa*.

**Club, Spiked (TL 0):** A wooden club around 80cm long, embedded with metal spikes.

**Club, Metal (TL 1):** A handy metal club or pole, around 80cm long.

**Mace (TL 0):** A short (30cm) haft topped with a weight designed for crushing and bludgeoning, the head may be metal or stone.

**War Flail (TL 1):** A metal handle with a couple of links of chain which attaches to the equivalent of a long mace head. Also known as: *nunchaku*.

**Staff (TL 0):** A wooden fighting staff, 1.5 – 2m long, and used two-handed. Also known as: *Bo Staff*.

**Warhammer (TL 1):** A heavy metal hammer designed for the battlefield. It typically features a spike on the reverse of the hammer head. Also known as: *pick*.

**Bayonet, Fixed (TL 6):** A dagger-like weapon, designed for attachment to a rifle or carbine. When *not* attached to a musket or rifle, the bayonet performs as a combat knife. At TL 6 onwards, the bayonet is basically a combat knife, and when not attached can serve as such. At TL 6+ the bayonet weighs only 0.25 kg and costs Cr30.

**Toothed Spear (TL 0):** A short 1.8m spear, tipped with a spearhead of flint, obsidian, sharpened bone, antler or the teeth of a local predator. Found in primitive TL 0 or metal-poor societies.

**Short Spear (TL 1):** A short 1.8m spear, tipped with a spearhead of metal. It can easily be thrown as a javelin. Also known as: *yari-nage*.

**Long Spear (TL 1):** A long 2.5m spear, tipped with a spearhead of metal. It cannot be thrown as a javelin. Also known as: *yari*.

**Broad Spear (TL 1):** A heavy, wide-bladed 2.5m-long spear. Also known as: *partisan, spontoon, ox-tongue, gintak, bohemian earspoon*.

**Halberd (TL 2):** A wide slashing blade and a long spike mounted atop a 2.5m-long shaft; the back of blade includes a hook. Also known as: *poleaxe, bill, guisarme*.

**War Scythe (TL 1):** A sword blade mounted atop a 2.5m-long shaft. The blade extends upright from the pole rather than at right angles. Also known as: *naginata, Dacian falx, swardstav, sovnya*.

**Punching Dagger (TL 1):** A stabbing dagger with an integral handguard.

**Combat Knife (TL 1):** A short weapon with a flat, two-edged blade approximately 20 cm in length. Combat knives are usually carried in a belt sheath, they are usually as much a tool as a last-resort weapon of defence, and worn constantly. Also known as *tanto, sai, Fairburn Sykes, dagger, ballock, pugio, stilleto*.

**Long Sword (TL 1):** The standard one-handed long-edged blade weapon, featuring a flat, two-edged blade. It may or may not have a basket hilt or hand protector. A scabbard to carry the sword may be attached to the belt, or to straps (or a sash) over the shoulder. Blade length may vary from 70 to 95 cm. Also known as: *spatha, Viking sword, arming sword, Celtic sword*.

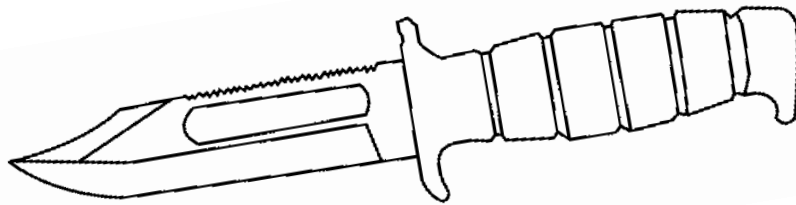
**Shortsword (TL 1):** A short, cut and thrust sword with a two-edged blade, 45-65 cm in length, often used in conjunction with a large shield. Also known as: *gladius, xiphos, kindjal*.

**Cutlass (TL 3):** A heavy, flat-bladed, single-edged weapon featuring a full basket hilt to protect the hand. Blade length varies from 60 to 90 cm. Popular on board ship due to its ability to cut ropes in emergencies, and allow fighting within the cluttered confines of a ship's deck. Also known as: *cavalry sabre*.

**Toothed Sword (TL 0):** Metal-poor societies can make effective slashing swords by fitting rows of obsidian blades or the teeth of some local predator along a metre-long wooden haft.

**Rapier (TL 3):** Also known as the foil, this weapon is a light, sword-like weapon with a pointed, edged blade 80 cm in length, and a basket or cup hilt to protect the hand. Rapiers are worn in scabbards attached to the belt. Also known as: *foil, sidesword*.

**Telescoping Sword (TL 9):** This high-tech retractable sword can easily be concealed before use.



**Broadsword (TL2):** A hand-and-a-half sword is designed to be used either one-handed or two-handed. It is a long, double-edged sword. Also known as: *miao-dao, broadsword, bastard sword*.

**Katana (TL 1):** A gently curved two-handed sword with a single-edged blade.

**Machete (TL 1):** Crude but effective one-handed single-edged chopping sword. Also known as *falchion, langemesser, machaira, kukri, barong*.

**Stun Baton (TL 9):** These weapons are designed to deal non-lethal damage and will incapacitate a living target instead of killing it. On a successful hit, the victim makes an Average (0) Endurance roll. If this is failed, the target is incapacitated and unable to perform any actions for 1D6+2 rounds. If the End roll succeeds, the victim is still stunned for one round.

**Force Sword (TL 14):** A metallic handgrip that, when activated, produces a blade of energy. The Force sword has an endurance of 100 hours before requiring a recharge. It can be hung from a belt when not in use.

**Shield, Small (TL 0):** A small one-handed shield used by skirmishers, swashbucklers and archers. Can be fastened to the waist-belt when not in use. Also known as: *buckler, targe*.

**Shield, Medium (TL 1):** A common fighting shield of wood and leather, usually with a metal boss, roughly one-metre in height. Also known as: *heater, pelta*.

**Shield, Large (TL1):** A large shield 1.2m in height that provides excellent protection in combat, but that is heavy and encumbering. It can provide decent protection against missile weapons and so is popular with soldiers conducting sieges. Cannot be used while mounted. Also known as: *hoplon, scutum*.

## PRIMITIVE RANGED WEAPONS

This section includes a variety of thrown weapons and primitive (TL 0 – 2) ranged weapons. Archery skill can be used with the bows and crossbows, but not slings. Only the character's Dexterity modifier can be used with the sling and staff sling. Rules for thrown weapons can be found on page 126.

| PRIMITIVE RANGED WEAPONS |    |      |          |      |        |
|--------------------------|----|------|----------|------|--------|
| Weapon                   | TL | Cost | Dmg      | Wgt  | Range  |
| Short Bow                | 0  | 50   | 2D6      | 1.0  | Medium |
| Long Bow                 | 0  | 75   | 2D6+1    | 1.5  | Medium |
| Composite Bow            | 1  | 150  | 2D6+2    | 1.0  | Medium |
| Crossbow, Light          | 2  | 200  | 2D6+1    | 3.0  | Medium |
| Crossbow, Heavy          | 1  | 250  | 3D6      | 5.0  | Medium |
| Bolas (E)                | 0  | 5    | Entangle | 1.0  | Thrown |
| Fighting Net (E)         | 0  | 40   | Entangle | 2.0  | Thrown |
| Throwing Club            | 0  | 10   | 1D6+2    | 0.5  | Thrown |
| Throwing Steel           | 2  | 10   | 1D6+2    | 0.1  | Thrown |
| Throwing Knife           | 1  | 35   | 2D6      | 0.25 | Thrown |
| Throwing Axe             | 1  | 40   | 2D6      | 0.75 | Thrown |
| Throwing Dart            | 1  | 3    | 2D6      | 0.25 | Thrown |
| Sling                    | 0  | 1    | 2D6      | -    | Short  |
| Staff Sling              | 1  | 2    | 3D6      | 1.5  | Medium |
| War Dart                 | 0  | 6    | 2D6+2    | -    | Short  |
| Javelin                  | 0  | 15   | 3D6      | 1.5  | Thrown |

**Short Bow (TL 0):** A short bow (less than 1.5m in length) that is also a 'self bow', i.e. it is a bow made of a single piece of wood or bamboo. The easiest bow to make, and using non-specialist types of timber. The bow is able to be used from horseback.

**Longbow (TL 0):** A long bow (typically 1.8 – 2m in length) that is also usually a 'self bow', i.e. it is a bow made of a single piece of wood. The archer is able to draw the string back much further than with a short bow, providing greater range and power. Cannot be used from horseback. Also known as: Japanese *higo-yumi*, *flatbow*.

**Composite Bow (TL 1):** Made from wood, animal horn and sinew laminated together, the drawn bow stores more energy than wood for the same length of bow, allowing a powerful bow in a shorter length. The bow is able to be used from horseback. Manufacture requires a range of components to source, takes longer than a wood bow and the finished bow is more sensitive to moisture. Leather cases are often made to carry the bows in. Also known as: *Mongol bow*, *recurve bow*, *Turkish bow*, *Manchu bow*.

**Crossbow, Light (TL 2):** Crossbows consist of a horizontal bow-like assembly mounted on a frame which is handheld in a similar fashion to the stock of a gun. It shoots arrow-like projectiles called bolts. It is easier to use than a bow (only DM -1 if unskilled), though heavier and more expensive. Between shots, a light crossbow string can be pulled back by wedging a foot inside an iron stirrup attached to the front of the crossbow. Reloading requires one full round.

**Crossbow, Heavy (TL 1):** Crossbows consist of a horizontal bow-like assembly mounted on a frame which is handheld in a similar fashion to the stock of a gun. It shoots arrow-like projectiles called bolts. It is easier to use than a bow (only DM -1 if unskilled), though heavier and more expensive. Between shots, a heavy crossbow string must be pulled back by temporarily attaching a geared mechanism such as a 'cranequin'

(cost includes cranequin, weight 3 kg). Such mechanisms are hung from the crossbowman's belt when not in action. Reloading requires one full round. Also known as: *gastraphetes*, *arcubalista*, *arbalest*.

**Bolas (TL 0):** Two weights connected by a cord or leather thong, the bolas are spun and then thrown at fleeing game to bring it down by entangling its legs. Humans constitute 'fleeing game', and can untangle themselves in two rounds.

**Fighting Net (TL 0):** A small combat net, with attached weights, that can be thrown from the left hand to entangle an opponent. It may have other uses too. A human can untangle themselves in one full round with an Average Dexterity roll.

**Throwing Club (TL 0):** A hardwood baton, usually slightly curved with a slight weight at one end that can be thrown to bring down small game such as rabbits, birds and so on. Early cultures, including the Egyptians, carried throwing clubs with them on to the battlefield. Also known as: *boomerang*, *rabbit stick*, *throw stick*.

**Throwing Steel (TL 2):** Coming in a myriad of designs, the throwing steel is a small but heavy metal blade that can be thrown. Often the blade is spun and so has a number of blades mounted on a circular body. Also known as *shuriken*, *kpinga*, *chakram*, *kleegat*. Cr10

**Throwing Knife (TL 1):** A combat knife optimized and balanced for throwing.

**Throwing Axe (TL 1):** A short handled hatchet that has a curved metal head, making it efficient to use as a throwing axe. Also known as: *francisca*.

**Throwing Dart (TL 1):** The dart is a heavy 40cm arrow fitted with an barbed iron head and a lead weight to add momentum. It is thrown either underhand, up into the air to come down vertically onto the head of a target, or thrown overhand like a throwing knife at shorter ranges. It is no toy. Typically carried within a leather quiver. Also known as *plumbata*.

**Sling (TL 0):** A woven textile or leather pouch with two long cords attached. A stone or lead bullet is placed in the pouch and the sling whirled rapidly, discharging the stone at considerable speed. In the hands of an expert, comparable with a short bow. Using cast lead sling bullets adds +1 to damage, otherwise ammunition can be picked up as smooth pebbles from a beach or riverbank.

**Staff Sling (TL 0):** This is a sling mounted on top of a 1.2m wooden shaft, the cords of the sling fastened to the tip of the pole. When slung in the manner of a trebuchet, a single cord automatically detaches and the stone is released. The staff sling allows smooth stones up to fist sized to be slung considerable distances.

**War Dart (TL 0):** The sling and war dart were once brought together in ancient Greece to create the *kestros*, allowing a metre-long heavy war dart to be slung out to much greater distances. Each war dart uses a leather sling to produce added momentum and distance to the throw. Also known as: *kestrosphendon*, *Swiss arrow*, *Yorkshire arrow*.

**Javelin/Short Spear (TL 0):** A short 1.8m spear, tipped with a spearhead of metal, flint, obsidian or teeth. It can easily be thrown as a javelin. Can be used as a short spear.

## AMMUNITION

**Quiver (TL 0):** Typically leather, textile or wood. Holds 20 arrows, crossbow bolts or throwing darts (not interchangeable). Weight: negligible. Cr20

**Quiver Load of Arrows (TL 0):** There are four types of arrow; barbed, broad-head, punch-head and incendiary:

**Barbed (TL 0)** – A standard military or hunting arrowhead, typically with barbs to ensure the arrow stays in the target. Weight for 20: 2 kg. Cost for 20: Cr40

**Broad-head (TL 0)** – A wide cutting head, doing +1 extra damage. Weight for 20: 2 kg. Cost for 20: Cr80

**Punch-head (TL 1)** – A needle-like piercing arrowhead, designed to penetrate armour designed to defeat melee damage. Ignores 4 points of armour. Also known as a bodkin. Weight for 20: 2 kg. Cost for 20: Cr80

**Incendiary (TL 1)** – The point is fashioned as a small cage into which a small oil-soaked rag is inserted and lit. If shot at something flammable (not humans) then allow that target to catch fire. Weight for 20: 2 kg. Cost for 20: Cr80.

**Quiver Load of Bolts (TL 1):** A load of 20 crossbow bolts. Weight: 2 kg. Cr40

**Pouch of Sling Stones (TL 0):** A bag of 20 river pebbles, carefully chosen for shape. Weight: 1.5 kg. Free

**Pouch of Lead Sling Bullets (TL 1):** A bag of 20 uniformly-cast lead sling bullets (+1 damage). Weight: 2 kg. Cr40

## FIREARMS – LOW TECH (TL 2–3)

'Flintlock Fantasy' settings often include magic and monsters, with gunpowder technology added to the mix. Examples of modern fantasy authors creating such a setting include Brandon Sanderson (the *Mistborn* series) and Brian McClellan (the *Powder Mage* trilogy). The author Naomi Novik wrote the *Temeraire* series, which is "a reimagining of the epic events of the Napoleonic Wars with an air force - an air force of dragons, manned by crews of aviators". Meanwhile, Django Wexler (author of the *Shadow Campaigns*) created a world resembling Europe and North Africa of the Napoleonic era, with its own version of the French Revolution, and a supernatural conspiracy fighting for control of the remnants of magic.

The composition of gunpowder on Earth was finally perfected in 1515. Typically, manufacturers used saltpetre 75 parts, sulphur 12.5 parts and charcoal 12.5 parts. The ingredients would have been of high purity. Sodium nitrate was sometimes used instead of potassium nitrate. It was cheaper than saltpetre, and made a slightly more powerful product, although it tended to absorb moisture from the air. In the Original Star Trek episode, *Arena*, Captain Kirk defeats his opponent by manufacturing gunpowder from the mineral deposits he finds on the planet. He utilises coal, potassium nitrate and sulphur and repurposes a hollow, bamboo-like stem as a barrel. His projectiles were diamonds. Some worlds will not have the correct elements, easily accessible, to allow for gunpowder to be created.



**Hand Cannons (TL 1-2)** – These might be available at the end of TL 1, and become more commonplace at TL 2, where they are quickly replaced by matchlock muskets. The primitive hand cannon consists simply of an iron barrel mounted on the end of a wooden handle. They include a flash-pan attached to the end of barrel and a touch hole drilled through the side-wall. The flash pan has a leather cover or hinged metal lid, to keep the priming powder dry until the moment of firing. These features are carried over to subsequent TL 2 firearms.

**Matchlock Guns (TL 2)** – The matchlock firing mechanisms is designed to free up the shooter's hands for steadying and aiming the heavy gun. An 'S'-shaped arm pivots forward when the trigger is pulled, touching a slow-burning match to an iron pan filled with gunpowder. This ignites and passes a flash into the barrel to detonate the main charge propelling the musket ball forward. The match has to be kept burning all the time, which is dangerous around powder and it is difficult to use in wet or windy weather.

**Flintlock Guns (TL 3)** – The flintlock firing mechanism is a more reliable system of firing a handgun. A sparking stone on the end of an arm is propelled forward by the trigger and strikes the face of an upright metal plate, sparks are created and as the plate is pushed back it exposes the pan containing gunpowder, which ignites and fires the musket ball within the barrel. This is a fairly weatherproof firing mechanism.

**Percussion Guns (TL 3)** – A late TL 3 development is the percussion cap, which replaces the sparking stone with a tiny cap of reliable primer that is struck by the hammer. To fire a shot, the user places a cup-shaped copper cap onto the top of a vent into the chamber. This cap contains a primer which explodes when struck by the falling hammer. The explosive cap also makes possible the first practical repeating arms, like the early revolvers. In addition, mass produced paper cartridges are now available, each containing a bullet and a measured amount of gunpowder that can be quickly torn open and poured into the barrel.

**Metal Cartridges (TL 4 onwards)** – This innovation brings the bullet, gunpowder and percussion cap together into a single metallic cartridge. Now, instead of pouring gunpowder and bullet down the barrel, a user can push a single cartridge into the barrel at the rear (the breech). These weapons are covered in the next section, *Firearms – Slugthrowers*.

| LOW TECH FIREARMS    |     |      |        |       |        |        |
|----------------------|-----|------|--------|-------|--------|--------|
| Weapon               | TL  | Cost | Range  | Dmg   | Recoil | Weight |
| Hand Cannon          | 1-2 | 500  | Short  | 2D6+2 | 11+/6+ | 8.0    |
| Pocket Pistol        | 2   | 100  | Close  | 2D6   | 6+     | 0.8    |
| Black Powder Pistol  | 2   | 100  | Short  | 2D6   | 8+     | 1.2    |
| Black Powder Carbine | 2   | 80   | Medium | 2D6+1 | 7+     | 3.0    |
| Black Powder Musket  | 2   | 100  | Medium | 3D6   | 9+     | 5.0    |
| Black Powder Rifle   | 2   | 160  | Long   | 3D6   | 8+     | 4.5    |
| Blunderbuss          | 2   | 120  | Short  | 4D6   | 9+     | 4.0    |
| Percussion Revolver  | 3   | 150  | Short  | 2D6   | 0      | 0.9    |
| Percussion Rifle     | 3   | 200  | Long   | 3D6   | 6+     | 4.0    |

**Hand Cannon (TL 2):** This has an iron barrel only 30-40cm long, attached to the end of a long wooden stock. The barrel has been forged with an integral iron hook on its underside. The hook enables the gun to grip a low wall or a wooden rail, preventing the recoil of the gun from kicking the weapon backwards. The overall length of the arquebus is 1.30m, and it has a caliber of 0.9' or less. Two Recoil values are given, the first is used when the hand cannon is carried in two hands, the second is used when the gun is hooked over an embrasure, wall or wooden fence. Length: 130cm. Also known as *arquebus*, *hackbut*.

**Pocket Pistol (TL 2):** A small, concealable pistol, constructed with wheellock or flintlock mechanism, as appropriate. Length: 18cm.

**Black Powder Pistol (TL 2):** A substantial black powder pistol, constructed with wheellock or flintlock mechanism, as appropriate. These pistols were initially developed for the cavalry, who required a weapon that they could reload while mounted. Even then, many cavalymen carried several loaded pistols into battle with them. Soon adopted by sailors, officers, highwaymen and duelists. When empty, the barrel can be grasped and the brass-weighted butt used as an effective club. Length 46cm.

**Black Powder Carbine (TL 2):** A cavalryman's weapon, providing the range that a pistol could not. The carbine is a shortened musket, and can be holstered next to the saddle and reloaded while mounted. Other troops have found this shorter musket of use, as have coach-drivers, pirates, highwaymen and sailors. Length: 90cm.

**Black Powder Musket (TL 2):** A direct development of the arquebus, the musket is longer for better accuracy, yet lighter and slightly more accurate. Without the rifling grooves within the barrel, the musket ball has no spin when it leaves the muzzle, making it inaccurate compared to later rifles. The blunderbuss, black powder pistols, pocket pistol and carbine are all, likewise, smooth-bore weapons. The musket is a soldier's standard weapon and he learns to drill with it, eat with it and sleep with it. Once fired he can (at TL 3) fit a spike bayonet to the end of the barrel. This bayonet has a ring which fits around the muzzle, allowing him to reload quickly if needed. The bayonet turns the unloaded musket into a short spear. Length: 150cm.

**Black Powder Rifle (TL 2):** The rifled barrel, where a spiral groove was cut onto the inner face, imparts spin to the bullet as it passes along, improving accuracy. Rifled muskets existed in the 17<sup>th</sup> century, but only became commonplace in the early 19<sup>th</sup> century. To follow the grooves, the ball has to fit snugly which makes ramming it down the barrel during reloading a more time consuming task. Because of this, the rifle was for many decades a weapon of the hunter and sportsman. Length: 150cm.

**Blunderbuss (TL 2):** The blunderbuss is a shotgun-style black powder weapon, often with a bell shaped, flared muzzle. It is loaded with shotgun pellets and is a short-range weapon of significant stopping power. Use the rules for Shotgun Spread (page 125). Length 50cm.

## LOW TECH FIREARMS ACCESSORIES

**Gunpowder (TL 2):** Enough gunpowder for 20 shots. Weight: 0.25 kg. Cr10

**Gunpowder Keg (TL 2):** A relatively small, easily transportable wooden keg of gunpowder, 33cm long and 28cm wide. It is bound with strappings of reed or sapling wood rather than iron bands to avoid sparks and is bulky, encumbering and requires two hands to carry. Contains 3 kg of gunpowder – enough for 240 shots. If it explodes it does 6D6 HE, each extra keg used to create an explosion adds 10 points of damage. Total weight: 5 kg. Cr130.

**Musket Balls (TL 2):** Comes as 0.5kg of musket balls for Cr10. How many you get depends on the calibre – Pistol 30 balls, Musket/Rifle/Carbine 20 balls.

**Paper Cartridges (TL 2):** Forty paper cartridges, complete with powder and shot. Cr40. Weight 0.5 kg.

**Bayonet (TL 3):** A long shortsword-like weapon, designed for attachment to a musket. When not attached to a musket or rifle, the bayonet performs as a shortsword. Early in TL the bayonet is instead rammed into the barrel of the musket, preventing it from being fired again. Damage 3D6 (fixed) Weight 1 kg, Cr60.

**Cartridge Box (TL 2):** A stiff leather box with a tinned and compartmentalised interior for holding up to 40 paper cartridges. Comes with a leather shoulder strap. Rain-proof. Cr2. Weight 0.5 kg (when full).

**Cleaning Tools (TL 2):** Four useful tools for cleaning and maintaining a black powder firearm. They are kept on a metal ring on a chain to prevent them being dropped or getting lost. Cr5. Weight 0.1 kg.

**Flints (TL 2):** Spare gun flints, shaped and prepared for use in a flintlock weapon. Cr3.

**Leather Belt Pouch (TL 0):** Keep your shot in here. Cr4.

**Leather Shoulder Bag (TL 0):** For shot, tools, powder horn and other accoutrements. Cr15.

**Powder Measure (TL 2):** A small brass gunpowder measure for the accurate pouring of powder from powder horn into the musket or pistol. Comes with a leather lanyard for hanging around the neck or from the wrist. Cr5. Weight negligible.

**Powder Horn (TL 2):** A waterproof, lidded organic container on a shoulder sling, when full it contains powder for up to 20 shots. Cr3, weight 0.25 kg.

**Double Barrelled (TL 2):** Some gunsmiths produce double-barrelled versions of muskets, pistols or blunderbusses, capable of two shots, but with weight and cost multiplied by 1.25. If both barrels are fired simultaneously, and the attack succeeds, both shots hit, however, the advantage of two barrels is that you can effectively have a second shot ready to fire immediately afterwards.

**Blade-Barrel (TL 2):** Because reload times at TL 2 are so long, all muskets and rifles have a barrel attachment for a bayonet (turning the musket into a short spear) See the entry for bayonet (above). Some pistol designs take a similar approach, building daggers or even hatchets *into* the gun's design, ready for use as soon as a shot is fired. A blackpowder pistol with an in-built dagger or hatchet costs Cr400 and weighs 1.6 kg, although they are not common due to the difficulty in safely carrying or holstering them.

## FIREARMS – SLUGTHROWERS

These 'slugthrower' firearms are dominated by the TL 4 innovation of the metal cartridge, although other types of ammunition are later introduced. In the Game Master's setting, these variant ammunition and propellant types might replace conventional cartridges or be used concurrently by different factions. This might provide a point of technological difference with forces at similar Tech Levels.

**Conventional Ammunition (TL 4+) –** Modern metal cartridge cases hold both projectile, propellant and primer.

**Caseless Ammunition (TL 7+) –** A projectile type that has a propellant mix bonded to it without need for a metal case. See Exotic Ammo (page 168).

**Binary Propellant Ammunition (TL 9+) –** Projectiles are loaded separately without cases, and are propelled by the mixing of two volatile gases from a weapon-mounted gas cylinder. See Exotic Ammo (page 168).

**Gyrojet Ammunition (TL 8+) –** Projectiles are effectively miniature rockets holding both propellant and projectile. They are detonated electrically, spin is imparted by angled rocket ports on the round, leaving the barrel smoothbore. Gyrojet weapons are recoilless and suited for use in zero gravity environments.

**Railgun (TL 12+) –** The weapon launches steel or tungsten projectiles via an electromagnetic linear rail (railgun), or magnetic coils (coilgun). There is no propellant and spin is imparted by the magnetic field. Either weapon might be known as *gauss gun*, derived from the gauss, a measurement of magnetic induction.

| SLUGTHROWER WEAPONS  |    |      |        |        |        |     |     |           |        |  |
|----------------------|----|------|--------|--------|--------|-----|-----|-----------|--------|--|
| Weapon               | TL | Cost | Range  | Dmg    | Recoil | Wgt | Mag | Ammo Cost | Notes  |  |
| Zip Gun              | 6  | 20   | Close  | 1D6+2  | 4+     | 0.5 | 1   | 1         |        |  |
| Autopistol           | 5  | 150  | Short  | 2D6    | 6+     | 1   | 16  | 7         |        |  |
| Body Pistol          | 8  | 500  | Short  | 2D6    | 6+     | 0.5 | 6   | 20        |        |  |
| Revolver             | 5  | 150  | Short  | 2D6+2  | 7+     | 1   | 6   | 5         |        |  |
| Thud Gun             | 8  | 300  | Short  | varies | 4+     | 1   | 5   | 10        |        |  |
| Shotgun              | 4  | 200  | Medium | 4D6    | 7+     | 3.5 | 6   | 6         | DM +1  |  |
| Auto Shotgun         | 7  | 700  | Medium | 4D6    | 8+     | 3.5 | 8   | 12        | ROF 4  |  |
| Machine Pistol       | 6  | 400  | Medium | 1D6+3  | 7+     | 1   | 15  | 7         | ROF 4  |  |
| Submachinegun        | 5  | 500  | Medium | 2D6    | 6+     | 3   | 30  | 20        | ROF 4  |  |
| PDW                  | 7  | 600  | Medium | 2D6+2  | 7+     | 2.5 | 30  | 40        | ROF 4  |  |
| Pneumo Rifle         | 4  | 400  | Medium | 1D6+2  | +      | 3   | 15  | 15        |        |  |
| Carbine, Bolt-Action | 5  | 300  | Medium | 3D6    | 7+     | 3   | 10  | 12        |        |  |
| Carbine, Auto        | 7  | 500  | Medium | 3D6    | 7+     | 3.5 | 30  | 20        | ROF 4  |  |
| Anti-Material Rifle  | 7  | 1200 | Long   | 6D6    | 8+     | 14  | 7   | 22        |        |  |
| Rifle, Single-Shot   | 4  | 300  | Long   | 3D6    | 6+     | 4   | 1   | 1         |        |  |
| Rifle, Bolt-Action   | 4  | 400  | Long   | 3D6    | 6+     | 4   | 5   | 5         |        |  |
| Rifle, Semi-Auto     | 5  | 500  | Long   | 3D6+3  | 8+     | 4   | 10  | 10        |        |  |
| Assault Rifle        | 6  | 500  | Long   | 3D6    | 7+     | 4   | 30  | 20        | ROF 4  |  |
| Battle Rifle         | 6  | 550  | Long   | 3D6+3  | 8+     | 4   | 20  | 10        | ROF 4  |  |
| Automatic Rifle      | 5  | 700  | Long   | 3D6+3  | 7+     | 7   | 30  | 20        | ROF 10 |  |
| Gyrojet Pistol       | 8  | 150  | Short  | 2D6    | 2+     | 1   | 6   | 8         |        |  |
| Riot Pistol          | 9  | 400  | Short  | 2D6+2  | 2+     | 1.2 | 4x5 | 20        |        |  |
| Gyrojet Carbine      | 9  | 900  | Long   | 3D6    | 2+     | 3   | 20  | 30        | ROF 4, |  |
| Railgun              | 12 | 1500 | Long   | 4D6    | 6+     | 4   | 60  | 15        | ROF 10 |  |

**Zip Gun (TL 6):** A home-made, one-shot pistol. It has low range, poor performance and high chance of misfire. Roll 1D6 each time it is fired. On a 1 it does not fire; roll again, on a 1 the user is shot. Fires small-calibre ammo also used by the machine pistol.

**Autopistol (TL 5):** Semi-automatic handgun that feeds from a magazine in the grip. It is single-shot. Example: *PKD Detective Special, Sig Sauer P320*.

**Body Pistol (TL 8):** A light, concealable hold-out weapon made from advanced polymers. Example: *Glock 26*.

**Revolver (TL 5):** A rugged handgun that will not jam, with a manually-loaded rotating drum to feed the bullets into the barrel. Example: *Colt Peacemaker, Taurus 444*.

**Thud Gun (TL 8):** A revolver-style dart gun that is powered by compressed air. The dart, usually of 12mm calibre, is a ballistic syringe loaded with an agent and tipped with a hypodermic needle. The dart is propelled from the gun by compressed gas. It is charged by using a manual lever to restore the gun's air pressure. This must be done after every 2 shots. Select a poison from those on page 95.

**Shotgun (TL 4):** Smoothbore weapon firing a spread of steel shot with greater chance of hitting at close range. This shotgun represents a pump-action model. Example: *Benelli M4*.

**Auto Shotgun (TL 7):** A semi-automatic shotgun, usually for military use, that is capable of burst fire. It has a box magazine. Example: *Saiga-12*.

**Machine Pistol (TL 6):** A fully automatic pistol-sized firearm in a small calibre, providing the firepower of a submachinegun in a concealable package. Example: *MAC-11, Steyr TMP*.

**Submachinegun (TL 5):** A light assault weapon, often used by special forces, police and rear-echelon troops. It uses pistol calibre ammunition. Rounds are interchangeable with the autopistol; at the GM's discretion, so are the magazines. Example: *Thompson M1A1, M3 'Grease Gun', Sterling L2A3, HK UMP*.

**PDW (TL 7):** A personal defence weapon. Like the SMG it is used by special forces, police and rear-echelon troops. It has a better range and bigger punch than an SMG because it fires intermediate rounds (sized-between pistol and rifle cartridges). Example: *FN P90*.

**Pneumo Rifle (TL 4):** An air-powered rifle, firing a heavy 13mm round shot at 180m/second. The rifle is smoothbore to reduce barrel friction, and has a detachable air reservoir for 20 shots. It can be manually recharged (taking 15 mins). The weapon has an integral tubular magazine. Useful on low-tech colonies, since lead ammunition can be quickly and easily handmade. Comes with 500g air pump and 250g bullet-casting kit. Example: *Pritchard or Giradoni Air Gun*.

**Carbine, Bolt-Action (TL 5):** A lighter, shorter and handier rifle, that still fires a powerful rifle cartridge from a box magazine. Example: *FR8, Enfield No.5 Mk1, SKS*.

**Carbine, Auto (TL 7):** A shortened version of a military assault rifle, using the same ammunition and capable of automatic fire. Often used by rear-echelon troops. Example: *Colt M4A1, AK-74, HK53, L22A2*.

**Anti-Material Rifle (TL 7):** A very large calibre rifle used primarily for attacking unarmoured vehicles, engines, unreinforced walls, sensitive equipment as well as personnel. It fires a 13mm cartridge from a box magazine. The weapon is large and unwieldy and is fitted with a bipod which will reduce recoil by 2. Example: *Barrett M82, Hécate II*.

**Rifle, Single-Shot (TL 4):** An early TL 4 rifle firing a single round. A fresh cartridge must be loaded each combat round.



**Rifle, Bolt-Action (TL 4):** A late TL 4 rifle with improved ballistics, high-speed rounds and internal magazine. At TL 5 it also becomes popular as a hunting rifle. Example: *Bergara B-14 Ridge, Lee Enfield No.4 Mk1, Springfield M1903.*

**Rifle, Semi-Auto (TL 5):** A semi-automatic rifle firing one round per pull of the trigger, without manually cycling a bolt. It is a standard military arm on many low-tech worlds and a reliable hunting rifle at TL 6-9. Rounds are loaded into an internal magazine at TL 5, but box magazines are used at TL 6 and higher. Example: *Remington Model 742, M1 Garand.*

**Assault Rifle (TL 6):** An automatic-firing military rifle in an intermediate rifle calibre for an intermediate shooting distance. Many assault rifles come with accessories or adaptations, such as folding stocks, scopes, and so on, to meet the needs of particular missions. Rounds are interchangeable with the carbine (auto) and the General Machinegun; at the GM's discretion, so are the box magazines. Example: *Steyr AUG.*

**Battle Rifle (TL 6):** An automatic-firing military rifle firing a heavier calibre for more power and range. Uses box magazines. Example: *FN SCAR.*

**Automatic Rifle (TL 5):** A squad support weapon that is essentially a very heavy rifle capable of automatic fire for sustained infantry support. It uses box magazines and is fitted with a bipod which will reduce recoil by 2. It uses the same rifle cartridges of the squad. Example: *Bren Gun MKII, Browning M1918 BAR, L86A2, RPK-74.*

**Gyrojet Pistol (TL 8):** A low-velocity, recoilless pistol designed for use in zero-G environments. Fires solid rounds or flechettes.

**Riot Pistol (TL 9):** An advanced single-shot gyrojet pistol with two (double-row) magazines, able to load up to four different types of ammunition at once. The user selects which magazine stack he wishes to extract from as the situation demands. Available ammunition choices include general purpose, armour piercing, high explosive, incendiary, heat-seeking and gas. Example: *Snub Pistol.*

**Gyrojet Carbine (TL 9):** A carbine version of the recoilless gyrojet, a longer barrel isn't needed because the rounds continue to accelerate away from the barrel. Useful in a zero-G environment. Fires solid rounds or flechettes. Example: *Accelerator Rifle.*

**Railgun (TL 12):** A military rifle using magnetic linear technology to accelerate a fin-stabilised flechette, from a 60-round magazine (with integral power cell). It is capable of automatic fire, is gyro-stabilised for lower recoil and has laser range-finding, a scope and targeting computer to improve long range shooting. The railgun is the ultimate development of the hand-held kinetic energy firearm. Example: *Gauss Rifle, EM-1 Railgun.*

## SLUGTHROWER NOTES

**ROF (Rate of Fire) 4 or 10:** These weapons fire a burst of rounds with every pull of the trigger which gives the user extra attacks. See page 123.

**Flame:** On a successful hit the target is on fire. See page 94.

## EXOTIC AMMO

Your setting may feature caseless (TL 7) or binary propellant (TL 9) firearms. The following firearms can be manufactured to take one of these exotic ammunition types: pistols, shotguns, SMG & PDW, carbines, the anti-material rifle, bolt-action & semi-auto rifles, the assault rifle, battle rifle and automatic rifle. Caseless ammo capacity is increased by x1.5, whilst binary propellant capacity is increased by x2.

## AMMUNITION TYPES

Some firearms can fire different types of ammunition.

| AMMO TYPES            |                                                                                                                                          |           |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Shotgun               | Effect                                                                                                                                   | Ammo Cost |
| <b>Buckshot</b>       | DM+1 short range; damage: 4D6 short, 3D6 medium, 1D6 long                                                                                | x1        |
| <b>Gas</b>            | Target affected by sedative or paralytic gas. See page 95.                                                                               | x1        |
| <b>Smoke</b>          | 2m cloud for 1 combat round                                                                                                              | x1        |
| <b>Baton</b>          | Damage: 1D6, plus knockdown                                                                                                              | x1        |
| <b>High Explosive</b> | Fin-stabilised, damage: 5D6, can be used against locks, vehicles                                                                         | x5        |
| <b>Tungsten Core</b>  | Anti-vehicle solid-slug, damage: 4D6 AP2                                                                                                 | x3        |
| <b>Tranq</b>          | Roll 1D6+2 damage, if this penetrates target's AV, the tranq takes effect: target affected by sedative or paralytic agent (see page 65). | X1        |
| <b>Paint</b>          | Used in a vacuum to mark targets in suits, rather than kill them                                                                         | X1        |
| Gyrojet               | Effect                                                                                                                                   | Ammo Cost |
| <b>GP</b>             | Pistol damage: 2D6; Rifle damage: 3D6                                                                                                    | X1        |
| <b>Flechette</b>      | DM+1 short range; pistol damage: 2D6; rifle damage: 3D6                                                                                  | X3        |
| Riot Pistol           | Effect                                                                                                                                   | Ammo Cost |
| <b>GP</b>             | Damage: 2D6+2                                                                                                                            | X1        |
| <b>Tungsten Core</b>  | Anti-vehicle solid-slug, damage: 4D6 AP2                                                                                                 | x3        |
| <b>High Explosive</b> | Fin-stabilised, damage: 5D6, can be used against locks, vehicles                                                                         | x5        |
| <b>Incendiary</b>     | Target and 1m radius area, now on fire 1D6. See page 94.                                                                                 | X3        |
| <b>Gas</b>            | Target affected by sedative or paralytic gas. See page 95.                                                                               | x1        |

## FIREARMS – ENERGY WEAPONS

Science fiction is replete with fantastic energy weapons, most having little scientific explanation, and a host of exotic-sounding names and titles. In these rules we will focus on three speculative technologies that might be utilised for hand-held energy weapons. The names used are placeholders, and the Game Master is encouraged to create a name that fits his own setting, or borrow a name from some other, more well-known setting.

**Stun Weapons (TL 9)** – Stun guns in speculative fiction often knock out a target using an energy blast. In many cases this seems to be a sonic pulse, although this means that wearing a vacc suit, particularly in a vacuum, will render such weapons useless. Another version might be a low-power laser ionising the air to allow a very high voltage charge to hit the target, affecting the nervous system, respiration and heartbeat. Microwave weapons are also feasible and modern versions emit short-wavelength microwaves that affect pain receptors 0.4mm under the skin, creating the feeling that one's skin is on fire, without the burns. A victim will be in shock and reflexively get out of the way of the beam. Stun weapons are recoilless and thus suited for use in zero gravity environments.

**Lasers (TL 9)** – Hand-held laser weapons create a short-duration, high-energy burst rather than a high-temperature, flashlight-style cutting beam, since the power required

for that would be phenomenal. The power cell instead feeds a fast-discharge capacitor which stores the energy, then releases it in a single, instant burst of power. Laser weapons are rated in megawatts of power released per one-one hundredth of a second. Early lasers require the use of a bulky power pack that must be carried on the back, or on a belt. Later laser weapons carry their own power cells. All lasers are recoilless and thus well suited for use in zero gravity environments.

**Plasma Weapons (TL 14)** – Plasma weapons fire a bolt of plasma that is created by the super-heating of a single hydrogen fuel pellet. The plasma is magnetically focussed, and the bolt itself will be prevented from dissipating too quickly by a tunnel of hot air surrounding the bolt that is created by the weapon's on-board soft laser. The fact that a laser is used in the weapon may cause terminology to reflect this. Unlike a pure laser, the plasma gun has recoil, and it will be reloaded with a magazine holding both fuel pellets and disposable power cells. One shot consumes one of each. Early plasma weapons are large, used alongside gauss or gyrojet weapons, as infantry support weapons. Some settings call these energy weapons 'blasters'.

**Disintegrator Weapons (TL 16)** – These weapons defy much scientific explanation, but appear to be a type of particle accelerator that inflicts devastating damage on a target. In some cases, the target is completely vaporised, as if the beam emitted by the weapon is able to break the atomic bonds that hold a target's cells together. This destruction and breakdown of atomic bonds is perhaps related to the same technology used by material and food replicators as well as the matter transporter, both of which also appear at this Tech Level. Examples from a setting 'where no man has gone before' show us that the disintegrator's effect on a target can be dialled down to more moderate levels, leaving the body relatively intact, or even to leave the target unharmed, but knocked out. This versatility means they can also be used as welding torches or cutting tools, and can create heat sources by firing at a large, solid object (like a rock). The stream can be adjusted to strike multiple targets at once. Some settings call these energy weapons 'phasers' or 'disruptors'.

| ENERGY WEAPONS        |    |       |        |         |        |     |      |           |                 |
|-----------------------|----|-------|--------|---------|--------|-----|------|-----------|-----------------|
| Weapon                | TL | Cost  | Range  | Dmg     | Recoil | Wgt | Mag  | Ammo Cost | Notes           |
| Agoniser              | 9  | 750   | Short  | special | -      | 1.5 | 8R   | -         | Stun            |
| Stun Gun              | 10 | 600   | Short  | special | -      | 1   | 10R  | -         | Stun            |
| Laser Pistol          | 10 | 1000  | Short  | 3D6     | -      | 1.5 | 50R  | 200       |                 |
| Laser Carbine         | 9  | 2000  | Long   | 4D6     | -      | 5   | 100R | 400       |                 |
| Laser Rifle           | 10 | 3000  | V/Long | 5D6     | -      | 6   | 100R | 400       |                 |
| Laser Pistol          | 11 | 1200  | Short  | 3D6     | -      | 1   | 50R  | 200       |                 |
| Laser Carbine         | 11 | 2200  | Long   | 4D6     | -      | 3   | 100R | 400       |                 |
| Laser Rifle           | 11 | 3520  | V/Long | 5D6     | -      | 5   | 100R | 400       |                 |
| Plasma Pistol         | 14 | 1400  | Short  | 3D6     | 5+     | 1   | 20   | 50        |                 |
| Auto Plasma Carbine   | 14 | 3000  | Long   | 4D6     | 6+     | 3   | 40   | 150       | ROF 4           |
| Plasma Rifle          | 14 | 4000  | V/Long | 5D6     | 7+     | 4   | 40   | 150       |                 |
| Plasma Cannon         | 15 | 9,000 | V/Long | 8D6     | 9+     | 10  | 60   | 300       | ROF 4           |
| Disintegrator Pistol  | 16 | 3400  | Short  | 10D6    | -      | 1   | 200R | 200       | Variable Energy |
| Disintegrator Carbine | 16 | 7000  | Long   | 10D6    | -      | 1   | 200R | 200       | Variable Energy |
| Disintegrator Rifle   | 16 | 10000 | V/Long | 10D6    | -      | 3.5 | 400R | 500       | Variable Energy |

**Agoniser (TL 9):** A hand-held pistol that uses microwaves or a high-voltage neural beam to induce pain on a target, rendering them temporarily immobile and senseless with agony.

**Stun Gun (TL 10):** A sonic-based stun weapon that renders a target unconscious for a short amount of time. Cannot be used where there is no atmosphere, or on a target within a pressurised suit.

**Laser Pistol (TL 10):** A laser pistol reliant on a 1 kg belt-mounted power cell, connected by a heavy duty cable. Once the cell is depleted it requires recharging for 3 hours.

**Laser Carbine (TL 9):** An early laser rifle, short and compact, that is reliant on a 3 kg backpack-mounted power cell. The power cell is connected by a heavy duty cable. Once the cell is depleted it requires recharging for 3 hours. The carbine is a military-style weapon.

**Laser Rifle (TL 10):** A heavy military laser rifle, used for supporting fire and long range marksmanship. It is reliant on a 4 kg backpack-mounted power cell, connected by a heavy duty cable. Once the cell is depleted it requires recharging for 3 hours.

**Laser Pistol (TL 11):** A handy laser pistol with integral power cell in the handgrip. Once the cell is depleted it requires recharging for 3 hours.

**Laser Carbine (TL 11):** A short military laser weapon used by rear echelon troops, special forces and security forces. It has an integral power cell forward of the trigger guard. Once the cell is depleted it requires recharging for 3 hours.

**Laser Rifle (TL 11):** A heavy laser rifle, with integral power cell forward of the trigger. The rifle includes a scope for use in long distance combat. Once the cell is depleted it requires recharging for 3 hours.

**Plasma Pistol (TL 14):** A pistol using plasma technology to fire a bolt of plasma. It does not require recharging, but does include a magazine for replenishing disposable power cells and hydrogen fuel pellets.

**Auto Plasma Carbine (TL 14):** A short, military repeating plasma blaster for use by infantry. It does not require recharging, but does include a magazine for replenishing disposable power cells and hydrogen fuel pellets.

**Plasma Rifle (TL 14):** A rugged military rifle used for devastating single shot attacks out to long range. The rifle includes a scope for use in long distance combat. It does not require recharging, but does include a magazine for replenishing disposable power cells and hydrogen fuel pellets.

**Disintegrator Pistol (TL 16):** A disintegrator in a pistol form, with handgrip and integral power cell.

**Disintegrator Carbine (TL 16):** Similar to the disintegrator pistol, this military variant provides for a greater range. It's compact design and folding stock is perfect for shipboard use. Once the cell is depleted it requires recharging for 4 hours. A top-mounted display allows control of the energy output, from 1D6 right up to 10D6.

**Disintegrator Rifle (TL 16):** A purely military rifle, designed for the infantryman. It features long range scope, laser range-finding and thermal imaging. It is rugged and designed for prolonged use, with a long duration power cell. Once the cell is depleted it requires recharging for 4 hours. A top-mounted display allows control of the energy output, from 1D6 right up to 10D6.

## ENERGY WEAPON NOTES

**ROF (Rate of Fire) 4:** These weapons fire a burst of plasma bolts with every pull of the trigger which gives the user extra attacks. See page 123.

**R:** Denotes an energy weapon that requires a recharge after this number of rounds has been expended; alternatively, a fresh powerpack (see Ammo Cost) can be used, instead.

**Stun:** These weapons are designed to deal non-lethal damage and will incapacitate a living target instead of killing it. On a successful hit, the victim makes a Difficult Endurance roll. If this is failed, the target is incapacitated and unable to perform any actions for 1D6+2 minutes. If the End roll succeeds, the victim is still stunned for one round.

**Variable Energy:** The disintegrator weapon can deal variable damage, up to 10D6 in one attack. Simply subtract the number of dice done in an attack from the Magazine. If used at the level of 6D6 or more, it can damage vehicles, but unlike slugthrower firearms, disintegrators *can* reduce a vehicle's Hull and Structure points.

## HEAVY WEAPONS

Heavy weapons are those infantry weapons that are able to cause large amounts of damage, and are typically deployed against vehicles, fortifications or enemy infantry positions. All of the weapons in this section require the use of the Heavy Weapons skill (although machineguns can also be used by someone with any level of Gun Combat skill, at a default level-0).

| HEAVY WEAPONS                       |    |        |           |          |        |     |       |           |                    |
|-------------------------------------|----|--------|-----------|----------|--------|-----|-------|-----------|--------------------|
| Weapon                              | TL | Cost   | Range     | Dmg      | Recoil | Wgt | Mag   | Ammo Cost | Notes              |
| <b>Squad Machinegun</b>             | 7  | 1000   | Long      | 3D6      | 8+     | 5   | 100   | 50        | ROF 10             |
| <b>General Machinegun</b>           | 6  | 1200   | Long      | 3D6+3    | 8+     | 10  | 100   | 60        | ROF 10             |
| <b>Heavy Machinegun</b>             | 6  | 4000   | Long      | 6D6      | -      | 12  | 200   | 100       | ROF 10             |
| <b>Grenade Launcher</b>             | 7  | 400    | Medium    | varies   | 4+     | 5   | 6     | -         | Indirect           |
| <b>Underbarrel Grenade Launcher</b> | 7  | 1000   | Medium    | varies   | 4+     | 1.5 | 1     | -         | Indirect           |
| <b>Flamethrower</b>                 | 5  | 1200   | Short     | 4D6      | 7+     | 6   | 16    | 50        | Flame<br>ROF 4     |
| <b>Commando Mortar</b>              | 6  | 300    | V/Long    | 6D6      | -      | 16  | 1     | 4         | Indirect           |
| <b>Rocket Launcher</b>              | 6  | 2000   | Medium    | 9D6 AP2  | -      | 10  | 1     | 500       |                    |
| <b>Disposable Rocket</b>            | 6  | 400    | Medium    | 9D6 AP2  | -      | 3   | 1     | -         |                    |
| <b>Missile Launcher</b>             | 7  | 10,000 | V/Long    | 10D6 AP2 | -      | 25  | 1     | -         |                    |
| <b>Disposable Missile</b>           | 8  | 4000   | V/Long    | 10D6 AP2 | -      | 10  | 1     | -         | Dispose<br>DM+1    |
| <b>Smart Missile</b>                | 10 | 12,000 | V/Long    | 11D6 AP2 | -      | 8   | 1     | -         | Dispose<br>DM+3    |
| <b>Energy Missile</b>               | 12 | 16,000 | V/Long    | 10D6 AP3 | -      | 6   | 1     | -         | Dispose            |
| <b>Munitions Launcher</b>           | 10 | 2500   | Medium    | varies   | -      | 5   | 1     | -         | Indirect<br>DM+3   |
| <b>Wrist Rockets</b>                | 10 | 2200   | Medium    | 4D6      | -      | 1   | 4     | 20        | ROF 4              |
| <b>Assault Railgun</b>              | 12 | 1600   | Very Long | 6D6      | 10+    | 12  | 20    | 40        | ROF 4              |
| <b>Plasma Cannon</b>                | 12 | 9,000  | Very Long | 8D6      | 9+     | 10  | 60    | 300       | ROF 4              |
| <b>Disintegrator Cannon</b>         | 16 | 50,000 | Distant   | 20D6     | -      | 14  | 2000R | -         | Variable<br>Energy |



**Squad Machinegun (TL 7):** A dedicated belt-fed machinegun designed to use the same ammunition as the squad's assault rifles. It provides direct fire support at the squad level. Fitted with a bipod which will reduce recoil by 2, and an ammo box for the belt. Example: *M249 SAW, HK123*.

**General Machinegun (TL 6):** A general purpose belt-fed machinegun used on vehicles and fixed emplacements. It uses a heavy rifle calibre, also used by the battle rifle, which is interchangeable if the GM decides. It requires a crew of two, one to carry and operate, the other to load and carry additional ammo belts. Fitted with a bipod which will reduce recoil by 2, and an ammo box for the belt. Example: *M60, GPMG*.

**Heavy Machinegun (TL 5):** A large calibre (12-15 mm) machinegun that is able to direct destructive fire on to lightly armoured vehicles, buildings or fortifications. Heavy and difficult to move. It is usually vehicle mounted, although infantry positions can mount the heavy machinegun on a tripod. Mounted on a tripod which eliminates recoil. Example: *M2 HB, DShKM*.

**Grenade Launcher (TL 7) –** This multi-purpose grenade launcher is a semi-automatic weapon firing one grenade for each pull of the trigger. It has a six-shot magazine forward of the trigger assembly, a retractable stock and forward grip under the barrel. It takes one full action to reload a grenade launcher. Grenades for the launcher are not interchangeable with hand grenades. Grenades are fired directly at a target if within range. Beyond range, the fire is Indirect. Example: *Milkor MGL*.

**Underslung Grenade Launcher (TL 7) –** An underslung grenade launcher can be added to any rifle. This grenade launcher is single-shot, holding one grenade. Grenades are fired directly at a target if within range. Beyond range, the fire is Indirect. Example: *M203, AG36*.

**Flamethrower (TL 5):** A portable, hand-held flame unit, with fuel bottle attached below the receiver. A pilot light at the barrel mouth must be on to fire the weapon.

**Commando Mortar (TL 6):** A portable, lightweight, close-support infantry weapon designed to be used by one man to provide indirect fire. The firing tube is held in place by the firer, and carried with an attached strap. Each shell (illumination, explosive or smoke) weights 0.5 kg. Soldiers can use it to provide immediate fire support for their own unit in defence or attack. The mortar can only make Indirect attacks. Example: *LGI Me F1, British L9A1 51mm Mortar*.

**Rocket Launcher (TL 6):** This shoulder-mounted launcher fires unguided anti-tank rockets. At TL 9 the launcher uses targeting sensors to provide a DM +1. It is dangerous to use in enclosed spaces. Anyone up to 2 meters behind a rocket launcher when it fires takes 3D6 damage from the exhaust. Example: *M1 Bazooka, Mk55 Tactical Launcher*.

**Disposable Rocket (TL 6):** A one-shot, unguided, disposable rocket that is used against armoured vehicles and fortifications. It is dangerous to use in enclosed spaces. Anyone up to 2 meters behind a rocket launcher when it fires takes 3D6 damage from the exhaust. Example: *M72 LAW, RPG-76*.

**Missile Launcher (TL 7):** A missile launcher that is shoulder-mounted, and used to lock on to a target. Set up and targeting requires three rounds. The user must guide the missile on to the target during its flight, using either a radio or wire-guidance system. If the user is wounded or distracted (by a nearby explosion, for example), the target lock is broken and the missile will only hit the target on a 1 on a rolled 1D6. It is dangerous to launch in enclosed spaces. Anyone up to 2 meters behind a rocket launcher when it fires takes 3D6 damage from the exhaust. Without the shoulder launcher, missiles loaded cannot be targeted or launched. Example: *HOT, TOW, AT-5 Spandrel*.

**Disposable Missile (TL 8):** A self-guiding, disposable, fire and forget anti-armour missile, attacks with a DM+1. It is dangerous to launch in enclosed spaces. Anyone up to 2 meters behind a rocket launcher when it fires takes 3D6 damage from the exhaust. Example: *FGM-148 Javelin*, *MBT-LAW*.

**Smart Missile (TL 10):** This disposable, fire-and-forget anti-tank missile is autonomous and self-guiding, attacking with a DM+3. It also gains a +1 for every TL below 10 at which the target was made. A TL 7 tank, for example, will give the smart missile a +3 on top of its +4 Die Modifier. The missile deploys its own counter-measures, and can carry out feint attacks, before striking its intended target. The smart missile is particularly hard to stop and is dangerous to launch in enclosed spaces. Anyone up to 2 meters behind a rocket launcher when it fires takes 3D6 damage from the exhaust.

**Munitions Launcher (TL 10):** A configurable, multi-purpose grenade launcher. The munitions launcher is a 5 kg weapon with a 6-round rotary magazine, that is capable of both direct fire and Indirect Fire. Its scope features an integrated laser rangefinder and a ballistic computer, which calculates where the grenade will land and will also guide the grenade (when using laser-guided HE grenades) or sets them to explode at a certain distance (when using fragmentation airburst grenades). This provides a +2 bonus to hit. Its propelled grenades can also alter their trajectory inflight to home in on a laser-painted target or on to specific co-ordinates (DM +3). It uses propelled grenades of all Tech Levels.

**Wrist Rockets (TL 10):** Micro-rockets mounted on a forearm launcher. When activated, one (or all four) of the unguided rockets are launched directly at a target. When not in use, the rockets are covered by a rugged forearm shield.

**Assault Railgun (TL 12):** A squad support gun initially firing a heavy 15mm caseless antipersonnel or explosive round from a 20-round magazine (with integral power cell). Once fired, it is then accelerated from mid-way down the barrel rails by magnetic linear induction. It is capable of automatic fire, is gyro-stabilised for lower recoil and has laser range-finding, a scope and targeting computer to improve long range shooting. Fitted with a bipod which will reduce recoil by 2.

**Plasma Cannon (TL 15):** Heavy plasma blaster used in infantry support roles. It is man portable, and includes a bipod which will reduce recoil by 2. It can alternatively be mounted on an 8 kg tripod to eliminate recoil entirely. It does not require recharging, but does include a magazine for replenishing disposable power cells and hydrogen fuel pellets.

**Disintegrator Cannon (TL 16):** A heavy disintegrator cannon, with longer range and greater power capacity than the disintegrator rifle. It can be fitted with a bipod for use in fortified positions, but is more frequently mounted on a tripod for ease of targeting. The cannon is a devastating anti-vehicle or anti-bunker weapon.

## HEAVY WEAPON NOTES

**ROF 4 or 10:** These weapons fire a burst of rounds with every pull of the trigger which gives the user extra attacks. See page 123.

**Indirect:** Beyond Medium range, the weapon is not aimed at the target, but is fired at a high trajectory to strike the target from above. It has a default DM of -4. If the strike of rounds can be seen, the modifier can be reduced by one each subsequent combat round up to a maximum of the user's Heavy Weapons skill. A character with Heavy Weapons-2, could reduce the penalty to -3 in the second round, and -2 in the third

round. All subsequent rounds in which the grenade launcher or mortar is fired will have a DM -2. The user himself might not see the fall of shot, but a forward observer who can, could radio the information to the user, allowing him, then, to adjust his fire appropriately.

**Flame:** On a successful hit the target is on fire. See page 94.

**Dispose:** A one-shot weapon, the launcher is disposed of after firing.

**Variable Energy:** The disintegrator cannon can deal variable damage, up to 20D6 in one attack. Simply subtract the number of dice done in an attack from the Magazine. If used at the level of 6D6 or more, it can damage vehicles, but unlike slugthrower firearms, disintegrators *can* reduce a vehicle's Hull and Structure points.

## PROPELLED GRENADES

There are two types of grenade: hand grenades that are thrown, and propelled grenades fired from grenade launchers. There are various types of propelled grenade.

| Weapon              | Cost | TL | Wgt | Damage                                  |
|---------------------|------|----|-----|-----------------------------------------|
| <b>Smoke</b>        | 15   | 6  | 0.5 | see description                         |
| <b>Riot Agent</b>   | 25   | 6  | 0.5 | see description                         |
| <b>Stun</b>         | 30   | 7  | 0.5 | see description                         |
| <b>Explosive</b>    | 30   | 6  | 0.5 | 5D6 (6m), 2D6 (10m)                     |
| <b>Anti-Armour</b>  | 30   | 6  | 0.5 | 5D6 AP2                                 |
| <b>Airburst</b>     | 30   | 10 | 0.5 | 5D6 (6m), 2D6 (10m)                     |
| <b>Damper Field</b> | 120  | 10 | 0.5 | Crashes electronics, robots & computers |
| <b>Time Drogue</b>  | 1300 | 17 | 0.5 | Stops time within burst radius          |
| <b>Incendiary</b>   | 30   | 6  | 0.5 | All targets within 3m are set on fire   |

**Smoke Grenade (TL 6):** The smoke grenade creates a thick cloud of coloured smoke 6 metres in radius, centred on the location of the grenade. This smoke imposes a -2 DM on all attacks within or through the cloud (doubled for laser weapons). Smoke dissipates in 1D6+2 rounds, although high winds and other extreme weather can sharply reduce this time.

**Riot Agent Grenade (TL 6):** The riot agent grenade creates a cloud of sedative or paralytic (see page 95).

**Stun Grenade (TL 7):** The stun grenade is non-lethal and does not inflict normal damage. A character within 6 meters of a stun grenade detonation must make a Difficult Endurance roll. If this is failed, the target is incapacitated and unable to perform any actions for 1D3+2 rounds (if in enclosed space, one round if out in the open). If the End roll succeeds, the victim is still stunned for one round.

**Explosive Grenade (TL 6):** Anti-personnel grenade, with damage decreasing with distance from the blast: 5D6 within 6 metres, 2D6 within 10 metres.

**Anti-Armour Grenade (TL 6):** Able to penetrate light armour and can inflict serious damage against lightly armoured and civilian vehicles as well as bunkers, buildings and fortifications. It will reduce the target's Armour points by 10, and then inflict 5D6 damage. It has a blast radius of 6m and anybody within that radius will suffer 2D6 damage. At TL 10, fired from the Munitions Launcher, this grenade is laser targeted and will try to home in on any target designated by the firer using the launcher's integral laser painter. This provides a +2 to hit.

**Airburst Grenade (TL 10):** The airburst grenade is used by the munitions launcher, and has similar properties to the explosive grenade, but it can be commanded to detonate in the air above a target, once the firer has established range to the target using the munition launcher's laser designator. This might negate the defender's -3 for shooting from cover, if he is out in the open, in a trench or behind a wall or other unroofed fortification.

**Damper Field (TL 10):** This grenade is used by the munitions launcher and creates a 10m area of electrical nullification, temporarily shutting down any electrical motors or operations, including vehicle engines and control systems, robots and computers. If in doubt, allow a saving throw on 10+ to resist the damper bomb. The effect is only temporary, and lasts 1D6+1 combat rounds.

**Time Drogue (TL 17):** Everything with 6m of the grenade is engulfed in a time pocket for 6 minutes, and for anyone and everything within it, it seems as though only half a second passes. Once the time drogue is deployed, friendly forces can move into the time pocket and will not be affected. When the effects wear off in 6 minutes, most targets will be disorientated and confused for a round.

**Incendiary (TL 7):** All targets in the 3 metre burst radius suffer 2D6 damage from burning. Targets (and everything else nearby) are also on fire; see Fire (page 94).

## HAND GRENADES

A hand grenade is a small explosive device designed to be thrown by hand (see page 126). Hand grenades typically have a 4-second delay. A variety of grenade types are described below. Each column is described as follows:

| Weapon                 | Cost | TL | Wgt | Damage                                        |
|------------------------|------|----|-----|-----------------------------------------------|
| <b>Smoke</b>           | 15   | 6  | 0.5 | see description                               |
| <b>Stun</b>            | 30   | 7  | 0.5 | see description                               |
| <b>Explosive</b>       | 30   | 6  | 0.5 | 5D6 (6m), 2D6 (10m)                           |
| <b>Seeker</b>          | 300  | 12 | 0.5 | 5D6 AP2                                       |
| <b>Damper Field</b>    | 120  | 10 | 0.5 | Crashes electronics, robots & computers       |
| <b>Time Drogue</b>     | 1200 | 17 | 0.5 | Stops time within burst radius                |
| <b>Incendiary</b>      | 30   | 6  | 0.5 | All targets within 3m are set on fire         |
| <b>Cryo Grenade</b>    | 250  | 10 | 0.5 | Freezes everything within 3m radius, 10D6     |
| <b>Shock Grenade</b>   | 70   | 11 | 0.5 | Electrical discharge strikes all targets, 4D6 |
| <b>Gravity Grenade</b> | 180  | 15 | 0.5 | Implodes, doing 12D6                          |

**Smoke Grenade (TL 6):** The smoke grenade creates a thick cloud of coloured smoke 6 meters in radius, centred on the location of the grenade. This smoke imposes a -2 DM on all attacks within or through the cloud (doubled for laser weapons). Smoke dissipates in 1D6+2 rounds, although high winds and other extreme weather can sharply reduce this time.

**Stun Grenade (TL 7):** The stun grenade is non-lethal and does not inflict normal damage. A character within 6 meters of a stun grenade detonation must make a Difficult Endurance roll. If this is failed, the target is incapacitated and unable to

perform any actions for 1D3+2 rounds (if in enclosed space, one round if out in the open). If the End roll succeeds, the victim is still stunned for one round.

**Explosive Grenade (TL 6):** Anti-personnel grenade, with damage decreasing with distance from the blast: 5D6 within 6 metres, 2D6 within 10 metres.

**Seeker (TL 12):** The seeker grenade is initially thrown, at which point air-foil surfaces unfold and a grav motor kicks in, launching the grenade toward the target which it can sense using thermal and motion-tracker sights. It is primarily designed to attack vehicles, since it allows the thrower to remain hidden during the throw. The grav motor allows it to twist and turn should the vehicle or person, try and dodge the grenade. The thrower gains a DM +2 when throwing the Seeker grenade and doubles the user's throw range.

**Damper Field (TL 10):** This grenade is used by the munitions launcher and creates a 10m area of electrical nullification, temporarily shutting down any electrical motors or operations, including vehicle engines and control systems, robots and computers. If in doubt, allow a saving throw on 10+ to resist the damper bomb. The effect is only temporary, and lasts 1D6+1 combat rounds.

**Time Drogue (TL 17):** Everything within 6m of the grenade is engulfed in a time pocket for 6 minutes, and for anyone and everything within it, it seems as though only half a second passes. Once the time drogue is deployed, friendly forces can move into the time pocket and will not be affected. When the effects wear off in 6 minutes, most targets will be disorientated and confused for one round.

**Incendiary (TL 7):** All targets in the 3 metre burst radius suffer 2D6 damage from burning. Targets (and everything else nearby) are also on fire; see Fire (page 94).

**Cryo Grenade (TL 10):** Everything within the 3m blast radius is subjected to cryogenic temperatures, freezing humans to death as if dropped in liquid nitrogen (10D6). This may be preferable to an explosion, depending on the circumstances. Soldiers can find other uses too, such as creating a frozen pathway across a river.

**Shock Grenade (TL 11):** An electronic discharge bomb, which creates multiple arcs of electricity that automatically strike anyone within the 6m burst radius, each doing 4D6 damage. Electrical items (including suit life support systems) will also be struck and be damaged on a result of 5-6 on a 1D6 roll. Player characters and important NPCs will not be able to duck to avoid the damage as described in Explosive Burst Radius (page 123).

**Gravity Grenade (TL 15):** This creates an intense 30G gravity point for one-hundredth of a second, creating an instant and devastating implosion, sucking all targets within 3m, as well as anything not bolted down, into the gravity well. Targets cannot avoid the damage as described in Explosive Burst Radius (page 123), and will suffer 12D6 damage. All that is left is a soggy, 20cm lump of compacted flesh, clothing, gravel and twisted metal lumped around the point where the grenade exploded. Implosion is often preferable during combat within a pressurized environment.



## WEAPON ACCESSORIES

The following accessories are commonly available for ranged weapons. Some high-tech firearms may already include scope or laser sight, see the relevant firearm description.

**Bipod (TL 5):** Some weapons boast integral bipods, but firearms sold without such accessories may also be fitted with them. Obviously, a bipod will only benefit a long weapon such as the rifle or carbine. A bipod lowers the recoil by 2 points when fired from a prone position. Weight 0.4 kg. Cost Cr75.

**Flashlight (TL 7):** Purpose-built flashlight attaching to the top, side or underside of a gun barrel. The flashlight is activated via a button on the trigger guard. Weight 0.1 kg. Cost Cr120.

**Gyro-stabilised (TL 9):** An internal stabilizer can be fitted to any weapon of weight 3 kg or more, and it reduces Recoil by 1 point. Weight 0.1. Cost Cr500.

**Laser Sight (TL 7):** The bonus for aimed shots increases to DM+2, but only at Short range. Weight negligible. Cost Cr100.

**Night Scope (TL 8):** A telescopic sight with full night vision capabilities, that can be attached to a carbine or rifle-sized weapon. It provides a +2 to hit with an aimed shot when Over Range if the user spends one combat round aiming, in daylight or darkness. Weight 0.5 kg. Cost Cr600.

**Palm Lock (TL 11):** An in-built security system reading the palm print, or requiring a DNA or iris check. It can either lock the weapon (Cr200), shock the thief with 1D6 damage (Cr400), or explode as a stun grenade (Cr700) or explosive grenade (Cr1200), the buyer decides on purchase. Weight negligible.

**Pistol Stock (TL 5):** Pistols and revolvers may be turned into light carbines by the attachment of a wire-framed folding shoulder stock. The range is increased to 50m, although the weapon can no longer be holstered. The overall length of the pistol is increased by 40 cm, although this can be folded if required. Attaching or detaching the stock takes 5 combat rounds. Weight 0.8 kg. Cost Cr75.

**Scope (TL6):** A telescopic sight can be attached to a carbine or rifle-sized weapon. It provides a +2 to hit with an aimed shot when Over Range if the user spends one combat round aiming. Scopes are delicate, however, and may be jarred out of alignment on an 8+ on 2D6, by any violent action (such as being left untended in a moving truck, a close explosion, or being dropped). Weight 0.2 kg. Cost Cr200.

**Smart Sight (TL 10):** The smart sight comes in the form of either smart goggles or an eyepiece screen, mounted on a combat helmet or on a headset arrangement. It is connected via cable to the weapon, where a targeting processor is located. The processor feeds ballistic data to the piece, placing a targeting reticule on the screen. The smart screen has low light capability, and gives a DM +1 to hit. At TL 12, the link is wireless, and can even be linked to a cyboptic if one is fitted to the user. Cost Cr1000.

**Sound Suppressor (TL 6):** May be attached to any slugthrower (but not thud gun, pneumo, gyrojet or railgun), and will partially mask the sound of the shot, but not silence the weapon entirely. Weight 0.1 kg. Cost Cr250.

**Speedloader (TL 6):** The speedloader is a preloaded frame of six revolver rounds that can be used to load all six chambers of a revolver simultaneously. The firer may have several speedloaders as part of his equipment to speed the reloading of his revolver. Reloading with a speedloader requires 1 minor action. Weight 0.12kg (loaded). Cost Cr10.

**Tough Scope (TL 9):** A ruggedized version of the Scope for use in hostile atmospheres, vacuum and extreme temperatures. They are almost unbreakable. AV 7, DP 1. Weight 0.5 kg. Cost Cr500.

**Underbarrel Grenade Launcher (TL 7):** Functions as the heavy weapon grenade launcher, but it fits under a rifle barrel and has 1 shot. Cost Cr1000. Weight +1 kg.

**Underslung Shotgun (TL 7):** A cut-down shotgun for use as a secondary weapon, attached beneath the barrel of a TL 7+ carbine or rifle. Use the data for the shotgun, except that this version uses a 5-shot magazine. Cost Cr600. Weight +1.3 kg.

## STANDARD VEHICLES

A brief description of some standard vehicle designs found across the works of science fiction, and suitable for use in a roleplaying setting, are provided here. The mass of these vehicles is given as displacement tons, as used in **Cepheus Universal**.

**APC (TL 6):** A military lightly-armoured tracked or wheeled vehicle, called an Armoured Personnel Carrier. It can carry 8 troops and has a turret, ready to be fitted with a vehicle support weapon of appropriate Tech Level to the world the APC was bought on. It has a maximum speed of 80 kph (off-road 40 kph), and a range of 600 km. At TL 12 range is 5,000 km. Mass is 10 tons, cost is Cr65,000 (without weapon). AV 16, 10/10.

**Air/Raft (TL 9):** An open-topped vehicle for 4 people and 500 kg of cargo, supported by anti-gravity motors. It is low-cost, ubiquitous, remarkably reliable and flexible, with a speed of 120 kph and an endurance of four weeks between recharges. Mass is 4 tons, cost is Cr120,000. AV 4, 4/4.

**ATV (TL 6):** A pressurized all-terrain ground vehicle, capable of floating on water, with bunks and kitchen to support eight people on an expedition, or 16 if the trip is 12 hours or less. An ATV has a hard point for a turret, but does not normally with a weapon. It has a maximum speed of 100 kph (off-road 40 kph), and a range of 600 km. At TL 12 range is 5,000 km. Mass is 10 tons, cost is Cr50,000. AV 4, 10/10.

**Battle Tank (TL 8):** A three-man combat tank, with a heavily armoured hull, and an armoured turret that houses a vehicle main weapon appropriate to the TL of the world on which it was bought. It also includes two machineguns or similar weapons for self-defence. It has tracks and a road speed of 60 kph, with an off-road speed of 30 kph. It has a range of 300 km. It masses 23 tons, and costs MCr1. AV 42, 15/15.

**Destroyer (TL 10):** A fast manoeuvrable long-endurance watercraft built for military action, intended to escort and patrolling. Powered by a fusion power plant, the destroyer carries two deck-mounted turrets fitted with suitable vehicle main weapons. Cargo capacity is limited to 40 tons, mostly used to carry ammunition and supplies. This has a crew of 10, it cruises at 40 kph and has an endurance of 10 weeks before refueling is required. Mass 800 tons, cost MCr5 (without heavy weapons). AV 12, 100/100.

**Excavator (TL 7):** An off-road vehicle equipped with a hydraulic tool (the type varying with the vehicle) that might be an articulated digging bucket, a heavy shovel, an agricultural plough, a drilling rig, a small crane, and so on. It can achieve 5 kph, and has an endurance of 100 hours continuous use. It has a crew of one and masses 6 tons. Cost is Cr50,000. AV 6, 6/6.

**Fast Boat (TL 7):** Small, fast-moving, semi-rigid, semi-inflatable watercraft. It cruises at 60 kph and has a range of 200 km. The fast boat is operated by one crewman, and can carry 6 passengers or 400 kg of cargo. Mass 2 tons, cost Cr6,000. AV 3, 2/3.

**G-Carrier (TL 12):** A military light-armoured grav APC with two crew and up to 6 troops carried as passengers. It includes a turret ready to be fitted with a vehicle support weapon appropriate to the TL of the world the G-Carrier was bought on. The G/Carrier can reach orbit (taking a number of hours equal to the world's Size code). It has a maximum speed of 400 kph and an endurance of four weeks between recharges. Mass is 8 tons, cost is MCr1 (without heavy weapon). AV 16, 10/10.

**Grav Belt (TL 12):** A grav belt is a harness fitted with artificial gravity modules allowing the wearer to fly. Endurance is four hours before a recharge. At TL 15, endurance is 24 hours. Weight is 10 kg, cost is Cr100,000.

**Grav Car (TL 11):** A fast moving grav car for four people, used on high tech worlds for personal transport, as well as by police forces. It can achieve 500 kph, has an endurance of 4 weeks and can carry one pilot and three passengers. It masses 3 tons. AV 4, 4/4.

**Grav Skiff (TL 12):** A sleek, open-topped grav craft with handrails, used for inspections, manual handling, light cargo transport and so on. It includes a pintle mount for a weapon at the front, and an underslung crane for technical work. It can fly up to 120 kph, and has a four week endurance. Mass 4 tons. Cr325,000. AV 4, 4/4.

**Grav Hauler (TL 10):** A large grav vehicle designed to lift heavy objects and carry them to new locations. It can be used as a flying crane, or as a cargo hauler, lifting cargo containers to fit flush with its lower hull. The grav hauler has a crew of 2 and can carry 20 tons of cargo. It has a speed of 400 kph and an endurance of four weeks between recharges. It can reach orbit like a G-Carrier. Mass is 20 tons, cost is MCr1. AV 4, 25/25.

**Ground Car (TL 6):** A ground car is a conventional wheeled automobile, with a speed of 120 kph and a range of 600 km. At TL 9, cars on established road networks are self-driving. An off-road variant costs 50% more. Mass 2 tons, Cr6,000. AV 4, 3/3.

**Helicopter (TL 6):** A rotor-driven aircraft capable of vertical take-off and landing. The helicopter can carry 500 kg of cargo, or 10 people, with a speed of 350 kph and a range of 1200km. It has a mass of 10 tons and costs Cr250,000. AV 4, 8/8.

**Hovercraft (TL 7):** A vehicle that travels over land or water on a cushion of air provided by a downward blast, the hovercraft is only usable on worlds with a Thin atmosphere or thicker. The hovercraft cruises at 60 kph and has a range of 2,000 km, and carries 3 tons of cargo space, or twelve passengers. Mass 8 tons, cost Cr200,000. AV 4, 8/8.

**Hydrofoil (TL 8):** Large watercraft using hydrofoils to achieve exceptional speed and performance. It cruises a 60 kph and has a range of 800 km. The hydrofoil can carry 12 passengers, three crew and 10 tons of cargo. Mass 60 tons, cost Cr80,000. AV 4, 16/16.

**Jeep (TL 7):** A rugged off-road vehicle, with roof, for carrying light cargo and four passengers on military missions or scientific expeditions. Mass 3 tons. Cargo capacity 300 kg. Road speed is 120 kph, off-road speed is 60 kph. It has a range of 600 km. Cr20,000. AV 4, 4/4.

**Skybus (TL 10):** A passenger and freight carrying grav vehicle, operating at starports and cities on high-tech worlds. The skybus has a crew of 1 and can carry 8 tons of cargo or 30 passengers. It has a speed of 400 kph and an endurance of four weeks between recharges. It can reach orbit like a G-Carrier. Mass is 14 tons, cost is Cr500,000. AV 4, 18/18.

**Small Steamship (TL 4):** A watercraft that is propelled by a steam engine. The steamship has a cargo capacity of 50 tons and cabins on-board for ten passengers. It has a crew of five. It has a speed 20 kph and a range of 4,000km. Mass 100 tons, cost Cr60,000. At TL 6 it uses petrochemicals. AV 4, 40/40.

**Submarine (TL 5):** A watercraft designed to operate under an ocean's surface. Submarines are often used as transport between domed cities on waterworlds and other planets with large fluid oceans. The submersible can carry 30 tons of cargo and 10 passengers. It has speed of 30 kph on the surface, half that underwater, with a range of 5000 km. Combat models swap the cargo space for 30 torpedoes and four torpedo tubes. At TL 11, a fusion power plant is fitted, with a ten week endurance, as well as a supercavitating drive that allows speeds of 500 kph underwater (doubling the cost). The submarine has a crew of 5 (civilian) or 25 (military). Mass 500 tons, cost MCr2. AV 18, 100/100.

**Tilt-Rotor (TL 9):** A turboprop utility aircraft that employs tilt-rotors on its wings to transition from fast, straight and level flight to vertical take-off and landing. The tilt-rotor can carry 6000 kg of cargo, or 20 people, with a speed of 500 kph and a range of 2000km. It requires a crew of 2, has a mass of 20 tons and costs MCr1.2. AV 4, 20/20.

**Truck (TL 6):** A truck is a conventional wheeled cargo hauler, with a speed of 90 kph and a range of 500 km. At TL 9, cargo trucks on established road networks are self-driving. An off-road variant of the truck (usually military) costs 50% more and has a mount for a 'light weapon' (see page 197). The truck has a crew of one, and carries two passengers and 1 ton (or 5 metric tonnes) of cargo. Mass 8 tons, Cr50,000. AV 4, 12/12.

**Twin Jet Aircraft (TL 6):** A fixed-wing aircraft propelled by jet engines, often used to transport cargo. It can carry 6 passengers or 5 tons of cargo at 600 kph for 4,000 km or six hours. At TL 9 the twin jet is capable of vertical take-off and landing (double the cost). Mass 12 tons, MCr1. AV 4, 64/64.

**Workloader (TL 10):** A cargo mech, built as an open-frame exoskeleton. An operator with Loader skill controls two hydraulic arms for lifting cargos boxes up to 4000 kg in weight. It has two rugged legs and a gyroscopic balance system. The workloader moves at walking speed and has an endurance of 72 hours. Mass 0.5 ton, Cr50,000. AV 4, 1/1.

## VEHICLE WEAPONS

Some vehicles are fitted with weaponry, and in **Cepheus Universal** we divide these weapons into three general categories that are used in the Vehicle chapter.

**Main Weapon** – A powerful, anti-tank or anti-vehicle weapon of great destruction.

**Support Weapon** – A less powerful weapon that is a vehicles' main armament.

**Light Weapon** – Probably one of several anti-personnel heavy weapons (page 172).

A main weapon could be a submarine's torpedo tubes or the 120mm gun on an M1A2 Abrams tank. A support weapon might be the 30mm Bushmaster cannon on the Stryker APC or the 73mm Light Gun on the BMP-1. Main weapons are the type of weapons in the turret of dedicated battle tanks, whilst support weapons are the type of weapon in the turret of APCs or Infantry Fighting Vehicles. Light weapons are used as anti-

personnel weapons on all kinds of vehicles. Included on the table are starship weapon equivalents that may be mounted on vehicles as main weapons (except for Starship Missiles and Torpedo which are included should they be used against surface targets).

Vehicle weapons can be fired with either Heavy Weapons or Gunnery skill.

#### **LIGHT WEAPONS**

Squad Machinegun (TL 7)  
General Machinegun (TL 6)  
Auto Grenade Launcher (TL 7)  
Flamethrower (TL 5)  
Assault Railgun (TL 12)  
Plasma Cannon (TL 12)  
Disintegrator Cannon (TL 16)

#### **SUPPORT WEAPONS**

Missile Launcher (TL 7)  
Light Gun (TL 5)  
Light Autocannon (TL 6)  
Hvy Autocannon (TL 7)  
Rapid Fire Railgun (TL 12)  
Gatling Laser Cannon (TL 8)

**Mortars (TL 5)** – A simple, lightweight, man-portable, muzzle-loaded weapon, consisting of a smooth-bore tube fixed to a base plate (to spread out the recoil), along with a lightweight bipod mount on which it leans, and a sight. Mortars launch explosive shells in high-arching trajectories and are typically used as indirect fire weapons for close fire support. They can be mounted on the backs of some vehicles to serve as mobile artillery support.

**Howitzers (TL 5)** – A type of artillery gun that is able to fire directly on to a target, as well as at high trajectories to strike targets that are not visible to the gun crew. Most are wheeled and towed into place, or can be mounted on a large chassis to become self-propelled artillery.

**Guns (TL 6)** – A type of cannon that is dedicated to firing directly on to a target, either to support infantry in the assault or defence, or to destroy vehicles or fortifications. They are not capable of firing at high trajectories to hit targets out of sight. Guns can be towed by other vehicles as assault guns or anti-tank guns, or can be mounted on a military vehicle as its main weapon (such as a tank).

**Hypervelocity Cannon (TL 9)** – The hypervelocity cannon is a large-scale coil gun that uses magnetic linear technology to accelerate rounds. They replace both howitzers and cannons.

**Missiles (TL 7)** – Tactical guided missiles use one of a variety of guidance methods to reach their target. Most are anti-armour or antiaircraft missiles. Set-up and targeting requires three rounds. An energy missile uses a stand-off attack, firing a plasma bolt only a metre or so before impact, destroying itself in the process. A gravitic missile is equipped with an anti-grav motor propelling it at tremendous speeds, it carries no explosive warhead, but instead destroys the target with a high-velocity tungsten rod.

**Nuclear Warheads (TL 6)** – Starship torpedoes carry anti-ship tactical nuclear warheads which do the same damage to a ground target as a Nuclear Demolition Charge (see page 189). However, strategic nuclear warheads are also available for attacking ground targets (cities) that have the power and destructive ability of a modern strategic nuclear warhead (300 – 1000 Kilotons). They cannot be used in space combat, in anything other than a mine capacity. A strategic warhead has a total destruction radius of 3 km doing 2000 points of damage to everything within that blast zone. Within 6 km, anyone and everything is subjected to 4D6 x 20 points of damage (and 1000 rads), unless they are inside a reinforced building, starship or armoured fighting



vehicle (in which case they can lower the damage to 1-6D6 on a 2D6 roll of 8+). Those within 12 km suffer 3D6 damage (and 50 rads). See the Radiation rules on page 98.

| VEHICLE WEAPONS (MAIN & SUPPORT)   |    |           |     |           |              |           |
|------------------------------------|----|-----------|-----|-----------|--------------|-----------|
| Vehicle-Mounted Weapon             | TL | Damage    | ROF | Max Range | Typical Ammo | Cost      |
| Light Autocannon (20-25mm)         | 6  | 6D6       | 6   | V/Long    | 200          | 7500      |
| Heavy Autocannon (30-40mm)         | 7  | 7D6       | 6   | V/Long    | 200          | 9500      |
| Lt Rotary Autocannon (20-25mm)     | 7  | 6D6       | 20  | V/Long    | 500          | 50,000    |
| Automatic Grenade Launcher         | 7  | 5D6 HE    | 4   | Long      | 40           | 4000      |
| Rapid Fire Railgun (15mm)          | 12 | 6D6       | 20  | Distant   | 1000         | 20,000    |
| Gatling Laser Cannon               | 8  | 6D6 AB    | 4   | Distant   | -            | 75,000    |
| Missile, Light Anti-Tank           | 7  | 9D6 AP2   | 1   | V/Distant | -            | 10,000    |
| Missile, Medium Anti-Tank          | 9  | 10D6 AP2  | 1   | V/Distant | -            | 20,000    |
| Missile, Medium Anti-Aircraft      | 8  | 9D6       | 1   | Extreme   | -            | 10,000    |
| Missile, Energy                    | 12 | 10D6 AP3  | 1   | Extreme   | -            | 16,000    |
| Missile, Gravitic                  | 13 | 9D6 AP5   | 1   | V/Distant | -            | 15,000    |
| Rocket Pod                         | 6  | 8D6 HE    | 6   | V/Long    | 6            | 4000      |
| Light Mortar (50mm)                | 5  | 3D6+2 HE2 | 1   | V/Long    | 12           | 1000      |
| Medium Mortar (60mm)               | 5  | 4D6 HE4   | 1   | Distant   | 10           | 2000      |
| Heavy Mortar(80mm)                 | 5  | 5D6 HE6   | 1   | Distant   | 8            | 4000      |
| Multiple Launched Rocket Artillery | 7  | 9D6 HE    | 10  | Distant   | 10           | 25,000    |
| Light Howitzer (100cm)             | 5  | 9D6 HE4   | 1   | V/Distant | 20           | 6000      |
| Medium Howitzer (150cm)            | 5  | 12D6 HE6  | 1   | V/Distant | 12           | 20,000    |
| Heavy Howitzer (200cm)             | 5  | 14D6 HE8  | 1   | Extreme   | 5            | 35,000    |
| Light Gun (40-60cm)                | 6  | 8D6 AP2   | 1   | Distant   | 40           | 50,000    |
| Medium Gun (70-90cm)               | 6  | 9D6 AP2   | 1   | Distant   | 40           | 80,000    |
| Heavy Gun (100-115cm)              | 6  | 10D6 AP2  | 1   | Distant   | 35           | 100,000   |
| Very Heavy Gun (120-130cm)         | 8  | 11D6 AP2  | 1   | Distant   | 30           | 120,000   |
| Light Hypervelocity Cannon         | 9  | 10D6 AP3  | 1   | Distant   | 50           | 200,000   |
| Medium Hypervelocity Cannon        | 10 | 11D6 AP3  | 1   | Distant   | 50           | 300,000   |
| Heavy Hypervelocity Cannon         | 11 | 13D6 AP3  | 1   | Distant   | 50           | 500,000   |
| Bomb, Standard                     | 5  | 12D6 HE2  | 1   | -         | -            | 800       |
| Bomb, Heavy                        | 5  | 14D6 HE6  | 1   | -         | -            | 1200      |
| Depth Charge                       | 5  | 10D6 HE   | 1   | Short     | 6            | 1500      |
| Light Fusion Gun                   | 12 | 14D6 AB   | 1   | Distant   | 100          | 300,00    |
| Medium Fusion Gun                  | 13 | 16D6 AB   | 1   | Distant   | 100          | 600,000   |
| Heavy Fusion Gun                   | 15 | 18D6 AB   | 1   | Distant   | 100          | 900,000   |
| Advanced Fusion Cannon             | 16 | 20D6 AB   | 1   | V/Distant | 250          | 1 million |
| Heavy Disintegrator Cannon         | 16 | 20D6 AP5  | 1   | V/Distant | -            | 3 million |
| Underwater Torpedo Lchr            | 5  | 12D6      | 1   | V/Distant | 30           | 500,000   |
| Underwater Smart Torpedo Lchr      | 8  | 14D6      | 1   | V/Distant | 30           | 500,000   |
| Heavy Laser Cannon                 | 9  | 12D6 AB   | 1   | Extreme   | -            | 500,000   |
| Auto-Railgun Cannon                | 8  | 10D6 AP3  | 4   | Extreme   | 200          | 750,000   |

## VEHICLE WEAPON NOTES

**Missile/Torpedo Guidance:** At **TL 7** the user must guide the missile/underwater torpedo or space torpedo on to the target during its flight, using either a radio or wire-guidance system. If the user is wounded or distracted (by a nearby explosion, for

example), the target lock is broken and the missile will only hit the target on a 1 on a rolled 1D6. At **TL 8**, the missile will guide itself once it leaves the launcher, the user must still make a Heavy Weapons roll, but with a DM +2. He does not steer the missile on to the target. At **TL 10**, the missile is smart, autonomous and self-guiding, attacking with a DM +3. It also gains a +1 for every TL below 10 at which the target was constructed. A target TL 7 tank, for example, will provide the smart missile a +3 on top of its +3 Die Modifier.

**Damage:** Damage inflicted and the effect that damage will have; AP, AB and HE and other rules related to vehicle combat are discussed in **Cepheus Universal**.

**ROF (Rate of Fire):** A measure of automatic fire, which expends a number of rounds each attack equal to the ROF value.

**Max. Range:** The *maximum* range of the weapon. This is different to all other weapon systems where the range band is the *effective* range. This is compatible with the combat and weapon rules published in Moon Toad's *Vehicle Design Guide*.

**Typical Ammo:** A typical load of ammunition carried by a military vehicle.

**Cost:** Provided as a guide. When designing vehicles, use the generic weapon costs in the Vehicle chapter, or swap them out for the costs in this list, if desired.

## ARCHAIC ARTILLERY

This section includes rules for both gunpowder cannons and low-tech torsion-powered artillery, for use on low tech worlds, or in low tech settings.

### EARLY CANNONS

Gunpowder artillery preceded the hand cannon, and although the hand-held firearm continually evolved, there was very little technological innovation regarding cannons. They remained essentially giant smoothbore muzzle-loading hand cannons until the late 1800s on Earth. The Heavy Weapons skill is used to aim, fire, maintain and reload a cannon efficiently.

**FIRING PROCEDURE** – After a shot the barrel is cleared of burning embers with a long corkscrew-like tool. The barrel is then sponged to put out any sparks, and then dried. The powder charge (a bag of gunpowder) is put into the barrel and pushed home. Then a wad is rammed into the gun before the cannon shot is rammed in after it. This separates the shot from the powder. The gun captain then inserts priming wire into the touch-hole and primes the hole with powder. He must also insert a 'pricker' into the touch-hole to tear open the bag containing the powder. On a ship the cannon is then run forward on its wheels so the barrel clears the hull of the vessel; the gun crew stand clear and the gun captain fires the gun by placing a smoldering match (on the end of a pole or 'linstock') to the touch-hole. At TL 3 a flintlock trigger is used.

**Cannon Size:** The cannon is referred to by the size of cannon shot it can fire.

**Dmg:** The damage a weapon inflicts on a vehicle, building or person.

**Wgt:** Weight in kilograms (kg), this includes gun carriage and related equipment.

**Cost:** Price in Credits (Cr), this includes gun carriage and related equipment.

**Crew:** The number of gunners required to operate the cannon. If the crew is reduced by 25% a DM -1 is incurred. Below 50% the gun cannot be fired. Crew also represents

the number of horses required to pull an army field gun of that size. At TL 2 (pre-1700) the carriage systems are inadequate and the number of horses must be doubled.

**Ammo Cost:** Cost for 1 cannon shot. Multiply x1.5 for anti-personnel or exploding shot.

| EARLY CANNONS     |        |           |      |      |      |           |
|-------------------|--------|-----------|------|------|------|-----------|
| Cannon Size       | Dmg    | Range     | Wgt  | Cost | Crew | Ammo Cost |
| <b>1 pounder</b>  | 6D6    | Very Long | 25   | 1200 | 2    | 2         |
| <b>3 pounder</b>  | 7D6    | Very Long | 400  | 1800 | 4    | 4         |
| <b>6 pounder</b>  | 8D6    | Very Long | 800  | 2300 | 4    | 5         |
| <b>9 pounder</b>  | 9D6    | Distant   | 1200 | 3600 | 6    | 6         |
| <b>12 pounder</b> | 9D6    | Long      | 1700 | 3200 | 6    | 6         |
| <b>24 pounder</b> | 9D6+3  | Very Long | 2500 | 4200 | 10   | 8         |
| <b>32 pounder</b> | 10D6   | Very Long | 2750 | 5000 | 12   | 10        |
| <b>42 pounder</b> | 10D6+3 | Very Long | 3200 | 6500 | 14*  | 12        |

\*A very large TL 3 cannon only usable at sea or within a fortification, therefore the crew rating only refers to its gun crew, not the number of horses used to pull it.

### RATE OF FIRE

Rate of Fire is 1 shot every 3 combat rounds. Should a commander want a volley, barrage or broadside all fired simultaneously, then the guns need to co-ordinate and may instead fire once every minute. These rates quickly slow down as the combat progresses. After every six shots, add another combat round to the ROF. The ROF cannot be slower than 1 per 20 rounds (2 minutes).

| Manpower | Rate of Fire        |
|----------|---------------------|
| 100%-76% | 1 per 3 full rounds |
| 75%-51%  | 1 per 6 full rounds |
| 50%      | Not allowed         |

### INFLICTING DAMAGE

Cannonballs shot on land act like a destructive bowling ball and the gunners aim for 'grazing shots', wherein the ball bounces off the ground and continues on its merrily destructive way. Against packed formations of men, the result is hideous. Assume that anyone directly behind the target must make a Difficult (-2) Dexterity roll or take full damage.

**Anti-Personnel Shot** – There are two main types of anti-personnel shot, Grapeshot and Canister. Both reduce the cannon's range by one band. Grapeshot is a canvas-wrapped sack of smaller round shot which fits down the barrel. The packages break open when fired and the balls scatter with deadly effect. Grape is often used against an enemy ship to kill or injure the officers, or against enemy boarding parties. Canister shot is composed of a can filled with dozens of musket balls. The can breaks open on firing to turn the gun into a giant shotgun for use against enemy personnel. Both of these types of shot do damage in a wide area: ½ metre across per damage dice. For example, a 6 pounder doing 8D6 damage ordinarily, has a grapeshot spread of 4m. Everyone within the zone suffers damage equal to 2D6.

**Exploding Shells** – At TL 3, fused shells can be used, although they are unreliable and rather dangerous. Two halves of a shell are filled with gunpowder, a fuse lit and then the shot fired at an enemy, having an area effect. When firing an explosive shell, roll 1D6 – on a result of 1 or 2 the shell breaks up, detonates early or fails to detonate resulting in no effect. TL 4 exploding shells are reliable and do not need to roll for failure. Fused shot explodes when it lands, doing damage in a wide area 1m across, per

damage dice (for example, a 6 pound cannon doing 6D6 damage ordinarily, has an explosive diameter of 6m). Everyone within the zone suffers damage equal to 4D6. Any building or vehicle hit by an exploding shell takes the full damage of the shell.

## TORSION ARTILLERY

At TL 1 and 2, powerful torsion or tension-powered artillery machines can be manufactured from wood and steel. They are typically large and relatively immobile mechanisms, though two at least can be fitted to carts for transportation.

| TORSION ARTILLERY |        |           |        |      |      |
|-------------------|--------|-----------|--------|------|------|
| Weapon            | Damage | Range     | Weight | Cost | Crew |
| Scorpion          | 4D6    | Medium    | 28     | 800  | 1    |
| Siege Ballista    | 6D6    | Long      | 160    | 1500 | 3    |
| Catapult          | 8D6 B  | Long      | 200    | 1500 | 3    |
| Springal          | 6D6+2  | Long      | 400    | 2200 | 3    |
| Trebuchet         | 14D6   | Very Long | 1500   | 3000 | 6    |

**Scorpion (TL 1):** Resembling an over-sized crossbow, mounted on a wooden tripod, the scorpion can shoot bolts at great speeds and with great accuracy (+1 due to the tripod and sighting mechanism). Can be disassembled into bow and tripod for transport by cart or mule, or fixed to a cart. Reloading requires two combat rounds. Cr800

**Siege Ballista (TL 1):** A gigantic crossbow on a tripod, built as a siege engine, and taller than a man in height. Using the same principles of the scorpion and crossbow, the ballista launches fist-sized stones at troops or buildings. Can be disassembled into multiple parts for transport by cart. Reloading requires four combat rounds. Cr1500

**Catapult (TL 1):** The famous single-armed catapult, mangonel or onager, that propels large stones forward as the arm is released from tension. Some catapult frames are fitted with wheels for mobility. Reloading requires six combat rounds. Cr1500

**Springal (TL 1):** Using the same mechanical concept as a crossbow, the springal is a descendant of the ballista, with inward swinging arms. It is a gigantic machine built inside a rectangular frame that which was used to fire either large darts or round shot. Switching between the two ammunition types takes four combat rounds. Reloading requires five combat rounds. Cr2200

**Trebuchet (TL 1):** A very large catapult using a heavy counterweight of stones that propels a central arm that drags along an attached sling. The sling arcs over the trebuchet and launches very large stones at the target. Reloading requires ten combat rounds. Cr3000

## AREA EFFECT

The scorpion, springal and siege ballista shoot bolts or rounded stones that can kill a man; the rounded stones also having a great use in sieges to pummel defences, gateways and enemy siege engines. The catapult and trebuchet are almost purely designed for the destruction of walls and fortifications, but their devastating stone falls can kill humans too. Assume that anyone directly behind the target must make a Difficult (-2) Dexterity roll or take half damage. Some settings may have use of a naphtha compound to create a fiery explosion. Anyone caught within 4 metres of it is on Fire (see page 94).

**AMMUNITION**

**Quiver Load of Bolts (TL 1):** A load of 20 bolts used in the scorpion or springal. Weight: 4 kg. Cr80.

**Basket of Round Shot (TL1):** A basket of 20 round stone shot used in the siege ballista and springal. Weight: 40 kg. Cr20.

**Large Stones (TL 0):** The catapult and trebuchet shoot large irregular boulders or sack-loads of rubble. These arrive by the wagon-load. Five stones weigh: 50 kg. Cr20.

**Cauldron of Naphtha (TL 1):** A cauldron of naphtha will provide 6 shots. A dangerous and difficult substance to create, transport and load. It takes 24 hours to brew a cauldron of the naphtha. Weight: 10 kg. Cr200.

**MEDICAL**

**Alpha Wave Unit (TL 13):** Analyses and identifies different brain waves, and personalities, and used to edit the memories of humans or clones. It is able to store a human or AI mind for up to 100 hours (after which a risk of mental illness sets in). It is mounted on wheels or uses a grav suspensor. Weight 160 kg. Cost MCr2.

**Autodoc (TL 12) –** An autodoc is a specialized, immobile robot with Medical-2 skill, which can be installed inside a hospital, large vehicle or a spacecraft. It weighs 500 kg and runs off the starship or building power supply, although in the event of a power-cut it contains 12 hours-worth of emergency battery power. The autodoc is a sealed unit, lowering the chances of cross-infection, and it can be pressurized for use as a decompression chamber if required. Weight 500 kg. Cost Cr500,000.

**Bioscanner (TL 13):** Handheld device that provides data on an organic body's composition, chemistry, physiology and life signs. Weight 1.5 kg. Cost Cr900.

**Cryoberth (TL 9):** A unit designed to store humans at low temperatures for long periods of time. In the interstellar transport industry they are used to ship passengers (and termed 'freezer berths'). Elsewhere they can be used to freeze medical patients who require treatment at a later date. Freezing/defrosting requires a 5 minute process. Weight 500 kg. Cost Cr50,000.

**Empathy Machine (TL 11):** A sophisticated lie-detector which analyses multiple physiological cues. It can also detect clones that have been 'force-grown' and are faking their emotions, as well as alien shape-changers. It requires a Difficult (-2) Comms roll and takes 1 hour. Weight 3 kg. Cost Cr3000.

**Folding Stretcher (TL 6):** Useful when extracting an immobile casualty. Weight 5 kg. Cost Cr180.

**First Aid Kit (TL 6):** A small medical kit carried on a pouch that fits onto a rucksack or belt. It contains basic items, bandages, a pressure dressing, antibiotics, painkillers, eye-wash, antihistamines and plasters. It contains enough supplies for two wound treatments. Weight 0.5 kg. Cost Cr50.

**Doctor's Kit (TL 7):** This doctor's medical kit contains diagnostic devices and scanners, surgical tools and a plethora of drugs and antibiotics, allowing a medic to practice his art in the field. Weight 10kg. cost Cr1,000.

**Medical Regenerator (TL 16):** A small handheld medical tool for repairing tissue, particularly during a procedure. It can repair a flesh wound immediately. Weight 0.8 kg. Cr450.

**Super Conducting Memory Unit (TL 13):** These are bulky mainframes kept in sub-zero basements that have the relevant computing architecture to hold a single AI or



human mind, or the copy of a human mind. The mind is dormant when occupying the SMU, it is not conscious and cannot take any action. Long term storage of an AI mind within an SMU (over several years) may lead to severe psychological problems and IQ degradation. Weight 250 kg. Cost MCr4.

## EXPLOSIVES

| EXPLOSIVES                       |                         |        |      |
|----------------------------------|-------------------------|--------|------|
| Weapon                           | Damage                  | Radius | Cost |
| <b>Petard</b>                    | 8D6                     | -      | 100  |
| <b>Detcord/3m</b>                | 3D6                     | 1D6+2  | 250  |
| <b>Detonator &amp; Line</b>      | 2D6                     | 0.5m   | 50   |
| <b>Dynamite Stick</b>            | 4D6                     | 3m     | 5    |
| <b>Plastic Explosive Block</b>   | 8D6                     | 10m    | 200  |
| <b>Time Charge</b>               | 8D6                     | 10m    | 500  |
| <b>Mine, Anti-Personnel</b>      | 4D6                     | -      | 75   |
| <b>Mine, Anti-Vehicle</b>        | 8D6 AP3                 | 3m     | 600  |
| <b>Nuclear Demolition Charge</b> | 4D6 x 20 plus 1000 rads | 800m   | MCr1 |
| <b>Shot Exploder</b>             | -                       | -      | 2100 |

**Petard (TL 2):** This breaching charge is a bell-shaped metal canister that directs a blast of gunpowder to blast a hole through a timber palisade, wall or gate. It is secured to the wall or palisade with timber props or nails. The metal canister weighs 5kg, the powder 3kg and the timber, rope, nails and tools to secure it weigh 5kg. . Cost Cr100

**Detcord (TL 7):** Long, flexible plastic tubes filled with explosive that have a wide variety of uses, including cutting pipes and trees, breaching doors, and so on. Miners feed detcord into pre-drilled holes in the rock face. When detonated, the rock face is shattered. For a 3m long length: Weight 0.5 kg. Cost Cr250.

**Detonator (TL 5):** A finger-sized cylindrical detonator (or blasting cap) used to ignite detcord or plastic explosive. The detonator is pushed into plastic explosive or into the end of detcord, and the attached firing line (100m long) is plugged into a Shot Exploder. Detonators can explode (doing 2D6 damage) if treated very harshly. Weight 1kg (With a 100m-long firing line). Cost Cr50.

**Dynamite (TL 4):** An early explosive often formed into sticks for portability. It is less prone to accidental explosion than gunpowder, but is still sensitive to shock, and it degrades over time. Damage of one stick is 4D6, blast radius 3m. Five sticks together does 8D6 HE, ten sticks does 9D6 HE2, twenty sticks does 10D6 HE3. Cost Cr5 per stick. Weight 0.2 kg per stick.

**Plastic Explosive (TL 6):** This generic, multi-purpose plastic explosive cannot be detonated by a gunshot or by dropping it onto a hard surface. It does not explode when set on fire or exposed to microwave radiation. Detonation can only be initiated by a shockwave, such as when a detonator inserted into it, is fired. Damage of one block is 8D6 HE2, two blocks together is 10D6 HE2, three blocks is 11D6 HE3, four blocks is 12D6 HE3. Weight 1 kg block. Cost Cr200 per block.

**Time Charge (TL 9):** A 1 kg block of plastic explosive in an acrylic case, with both a digital timer (up to 36 hours) and radio activation (up to Long range), for versatility. The case has a magnetic side, and a peel-off sticky side, for mounting on walls, ceilings

or machinery. It takes one round to set the timer (unless done in advance) and one round to set the time charge in place. Weight 1 kg. Cost Cr500.

**Mine, Anti-Personnel (TL 6):** A small mine, that when trodden on, is designed to wound or kill an infantryman, with the leg taking the blast. Some can be activated by a tripwire as well as by pressure. If characters are walking through the location of the mine, each rolls 1D6 with a '1' indicating detonation. Weight 2 kg. Cost Cr75.

**Mine, Anti-Vehicle (TL 6):** A heavy mine (shaped as a bar, or a plate) designed to be laid on trackways and unpaved roads to disable tanks, or destroy smaller vehicles. It has a pressure sensor, but at TL 8 also includes a magnetic sensor and can be detonated remotely if needed. If a vehicle is moving through the location of the mine, roll 1D6 with a '1-2' indicating detonation (electronic or weapon components will not be damaged). Weight 8 kg. Cost Cr600.

**Nuclear Demolition Charge (TL 7):** This military demolition charge can be used to blow holes in mountains and destroy large, immobile constructions. It is equal in power to 250 tons of TNT and gives military engineers the ultimate tool for moving, destroying or clearing obstacles. Lethal blast radius is 400m, with a wounding and sickness radius of 800m (3D6 damage, 50 rads, see page 98). It is supplied in a cylindrical hard case inside a carry bag that can be strapped to a soldier's back. Weight 20 kg. Cost MCr1.

**Shot Exploder (TL 5):** An electrical unit on a carry sling, used to send a current along a firing line to ignite a detonator. Up to six firing lines can be plugged into the unit at one time for a simultaneous blasting. Operated by twisting the charging handle; this discharges the unit's capacitor, that had been previously charged by the integral battery. Charging the capacitor takes 2 minutes. Weight 4 kg. Cost Cr750.

## SECURITY & DECEPTION

**Bug (TL 6):** These tiny audio bugs have a range of 100m, and transmit for 24 hours. Cost Cr20.

**Bug, Video (TL 7):** With a fisheye lens, these micro-cameras give a distorted 180 degree view as well as recording sound. They are often disguised as a button or part of furniture. They transmit out to 100m for 24 hours. Cost Cr50.

**Bug Unit (TL 7):** This pocket-sized unit is a receiver and a recorder of bug transmissions and cable taps, it can also be used as a bug detector to sweep rooms and search for bugs. The Comms skill can be used to sweep a room for bugs. Weight 0.2 kg. Cost Cr450.

**Cable Tap (TL 7):** Transmits information that it picks up from a communication cable within 2cm, transmitting the signal out to 100m indefinitely. Weight 0.5 kg. Cost Cr50.

**Chameleon Suit (TL 13):** This full body suit can shift its colour patterns to blend in with the surroundings. It does not make a wearer invisible, but at medium and long ranges the wearer will be almost impossible to see. It includes infra-red cloaking. Weight 4 kg. Cost Cr12,000. The same technology can be applied to Combat Armour, doubling its cost.

**Cloak Generator (TL 16):** A development of the chameleon suit, this heavy generator can throw out a perfect visible light wave field that effectively renders anything within it invisible, from all angles. The object still exists, and can be touched and interacted with normally. Rain, fog, smoke and the effects of lights shone into the field may outline the object at the GM's discretion. The field has a radius of 100m, and if unconnected to a power source, has an endurance of 12 hours. Weight 12 kg. Cost MCr2.

**Disguise Kit (TL 6):** A case full of polycarbonate skin, hair grafts, contact lenses, and other gear (such as false teeth etc.) which can be used by anyone to transform their appearance into something totally different. AT TL 10 it includes a variety of living 'false faces' grown on collagen and polysaccharides. These seem utterly real but can only be worn for 3-5 hours at a time before they begin they deform. Weight 3 kg. Cost Cr600.

**Force Shield Generator (TL 16):** A large, but portable unit that can generate a phased, resonance repulsor field with a radius of 30m. This has the effect of negating all small arms attacks, and reducing heavy weapons and vehicle mounted weapons by 10D6 and starship damage by 2D6. Each starship or anti-vehicle weapon attack that hits the force shield will disable it on 12+ on 2D6. The field can be extended to 100m, but this reduces the defensive level from -10D6 to -4D6. Weight 35 kg. Cost MCr1.

**Holo Belt (TL 14):** Holo projector worn on a belt that can throw a field around the wearer, creating the illusion is is someone or something else. At close range, or in darkness, the nature of the hologram is very obvious. Make an Average Int roll to detect a hologram, seen briefly or at distance, DM+4 if within 3m or at night. The projector has an endurance of 20 minutes. Weight 0.5 kg. Cost Cr3,500.

**Infra-Red Motion Sensor (TL 7):** Portable unit that can be stuck to a wall or door. It sends a unique signal if tripped, to a bug detector or hand radio, or it can just give off an audible alarm. Can pick up motion through the door or wall if not too thick. Weight 0.25 kg. Cost Cr50.

**Laser Ear (TL 8):** This rifle-like piece of gear can be pointed at a wall or pane of glass and it will pick up any conversations on the other side. It's range is line of sight. It comes with a table-top tripod that can be folded within-it for ease of mobility and a black acrylic case. Weight 2.5 kg. Cost Cr1000.

**Optic Probe (TL 8):** Fibre-optic lens and cable for cameras that can be pushed under doors, round corners etc. for surveillance purposes. Weight 1 kg. Cost Cr100.

**Security Kit (TL 7):** Allows an attempt to be made to crack mechanical locks as well as card and other electronic locks. This kit is illegal on worlds of law level 8+; on such worlds the cost rises to Cr2,000 or more. Weight 0.5kg. Cost Cr500.

**Timescanner (TL 17):** With a Difficult Comms roll, this hand-held device will detect echoes of the past in a specific location, requiring 10 minutes of study. The echoes may be indistinct, such as sounds, ghost images, and so on. Spending 12 hours is an Average task, and may produce more detailed imagery from a point in the past. The user must specify the point in time to be studied, or spend an hour scanning the timeline looking for a possible event. Weight 3 kg. Cost Cr50,000.

## TOOLS

**Broadcast Power Field (TL 16):** A large, vehicle-mounted antimatter power plant that is able to beam its power out across a wide area, to be used by any TL 16 or 17 equipment that requires power to operate. Its operating field has a range of 100m indefinitely, or 1 km if it is refueled after one week. Weight 20 kg (base station), surrounded by nine 6 kg repeaters that can be staked into the ground, or stand on integral tripods. Cost MCr2.

**Capture Net (TL 4):** A tough net, 4 sq. meters in size, used to catch animals. It has other uses, too. Weight 2 kg. Cost Cr50.

**Construction Kit (TL 4):** Wood and metal working tools for construction. At TL7 most of the tools are powered. Weight 12 kg. Cost Cr1500.

**Electronics Toolkit (TL 5):** For repairing, constructing and modifying complex electronic devices. Weight 8 kg. Cost Cr1000.

**Engineering Toolkit (TL 7):** For the maintenance and repair of power plants and ship-board systems. Weight 12 kg. Cost Cr2000.

**Forensics Toolkit (TL 8):** Used for investigating crime scenes and testing samples. Weight 6 kg. Cost Cr1000.

**Generator (TL 6):** A portable power unit that provides continuous electricity for 48 hours and includes a charging adaptor for most types of power cells. It runs on petrochemicals. At TL 9 it is hydrogen powered, at TL 12 it is fusion powered and needs refueling with hydrogen after 60 days. Weight 70 kg. Cost Cr800 (TL 6), Cr1000 (TL 9), Cr9000 (TL 12).

**Grapnel (TL 4):** A small, light-weight, folding grapnel used for climbing, but with many other uses too, as a pick or even an anchor. Weight 0.2 kg. Cost Cr30.

**Harpoon Grapnel (TL 9):** Rifle-sized grapnel launcher, using compressed air to launch a spring-loaded grapnel up to a 10 metres height. It includes a small powered winch to haul the user (or up to 100 kg) up. The grapnel is often part of a well-equipped survival kit onboard starship lifeboats, as part of the kit's climbing gear. Weight 4 kg. Cost Cr680. Combat: Range 10m, DM 0, 1D6+2, 1 shot.

**Materials Replicator (TL 16):** Using matter-energy conversion technology, this mobile unit recycles inorganic materials in order to fabricate a finished item, at the molecular level. It can create small technical parts, tools, small amounts of construction materials, items of clothing and so on. It must contain the pattern in its database and be regularly topped up with specialised inorganic matter. Weight 25 kg. Cost MCr2

**Mechanical Toolkit (TL 4):** Required for repairs and construction. This kit contains diagnostic sensors, hand tools, computer analysis programs (at appropriate tech levels) and spare parts. Weight 12kg. Cost Cr1000.

**Mobile Tractor Unit (TL 16):** A large, but portable unit floating on an anti-grav field that, once set up, can be used to move objects around using focused gravity beams. It has an operating range of 30m and is used in mining, in construction and in cargo loading. The tractor beams can lift 10,000 kg in one go; half this if extending the range out to 60m. Weight 250 kg. Cost MCr1.2.

**Multi-Tool (TL 5):** A compact single tool with extendable tool heads and blades. Comes in its own belt-mounted wallet. Weight negligible. Cost Cr30.

**Net Gun (TL 10):** This bulky gun is designed to launch small capture nets out to a range of 10m and is used to capture small, moving creatures. DM+1

| Net Gun |     |       |         |        |     |
|---------|-----|-------|---------|--------|-----|
| Wgt     | ROF | Range | Dmg     | Recoil | Rds |
| 4 kg    | 1   | Short | special | 8+     | 1   |

to hit if within range. The net takes around 5 minutes to repack into its launch canister. It comes with 4 pre-packed net canisters. Weight 4 kg. (loaded). Cost Cr900.

**Plasma Torch (TL 9):** A compact hand welder used either for welding together metals or for cutting through metal sheet. Burn time is 30 minutes before a refill gas bottle is required. Weight 0.5 kg. Cost Cr350. Combat: Personal Range, DM -1, 1D6+3, 5 shots.

**Survival Watch (TL 8):** Allows the user to tell the local time, can also be configured to multiple worlds, it includes alarms and stop-watch. Its main feature is its emergency beacon with a range (once activated) of up to 10 km. The beacon uses a standard emergency frequency which can be tracked by any modern avionics system, or either a Hand Radio or a Scanner. The signal will last for 24 hours. Weight negl. Cost Cr300.

**Scientific Toolkit (TL 9):** Required for scientific testing and analysis. This kit contains diagnostic sensors, hand tools, computer analysis programs (at appropriate tech levels) and spare parts. Weight 10 kg. Cost Cr1000.

**Slipshape Tool (TL 16):** A memetic metallic tool that can change its shape upon voice or key command. It can take the shape of any hand-held, unpowered tool, that it has stored in its memory, from axe to wrench, screwdriver to saw, and more. Weight 1.5 kg. Cost Cr120.

**Solar Base (TL 8):** A fold-out solar power charger, only 15 cm wide when folded. Able to charge flashlights, lamps, radios, nav units, night-vision goggles etc. but time to recharge depends on the typical power usage of the device being charged and the available daylight. Weight: 0.3 kg Cost: Cr50.

## SURVIVAL GEAR

**Atmosphere Generator (TL 16):** A large, but portable unit that creates a gravity bubble within which it can then replace the atmosphere with a type required by the user for survival. The bubble will be 50m metres in diameter, perfect for setting up a camp, or for a site where characters are at work. The bubble has an endurance of 1 week, before requiring a recharge. Weight 80 kg. Cr12,000.

**Bio-Box (TL 9):** Some things need protecting from vacuum or hostile atmosphere, biological samples, delicate materials and even small animals. The Bio-Box is similar in size to a pet carrier, but is pressurized and comes with its own 6 hour air-supply. Weight 2 kg. Cost Cr200.

**Canteen (TL 5):** The 1 litre canteen cup includes folding handle with cup/lid. It can be used to boil water. Weight 1 kg when full. Cost Cr10.

**Cold Weather Clothing (TL 1):** For arctic-style climates. Weight 2 kg. Cost Cr200.

**Compass (TL 3):** Some worlds with hot active cores will produce a magnetic field, the compass indicates magnetic north. Many inactive moons and planets have no magnetic field. Weight negl. Cost Cr10.

**Cooking Kit (TL 5):** This cooking kit includes mess tins, cups and cutlery for six people. Weight 3 kg. Cost Cr60.

**Cooking Stove (TL 6):** A portable cooking stove using refillable gas canister at TL 6-9 or flameless heat blocks at TL 10+. Weight 0.2 kg. Cost Cr40.

**Desert Clothing (TL 1):** Clothing suitable for hot desert conditions is not expensive and usually consists of a hat or head scarf to shade the head, sunglasses and loose-fitting trousers and a burnoose or shirt designed to cool the body. Exposed skin must be covered and protected. Includes sturdy desert boots. Weight negligible. Cost Cr80.

**Desert Survival Kit (TL 5):** A desert survival kit found often carried in vehicles. It includes a folding shovel, water purification tablets, two rocket flares, a signal mirror, a knife, first aid kit, moisture-traps, 2L canteen of water and several empty water bags. Weight 3.5kg. Cost Cr450.



**Diving Rebreather (TL 8):** Extracts oxygen from water to allow the wearer to breathe for an unlimited time underwater. Functions only on worlds with thin, standard, or dense (type 4 through 9) atmospheres. Weight 4 kg. Cost Cr4000.

**Diving Set (TL 6):** Includes swim fins, wet suit, face mask. Protects against the effects of cold (5° Celsius or below), along with improving speed and manoeuvre ability underwater; add DM +1 to all movement skill checks in these situations when wearing proper swimming equipment. Weight 6 kg. Cost Cr200.

**Expedition Tent:** A configurable canvas tent for up to three people. At TL 11 it is self-pitching and self-striking. Weight 3 kg. Cost Cr200.

**Flashlight (TL 5):** An electric torch with a 6 hour endurance. AT TL 10 it has unlimited endurance. Weight 0.5 kg, Cost Cr10. A head-torch costs Cr20.

**Flares (TL 6):** Four flares of various colors used for signaling or illumination. Weight: negligible. Cost Cr10.

**Food Replicator (TL16):** Using matter-energy conversion technology, this mobile unit can be filled with any biological matter (even toxic) and then used to produce all kinds of tasty and nutritional meals and foodstuffs, limited only by the types of food that it contains the patterns for. Weight 6 kg. Cost Cr120,000.

**Glowsticks (TL 7):** A pack of 20 glowsticks filled with a green chemiluminescent substance, each providing up to 12 hours of 360 degree illumination when 'snapped' and shaken. Weight: 0.5kg. Cost Cr10.

**Goggles (TL 4):** Light-weight goggles provide protection from wind-blown sand and sun glare. Weight negligible. Cost Cr20,

**Gravchute (TL 12):** A small backpack grav unit with enough power to allow a person to float down to earth, with some directional control. It can be reused up to ten times before requiring a recharge. Weight 4 kg. Cost Cr8000.

**Hydro-Pack (TL 7):** An artificial bladder containing up to 4 litres of drinking water, accessed via a drinking tube. Hydro-Packs have shoulder straps. Hikers must drink 1 litre of water/day, high-energy activities double this requirement. Weight: 4 kg (1 liter = 1 kg.). Cost Cr30.

**Lantern (TL 3):** Wind-proof, oil burning lamp with 6 hour endurance. At TL 6 this is electric. At TL 10 it has unlimited endurance. Weight 0.5 kg. Cost Cr20.

**Mess Tin (TL 4):** Folding mess tin, cup, cutlery. Weight 0.2 kg. Cost Cr10.

**Night-Vision Goggles (TL 7):** These goggles provide vision in dark conditions as long as there is some ambient light. At TL 9 they can operate in total darkness. Weight 0.25 kg. Cost Cr500.

**Oxygen Tanks (TL 5):** A set of compressed oxygen tanks with a breathing mask and carry straps, for use underwater, in smoke, dust, gas, or in an exotic (type A) atmosphere. Two tanks last 6 hours. Weight 4 kg. Cost Cr500.

**Parafoil (TL 7):** A steerable parafoil-style lightweight parachute, with altimeter, reserve chute, helmet and goggles. Weight 12 kg. Cost Cr1000.

**Poncho (TL 5):** A useful hooded waterproof coverall, with a reversible sun-reflective inner. Weight 0.5 kg. Cost Cr30.

**Pressure Tent (TL 10):** A tent for three people with filters able to screen a tainted atmosphere. It has an airlock, and can withstand vacuum with its own air supply for 3 days. Weight 24 kg (in three parts). Cost Cr4000.

**Raft (TL 5):** A one-man survival raft used to cross rivers. It is inflatable with a small foot-pump and will carry either one person or 100 kg of equipment. Weight 2 kg. Cost Cr60.

**Ration Pack (TL 3):** Three days food for one person. Weight 2 kg. Cost Cr10.

**Rope (TL 6):** A tough, nylon climbing rope, 36m in length. Weight 5 kg. Cost Cr60.

**Rucksack (TL 4):** A waterproof 40 litre rucksack, with side and front pockets and equipment straps below and above. It includes attachments for a Hydro-Pack. Holds roughly 25 kg of kit. Weight negligible. Cost Cr40.

**Sand Suit (TL 10):** A full body suit with an inner water-cooled layer, that is designed to protect the wearer from the heat of the desert. It also recycles urine as drinking water. It serves as a 'sealed suit' in Hot conditions (see page 96), effectively cancelling out any damage received. Typically worn by desert dwellers on long-distance desert journeys. Armour Value 2. Weight 5 kg. Cost Cr1200.

**Sand Tent (TL 8):** A two-man tent designed for desert conditions, it screens out sand and dust and also collects and recycles condensation for use as drinking water. It cools the air within, making it a comfortable shelter, and will count as 'shade' for the purposes of recovering from heat damage. With its long, sloping profile and its 50cm-long tent pegs, it can stand up to powerful sandstorms well. Weight 5 kg. Cost Cr2000.

**Sleeping Bag (TL 6):** A warm and compact sleeping bag. Weight 1.5 kg. Cost Cr20.

**Shovel (TL 5):** A folding military shovel that packs down to the size of a baseball glove and fits into a snug protective case. One edge of the blade is serrated, allowing it to be used as a crude saw. This is an extremely useful piece of equipment. Weight 1.5 kg. Cost Cr30. Combat: Close Range, DM -1, 2D6.

**Speargun (TL 7):** A one-shot, tension-fired underwater speargun used to hunt large fish. The 80cm spear is connected to the gun by a 30m cord, and it has a

| Speargun |     |       |       |        |     |
|----------|-----|-------|-------|--------|-----|
| Wgt      | ROF | Range | Dmg   | Recoil | Rds |
| 2.5 kg   | 1   | Short | 1D6+2 | 6+     | 1   |

Short range. It is loaded and fired in a similar manner to a crossbow, requiring one full round. Weight 2.5 kg. Cost Cr200. Additional spears cost Cr25 each.

**Survival Mask (TL 5):** A survival mask can filter contaminants, taints, smoke and dust (atmosphere types 4, 7, and 9) and can also compress breathed air to allow survival in a low pressure atmosphere (types 2 and 3). Useful in Very Thin or Tainted atmospheres. Weight negligible. Cost Cr150.

**Tarpaulin (TL 1):** A versatile waterproof sheet with eyelets and cordage. At TL12, (for Cr2,000) it has chameleon-pattern programming, and can mask the heat of anything beneath it. Weight 2 kg. Cost Cr10.

**Thruster Pack (TL 9):** A manoeuvring pack worn with a vacc suit, allowing the user to move freely in zero gravity. Weight 4 kg. Cost Cr1,000.

**Underwater Oxygen Tanks (TL 5):** A complete set of compressed oxygen tanks with regulator and breathing connections. Two tanks last 6 hours. Weight 4 kg. Cost Cr800.

**Vaporator (TL 10):** A backpack carried device which collects moisture from the air, especially at night. It unfolds and extends for operation, sitting on its integral tripod. It can collect 2 litres of water per 25 hours (usually 0.5L during the day and 1.5L during the night). There is a 1 in 6 chance that no water is collected. It is powered by integral solar panels. Weight 8 kg. Cost Cr2500.

**Wire Saw (TL6):** A survival saw made up of heavy cutting wire, with loop handles on either end. Will cut through branches with time and effort. Weight negligible. Cost Cr10.

## ELECTRONIC DEVICES

**Binoculars (TL 4):** These optical aids are electronic at TL 8, with low-light and recording capability. Weight 1 kg. Cost Cr100.

**Camera (TL 8):** A hand-held digital camera capable of still photography as well as video. It is capable of low-light photography, and includes an in-built timer. The camera can take 10,000 still shots, or up to 100 hours of video. It can be plugged directly into any computer for the download of images and video. Weight 1 kg. Cost Cr500.

**Computer, Portable (TL 7):** A portable computer with a folding screen for use in business, science or as a vehicle workstation. Computer rating varies with TL at a magnitude lower than the computer ratings of starship mainframes (hence the different numeric values: 01, 02, 03, etc.). Weight 4 kg. At TL 12 it includes holographic display. At TL 10 weight is 2 kg, at TL 12 weight is 1kg. For cost and rating, see table.

| PORTABLE COMPUTERS |    |          |
|--------------------|----|----------|
| Rating             | TL | Cost     |
| Model/01           | 7  | Cr 250   |
| Model/02           | 9  | Cr 500   |
| Model/03           | 10 | Cr1,000  |
| Model/04           | 12 | Cr 2,500 |
| Model/05           | 13 | Cr 4,000 |
| Model/06           | 15 | Cr 6,000 |
| Model/07           | 16 | Cr 8,000 |

**Computer, Hand (TL 8):** Handheld computer (or 'com') with a touch screen, and connection to a local network. The TL 8 smart phone is a hand computer. Computer ratings vary with TL. At TL 12 it includes holographic display. Weight 0.2 kg. Cost is twice that of a portable computer.

**Drone (TL 8):** A palm-sized prop-powered flying drone used for remote viewing of a location. It requires connection via a portable computer, and is fitted with low-light camera. It has an endurance of 10 minutes and a maximum range of 500m. At TL 10 it is grav-powered and has endurance of 1 hour and range of 5 km. At TL 14 it is fitted with a multi-sensor (cost is ten times more). Weight 1 kg. Cost Cr500.

**Inertial Nav System (TL 9):** A hand-held or belt-mounted device which indicates direction and distance traveled from the starting location, usually imaging directly in to a preloaded map set for precision navigation. Weight 1.5kg. Cost Cr500.

**Jammer (TL 9):** A strictly military frequency jammer, with a range of 400m. At TL 11 it has a range of 1 km and with a successful Average Computer roll, can identify and locate all transmitters within that range. Weight 0.5 kg. Cr 10,000.

**Motion Tracker (TL 10):** A hand-held unit that uses discriminatory Doppler shift ultra-sound scanning to detect movement within a 100 degree cone ahead of the operator. Range varies, from 10-20m indoors to almost 600-800m in open country. It is subject to varied limitations, anything that moves can show up on the reading if not properly tuned out, extremely slow-moving objects may not show up at all, and of course any enemy listening on radio will probably detect the loud ultra-sound bursts being emitted from the unit. To operate a motion tracker, make an Average Comms roll, with a failure lots of clutter or false data is picked up. Weight 2 kg. Cost: Cr1600.

**Multi-sensor (TL 12):** Handheld device used for exploration and survey work. It detects and records atmospheric makeup, radiation, magnetism, electrical activity, chemicals and biological molecules, as well as nearby metals. It includes digital camera and sound recorder. Weight 1 kg. Cost Cr5000.

**Night-Vision Goggles (TL 7):** These goggles allow limited field of view in near total darkness, as long as there is some starlight or other ambient light to enhance. At TL 9, field of view is human-normal, at TL 12, the goggle's weight is negligible. Weight 0.8 kg. Cost Cr500.

**Radio, Headset Radio (TL 7):** Has a 100m range, for use in vehicles and buildings. At TL 10 the range is 1 km. At TL 10 the radio is no bigger than a badge, button or wristband. Weight negligible. Cost Cr100.

**Radio, Hand Radio (TL 6):** Hand-held radio with 1km range. At TL 10 range is 5km, at TL 12 range is 50km. Weight 0.25 kg. Cost Cr 200.

**Radio, Base Radio (TL 5):** A vehicle or base station radio with a range of 5 km. At TL 7 range is 25 km, at TL 10 range is 500 km. Weight 5 kg. Cost Cr 2000.

**Radio, Long-Range Radio (TL 12):** A high-tech long distance radio with range of 3,000 km. Weight 2 kg. Cost Cr 3000.

**Radio, Grav Communicator (TL 16):** A powerful communications system using a gravity ripple effect for communications, able to pass through solid bodies, water or underground and out into space. The base station (weight 4 kg) has a range of 50,000 km, handheld transceivers (weight 0.5 kg) have a range of 1,000 km. Base station Cost Cr12,000. Handheld transceiver Cost Cr1000.

**Radio, FTL Communicator (TL 16):** This portable base station is capable of communication between worlds in the same or distant star systems. Range is 1-6 parsecs, and speed of communication varies, depending on the GM's setting (1 parsec per hour, or day?). Weight 24 kg. Cost is Cr500,000 per parsec range.

**Robot (TL 9):** A waist-high utility robot, that comes in many different configurations, wheeled or tracked, with two utility arms that can be swapped out for one or two tools useful for the robot's primary purpose. The robot has a portable computer at the TL of the world it was purchased on. At TL 12, tracks will be swapped for a ground-effect grav motor. Weight 100 kg. Cost Cr15,000. Robots can be custom designed in the Robots chapter.

**Tech-Scanner (TL 9):** This device detects the electromagnetic emissions of technological devices, and can be used as a diagnostic tool when examining equipment (+1 DM to work out what's wrong with it) or when searching for hidden bugs or devices. The Comms skill can be used to sweep a room for bugs. Weight 0.2 kg. Cost Cr700.

**Survey Drones (TL 13):** A set of four grav-powered spherical survey drones. Between them they can conduct a rapid, automated aerial survey of 1 sq km, in 5 minutes. The survey is topographical and includes camera, radar and laser imaging data to create a full 3D survey map of the area. Scale this up as needed. The drones can also split up to explore tunnels, buildings or parts of a starship interior that are accessible. Requires a 6 hour recharge after 3 hours of use. Weight of Set 4.8 kg. Set of four drones Cr12,000. Replacement drone: Weight 1.2 kg. Cost Cr4000.

**Universal Translator (TL 15):** A hand-held, intelligent, learning translation system which translates in real time with all known languages. At TL 16 the translator is no bigger than a brooch or wristband. Using the translator, a scientist can spend some time translating the speech of an alien language. Weight 1.5 kg. Cr1000.



## COSTS OF LIVING

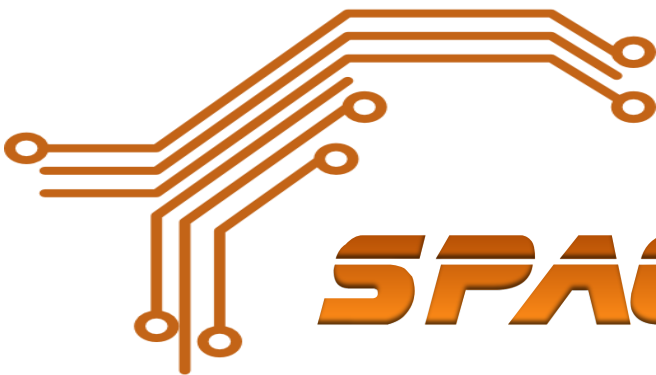
When not on a starship, characters must pay for lodging and food. The following table provides costs of living. One column shows a monthly cost of living – assuming long-term purchase of lodging and groceries. The second shows daily living, in case of travelers and tourists paying for a motel or hotel room and buying prepared food. Costs also include various expenses such as taxes and transportation.

| Lifestyle    | Monthly Cost | Daily Cost |
|--------------|--------------|------------|
| Subsistence  | Cr500        | Cr25       |
| Common       | Cr1000       | Cr50       |
| High Status  | Cr5000       | Cr300      |
| Elite Status | Cr15000      | Cr1000     |









# SPACESUITS

If you are a professional spacer, in a setting that takes the dangers of space seriously, then your spacesuit is more important to you than any mere piece of clothing. You spend hours in it every week and some people spend most of their working day inside a suit. Your suit is personal, customised to the way you work. Each is a self-contained space vehicle in its own right, with a breathable atmosphere, pressure regulation, heating and cooling controls, drinking water, radio (equivalent to a hand radio) and TV camera communications, 5 metre-range flashlight, limited micrometeoroid and radiation protection.

## SPACESUIT TYPES

There are three main types of spacesuit. The soft suit is introduced at TL 6 and is bulky and difficult to use when fully pressurised. At TL 9, the modular vacc suit is introduced, with semi-rigid sections to allow for freedom of use in a vacuum. An improved version appears at TL 12 that is much more comfortable to wear.

### SPACESUITS & ACTION PENALTIES

Suits are quite bulky and constraining. All suits begin with a basic penalty to all physical actions taken while in the suit. Added extras may further increase general encumbrance, and can further increase this action penalty. Training and experience counts for a lot, though. The wearer can use his Vacc Suit skill to cancel out the action penalty DMs up to his level in that skill. For example, a wearer with Vacc Suit 2 inside a soft suit (action penalty -3) suffers only a -1 penalty when attempting physical tasks.

| SPACESUIT TYPES    |    |                |       |          |              |
|--------------------|----|----------------|-------|----------|--------------|
| Spacesuit          | TL | Action Penalty | Mass  | Cost     | Armour Value |
| Soft Suit          | 6  | -3             | 24 kg | Cr7,000  | 4            |
| Vacc Suit          | 9  | -2             | 12 kg | Cr9,000  | 5            |
| Vacc Suit          | 12 | 0              | 4 kg  | Cr10,000 | 5            |
| Hostile Envirosuit | 9  | -3             | 40 kg | Cr20,000 | 18           |

**Soft Suit (TL 6)** – The soft suit is the standard vacc suit available at TL 6-8. Although it is possible to carry out EVAs in the pressurized soft suit, the -3 Action Penalty (AP) makes it inferior to the TL 9 vacc suit. The soft suit includes a personal life support system backpack that provides air and power for 6 hours. It has AV 4, masses 24 kg and Costs Cr7,000. All movements carried out in a pressurized soft suit are done so with a -3 action penalty. It cannot be customised.

**Vacc Suit (TL 9)** – The vacc suit is capable of maintaining its shape and flexibility while pressurized (unlike the soft suit). This makes it ideal for all EVAs in zero gravity and on a world’s surface. Most crews dress casually while on board ship, resorting to their suits during any period of danger in which there may be an unexpected decompression through hull puncture. Select your PLSS.

**Vacc Suit (TL 12)** – The TL 12 version of the vacc suit is sleek and comfortable enough to wear easily as a shipboard uniform. Select your PLSS.

**Hostile Envirosuit (TL 9)** – These are huge, heavily armoured suits are designed for intense temperatures, flame, high radiation and even deep sea and high atmospheric pressures. They cannot be customised, but already come with an air tester, radiation counter, crampons, flare damper, flashlights, inertial locator and a thruster pack for use in space or underwater. It has a hoist point on the crown of the helmet for, lifting out of the water, or onto a world surface, if required. The hostile envirosuit provides Armour Value 18, masses 40 kg and Costs Cr20,000. A difficult beast to operate, the hostile environment requires a minimum skill of Vacc Suit 1 and has an action penalty of -3. The PLSS operates for 6 hours.

CUSTOMISE YOUR VACC SUIT

Those characters who virtually live in their suits may want to buy their own and customise them to suit the types of missions they will be carrying out. The vacc suit is a modular design, and although once built cannot be altered, it can be tailored by the manufacturer to suit the needs of the customer. Any vacc suit is more expensive than a soft model, but the player character gets what they *want*. To customise a vacc suit, decide what Tech Level is appropriate, select a PLSS (personal life support system) and then choose up to **three options** that make your suit unique. Total the weight, the cost and make a note of any penalty assigned to any actions taken when in the suit. A suit is made up of several components: a torso assembly, a helmet and limbs. Select the basic suit design from the table below.

LIFE SUPPORT

The Personal Life Support System (PLSS) includes oxygen, pressure regulation, temperature control and carbon dioxide venting system. At TL 11, halve the mass; at TL 14, quarter the mass.

| PLSS | Duration (hrs) | Mass (kg) | Cost (Cr) |
|------|----------------|-----------|-----------|
| A    | 6 hrs          | 6         | 2,000     |
| B    | 9 hrs          | 8         | 4,000     |
| C    | 24 hrs         | 12        | 10,000    |

The PLSS can be recharged (air, water and electricity) at dedicated connectors within most airlocks. Other places, such as mining facilities, may have large banks of connectors in an EVA prep hall for many users to top up their PLSS at once. Every acceleration couch in a spacecraft includes an umbilical allowing the wearer to take-off the PLSS (it gets in the way when laying in the couch) and rely on air from the vehicle if needed. Most crewman take off the helmet once aboard, but keep the suit on.

## ADDING COMPONENTS

All vacc suits have **three** customizable slots that can be used to build in extra Components. If desired, select up to three Components from the table below. Once chosen add the combined weights, this total may adversely affect the suit's action penalty: if adding 5 kg+ then apply a DM of -1 to the action penalty, if adding +8 kg then apply a DM of -2 to the action penalty. All Components are TL 9.

| VACC SUIT COMPONENTS   |                                                                                                                                |           |           |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| Feature                | Component                                                                                                                      | Cost (Cr) | Mass (kg) |
| Ablative Layer         | Armour Value 6, from energy weapons only                                                                                       | 1200      | 2         |
| Air Tester             | Provides detailed chemical analyses                                                                                            | 500       | 1         |
| Binoculars             | Fold-down vision aids, with laser range finder                                                                                 | 800       | 1         |
| Climbing Harness       | Integral harness with rings and attachment points for rope climbing and rappelling                                             | 250       | -         |
| Collapsible Helmet     | Helmet folds back and slots away into the suit                                                                                 | 500       | -         |
| Comms Upgrade          | Upgrade radio to a base radio (see page 196)                                                                                   | 250       | -         |
| Crampons               | Extending boot spikes for stability on icy terrain                                                                             | 200       | 1         |
| Flashlights            | Helmet-mounted flashlights, illuminating up to 20m                                                                             | 200       | -         |
| Flare Damper           | Magnetic field generator that protects against solar flares, it saps power from the PLSS at 10 minutes of power per hour's use | 1000      | 3         |
| Heads-Up Display       | Suit data is fed to the suit's faceplate                                                                                       |           |           |
| Inertial Locator       | Shows direction and distance from a starting location                                                                          | 1,200     | 1.5       |
| Mag Boots              | For use on metal hulls and deck plates in zero-G                                                                               | 200       | 2         |
| Power Pack             | Power coupling, for running equipment etc. as an emergency back-up. Saps power like the flare damper                           | 1000      | 2         |
| Radiation Counter      | Provides warnings of radiation threats                                                                                         | 200       | -         |
| Resistant Outer Layer  | Adds +2 to suit Armour Value                                                                                                   | 1000      | 3         |
| Zero-G Tool-Belt       | With a full set of zero-G tools attached                                                                                       | 500       | 2         |
| Spotlight              | Powerful shoulder spotlight, illuminating up to 80m                                                                            | 1000      | 1         |
| Thruster Unit          | PLSS-mounted thrusters providing free movement in zero gravity or underwater, for a total of 60 minutes                        | 1000      | 4         |
| Wrist-Mounted Computer | as hand computer Model/001                                                                                                     | 1000      | 1         |

## PUTTING A SUIT ON

Putting on a suit requires 5 minutes if help is available, 8 minutes if it isn't. Taking the suit off takes 2 minutes. Halve these times if the suit is TL 11+. It is possible to do either of these a lot faster in an emergency! To speed up the process use the tasks below (do **not** factor in the suit's action penalty).

*Putting on a vacc suit while under stress:* Difficult (-2) Vacc Suit roll, 1 min

*Taking off a vacc suit while under stress:* Difficult (-2) Vacc Suit roll, 2 rounds

## BREACH!

Once a suit is breached (perhaps from colliding with a piece of sharp debris while the wearer is repairing a drive) the suit loses air. At TL 10 the suit can self-seal in two rounds. Otherwise, refer to the section on Operating Vacc Suits (page 102).



# APPENDIX

## ACADEMIC SUBJECTS

Education in Cepheus Universal covers all kinds of academic disciplines. A professor of Medieval History, for example, will have an Education of at least 12. So what is the relationship between skills and Education? The recommended rule of thumb is that a skill can be observed by a man with a clipboard for a few hours or a day, who could then assess you; whilst Education covers academic disciplines that would require a 10,000 word essay for a proper assessment. Some disciplines, such as Electrical Engineering will probably also require some levels in one of the skills.

Some settings rely heavily on academic disciplines, such as those in which No Man Has Gone Before, or the sanity-smashing investigations into Things Man Was Not Meant To Know. To add more weight to the academic fields, use the rules below.

### LEVELS OF SPECIALISATION

A character gains two levels of academic speciality for every point of Edu over 8. A level of speciality is similar to a skill level. A student with Edu 9 could have Psychology-2, or perhaps Religious Studies-1 and Literature-1. A senior lecturer at MIT with Edu 12 might have Computer Science-5, Machine Learning-2 and Deep Sea Robotics-1.

Make a task roll to carry out research, conduct investigations, recall knowledge or apply learning to a situation. A characteristic roll using Education would remain at 6+ for general knowledge, but shift up to 8+ or higher, for an academic task. Your level in an academic speciality can be added to the roll. A non-academic attempting the task will suffer a -3 untrained penalty. So, an exploration scientist with Edu 10 might have Planetary Geology-2 and Xenobiology-2, and when asked to calculate how many photon torpedo blasts would be needed to rupture a planet's crust needs to succeed with an 8+ roll, with a +2 specialisation bonus.

### FIELDS OF STUDY

The fields of study available to your characters should depend on the type of science fiction setting they inhabit. Victorian heroes will have a different list of specialities compared to the crew of a warp-capable cruiser on a 5-year voyage. Find a balance between too broad (physical sciences) and too narrow (Bronze-Age Bell Beaker pottery).



Examples of academic specialities are given here:

|                     |                                                    |
|---------------------|----------------------------------------------------|
| Accounting          | History ( <i>ancient, modern, medieval, etc.</i> ) |
| Anthropology        | Linguistics                                        |
| Archaeology         | Literature ( <i>by type or language</i> )          |
| Architecture        | Mathematics                                        |
| Astronomy           | Meteorology                                        |
| Astrophysics        | Nuclear Physics                                    |
| Biochemistry        | Occultism                                          |
| Botany              | Palaeontology                                      |
| Business Management | Physics                                            |
| Chemistry           | Physiology                                         |
| Computer Science    | Political Theory                                   |
| Cryptozoology       | Psychology                                         |
| Cybernetics         | Sociology                                          |
| Ecology             | Stellar Cartography                                |
| Economics           | Stratigraphy                                       |
| Genetics            | Temporal Mechanics                                 |
| Geography           | Theology                                           |
| Geology             | Warp Theory                                        |
| Glaciology          | Zoology                                            |

Referees can play with around with these subjects when creating their own setting. An occult steampunk setting, for example, might have Folklore, Occultism, Martian History and Hollow Earth Geography. An 'Ape City' setting might have Human Studies and Human Physiology as well as Ape Sociology and Ape Physiology.

# COMBAT DATA SHEET

## COMBAT RANGE BANDS

| Range                         | Distance to Target       |
|-------------------------------|--------------------------|
| <b>Personal</b>               | Touching, less than 1.5m |
| <b>Close</b>                  | 1.5 to 3 m               |
| <b>Short (Pistol, Thrown)</b> | 3 to 15 m                |
| <b>Medium (Shotgun, SMG)</b>  | 15 to 50 m               |
| <b>Long (Rifle, MG)</b>       | 51 m to 250 m            |
| <b>Very Long</b>              | 251 m to 500 m           |
| <b>Distant</b>                | 501 metres+              |

## AUTOMATIC FIRE

| Rate of Fire   | Number of Chances to Hit |
|----------------|--------------------------|
| 1 shot         | x 1 attack               |
| 4-round burst  | x 2 attacks              |
| 10-round burst | x 3 attacks              |

**Unreliability:** On a 'double-2' roll, check to avoid a jam or misfire, 8+ on 2D6.

## AREA FIRE DMS

| Weaponry             | Bonus to Pool                         |
|----------------------|---------------------------------------|
| Single shot          | -                                     |
| 4 round burst        | +1D6                                  |
| 10 round burst       | +2D6                                  |
| 20 round burst       | +3D6                                  |
| Grenade              | +2D6                                  |
| Vehicle Support Wpn  | +3D6                                  |
| Vehicle Main Wpn     | +4D6                                  |
| Airstrike/ Artillery | See Fire Support, in the next section |

Every success adds 1D6 to pool, add add-optional DMS for table. Half the total of dice if enemy entrenched.

## Explosions

Diff (-2) Dex roll to avoid damage if in the radius of an explosion. DMS: going prone (+1), ducking behind hard cover (+2) or occupying a trench or bunker (+4). Diving for cover means being prone and losing next action.

## MELEE CBT 8+

+ Melee Cbt skill  
+1 Strength or Dexterity DM (player choice)  
+ Dex DM if animal  
-1 if not fighting at weapon's range  
Defender can parry: 8+ Melee Cbt (+Dex DM)  
Success? Weapon AV takes damage

## ORDER OF COMBAT

1st: Melee Combat incl. parrying  
2nd: Aimed Fire  
3rd: Area Fire  
4th: Non-Combat Actions

## MOVE

3m/rd if attacking/parrying  
6m/rd if walking  
9m/rd if jogging (can dodge, try a perception roll)  
18m/rd if sprinting (no defence, no perception rolls)

## GUNFIRE 8+

| Advantage                              | Modifier |
|----------------------------------------|----------|
| Shooter's Dexterity Modifier           | +DM      |
| Target in Weapon's Range Band          | 0        |
| Target 1 band beyond Range Band        | -2       |
| Target 2 bands beyond Range Band       | -4       |
| Point Blank (3m or less)               | +2       |
| Aiming for one round                   | +1       |
| Aiming for one round with Scope        | +2       |
| Propelled Grenade with burst effect    | +1       |
| Shotgun Short Range (4D6 damage)       | +1       |
| Medium Range (2D6 damage)              | 0        |
| Long Range (1D6 damage)                | 0        |
| Target Moving, Dodging, Rushing        | -1       |
| Target in Cover, Partial               | -3       |
| Firer's Str is lower than gun's Recoil | -1       |
| Firing into Hand-to-hand Combat        | -2       |

## GRAPPLE

Melee Combat Roll (add Str DMS) Unarmed Parry allowed

# Continue pinning the victim with no other effects.  
# Disarm his opponent. If he succeeds by 4+ he can take the weapon away; otherwise ends up on floor.  
# Drag his opponent up to three meters.  
# Escape the grapple and move away.  
# Inflict damage equal to 2 + the Effect.  
# Knock his opponent prone.  
# Throw opponent up to three metres for 1D6+2 damage.

## Injury Type

## Effect

|                                                        |                                                                                                                                                            |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wounded</b>                                         | Cut/bruised, but unaffected                                                                                                                                |
| <b>Injured</b><br>(End 0)                              | Stunned for 1 round, but still fighting. A hit location may be temporarily disabled.                                                                       |
| <b>Seriously Wounded</b><br>(Two characteristics at 0) | Unconscious, roll Av End each minute to wake. Roll Av End each hour, failure means 1pt dmg. Lose 1 pt each hour if hourly Av End failed. Requires surgery. |
| <b>Dead</b> (All at 0)                                 | Dead is dead.                                                                                                                                              |

# GAME PLAY SHEET

|                      |           |
|----------------------|-----------|
| Skill rolls          | <b>8+</b> |
| Characteristic rolls | <b>6+</b> |
| Perception Roll      | <b>8+</b> |
| Attack rolls         | <b>8+</b> |

## AIDING ANOTHER

Task leader makes skill roll. Each helper makes the same roll. The Effect of a helper's result provides a bonus:

|                     |    |
|---------------------|----|
| Successful          | +1 |
| Exceptional Success | +2 |
| Failed              | 0  |
| Exceptional Failure | -2 |

## FALLING DAMAGE

1D6 damage every 3m, apply DM for world size:

|                |        |
|----------------|--------|
| World Size 3-  | +2 dmg |
| World Size 5-  | +1 dmg |
| World Size 10+ | +1 dmg |
| World Size 12+ | +2 dmg |

## Circumstance Modifiers

Useful tools (+1 or +2); lack of tools (-1 or -2); darkness/fog/heavy rain etc.(-2 to -5)

## CHARACTERISTIC MODIFIERS

| Characteristic | 0-2 | 3-5 | 6-8 | 9-11 | 12-14 | 15-17 | 18-20 | 21-23 |
|----------------|-----|-----|-----|------|-------|-------|-------|-------|
| Modifier       | -2  | -1  | -   | +1   | +2    | +3    | +4    | +5    |

| Difficulty            | DM | Example: Driving a Car                   |
|-----------------------|----|------------------------------------------|
| <b>Easy</b>           | +4 | Starting it while under stress           |
| <b>Routine</b>        | +2 | Reversing at speed                       |
| <b>Average</b>        | +0 | Swerving to avoid a crash                |
| <b>Difficult</b>      | -2 | Handbrake turn to throw off pursuer      |
| <b>Very Difficult</b> | -4 | Putting the car on two wheels            |
| <b>Formidable</b>     | -6 | Jumping the car from one roof to another |

| Effect              | Degree of Success   |
|---------------------|---------------------|
| <b>-6 or lower</b>  | Exceptional Failure |
| <b>-1 to -5</b>     | Failure             |
| <b>0 to +5</b>      | Success             |
| <b>+6 or higher</b> | Exceptional Success |

## MULTIPLE ACTIONS

A character can try to do two or more things at once, For every extra thing that the character is doing, he suffers a -2 DM to all skill checks.

## OPPOSED CHECKS

If two characters are opposing each other directly in a task, then the character who obtains the highest Effect wins. For ties on opposed checks, the character with the highest relevant characteristic score wins. If the characters tie on characteristic scores, they reroll.

## Hiking Cross-Country

Going: *Easy (5 kph); Hard (2.5 kph); Slow (1.2 kph)*

Stress Roll (Int) **10+**

*Fail? Stunned for 1 rd and temporarily lose 1 pt of Int*

**1st point of Int lost** – Stunned for one round.

**Int reduced to Half** – Make an Average (0) Int Panic check; failure means rolling on the Panic Effects table.

**Int reduced to 0** – The player makes Difficult (-2) Int Panic check. If successful, Int is returned to 1. If failed, roll on the Panic Effects table with a +2 DM.

## Time Frame

**1D6 seconds**  
**1D6 rounds**  
**1D6 minutes**  
**1D6 kiloseconds/15mins**  
**1D6 hours**  
**1D6 days**  
**1D6 weeks**  
**1D6 months**  
**1D6 quarter**

## PANIC EFFECTS

| 2D6 | Description                                                                                                                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-4 | <b>Stunned</b> – 1D6+2 rounds                                                                                                                                                                                                           |
| 5-6 | <b>Flee</b> – Overwhelming drive to escape the situation for 1D6+2 rounds                                                                                                                                                               |
| 7-8 | <b>Stunned</b> – Stunned for 1D6+2 minutes                                                                                                                                                                                              |
| 9   | <b>Phobia</b> – Stunned for 1D6+2 rounds, and also develop a phobia. Pick something appropriate; each time you are exposed to it take a -2 to all Stress checks.                                                                        |
| 10  | <b>Flee</b> – Overwhelming drive to escape the situation for 2D6 minutes                                                                                                                                                                |
| 11  | <b>Stupefied</b> – Blank, uncommunicative, unresponsive and oblivious for 1D3 days                                                                                                                                                      |
| 12  | <b>Psychotic Break</b> – Reality has no meaning, the character turns illogically on his comrades by sabotaging equipment and hampering any chance of escape or survival, even if it means the character's own death. Lasts until cured. |

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